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PREPARATION OF DIGITAL TECHNICAL INFORMATION FOR ARMY AVIATION OPERATOR'S MANUALS AND CHECKLISTS FOR CREWED AND UNCREWED AIRCRAFT SYSTEMS

AMSC 3083 **AREA TMSS**

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FOREWORD

This standard is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense (DoD).

This standard establishes the technical content requirements and mandatory style and format requirements for the preparation of Technical Manuals (TMs) and subsequent revisions required to support the aviation weapon systems within the Department of the Army. The requirements contained in this standard cover aviation operation.

Comments, suggestions, or questions should be addressed to AMCOM Logistics Center, ATTN: AMAM-LLP-S Bldg 5301, Martin Road, Redstone Arsenal, AL 35898 or e-mailed to usarmy.redstone.amcom.list.immc-pub-epco@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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1 SCOPE

1.1 General.

This standard establishes the requirements needed to prepare digital Technical Manuals (TMs) outputted into ETM or IETM format describing operating procedures, Checklists (CL), and Maintenance Test Flight (MTF) manuals for operators of Army Aircraft and Uncrewed Aircraft Systems (UAS).

1.2 Classification.

The TMs to be prepared in accordance with this standard include:

-10 - Rotary Wing Aircraft Operator's Manual or Fixed Wing/UAS Operator's Reference Manual

-CL - Operator's and Crewmember's Checklist

-EV - Enhanced Version of Operator's and Crewmember's Checklist and/or the Maintenance Test Flight Manual

-MTF - Maintenance Test Flight Manual

1.3 Media Type.

Aviation operator manuals may be developed in ETM or IETM format, and this standard supports the development of both types. If a paper copy of the manual is required, the development will follow all ETM requirements. If it is determined that a paper manual is not required, an IETM may be developed and will follow all IETM requirements.

1.4 Examples/Figures.

The figures used in this standard are examples only. In the event of a conflict between the text and figures, the text of this standard takes precedence over the figures. The figures in this standard represent compliant material, but they may not represent all variations of standard compliant material.

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2 APPLICABLE PUBLICATIONS

2.1 General.

The publications listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include publications cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARD/DEPARTMENT OF DEFENSE HANDBOOK

MIL-HDBK-310	Global Climatic Data for Developing Military Products
MIL-HDBK-2361	Army Digital Publications Development Implementation Guide
MIL-STD-3013	Glossary of Definitions, Ground Rules, and Mission Profiles to Define Air Vehicle Performance Capability
MIL-STD-40051	Preparation of Digital Technical Information for Technical Manuals (TMs)

(Copies of these documents are available online at https://quicksearch.dla.mil)

2.2.2 Other government documents and publications.

The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

ARMY REGULATIONS (AR)/DEPARTMENT OF ARMY PAMPHLETS (DA PAM)

AR 25-30	Army Publishing Program
AR 25-38	Army Printing and Distribution Program
AR 95-1	Flight Regulations
AR 385-10	The Army Safety And Occupational Health Program Procedures
DA PAM 25–38	Army Printing and Distribution Procedures
DA PAM 25-40	Army Publishing Program Procedures
DA PAM 385-10	The Army Safety And Occupational Health Program
DA PAM 738-751	Functional User's Manual for the Army Maintenance Management System - Aviation

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(Copies of these publications are available in electronic format through the Army Publishing Directorate (APD) website at https://armypubs.army.mil/).

DOD INSTRUCTIONS

DoDI 5200.48	Controlled Unclassified Information (CUI)
DoDI 5230.24	Distribution Statements on DoD Technical Information

(Copies of these publications are available in electronic format through the DOD Issuances website at https://www.esd.whs.mil/).

FORMS

DA Form 2408-13-1	Aircraft Inspection and Maintenance Record
DD Form 365 series	Record of Weight and Balance Personnel
DD Form 1423	Contract Data Requirements List

(Copies of these publications are available in electronic format through APD website at https://armypubs.army.mil/).

TECHNICAL MANUALS (TM)/TECHNICAL BULLETINS (TB)

TB 1-1500-1	Technical Bulletin Supplemental Divider Package for Aviation Checklists
TB 1-9150-200-24	Engine and Transmission Oils, Fuels, and Additives for Army Aircraft
TM 1-1500-328-23	Aeronautical Equipment Maintenance Management Policies and Procedures
TM 55-1500-342-23	Aircraft Weight and Balance
TM 750-244-1-5	Procedures for the Destruction of Aircraft and Associated Equipment to Prevent Enemy Use

(Copies of these publications are available in electronic format through the Army Publishing Directorate website at https://armypubs.army.mil/).

TRAINING CIRCULARS

TC 3-04.5	Instrument Flight for Army Aviators

(Copies of these publications are available in electronic format through the Army Publishing Directorate website at https://armypubs.army.mil/).

WEBSITES

AMCOM Publications Online	https://pubsweb.redstone.army.mil/PubsOnline/

2.3 Order of precedence.

Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the reference cited here (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3 DEFINITIONS

3.1 Acronyms.

The acronyms used in this standard are defined as follows:

AC	Alternating Current
ADF	Automatic Direction Finder
AHRS	Attitude Heading Reference System
ALSE	Aviation Life Support Equipment
AMCOM	US Army Aviation and Missile Command
AMSC	Acquisition Management Systems Control
APD	Army Publishing Directorate
APU	Auxiliary Power Unit
AR	As Required
ATC	Air Traffic Control
AV	Air Vehicle
CG	Center-of-Gravity
CL	Checklist
CR	Conditionally Required
DA	Department of the Army
DC	Direct Current
DME	Distance Measuring Equipment
DoD	Department of Defense
DTD	Document Type Definition
EIC	End Item Code
EMO	Electronic Media Only
ETM	Electronic Technical Manual
EV	Enhanced Version
FADEC-F	The Emergency Response Method
FAT	Free Air Temperature
FOSI	Formatted Output Specification Instance
FRC	Final Reproducible Copy
FRC	Flight Reference Card
FW	Fixed Wing
G	Gravity
GBAS	Ground Based Augmentation System

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GLS	Ground Based Augmentation System Landing System
GPO	Government Publishing Office
GPS	Global Positioning System
IADS	Interactive Authoring and Display Software
IAS	Indicated Airspeed
IAW	In Accordance With
ICAO	International Civil Aviation Organization
IETM	Interactive Electronic Technical Manual
IFR	Instrument Flight Rules
IGE	In-Ground Effect
ILS	Instrument Landing System
IMC	Instrument Meteorological Conditions
IMN	Indicated Mach Number
INS	Inertial Navigation System
IR	Infrared
IRU	Inertial Reference Unit
ISA	International Standard Atmosphere
KIAS	Knots Indicated Airspeed
LORAN	Long Range Navigation
MFD	Multifunction Display
MLS	Microwave Landing System
MTF	Maintenance Test Flight
MWO	Modification Work Order
NATO	North Atlantic Treaty Organization
NAVAID	Navigational Aid
NDB	Non-directional Radio Beacon
NSN	National Stock Number
OAT	Outside Air Temperature
P	Prohibited
PDF	Portable Document Format
PFD	Primary Function/Flight Display
PSI	Pound Per Square Inch
R	Required
RATO	Rocket-Assisted Takeoff
RPM	Revolutions Per Minute

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RW	Rotary Wing
SDF	Simplified Directional Facility
SGML	Standard Generalized Markup Language
SLAR	Side Looking Airborne Radar
TACAN	Tactical Air Navigation
TEAC	Turbine Entire Analysis Check
TM	Technical Manual
TOC	Table of Contents
UAS	Uncrewed Aircraft Systems
VFR	Visual Flight Rules
VHF	Very High Frequency
VOR	Very High Frequency Omni-directional Range
Vmc	Minimal Controllable Airspeed
VORTAC	Very High Frequency Omni-directional Range Tactical Air Navigation
W/C/A	Warnings, Cautions, Advisories
WAAS	Wide Area Augmentation System
WP	Work Package
XML	eXtensible Markup Language
XSL	XML Stylesheet Language

3.2 Terminology.

Acquiring Activity	The DoD component, activity, or organization of a using military service, or that organization delegated by a using service, that is responsible for the selection and determination of requirements for TMs.
Callout	A letter, number, or other device for identifying or calling attention to a particular part of an illustration.
Change	A set of the corrected pages to the basic manual. It consists of information that improves or clarifies the basic manual without requiring or reorganization of the technical content of the basic manual.
Change designator	A unique number or letter used to differentiate each change page from the original pages and to differentiate among changes.
Department of Defense (DoD)	The Office of the Secretary of Defense (OSD) (DoD) (including all boards and councils), the Military Departments (Army, Navy, Space Force, and Air Force), the Organization of the Joint Chiefs of Staff (OJCS), the Unified and Specified Commands, the National Security Agency (NSA), and the Defense Agencies.
Document Type Definition (DTD)	The definition of the markup rules for a given class of documents. A DTD or reference to one should be contained in any SGML/XML conforming document.

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Electronic Technical Manual (ETM)	A page-oriented file usually based on a paper original that may or may not be prepared from a digital database. An ETM usually has hyperlinks added. ETMs can be distributed as digital media (for example, Portable Document Format (PDF)) or printed on paper.
End Item Code (EIC)	A final combination of end products, component parts, or materials that is ready for its intended use (e.g., tank, mobile machine ship, aircraft, receiver, rifle, recorder).
Equipment Publication	A publication that deals with the installation, operation, maintenance, training, and parts supported of Army materiel, including Firing Tables (FTs), Equipment Technical Manuals (TMs), Technical Bulletins (TBs), and Modification Work Orders (MWOs) are examples of DA publications media used to provide these essential instructions.
eXtensible Markup Language (XML)	A set of rules for encoding documents electronically through the use of markup. Its primary purpose is to facilitate the sharing of structured data across different information systems. It is a product of the World Wide Web Consortium (W3C).
Final reproducible copy	The final approved manuscript with illustrations ready for printing and distribution as a DA equipment publication and the source files that created it: XML, graphics, etc..
Flight Reference Card (FRC)	A functional emergency procedure checklist in colored card format that enables crewmembers to respond to emergencies in context with the situation, delaying non-critical actions while prioritizing aircraft control, crew coordination, and deliberate action.
Foldout page	A page the same height as, but is wider than, a standard page. Foldout pages are folded 2, 4, or 6 times (depending on width) to assume the dimensions of a standard page.
Icon	Pictorial images which may be used in lieu of words. For example: the icon is used to represent “Electrostatic Discharge Sensitive.” See MIL-STD-40051 for authorized icons.
Illustration	A general term meaning graphic presentations of all types. Illustrations include pictorials, functional diagrams, and line graphs. This term is used synonymously with figure, graphic, drawing, diagram, and artwork.
Interactive Electronic Technical Manual (IETM)	A technical manual designed for electronic window display and possessing the following three characteristics: **a.** Format and style are optimized for window presentation; that is, the presentation format is frame-oriented, not page-oriented. **b.** The elements of technical data constituting an IETM are so interrelated that a user's access is facilitated and is achievable by a variety of paths. **c.** An IETM can function interactively (as a result of user requests and information input).
Key	A tabular listing of the callout or item numbers/letters and their meanings.
Leader lines	A line with or without arrowhead extending from index number or letter/nomenclature to item.
Leading	The vertical spacing between lines of type measured from baseline to baseline (bottom of line to bottom of next line below). Leading is measured in points.
Legend	Explanatory matter on a map, chart, illustration or the like, listing and explaining the symbols or other information used.

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Module	A major functional division of a publication.
National Stock Number (NSN)	13-digit number assigned to a repair part to be used for requisitioning purposes.
Nomenclature	The approved name or alphanumeric identifier assigned to an item, equipment, or component in agreement with an organized designation system.
Part	The next lower division of a publication below volume. Parts should normally be separately bound.
Placard	Item names shown on controls, switches, panels, etc.
Revision	A second or subsequent edition of a manual which normally supersedes the proceeding edition. A revision is necessary when more than 50% of the data is being changed.
Set	A number of individual manuals or volumes which comprises a complete TM package of information for an item.
System	A group of items united or regulated by interaction or interdependence to accomplish a set of specific functions.
Technical manual	A manual that contains instructions for the installation, operation, maintenance, and support of weapon systems, weapon system components, and support equipment. TM information may be presented, according to prior agreement between the contractor and the Government, in any form or characteristic, including hard printed copy, audio and visual displays, electronic embedded media, disks, other electronic devices, or other media. They normally include operational and maintenance instructions, parts lists, and related technical information or procedures exclusive or administrative procedures.
Validation	The process by which a contractor (or other agency, as directed by the contracting activity) tests a technical publication for completeness, compliance with contractual requirements, and technical accuracy. It is conducted at the contractor's facility or at the operational site, and entails the actual performance of operating and maintenance procedures on the equipment for which the publication was written. Data such as stock/part numbers and SMR codes in RPSTL, hardware depiction, schematic diagrams, and wiring data contained in the publication are checked against current source data.
Verification	The process by which an equipment publication is tested for completeness, accuracy, and adequacy of content, and usability by government personnel (not contractors) against the production configurations of the related system or equipment. The proponent's technical publications element will be responsible for ensuring all technical and equipment publications are fully verified; no publications will be submitted for authentication until they have been fully verified and proved (under Army or other DoD component jurisdiction) to be adequate (comprehensive and usable) for the operation and maintenance of equipment procured for operational units.
Volume	The first separately bound subdivision of a publication.
XML Stylesheet Language (XSL)	A stylesheet language that can be used for rendering XML documents.

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4 GENERAL REQUIREMENTS

4.1 General.

This section provides the technical content requirements for the operator's manuals, checklists and MTF manuals. The specific content, style, and format requirements for the development of operator's manuals, checklists, and MTF manuals are provided in the following appendixes:

APPENDIX A	-	REQUIREMENTS MATRICES
APPENDIX B	-	AVIATION INTRODUCTION AND GENERAL INFORMATION (MODULE 1)
APPENDIX C	-	AIRCRAFT AND SYSTEMS DESCRIPTION AND OPERATION (MODULE 2)
APPENDIX D	-	AVIONICS (MODULE 3)
APPENDIX E	-	MISSION EQUIPMENT (MODULE 4)
APPENDIX F	-	OPERATING LIMITS AND RESTRICTIONS (MODULE 5)
APPENDIX G	-	WEIGHT/BALANCE AND LOADING (MODULE 6)
APPENDIX H	-	PERFORMANCE DATA (MODULE 7)
APPENDIX I	-	NORMAL PROCEDURES (MODULE 8)
APPENDIX J	-	EMERGENCY PROCEDURES (MODULE 9)
APPENDIX K	-	SUPPORTING INFORMATION (MODULE 10)
APPENDIX L	-	CONTENT REQUIREMENTS FOR OPERATOR'S AND CREWMEMBER'S CHECKLISTS (-CL)
APPENDIX M	-	CONTENT REQUIREMENTS FOR MAINTENANCE TEST FLIGHT MANUALS (-MTF)
APPENDIX N	-	STYLE AND FORMAT REQUIREMENTS FOR ENHANCED VERSION (-EV) CHECKLIST OR MTF

4.2 Types of aviation technical publications.

APPENDIX A lists specific technical content requirements for each type of publication covered by this standard.

4.2.1 Operator's manual (-10).

The operator's manual shall describe briefly and concisely the operation of the complete aircraft. UAS shall include ground stations and other elements in addition to the actual air vehicles. The description of aircraft, aircraft systems, sub-systems, and components shall contain only that detail required to explain the operation, operational procedures, and checks necessary for the pilot to safely and efficiently operate the aircraft, aircraft systems, and mission equipment during flight and ground operation.

Unless otherwise specified by the acquiring activity, the publication number shall be the same as the TM number of the aircraft system, followed by "-10" for operator's manuals.

4.2.2 The operator's checklist (-CL).

The operator's checklist (-CL) is a condensed or simplified version of Module 8 and 9 of the operator's manual which consist of a series of controls (or checks) and the required actions. The sequence of items appearing in the

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checklist normal procedures shall follow the same sequence as found in Module 8 of the operator's manual. Emergency procedures shall follow the applicable requirements matrix. In unusual circumstances, explanatory material shall be used in the -CL in the form of warnings, cautions, and notes, only if specified by the acquiring activity. The contents of the -CL shall be as described in the paragraphs below and in APPENDIX L.

Unless otherwise specified by the acquiring activity, the publication number shall be the same as the TM number of the aircraft system, followed by "-CL" for operator's checklists, followed by volume 1 or 2 accordingly. Examples: TM 1-1520-263-CL-1 or TM 1-1520-263-CL-2. Software publication versions shall be the same as the TM number of the aircraft system, followed by "-CL", followed by the software version (V1, V2, or V3, etc.), followed by the volume number. Examples: TM 1-1520-263-V2-1 or TM 1-1520-263-V2-2.

4.2.3 The operator's maintenance test flight manual (-MTF).

The operator's maintenance test flight (-MTF) manual shall provide complete instructions for performing an MTF for a specific model, type, and series of aircraft. For specific criteria which require a general or limited MTF, reference shall be made to TM 1-1500-328-23 and the applicable aviation field and sustainment maintenance manual. The contents of the -MTF shall be as described in the paragraphs below and in APPENDIX M.

Unless otherwise specified by the acquiring activity, the publication number shall be the same as the TM number of the aircraft system, followed by "-MTF" for MTF manuals.

4.3 Selective application and tailoring.

MIL-STD-3083(AV) contains some requirements that may not be applicable to the preparation of all technical manuals. Selective application and tailoring of requirements contained in MIL-STD-3083(AV) are the responsibility of the acquiring activity and shall be accomplished through the use of the matrices provided in APPENDIX A. The applicability of some requirements is also designated by one of the following statements: unless specified otherwise by the acquiring activity; as specified by the acquiring activity; or when specified by the acquiring or proponent activity.

4.3.1 Tailoring of content.

This standard shall be tailored using the matrices in APPENDIX A. The content of this standard can also be tailored based on the following three designations:

a. Required. These content items that are designated as required by the use of the word "shall" will be included in the publication and shall not be tailored out. If a required item is not applicable to your system, the item shall be included but state that it is "not applicable."

b. Optional. The content items usually are caveated by "as applicable," "as specified by the acquiring activity," "unless specified by the contracting activity." They may contain use of the word "may" or "should," and they are areas that are flexible and can be tailored in or out.

c. Prohibited. In some cases, this standard states "shall not" or states that something is prohibited. These shall not be included or used for the publication covered.

The standard cannot cover every possible content or format issue and state whether it is required, optional, or prohibited. If the standard does not specifically say that a content item or particular format can be used/included, determination shall be made through discussions with all stakeholders and publication authentication approval authorities as to whether the content or format issue can be permitted for a particular publication.

4.4 Preparation of digital data for electronic delivery.

Data prepared and delivered digitally in accordance with this standard shall be Extensible Markup Language (XML) tagged using the Army Document Type Definition (DTD). The Army developed stylesheets shall be used. Refer to

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4.4.4 for information on obtaining or accessing the Army DTD and stylesheets. XML tags used in the Army DTD are noted throughout the text of this standard in bracketed, bold characters (e.g., **<descwp>**) as a convenience for the author and to ensure that the tags are used correctly when developing a document instance. Refer to MIL-HDBK-2361 for additional guidance on XML.

4.4.1 Metadata <metadata>.

Metadata shall be required in development of all work packages using this standard.

4.4.2 ID attributes.

The ID attribute is mandatory throughout the technical content to enable the Army to take full advantage of XML. The use of IDs is mandatory as defined in the DTD.

4.4.3 Use of the DTDs/stylesheets.

The DTDs referenced in this standard interpret the technical content and structure for the functional requirements contained in this standard and are mandatory for use. The stylesheets referenced herein interpret the style and format. Development of publications covered in this standard shall be accomplished through the use of this standard, the Army DTD, and the Army developed stylesheets. Within this standard, when a specific tag set is to be used for data, that tag set name is identified between "< >" marks following the title of the paragraph.

4.4.4 Obtaining the DTDs/stylesheets.

The DTDs, stylesheets, associated tag and attribute descriptions, which are XML constructs, may be obtained from the AMCOM Publications Online website https://pubsweb.redstone.army.mil/DTD-FOSI/DTD-N-FOSI.html. Additionally, available on the website are files containing documentation of the deleted, changed, and added elements between the legacy XML DTD and the current XML DTD. AMCOM Publications Online assets may be obtained from https://pubsweb.redstone.army.mil/PubsOnline/Default.aspx.

4.5 Style and format.

This standard provides style and format requirements for the technical content requirements described in this standard. These requirements are considered mandatory and are intended for compliance. The U.S. Government Publishing Office (GPO) Style Manual shall be used as a general guide for capitalization, punctuation, compounding of words, numerals in text, and spelling of nontechnical words.

4.5.1 Examples of style and format.

The examples provided in this standard are an accurate interpretation of the technical content, style, and format requirements contained in this standard and are provided to permit the effective use of the DTD. Any conflicts between examples and the text of the standard shall be resolved in favor of the text.

4.5.2 TM divisions.

The hierarchical breakdown of a TM is volumes (if required and only for ETMs), modules (if applicable), sections (if applicable), work packages, paragraphs, subparagraphs, and steps. Each division used should have at least two occurrences (for example, where there is a Volume 1, there should be a Volume 2; where there is a Module 1, there should be a Module 2; etc.). Except for the ETM -CL, volumes shall be partitioned only between modules. See APPENDIX L for guidance on implementing volumes for the ETM -CL.

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4.5.2.1 ETM Volume and size content.

a. Division of -10 Operator Manuals into volumes shall occur when the number of printed pages exceeds 1,500 pages or 750 sheets. Each volume shall not exceed 1,500 pages or 750 sheets. When a change causes a manual to exceed page limits, the manual shall be divided into volumes.

b. -CL and -MTF manuals shall not exceed 500 pages or 250 sheets.

c. Rotary Wing checklists shall be developed into two separate volumes. Refer to APPENDIX L.

d. For Electronic Media Only (EMO) manuals, the 1500 page limit may be exceeded; however, the file size shall not exceed 500 MB. Documents that might/will be printed at some point shall not exceed the 1500 page limit.

e. For -10 Operator and -MTF Manuals, refer to MIL-STD-40051 for front and rear matter requirements for multivolume manuals. For -CL, refer to APPENDIX L.

f. Separate volumes shall not be used to distinguish between models of equipment (e.g., -10 for basic model, -10-1 for model A, -10-2 for model B, etc.).

4.5.2.2 Modules.

Each ETM and IETM -10 Operator Manual prepared in accordance with this standard shall be divided as found in the applicable Requirements Matrix, refer to APPENDIX A. Each module shall be made up of one or more work packages. Module titles and content shall be based on the titles given in the matrices in APPENDIX A.

4.5.2.3 Sections.

Sections shall be used to divide content within the ETM -CL and -MTF manuals. Sections shall be identified by a tabbed index page. Sections are prohibited in a -10 Operator Manual.

4.5.2.4 Work packages.

Work packages shall be used to logically divide TM data into functional descriptive or task-oriented information. Work packages shall begin on a right-hand page. Work packages may contain one or more tasks. Refer to APPENDIX B through APPENDIX M for the specific content requirements for each of the functional work package types.

4.5.2.4.1 ETM Work package size.

To facilitate the change/revision process, all work packages for -10 Operator Manuals shall be kept small (50 or fewer pages or 25 or fewer sheets). Due to the pocket size nature of these manuals, work package size for -CL and -MTF manuals shall not exceed 10 pages or 5 sheets. This standard allows the acquiring activity to tailor content in multiple ways, either through a single work package or the use of multiple work packages.

4.5.2.4.2 Work package identification number <wpno>.

A unique number shall be assigned to each work package. This identifier may be used for database retrieval purposes. The work package identification number shall not appear on the printed page and should not be confused with the work package sequence number in 4.5.2.4.3. It shall be assigned when preparing the document instance in accordance with the DTD and shall not be changed throughout the life of the work package. The work package identification number shall consist of an alpha designation for the type of information contained in the work package, a five-digit block number assigned by the acquiring activity, and the TM number. The TM number is used

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only to provide uniqueness and avoid duplication of a work package identification number. Other than that it shall not have significance.

a. The following alpha designators shall be assigned to the specific types of information contained within the work packages.

TABLE I. Alpha designators.

Alpha Designator	Information Type	-10	-CL	-MTF
G	General Information	ALL	N/A	N/A
O	Operator Instructions	ALL	N/A	N/A
D	Detailed Procedures	N/A	ALL	FW, RW
N	Normal Procedures	ALL	ALL	FW, RW
E	Emergency Procedures	ALL	FW	N/A
W	Warning Emergency Procedures	N/A	RW, UAS	N/A
C	Caution Emergency Procedures	N/A	RW, UAS	N/A
A	Advisory Emergency Procedures	N/A	RW, UAS	N/A
	Abnormal Emergency Procedures (FW)	N/A	FW	N/A
M	Mission Equipment Emergency Procedures	ALL	ALL	N/A
P	Performance Data and Forms	ALL	ALL	FW, RW
T	Troubleshooting Guides	N/A	N/A	FW, RW
F	Foldout Information	ALL	ALL	FW, RW
S	Supporting Information	ALL	ALL	RW, MTF

b. The following examples explain work package database identification numbering:

TABLE II. Example work package database identification numbering.

O00432-1-1520-263-10-V2	
O	Identifies a work package containing operating instructions.
00432	Identifies the 432nd work package containing specific operator instructions that was initially developed for the AH-64E helicopter.
1-1520-263	Identifies the AH-64E helicopter TM. This is the TM under which this work package was initially developed.
10-V2*	Identifies the specific manual the file is needed for and the software version, as applicable. This identifies the operator's manual containing version 2 software for the AH-64E.
W00034-1-1520-263-CL-V4-1	
W	Identifies work package or task containing warning emergency procedures.
00034	Identifies the 34th work package containing specific emergency procedures that was initially developed.
1-1520-263	Identifies the AH-64E helicopter TM. This is the TM under which this work package was initially developed.

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TABLE II. Example work package database identification numbering. (continued)

CL-V4-1*	Identifies the specific manual the file is needed for and the software version, as applicable. This identifies the operator's checklist containing version 4 software for the AH-64E.
* If software version is not applicable to the TM number, then it shall not be used.	

4.5.2.4.3 Work package sequence number.

For -10 Operator Manuals, refer to MIL-STD-40051. For -CL and -MTF Operator Manuals, work package sequence numbers shall follow the guidance in 4.5.2.4.3.1 through 4.5.2.4.3.4.

4.5.2.4.3.1 Work package sequence numbers for -CL manuals.

To maintain a sequential order in the TM and to facilitate referencing, each Work Package (WP) shall initially be assigned a two-digit number beginning with the section letter and 1. For example, the first work package of the first Normal Procedures section would be assigned the two-digit number N1. A new work package that is inserted between two work packages shall use a point numbering scheme to create a new sequence number that logically fits between the two existing work package numbers. Point numbers shall start with "1" and continue in numerical sequence as needed. If the work packages already have point sequence numbers, an additional point level shall be added to create a new sequence number that follows the same criteria. For example, to insert three work packages (WP) between WPs N10 and N11, the numbers N10.1, N10.2, and N10.3 shall be used. See TABLE III for checklist work package numbering format.

TABLE III. Checklist work package numbering.

Page Type	Numbering Format
Normal Procedures	Work packages shall begin with "N" followed by a sequential series of numbers beginning with 1; for example, N1, N2, N3, N4, etc.
Detailed Procedures	Work packages shall begin with "D" followed by a sequential series of numbers beginning with 1; for example, D1, D2, D3, D4, etc.
Performance Charts	Work packages shall begin with "P" followed by a sequential series of numbers beginning with 1; for example, P1, P2, P3, P4, etc.

For the Emergency Procedures section, work packages shall not be identified by sequence numbers.

4.5.2.4.3.2 Assignment of new work package sequence numbers for a change.

When a new work package is added to a module, it will be added to the end of the module and use the next available sequential work package number. For example, if N98 is the number of the last work package in the Normal Procedures module, N99 shall be the number of the new work package. A new work package that is inserted between two work packages shall use a point numbering scheme to create a new sequence number that logically fits between the two existing work package numbers. Point numbers shall start with "1" and continue in numerical sequence as needed. For example, to insert three work packages between WPs N10 and N11, the numbers N10.1, N10.2, and N10.3 shall be used. If the work packages already have point sequence numbers, an additional point level shall be added to create a new sequence number that follows the same criteria. For example, to insert a work package between P10 and P10.1, the number P10.0.1 shall be used; to insert two work packages between P10.2 and P10.3, the numbers P10.2.1 and P10.2.2 shall be used.

4.5.2.4.3.3 Deletion of work packages in a change.

When a work package is deleted in a change, the work package shall be removed and a page inserted with a statement that says "WP XXXX was deleted" and a vertical bar shall be placed next to this statement. The deleted

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work package shall be listed on the change transmittal sheet. All work packages following the deleted work package shall retain their original work package sequence number.

4.5.2.4.3.4 Work package sequential numbering for -MTF manuals.

To maintain a sequential order in the TM and to facilitate referencing, each Work Package (WP) shall initially be assigned a sequential number beginning with the module letter and 1. For example, the first work package of the Normal Procedures module would be assigned the sequential number N1. A new work package that is inserted between two work packages shall use a point numbering scheme to create a new sequence number that logically fits between the two existing work package numbers. Point numbers shall start with "1" and continue in numerical sequence as needed. If the work packages already have point sequence numbers, an additional point level shall be added to create a new sequence number that follows the same criteria. For example, to insert three work packages (WP) between WPs N10 and N11, the numbers N10.1, N10.2, and N10.3 shall be used. See TABLE IV for -MTF work package numbering.

TABLE IV. Maintenance test flight work package numbering.

Page Type	Numbering Format
Normal Procedures	Work packages shall begin with “N” followed by a sequential series of numbers beginning with 1; for example, N1, N2, N3, N4, etc.
Troubleshooting Guides	Work packages shall begin with “T” followed by a sequential series of numbers beginning with 1; for example, T1, T2, T3, T4, etc.
Detailed Procedures	Work packages shall begin with “D” followed by a sequential series of numbers beginning with 1; for example, D1, D2, D3, D4, etc.
Performance Charts	Work packages shall begin with “P” followed by a sequential series of numbers beginning with 1; for example, P1, P2, P3, P4, etc.

4.5.2.4.4 Work package identification information <wpidinfo>.

All work packages shall include the identification information entries in the following sequential order, as applicable.

4.5.2.4.4.1 Work package title <title>.

The title of the individual work package shall clearly identify the content of the work package.

4.5.2.4.4.2 Effectivity/configuration notice <config>.

If applicable, an effectivity notice shall be included. When the work package does not apply to all configurations of the weapon system/equipment, the applicable configurations covered by the work package shall be listed. Omit this requirement if the same task/procedure applies to all configurations. (If certain configurations require different tasks/procedures, separate work packages shall be prepared.)

4.5.2.4.5 End of task/work package marking.

For -10 Operator Manuals, refer to MIL-STD-40051. For -CL and -MTF Operator Manuals, End of Task/Work Package marking shall follow the below guidance.

a. A 5-pt solid black line shall be used to indicate the end of a task. It shall span the entire page/screen. The line shall appear 4pts below the last text of the procedure.

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b. A 1-pt solid black line shall be used to indicate the end of a procedure. It shall span the entire page/screen. The line shall appear 4pts below the last text of the procedure.

c. The end of procedure line shall not be used in conjunction with the end of task line. When the end of procedure corresponds with the end of task, the end of task line shall be used.

4.5.2.5 ETM Page numbering.

For -10 Operator Manuals, refer to MIL-STD-40051. For -CL and -MTF Operator Manuals, work package page numbering shall follow the below guidance. Refer to 4.5.7.2.1 for page number location requirements.

Page numbers in the operator's checklist TMs shall consist of a capital letter and an Arabic number separated by a dash. The letter shall correspond to specific parts of the manuals as found below. Page numbers for standard manual pages shall be justified to the outside corner of the page. Page numbers in the alternate operator's CL shall be numeric. See TABLE V for checklist page numbering.

TABLE V. Checklist and MTF page numbering.

Page Type	Aircraft	Numbering Format
Front Matter pages	RW, FW, and UAS	Lower case Roman numerals: i, ii, etc.
Normal Procedures	RW, FW, and UAS	Work packages shall begin with "N" followed by a number, then followed by a dash and number; for example, N1-1, N1-2, N2-1, N2-2, etc.
Detailed Procedures	RW, FW, and UAS	Work packages shall begin with "D" followed by a number, then followed by a dash and number; for example, D1-1, D1-2, D2-1, D2-2, etc.
Performance Charts	RW, FW, and UAS	Work packages shall begin with "P" followed by a number, then followed by a dash and number; for example, P1-1, P1-2, P2-1, P2-2, etc.
Warning Section	RW	The front of each card shall begin with "W" followed by a "-" and number. For the reverse side of the card, add the letter "R" after the number. The first warning page would appear as "W-1" and its reverse side would be "W-1R"
	UAS	UAS: Work packages shall begin with "W" followed by a number, then followed by a dash and number; for example, W1-1, W1-2, W2-1, W2-2, etc.
Emergency Procedure Section	FW	Work packages shall begin with "E" followed by a number, then followed by a dash and number; for example, E1-1, E1-2, E2-1, E2-2, etc.
Caution Section	RW	The front of each card shall begin with "C" followed by a "-" and number. For the reverse side of the card, add the letter "R" after the number. The first caution page would be "C-1" and its reverse side would be "C-1R"
	UAS	Work packages shall begin with "C" followed by a number, then followed by a dash and number; for example, C1-1, C1-2, C2-1, C2-2, etc.
Abnormal Procedure Section	FW	Work packages shall begin with "AB" followed by a number, then followed by a dash and number; for example, AB1-1, AB1-2, AB2-1, AB2-2, etc.

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TABLE V. Checklist and MTF page numbering. (continued)

Page Type	Aircraft	Numbering Format
Advisory Section	RW	The front of each card shall begin with “A” followed by a “-” and number. For the reverse side of the card, add the letter “R” after the number. The first advisory page would be “A-1” and its reverse side would be “A-1R”
	FW and UAS	Work packages shall begin with “A” followed by a number, then followed by a dash and number; for example, A1-1, A1-2, A2-1, A2-2, etc.
Mission Equipment Section	RW	The front of each card shall begin with “E” followed by a “-” and number. For the reverse side of the card, add the letter “R” after the number. The first mission equipment page would be “E-1” and its reverse side would be “E-1R”
	FW and UAS	Work packages shall begin with “M” followed by a number, then followed by a dash and number; for example, M1-1, M1-2, M2-1, M2-2, etc.
Troubleshooting Section	RW and FW	Work packages shall begin with “T” followed by a number, then followed by a dash and number; for example, T1-1, T1-2, T2-1, T2-2, etc.
Limitations Section	RW, FW, and UAS	Limitation pages shall be numbered consecutively as “L-1, L-2, L-3, etc.”
Foldout Section	RW, FW, and UAS	Foldout pages shall be numbered consecutively as “FP-1, FP-3, etc.”

4.5.2.6 ETM Tabs.

Tab usage is determined both by the type of manual and the type of aircraft.

a. All: Normal/Troubleshooting/Detailed/Charts Sections shall have no more than 5 tabs in a set from top to bottom of the publication. Tabs may be repeated as necessary to divide the checklist as desired, so long as they follow the set sequence not to exceed 5 tabs in a set. For example, tab 6 would be at the same height as tab 1. Refer to FIGURE 1.

b. RW CL: Emergency Section shall have no more than 10 tabs spanning the height of the checklist. Tabs may be repeated as necessary to divide the checklist as desired, so long as they follow the set sequence not to exceed 10 tabs in a set. For example, tab 11 would be at the same height as tab 1. Refer to FIGURE 2.

c. FW and UAS CL: Emergency Section shall have no more than 5 tabs in a set from top to bottom of the publication. Tabs may be repeated as necessary to divide the checklist as desired, so long as they follow the set sequence not to exceed 5 tabs in a set. For example, tab 6 would be at the same height as tab 1. Tabs will have text that is determined by the applicable matrix in APPENDIX A. Refer to FIGURE 3.

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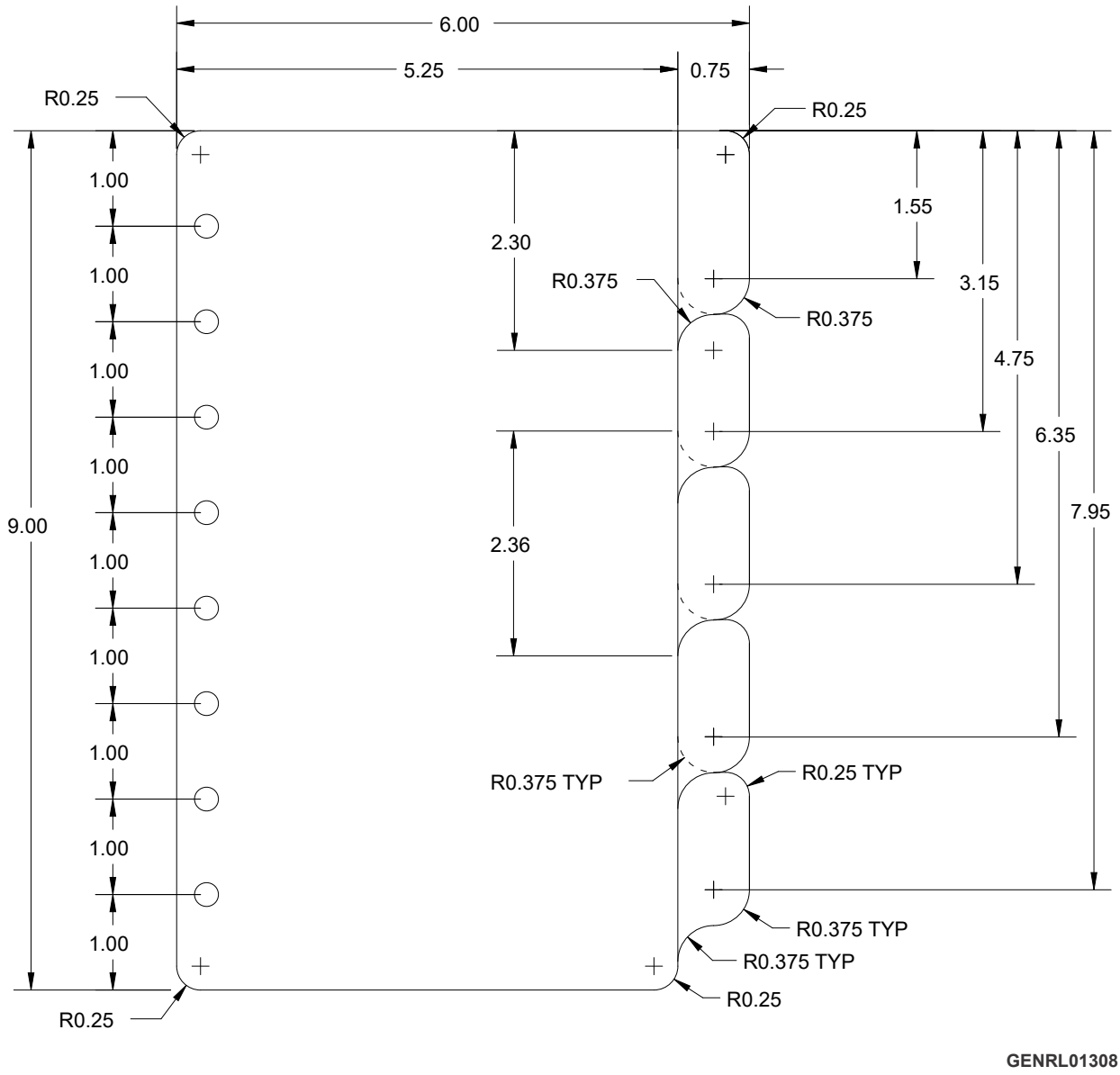


FIGURE 1. RW Normal-detailed-charts sections page and tab layout.

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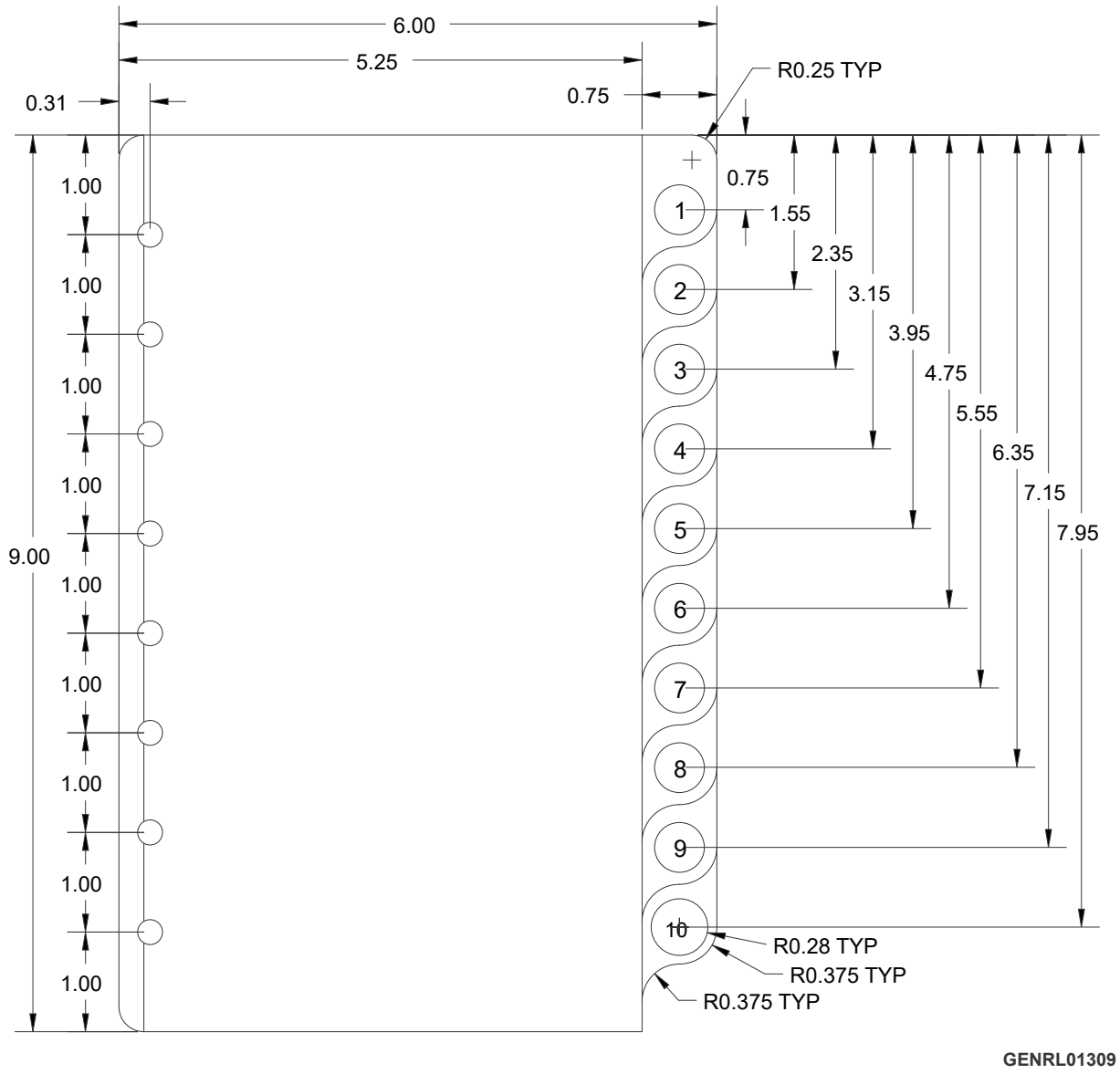


FIGURE 2. RW Emergency section page size and tab layout.

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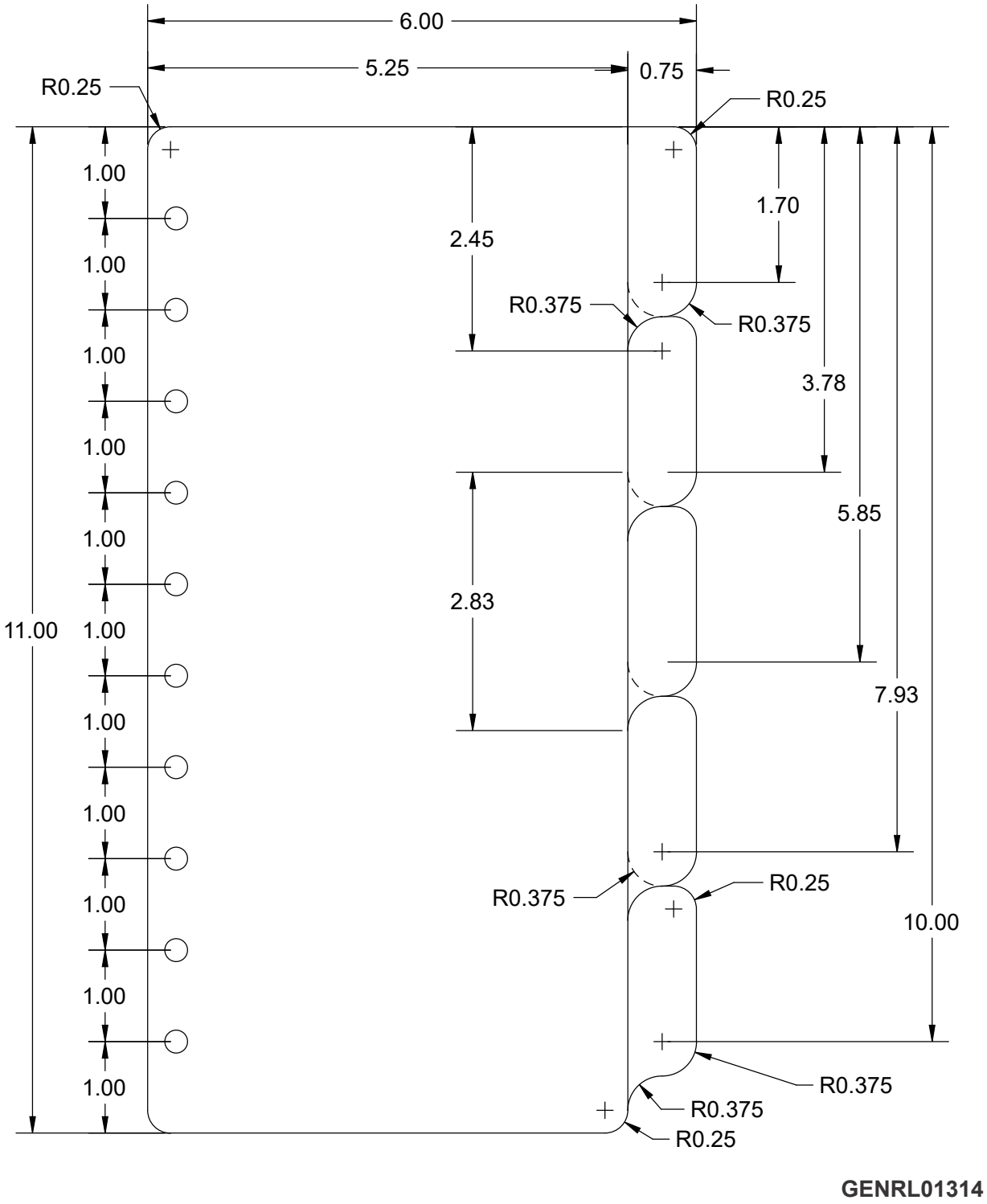


FIGURE 3. FW and UAS Section page size and tab layout.

4.5.2.6.1 Use of tabs.

Tabs shall be used for the following pages:

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a. Normal procedure section divider pages

b. Detailed procedure section divider pages

c. Performance Chart divider pages

d. Foldout section divider pages

e. RW CL emergency procedure pages shall be tabbed

f. FW and UAS CL emergency procedure pages shall be tabbed as section dividers

g. Each limitations page shall be tabbed

h. MTF Troubleshooting Guide section divider pages

The following pages shall not be tabbed:

a. Front and rear cover

b. Introduction pages

c. Normal Procedure pages

d. Detailed Procedure pages

e. Performance Data pages (Charts)

f. MTF Troubleshooting Guide pages

g. Rear matter (foldouts are the exception)

4.5.2.6.2 Tab size.

For Rotary Wing Normal-Detailed-Charts Sections Page and Tab Layouts, refer to FIGURE 1. For Rotary Wing Emergency Page Size and Tab Layout specifications pages, refer to FIGURE 2. For Fixed Wing and UAS Page Size and Tab Layout specifications, refer to FIGURE 3.

4.5.2.7 Checklist emergency section colors.

Each section of the emergency procedures shall have the following bleed border colors. Bleed borders shall span 1/4” of the inside edge of the page. The color shall also bleed past the tab of the pages. Refer to FIGURE L-2 for an example of the page. Refer to TABLE VI for tab coloring specifications.

TABLE VI. Checklist page and tab coloring.

Section	Color	PMS/Pantone Color code or equivalent may be used
Normal Section	None	*
Detailed Section	None	*
Performance Data Section	None	*
Warnings	Red	1788C
Emergency (FW only)	Red	1788C
Cautions	Yellow	7548C
Abnormal (FW only)	Yellow	7548C
Advisories	Green	7737C
Mission Equipment	Blue	298C

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TABLE VI. Checklist page and tab coloring. (continued)

Section	Color	PMS/Pantone Color code or equivalent may be used
Limitations Section	None	*
*These sections and tabs shall be printed on the same paper as the entire checklist with no color.		

4.5.2.8 Checklist emergency procedure styling and format.

The structure of the emergency procedures is the same, regardless of the section in which the procedures are found. This structure is detailed in 4.5.2.8.1 through 4.5.2.8.5.

4.5.2.8.1 Section index.

Section indexes shall be presented as tabular data. Entries shall be spaced for readability, functional linking, and target audience usability.

4.5.2.8.2 Task title.

a. The title shall appear in all caps and bold at the top of the page, left justified, just below the header.

b. When the procedure continues to another page, the title shall appear at the top of the subsequent page, in all caps, followed by "- Continued".

4.5.2.8.3 Section formatting.

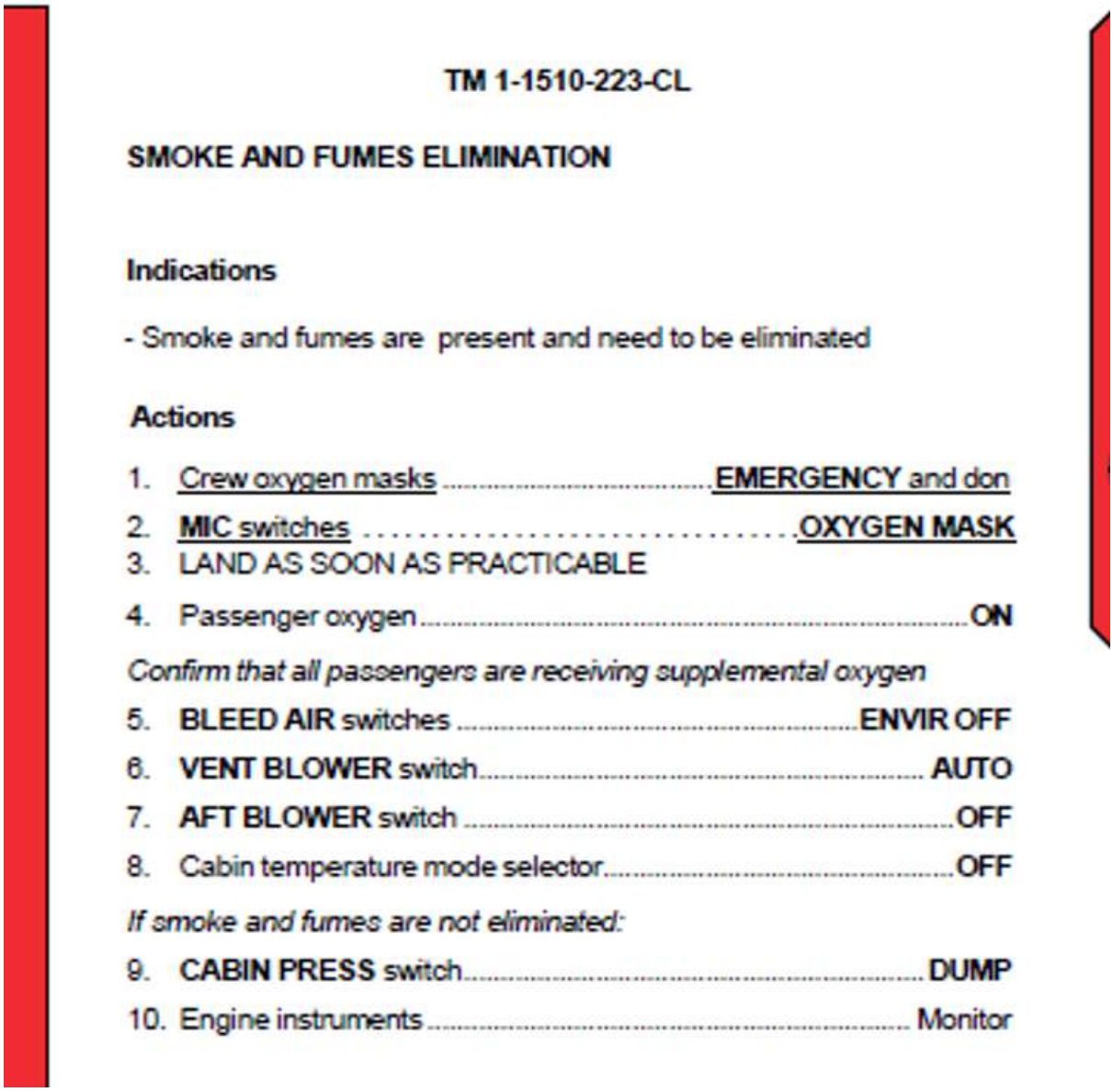
All sections within the emergency procedures shall be styled the same way (Indications, Action Response, Urgency to Land, Subsequent Actions, and Considerations). Refer to FIGURE 4.

a. Title shall appear in bold, left justified below previous section.

b. Condition statement **<condition>** shall be left justified and italicized.

c. The condition and response of a procedural step shall be separated by a long dash for RW and leader dots for FW and UAS.

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TM 1-1510-223-CL

SMOKE AND FUMES ELIMINATION

Indications

- Smoke and fumes are present and need to be eliminated

Actions

1. Crew oxygen masks ..... EMERGENCY and don
2. MIC switches ..... OXYGEN MASK
3. LAND AS SOON AS PRACTICABLE
4. Passenger oxygen ..... ON

Confirm that all passengers are receiving supplemental oxygen

5. BLEED AIR switches ..... ENVIR OFF
6. VENT BLOWER switch ..... AUTO
7. AFT BLOWER switch ..... OFF
8. Cabin temperature mode selector ..... OFF

If smoke and fumes are not eliminated:

9. CABIN PRESS switch ..... DUMP
10. Engine instruments ..... Monitor

FIGURE 4. Example of checklist emergency procedure section formatting.

4.5.2.8.3.1 Procedural steps for rotary wing.

Procedural steps, **<para>** or **<parareponse>**, shall be preceded by a dash. Text shall not wrap under the bullet. The action response, either immediate actions or actions, shall be outlined by a black box. Refer to FIGURE 5.

4.5.2.8.3.2 Procedural steps for fixed wing and UAS.

Procedural steps, **<para>** or **<parareponse>**, shall be preceded by a number for FW and UAS. Text shall not wrap under the number.

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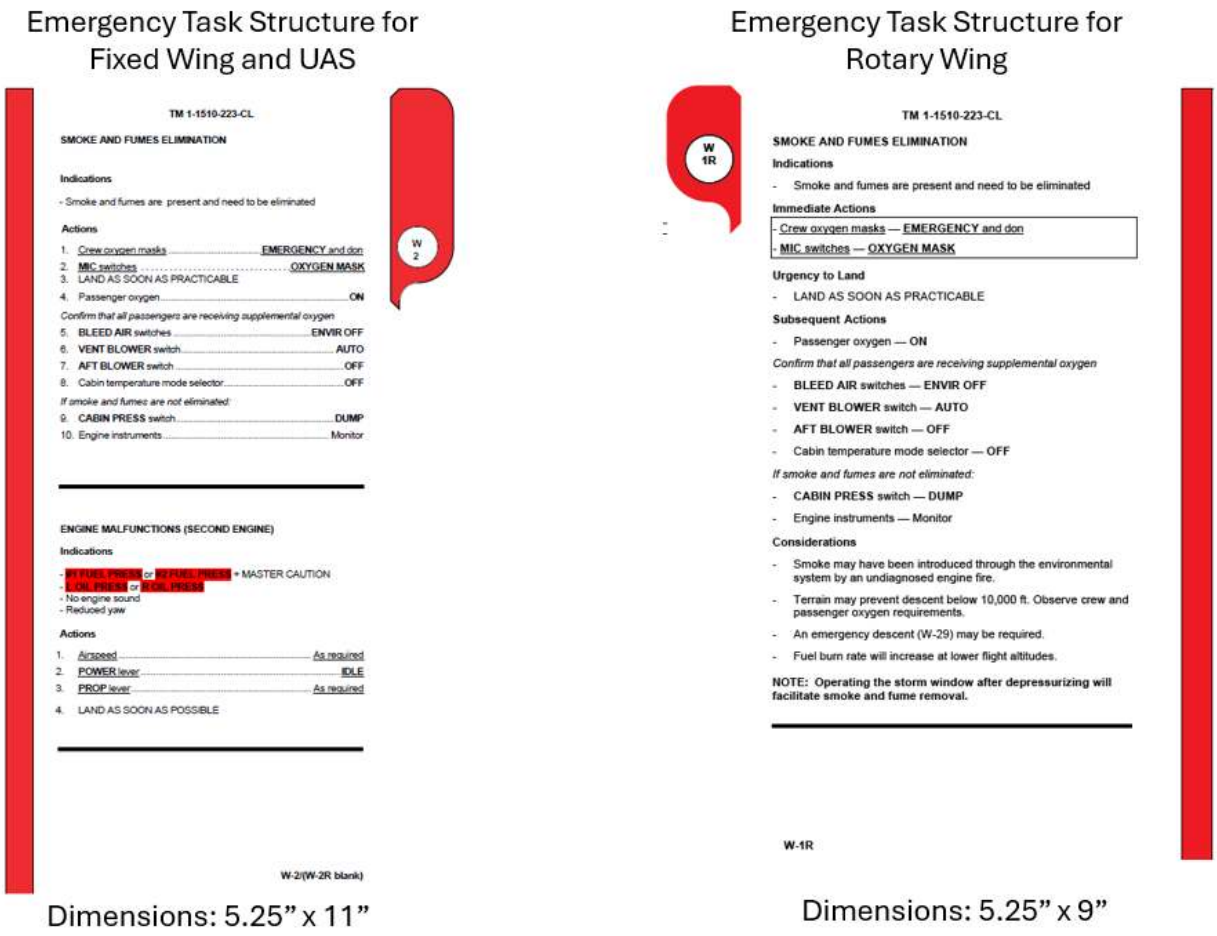


FIGURE 5. Example of emergency task structure.

4.5.2.8.4 References.

References shall be made to the applicable page number: e.g. C-5R, P1-1, N4-2, etc.

4.5.2.8.5 Page arrangement.

a. Whenever possible, in-flight emergency procedures should be contained on a single page. If this is not possible, the following criteria shall be used:

(1) Indications, action response, and urgency to land shall appear on the same page.

(2) Warnings, Cautions, and procedural steps shall not be divided across pages. For example, a single Warning shall not span two pages, but a "Warning 2" may appear on the second page.

(3) The TRADOC operator representative shall approve the use of multi-page emergency procedures.

b. Warnings, cautions, and notes shall not be broken across normal and detailed procedures.

c. Procedural steps shall not be broken across normal and detailed procedures; the entire step shall be completed on the page.

d. When any task spans more than a single page, a "Continued" icon shall be placed below the last text and right justified on the page. Refer to FIGURE 6.

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e. When a task spans more than one page, the task title shall be repeated at the top of the next page followed by "– Continued".

f. When a subsection within a task is divided, the subsection title shall immediately follow the task title and also have "– Continued" included.

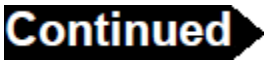


FIGURE 6. Continued icon.

4.5.2.9 CL and MTF limitations work packages.

Limitations shall be presented as tabular data, spanning the entire page. Headers to differentiate separate limits shall be white text on black background. Refer to FIGURE L-1.

4.5.2.10 Text.

The text shall be written in clear, simple, and concise language. Technical terms requiring special knowledge shall be avoided, except where no other wording shall convey the intended meaning. Procedures shall be broken down into distinct steps for accomplishment. All procedures called out shall be fully explained in logical completion sequence.

4.5.2.10.1 Task titles.

Each operator task shall have a title.

4.5.2.10.2 Procedural steps.

Refer to MIL-STD-40051 for requirements unless otherwise stated in this standard.

Action steps shall be developed as actionable steps or as "Challenge — Response" format. RW shall utilize a long dash, and FW and UAS shall utilize leader dots. Refer to FIGURE 4 and below for examples:

Actionable step **<para>**:	- Close cockpit windows, air control handles, and cockpit knobs
"Challenge — Response" **<pararesponse>**:	- ENG POWER CONT lever — OFF

4.5.3 Font size and style.

Font style, size, and spacing shall be in accordance with best commercial practices for technical publications. However, the minimum font size for a -10 Operator Manual is 8 point, -CL/-MTF is 10 point, and the minimum for a graphic is 6 point. Type shall be proportionally spaced (non-monospaced). Fonts shall be selected for a balance between readability and economy of space. Setting text in all capital letters shall be limited to appropriate uses, such as major headings, acronyms, and equipment markings.

4.5.4 ETM Page size and orientation.

The TM shall be prepared in a size selected from TABLE VII as specified by the acquiring activity. Orientation of pages, either vertical (portrait) or horizontal (landscape), shall be consistent throughout a given manual for ease of use. The growing prevalence of TMs used in electronic display mode (instead of paper) makes this consistency extremely important. Exceptions may be made only if essential for proper grouping of information for the user's

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benefit. Otherwise, information shall be formatted or reformatted so that all pages have the same orientation. Refer to FIGURE 7 and FIGURE 8.

TABLE VII. Manual styles and trim sizes.

Publication Type	Trim Size without Tabs	Trim Size with Tabs	Orientation	Maximum Printing Area
Operator's manual (-10)	8-1/2" x 11"	- - -	Vertical	7" x 10"
Rotary Wing Checklist (-CL) and (-MTF)	5-1/4" x 9"	6" x 9"	Vertical	4-1/2" x 8"
Fixed Wing/UAS Checklist (-CL) and (-MTF)	5-1/4" x 11"	6" x 11"	Vertical	4-1/2" x 10"
Alternate Checklist	8-1/2" x 11"	- - -	Horizontal	7-1/2" x 10"

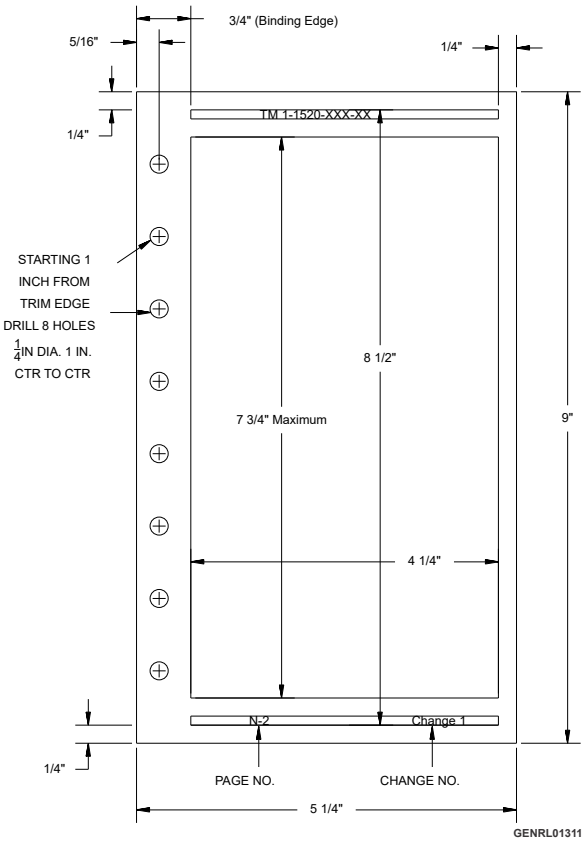


FIGURE 7. Page dimensions for RW condensed checklist and MTF manual.

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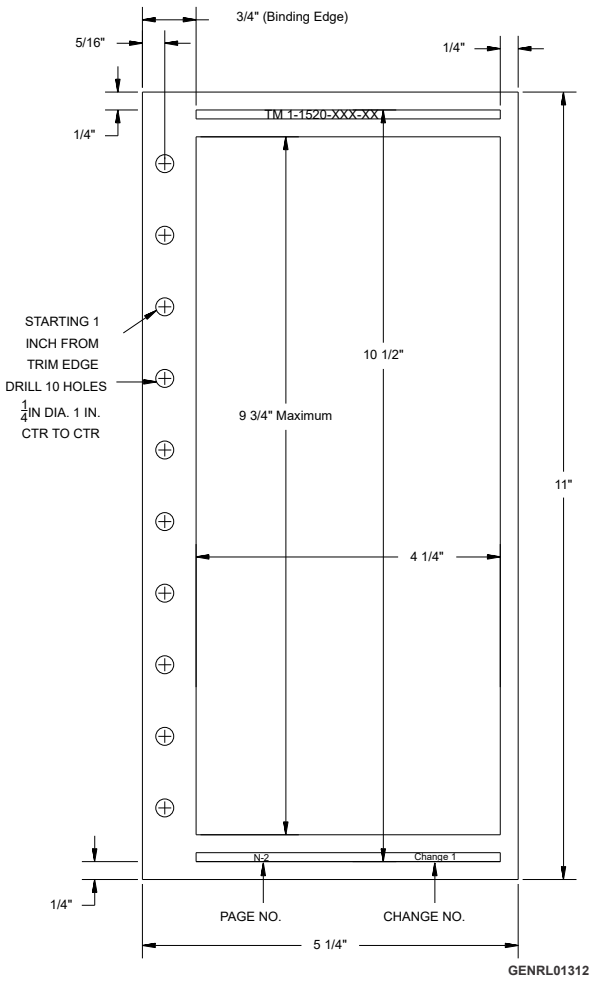


FIGURE 8. Page dimensions for FW and UAS condensed checklist and MTF manual.

4.5.4.1 Paper type.

For -CL and -MTF Operator Manuals, pages shall be printed on Nekoosa XM 10 mil/GSM 300 with Matte finish or equivalent.

4.5.4.2 Alternate checklist style and format.

The alternate checklist shall be prepared in three columns equally spaced across the 11 inch page which is turned sideways. The alternate checklist shall be printed on card stock and consist of normal procedures on one side and emergency procedures on the opposite side. Alternate checklists shall be no longer than 4 pages (2 sheets front and back).

4.5.5 ETM Blank pages.

For -10 Operator and MTF Manuals, refer to MIL-STD-40051. For -CL and emergency tabs, blank pages shall contain the same tab information, including color requirements, as the preceding page. The page number shall also be listed at the bottom followed by the word "Blank" to signify to the user that there is no missing technical information. For all other -CL blank pages, refer to MIL-STD-40051.

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4.5.6 ETM Foldout pages.

a. Foldout pages, if needed, shall be the same height as regular pages in the standard manual only, and shall be folded 2, 4, or 6 times, depending on the width necessary. If the publication will be printed, each foldout shall have a blank apron wide enough for the user to look at the data while reading text elsewhere in the TM. For electronic manuals only, the apron is not required. The foldout shall fold up narrower than the apron to accommodate binding.

b. For -CL and -MTF manuals, only two folds shall be used on foldout pages.

c. TABLE VIII lists the foldout maximum trim sizes and foldout maximum printing area for foldout pages. The minimum margin is 1/2 inch top and bottom and 1/2 inch on the side opposite the binding edge.

d. Foldout pages shall be the last printed material in the TM or volume before the DA Form 2028 and Authentication pages.

TABLE VIII. Foldout maximum page sizes (in inches).

Manual Trim Size	Foldout Maximum Page Trim Size (Including Apron)	Foldout Maximum Printing Area
8-1/2” x 11”	45” x 11”	36” x 10”
5-1/4” x 9”*	14-1/2” x 9”	7-7/8” x 8”
5-1/4” x 11”*	14-1/2” x 11”	7-7/8” x 10”
Note *The apron shall be a minimum of 6-1/8” to allow for full reading of the foldout pages and not covered by tabs.		

4.5.7 ETM Margin data.

Refer to MIL-STD-40051 for requirements unless otherwise stated in this standard.

4.5.7.1 Headers.

Headers shall consist of the security classification markings if any, the TM number, and the work package sequence number.

The security classification shall be centered at the top of the page. The TM number shall be centered beneath the security classification. The work package sequence number shall be on the same line as the TM number but shall be all the way to the outside of the page.

4.5.7.2 Footers.

Footers shall include the security classification markings if any, the page numbers, and other information as specified by the acquiring activity (for example, change designator).

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4.5.7.2.1 Page number location.

Refer 4.5.2.5 for page numbering format. Except for the front cover and foldout pages, all TM page numbers shall be in the outside corner at the bottom of the page. Front cover shall have no page number. For foldouts only, odd numbers will be assigned to the right-hand pages. Page numbers shall be in boldface type.

4.5.7.3 Protective markings.

When specified by the acquiring activity in accordance with DoDI 5230.24, a CONTROLLED UNCLASSIFIED INFORMATION (CUI) protectively marked TM shall be prepared. Any TM containing CUI information shall contain the required protective markings. Refer to DoDI 5200.48 for specific requirements on using the CUI protective marking. Classified information shall not be included in any TMs.

4.5.8 Emergency procedure pages.

Emergency procedure pages (Emergency Procedures Module) of the operator's manual that contain emergency procedure information/steps shall have heavy black diagonal lines around three edges IAW MIL-STD-40051.

4.5.9 Warnings, cautions, and notes.

Refer to MIL-STD-40051 for definitions, use, and display of warnings, cautions, and notes unless otherwise stated in this standard.

4.5.9.1 Display of warnings, cautions, and notes within the -CL and -MTF manuals.

Warnings, Cautions, and Notes in -CL and -MTF manuals shall comply with the additional requirements in 4.5.9.1.1 through 4.5.9.1.2.

4.5.9.1.1 Display within normal and detailed sections.

a. Warning, cautions, and notes shall be placed prior to the applicable procedure or step.

b. If multiple warnings, cautions, and notes apply to the same text, warnings shall appear first, cautions shall appear second, and notes shall appear last.

c. The header **`WARNING, CAUTION,`** or **`NOTE`** shall appear as a sidehead, bold, and followed by ":" and applicable text.

"**`WARNING:`** This is a warning."

"**`CAUTION:`** This is a caution."

"**`NOTE:`** This is a note."

d. Text shall wrap fully under the header.

e. Warnings, cautions, and notes shall be justified and span the entire page.

4.5.9.1.2 Display within emergency sections.

a. For RW, warnings, cautions, and notes shall be displayed in the order as laid out in J.5.1.4.2.1.

b. For FW and UAS, warning, cautions, and notes shall follow the guidance in 4.5.9.1.1.

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c. The header **WARNING, CAUTION,** or **NOTE** shall appear as a sidehead, bold, and followed by ":" and applicable text.

"**WARNING:** This is a warning."

"**CAUTION:** This is a caution."

"**NOTE:** This is a note."

d. For RW: when there is more than a single warning, caution, or note, they shall be numbered to allow for easier referencing when required:

"**WARNING 1:** This is the first warning."

"**WARNING 2:** This is the second warning."

4.5.9.2 Icons.

For -10 Operator Manuals, refer to MIL-STD-40051 for the use of icons within Warnings and Cautions. For -CL and -MTF manuals, icons shall not be used.

4.5.10 Abbreviations and acronyms.

All abbreviations and acronyms used within the TM shall be listed and defined in the abbreviations listing found in the general information module. Use of abbreviations and acronyms shall follow the following criteria:

a. Acronyms, abbreviations, and unusual terms may be used in any text, when applicable.

b. Abbreviations and acronyms, which are accepted as words (radar, sonar, laser, etc.,) need not be included.

c. All abbreviations and acronyms used in the manual, including those in tables or figures, shall be defined in the "list of abbreviations/acronyms" section of the manual.

d. Abbreviations and acronyms shall not be spelled out in work package titles, but if used in a work package title, it shall also be used inside the work package and spelled out upon first use.

e. Acronyms used in figures and tables shall be spelled out in a footnote to the applicable figure or table.

f. Acronyms such as PMCS shall be in all capital letters and shall contain no spaces or periods but abbreviations such as U.S., etc., e.g., i.e., and so forth may contain periods.

g. Army abbreviations and acronyms shall be taken from https://armypubs.army.mil/abca/SearchABCA.aspx .

h. Any new abbreviations and acronyms shall be developed in accordance with AR 25-30.

i. When abbreviations or acronyms are used as markings on the equipment (placarding), the same abbreviations or acronyms shall be used in the TM.

j. For -10 Operator Manuals, the first use of abbreviations and acronyms within a work package shall be spelled out.

k. For -CL and -MTF Operator Manuals, the first use of abbreviations and acronyms within a work package may be spelled out as determined by the acquiring activity, and if spelled out within a work package, it shall be spelled out completely with the abbreviation or acronym in parentheses immediately after the word(s).

4.5.11 Designator symbols/configurations.

An explanation of designator symbols, along with a table of symbols used in the TM, shall be provided. Designator symbols such as **B** shall be used in conjunction with text headings, text contents, and illustrations to show limited applicability of the material. If more than one model is described or the aircraft has a variety of configurations, one

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or more symbols may follow a text heading or illustration title to highlight that part of the text that pertains to the aircraft or systems in question. If the material applies to all series and configurations, no designator symbols shall be used. Where practicable, descriptive information shall be condensed and combined for all series to avoid duplication. A table showing designator symbols shall be included.

4.5.12 Dimensional data.

Except for weight and balance values found in an Operator's manual, linear dimensions shall be stated in feet and inches or in inches and decimal fractions, unless otherwise specified by the acquiring activity. No more than 3 decimal places shall be used. When dimensions are less than a foot, they shall be expressed in inches and decimal fractions. All dimensions, tolerances, clearances, measurements, and decimal equivalents appearing in Modules 8 and 9 of an Operator's manual shall be stated in bold capital lettering in the text and on illustrations.

4.5.13 Equations.

The use of equations shall be held to the minimum use required by the needs of the TM user.

4.5.14 Nomenclature.

The nomenclature of items shall be the short name used in the applicable aircraft parts manuals, TM 1-XXXX-XXX-20&P or TM 1-XXXX-XXX-20P. The only exception shall be the use of placard item names shown on controls, switches, panels, etc. These items shall be expressed as shown on the placards. These items shall appear in text and procedural steps in boldfaced capital letters.

4.5.15 Illustrations/graphics.

Supporting illustrations/graphics shall be developed for aviation manuals as described in 4.5.15.1 through 4.5.15.2.

4.5.15.1 General.

Illustration formats shall use Computer Graphics Metafile (CGM) or Scalable Vector Graphic (SVG) in an intelligent vector format for drawings used in technical publications. Line drawings, including exploded views, locator views, and detailed views, shall be used to support operating procedures. Raster images shall not be embedded into CGM/SVG vector graphics with the exception of using a raster image as a locator view of the CGM/SVG. Images of screen captures used to illustrate software tasks may be in JPG format, or in line art if more effective for the user.

a. Line drawings (black lines on white background) shall be used throughout the TM.

b. Photographic illustrations may be used only when prior approval has been obtained from the acquiring activity.

c. Illustrations, including diagrams and schematics, shall be clear, simple, and complete, and shall contain all necessary callouts to support the text.

d. The number of callouts on a single illustration or a single sheet of a multi-sheet illustration shall be 25 or less. If more than 25 callouts are required, the total number required shall be equally divided between two identical or similar illustrations (FIGURE C-1).

e. Broadsides (illustrations that have been rotated 90 degrees on the page) shall not be used.

f. In accordance with AR 25-30, AR 25-38, DA PAM 25-40, and DA PAM 25-38, use of color in illustrations/graphics shall not be merely for decorative effect but shall be justified for the purposes of safety and clarity to the operator.

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(1) All new requirements for color shall be processed through publication authentication approving authorities.

(2) For EMO publications, use of color should be judicious as it impacts file size and extensive use may result in requiring a volume break.

(3) Cockpit indications, including Multi-Function Displays (MFD), shall be presented as found in the aircraft, to include color.

(4) If required by the acquiring activity, performance charts may add color to aid in operator understanding and usage of the charts.

(5) For -CL manuals, color shall be restricted to the emergency sections of the operator's checklist.

(6) -MTF manuals are prohibited from color printing.

4.5.15.1.1 Lettering.

Lettering and type on original artwork shall be well-defined and large enough to be easily read when the illustration is reproduced at page size. Lettering and type shall be in capital letters. The minimum type size shall be eight points. Spacing of letters and words shall be controlled to insure clear, legible copy.

4.5.15.1.2 Keys for illustrations.

Keys may be included on the illustration. Where keys are too numerous or the explanations too lengthy to fit within the illustration cropped area without crowding, they shall be placed in tabular form immediately below the illustration or on the facing page. These tables shall be considered as a text function.

4.5.15.1.3 Foldouts.

Foldouts shall be kept to a minimum. When foldouts are provided they shall be placed at the end of the manual. Foldouts shall have a blank apron. Foldouts shall be page numbered consecutively starting with FP-1, FP-3, etc. Foldouts shall be identified with figure titles and figure numbers and be consecutively numbered starting with FO-1, FO-2, etc.

4.5.15.1.4 Index number callouts.

Index numbers are the preferred callout and shall start with Arabic numeral 1 and continue consecutively within an illustration. For multi-sheet illustrations, index numbers shall continue in sequence from one sheet to another.

a. Index numbers shall be presented in one of the following manners:

(1) In clockwise sequence, beginning at 11 o'clock. This is the preferred method.

(2) In the order mentioned in the text.

b. Within a multi-sheet illustration, if an item that has already been assigned an index number is used in more than one illustration in that multi-sheet illustration, it shall retain the same index number.

c. Index number callouts and nomenclature callouts shall not be used on the same figure.

4.5.15.1.5 Nomenclature callouts.

Nomenclature callouts should be avoided, but when used, reference to those callouts with the text shall be by figure only and the nomenclature in the text shall match the nomenclature in the figure. For example:

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"Remove safety disc (Figure 1) and safety disc washer (Figure 1) from valve body (Figure 1)"

4.5.15.1.6 Referencing illustrations/graphics.

Illustrations may be referenced within another work package. When an illustration is found within another work package, it shall be preceded by the WP sequence number, followed by the figure number (WP 0056, Figure 1). Illustrations shall not be duplicated in the manual without safety or air worthiness justification. Minor differences in illustrations, such as in callouts, are not duplications.

4.5.15.2 Graphical data.

4.5.15.2.1 General requirements.

Unless otherwise specified by the acquiring activity, data that includes more than three variables shall be presented graphically. Data with three variables shall be presented graphically if it represents continuous data (for example, torque available as a function of altitude and temperature).

4.5.15.2.2 Order of precedence.

In the event of a conflict between the graphical data presentation requirements in the text of this standard and the sample graphs provided, the text of this standard shall take precedence.

4.5.15.2.3 Explanatory text.

A brief explanation shall be provided for each graphic presentation including, but not limited to, description, purpose, and procedure for use, applicable conditions, and effects of their variations.

4.5.15.2.4 Priorities.

Unless otherwise specified by the acquiring activity, the following order of priorities shall be followed while preparing graphical presentations:

a. Minimize the possibility of user mistakes.

b. Cover the full applicable range of data. Unless data ranges are specified in the illustration requirements of this standard, the maximum probable ranges to be expected in operation shall be used. MIL-HDBK-310 can be used for reference for ranges of climatic data.

c. Provide adequate accuracy. The graphical presentation shall be readable over all ranges of the data. It shall also duplicate the source data to at least one percent of the applicable range of the parameter (for example, a free air temperature range from -60 °C to +50 °C should be readable to at least 1 °C).

d. Clarity and ease of use. Each graph shall be designed to directly provide the most commonly used parameters (for example, torque required to hover at known conditions of altitude, temperature, weight, and skid/wheel height). Less often used information, such as maximum temperature to hover at a given weight and altitude, shall be obtainable with additional effort.

e. Ensure standardization. Standardization tables are provided to ensure standardization of graphic illustrations. Type and spacing requirements are summarized in TABLE IX. Line requirements are summarized in TABLE X. The tables of standardization shall be used during preparation of basic, changed, or revised illustrations. The requirements in the tables of standardization are applicable to the final product. If graphic presentation is other than final size, adjustments shall be made to ensure that final size graphs meet the stated criteria.

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f. Place the graphs on the minimal number of pages, consistent with the importance of clarity and ease of use.

g. General appearance, cost, and ease of production shall be given consideration, but only as three of the lesser priorities.

4.5.15.2.5 Specific requirements for graphical data.

4.5.15.2.5.1 Titles.

Titles for graphs shall be the most succinct title that adequately indicates the nature of the graphical data.

4.5.15.2.5.2 Condition heading.

The range, parameter name, and units of each condition that apply to the data shall be listed with each condition separated. When abstract conditions (for example, clean configuration forward cg) are used, they shall be described in detail and/or quantified in the accompanying text. Conditions that apply to more than three similar graphs shall be listed only on the first example and shall be referred to on all subsequent graphs in the series. General aircraft or system limits shall not be listed. Any condition known not to affect the data shall not be listed. The effect of variation of each listed condition on the data shall be discussed in the text. If the effect of condition variation is not known and cannot be estimated, it shall be so stated in the text. General conditions (for example, rigging, instrument errors, fuel types, etc.) applicable to all data in a module shall be discussed in a paragraph titled “General Conditions” which shall appear near the beginning of the module: The information in the “General Conditions” paragraph shall not be repeated on the graphs within the module.

4.5.15.2.5.3 Sub-graphs.

For some graphical data, it may be desirable to include separate sub-graphs with data on the same general subject. Titles and conditions different from the main conditions shall be given for the sub-graphs.

4.5.15.2.5.4 Notes.

Notes should not be used on graphs. Notes may be placed on areas adjacent to charts, when absolutely necessary, in order to prevent misuse or misinterpretation of the data. If the note does not fit this condition, it should appear in the text.

4.5.15.2.5.5 Data basis.

Data basis information shall include data type (for example, flight test, estimated, etc.) and each actual data source document used to compute the data presented.

4.5.15.2.5.6 Examples.

An example shall be provided on the graphical data to demonstrate primary use of each type of graph. If there are two equally important uses of the charts, a maximum of two examples may be presented on the graph. Additional examples (text only) of other uses or methods of use of the data, where applicable, shall be included in explanatory text. These examples shall be in the same format as those with the graphical data.

a. The example text shall be clear yet succinct. Omit articles, conjunctions; prepositions, etc. Wanted parameter names only shall be used. A maximum of three parameters shall be used. If more wanted parameters are available, use additional examples in the explanatory text to explain them. Use one line each to list known

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parameters and values. If the known parameter value is obtained from elsewhere in the manual, or the source is not evident, parenthetically (below known parameter line) describe the most probable source, such as (from example 1) or (computed from winds aloft). The method for using the graph shall be described using one line per distinct step. Known values shall not be repeated in the method. If needed or useful intermediate values are obtained using the method, these values shall be stated. The example text shall be located on the left side of the graphical data. If multiple examples are used, each example shall be sequentially numbered using Roman numerals (for example, EXAMPLE I, EXAMPLE II, etc.). If a single example is used, it shall be identified by the heading "EXAMPLE."

b. Example values shall be chosen to represent reasonably critical conditions. Standard and absolute extreme conditions shall not be used. If restricted or special conditions are shown on the chart, the example values shall be chosen to illustrate their effect. Values shall be chosen to require graphical interpolation on every parameter.

4.5.15.2.5.7 Data range.

The data range presented shall cover the full applicable range of data. Scales shall extend to the next major division beyond the extreme or limit value(s) and no further, unless specified by the acquiring activity.

4.5.15.2.5.8 Grid.

The grid shall correspond to the primary scales. Grids shall be prepared to the graphical line standards (TABLE X).

4.5.15.2.5.9 Scales.

The scale title shall include the parameter name and units of measure. When used, multipliers shall be included with the units (for example, GROSS WEIGHT - pounds *1000). Multipliers shall be used only to meet specific illustration requirements in this standard for values with three zeros or more, or when significant improvement in the appearance of the graph would result. Resulting fractional values (for example, GROSS WEIGHT - 1000 pounds = 20.2) shall be avoided. Secondary scales should be located on the opposite side of the grid from the primary scale. Scale numbers shall be used for each major, or every other (most even value) major, scale increment, unless the secondary scale corresponds to markings on an aircraft indicator. In this case, the increment and value labeling shall be the same as those on the indicator.

4.5.15.2.5.10 Data line labels and values.

Labels for data lines shall include the parameter name, multiplier, if any, units, and corresponding value. They shall be located approximately at the midpoint of, and oriented parallel to, the data line, as read from the bottom of graph. Data line labels and values shall be prepared in accordance with the graphical standards as shown in TABLE IX and TABLE X. Labels shall minimally obscure the grids. Data line labels and values shall be located according to the following order of preference:

a. Parallel centered interrupting the line, alternately staggered to avoid masking a continuous area of the grid (shall be used for primary data line numbers).

b. At the end of, and parallel to, the data line (suitable for secondary data lines).

c. Adjacent and parallel to the data line (suitable for secondary data lines).

d. Outside the data lines with leader lines to each data line (suitable for secondary data lines).

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4.5.15.2.5.11 Primary data lines.

Primary data lines shall be prepared in accordance with TABLE X. Scales shall be chosen so that the mid-range of approximately linear data are oriented at approximately 45°. Increments shall be chosen so that the majority of the data lines are separated by at least one minor grid width and no more than one major grid width. Converging data lines shall be truncated (alternately) when the separation decreases to 1/2-1 minor grid spacing, so that actual convergence does not occur.

4.5.15.2.5.12 Secondary data lines.

Operating limits, restricted operating conditions, and optimum, recommended, or critical operating conditions shall be depicted, as applicable, on each graph. Secondary data lines shall be prepared in accordance with TABLE X.

4.5.15.2.5.13 Layout and sizing.

Scales and grid size shall be chosen to take maximum advantage of the available space to provide the most easily read graph, consistent with the previously specified range and readability requirements. Several single graphs on the same general subject may be included on a single page. For sequential graphs the following requirements apply. The general layout shall have the example text near the upper left corner of the page. The first step graph shall be near the upper right corner. The sequence shall be for the user to enter on left of first graph, move right, reflect down at right angles, reflect left, and reflect down, etc. until the primary wanted parameter is read out on the final scale. A transfer grid (in the direction of transfer only) shall be provided between each step graph. Intermediate parameters may be provided on secondary scales by continuing through the reflector data lines or by reflecting in the opposite direction to the primary direction.

4.5.15.2.5.14 Scaling.

Scale and data line increments shall conform to the rule of 1, 2, 5, or 10 minor divisions per major division, except as noted here. The preferred scale grid shall be five minor divisions per major division along each axis. Ten division grids are undesirable and shall be used only when absolutely necessary. Four division grids shall be used only with the permission of the acquiring activity. Asymmetrical (4 * 5) grids are permitted. For highly nonlinear variations approximately equal increments of the dependent variable(s) shall be used. The minimal minor grid spacing shall be six points, unless otherwise specified by the acquiring activity.

4.5.15.2.5.15 Units.

Each parameter on the graph and its corresponding unit of measure shall be those most commonly used for the subject aircraft. If the parameter is available on an aircraft indicator, the units used on the graph shall be the same as those on the indicator. If the parameter is not on an aircraft indicator, the units used shall be the same as those of the most often used source of the data. In some instances, two nearly equal common units may be in use or a transition may be in progress from an older model to a newer model. When this occurs, the primary unit of measure shall be that associated with the new model. Where practicable, the primary unit shall be used on the primary scale and the unit associated with the older model shall be presented on a (redundant) secondary scale. When scales or data include negative values, + and - prefixes shall be used with all numbers for that parameter. For data values on the graph, brackets shall be used around the prefixes.

4.5.15.2.5.16 Original graphical data designs.

For original (sequential) graphical designs, the following requirements also apply. See TABLE IX, TABLE X, and TABLE XI.

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a. Each “known” parameter shall be used only once in the sequence, unless its use will simplify a procedure.

b. The sequence shall proceed from the best-known (or most certain) parameter to the least certain parameter consistent with technical requirements.

c. Each sequential stop shall reflect at right angles (90° parameter transfers only). “Paralleling” data transfers shall be avoided.

TABLE IX. Graphical type standards.

	USE	PREFERRED FONT TYPE/ SIZE[1]	CAPITALI-ZATION	VERTICAL SPACING/ LOCATION	REMARKS
1.	Title	Sans Serif Bold 12 (2pts smaller for sub-graphs)	Upper Case	Centered at top of graphic[2]	Includes model designator symbol if applicable
2.	Sub-Title	Sans Serif Bold 10	Upper Case	Centered, 6 points below title[2]	Use only if required to distinguish similar graphs
3.	Condition heading	Sans Serif Bold 8 (2pts smaller for sub-graphs)	Upper Case	Centered, 6 points below title or subtitle[2]	Separate conditions; one line desirable
4.	NOTE	Sans Serif Bold 8	Upper Case		As required – Notes on graph page very undesirable – text preferred
5.	Note information	Sans Serif 7	Upper and lower case	Left justified; 6 points below NOTE	Use only to prevent misuse
6.	EXAMPLE	Sans Serif Bold 10	Upper Case	Centered over example text	Usually located near upper left corner of chart
7.	WANTED, KNOWN and METHOD Titles	Sans Serif Bold 8	Upper Case	Indented 18 points; 12 points below previous line	
8.	Wanted, Known and Method text	Sans Serif 7	Upper Case	Left justified; 6 points below title text	WANTED - Parameter names only maximum of three
					KNOWN – Parameter: Value, Units
					METHOD - One line per step
					Do not repeat known values
9.	Primary scale values	Sans Serif Bold 8	Upper Case	One minor grid division outside grid border	Oriented to read from bottom of page
10.	Primary scale titles	Sans Serif Bold 8	Upper Case	One minor grid division outside scale value numbers	Oriented to read from right (vertical scales) or bottom (horizontal scales) of page

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TABLE IX. Graphical type standards. (continued)

	USE	PREFERRED FONT TYPE/ SIZE[1]	CAPITALI-ZATION	VERTICAL SPACING/ LOCATION	REMARKS
11.	Secondary scale values	Sans Serif 7	Upper Case	Two minor grid divisions outside grid border	Additional secondary scales may be added
12.	Secondary scale titles	Sans Serif 7	Upper Case	One minor grid division outside scale values	
13.	Primary data line labels	Sans Serif Bold X (X = 1 minor grid division)	Upper Case	At minor grid division outside scale values	Parallel to data lines oriented to read from bottom
14.	Primary data line values	Sans Serif Bold X (X = 1 minor grid division)	Upper Case	Interrupting data lines near mid-point	Alternate values staggered: if negative use (-) and (+)
15.	Limit line labels	Sans Serif Bold X (X = 1 minor grid division)	Upper Case	Parallel to limit line, near midpoint, in prohibited area	
16.	Maximum Performance or Recommended Operation	Sans Serif Bold X (X = 1 minor grid division)	Upper Case	Location and orientation dependent on layout	
17.	Time limited or restricted operation	Sans Serif Bold X (X = 1 minor grid division)	Upper Case	Parallel to limit line, at midpoint in restricted area	
18.	DATA BASIS: title	Sans Serif Bold 8	Upper Case	Left justified, 6 points above figure title[2]	
19.	Data basis text	Sans Serif 7	Upper Case	Following DATA BASIS[2]	Includes data type and source data document(s)
NOTES:		[1]Size requirements apply to the final printed product. A 10% tolerance is allowed, however, deviation on any page should be in the same direction.			
		[2]Spacing is designed for heading information, Data Basis, and Figure information to be marginal copy to provide the full print area for the graph. If the full layout area is not required, or used, increase spacing to provide a balanced appearance.			

TABLE X. Graphical line standards.

	USE	COLOR	LENGTH	WEIGHT	REMARKS
1.	Primary Data	Black	To limits or operational range	heavy	Most even value
				medium	Alternate lines
				fine	Use if increments change
2.	Grid Lines	Grey	Correspond to Primary scales	medium	Major increments
				fine	Minor increments

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TABLE X. Graphical line standards. (continued)

	USE	COLOR	LENGTH	WEIGHT	REMARKS
3.	Transfer Grid	Grey	1/3 to 1 major grid	very fine	Direction of transfer only
4.	Grid Border	Black	Primary scale length	heavy	Over outside grid
5.	Primary Scale Tick Marks	Black	1/2 to 1 minor grid division	heavy	Inside grid border major grid only
6.	Secondary Scale Tick Marks	Black	As required	medium (Major)	Outside grid border
				fine (Minor)	
7.	Limit Lines	Black	As required	heavy	
8.	Maximum Performance or Recommended Operation	Black	As required	heavy	Major lines
				medium	Use if multiple lines
9.	Restricted or Time Limited Operation	Grey	As required	fine (Border shaded)	Shaded area with black border line
10.	Extrapolated Data	Black Dashed	As required	heavy, medium, fine	Use for data beyond source data conditions
11.	Beyond Limit Data	Black Dashed	As required	heavy, medium, fine	Use for data beyond operating limits to aid interpolation

TABLE XI. Line definitions.

Weight[1]	Width		
	inches	millimeters	points
Very Fine	0.004	0.1	0.29
Fine	0.008	0.2	0.58
Medium	0.012	0.3	0.86
Heavy	0.016	0.4	1.15
Very Heavy	0.020	0.5	1.44
Dashed: 5 to 10 x width line lengths, 3 to 5 x width gap space			
Dotted: 1 to 2 x width line lengths, 2 to 3 x width gap space			
[1]NOTE: Line weight requirements apply to the final printed product. A 20% deviation is allowed, however, deviation on any page should be in the same direction.			

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4.6 Changes/revisions.

A change/revision is used to incorporate appropriate new information (e.g., Safety messages, Modification Work Orders (MWOs), engineering drawing changes, DA Form 2028, etc.) into the basic TM (or previous edition) or clarifies, corrects, or improves existing information in the TM. A change shall be written in the same style and format as the basic manual. When updates to TMs are ordered, the deliverable product shall be changed pages/work packages or a complete revision of the TM. Changed information in changes and revisions shall be documented using the appropriate change level attributes and retained in the document instance for a historical record of changes.

4.6.1 Tagging changes/revisions.

Changes include alterations to an original text including insertions, modifications, deletions, etc. Changes shall be integrated into XML utilizing two available options: change elements and change attributes.

4.6.1.1 Change elements.

For changes involving only text (only a few words), the **<change>** (or element) may be used. The scope and type of changed information is indicated by being enclosed within the start and end tag. This type of change element is used only for changes when it is necessary to isolate the changed text to specifically identify a changed location, i.e. the change bar will only display next to the sentence changed (contained within the start and end tags) rather than display next to the entire paragraph. The scope and type of changed information shall be identified by assigning attributes to the **<change>** tag. Change tags shall include comment data as detailed in section 4.6.1.2.2.

4.6.1.2 Change attributes.

For changes involving larger amounts of text (steps or paragraphs), changes shall be integrated into XML by updating the attribute of the step/paragraph tag (element) itself (rather than inserting **<change>** tags).

4.6.1.2.1 Change attribute functions.

There are five attributes related to change documentation:

- **a.** inschlvl - contains numeric change designator, shall match chgno attribute, for use on any change, addition or deletion.
- **b.** delchlvl - contains numeric change designator, shall match chgno attribute, should only be used if there is a concurrent paper output.
- **c.** comment - very brief message to soldier about change.
- **d.** changeref - link to applicable **<change.history>** giving authority for the change.
- **e.** id - (if available on the element) allows revision summary to link directly to the change.

4.6.1.2.2 Comment data in change attributes.

Change attribute comment data shall be integrated into the attributes of not only **<change>** tags but also any other element (tag) containing change data. Comment data is important because it provides context for change and provides change detail that would not be readily apparent in the updated version of the text. For an IETM, the content of the comment attribute will be displayed when the pointer is hovered over the change bar.

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4.6.2 Changes/revisions for ETMs.

A revision shall be written to the latest version of this standard. Refer to MIL-STD-40051 for change and revision requirements for ETM publications.

4.6.3 Revisions for EMO ETMs.

a. EMO ETMs shall be revised. Change packages are prohibited for EMO ETM publications.

b. When an EMO ETM is revised, a revision summary shall be provided.

c. Change/revision bars may be used to indicate to the target audience where changes have occurred.

d. When 50 percent or more of the data is affected, revisions shall be done to the latest version of this standard. When less than 50 percent or more of the data is affected, revisions may remain in the current verision of the standard.

4.6.4 Revisions for Interactive Electronic Technical Manuals (IETMs).

All updates to an IETM shall be made through revisions. Change packages are prohibited for IETMs.

a. IETMs may be published as web-only or as a distributable disk. Refer to AR 25-30 and AR 25-38 for correct processes to request web-only distribution.

b. Each revision to an IETM shall be identified by a revision date and identifier (e.g. Revision 1, Revision 2). Revision identifiers shall never reset to 0.

c. When updates to an IETM are made, the entire IETM shall be reissued. An update package shall also be created, refer to Interactive Authoring and Display Software (IADS) help files on how to create an update package.

d. When an IETM is revised and reissued, a revision summary shall be included, and change markings shall be included for the revised material.

e. When 50 percent or more of the data is affected, revisions shall be done to the latest version of this standard. When less than 50 percent of the data is affected, revisions may remain in the current version of the standard. A revision summary shall be included in a revision.

f. Changed information in revisions should be documented using the appropriate change level attributes and retained in the document instance for a historical record of changes.

4.6.5 Change numbers and symbols.

Refer to MIL-STD-40051 for requirements unless otherwise stated in this standard. Standard -10 Operator Manuals shall contain an explanation of change symbols in the "About this Manual" section of the publication.

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5 DETAILED REQUIREMENTS

5.1 Publication development.

TM data developed in accordance with this standard shall be task-oriented and fully consistent with the operating concepts derived from the baseline documents in the technical data package.

5.2 Preparation of front and rear matter.

Refer to MIL-STD-40051 for all requirements unless otherwise specified.

5.2.1 Front matter <av_paperfrnt>.

As applicable, material preceding the first text page shall consist of the following in the order specified:

a. Front cover **<frntcover>**

b. Warning Summary **<warnsum>**

c. Change transmittal page **<chgsheet>**, if applicable

d. Revision summary **<revisionsummary>**, if applicable

e. Table of contents **<contents>**. Operator (-10) manual and -MTF manual only.

f. "About this Manual" **<about>**

5.2.1.1 Revision summary <revisionsummary>.

When a revision to a ETM or IETM is issued, a revision summary shall be prepared. The revision summary shall be a summary of the major changes that have been made to the publication. This shall include software updates, aviation safety or maintenance message changes, new system functionality, deletion of major systems or configurations, etc.

5.2.1.1.1 Safety message summary <safesum>.

A separate table **<table>** of all aviation messages incorporated into the TM shall be prepared and included in the revision summary. If there are no safety messages incorporated into the manual, then the following statement shall be included verbatim: “There are no new safety messages incorporated into the manual with this revision.”

5.2.1.2 Table of contents <contents>, operator (-10) manual and -MTF manual only.

A table of contents shall be developed for all Operator (-10) manuals. As determined by the acquiring activity, an -MTF manual may develop a table of contents. It is prohibited for a -CL manual.

5.2.2 "About this manual" <about>.

"About this Manual" information shall be included as the last item in the front matter. It shall be prepared for all TMs. Only one "About this manual" section shall be present within a TM. This section is intended to instruct the user on how to use the technical contents of the manual. It shall not be used to instruct the user on how to use the viewer (e.g. IADS, PDF, etc.) of the publication.

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5.2.2.1 Required Content.

"About this Manual" shall include the following sections:

a. Change History

b. Reporting of Errors block

c. Explanation of the use of Warnings, Cautions, and Notes, if applicable

d. Explanation of the use of Shall, Will, Should, and May, if applicable

e. Explanation of change symbols, if applicable

f. Explanation of TM Structure

g. Applicability/Configurations/Symbology

h. Additional Introductory Information, as required

5.2.2.1.1 Change history <changedates>.

When a change is done, a list of all changes with their authentication dates shall be placed at the top of the "About this Manual". For the initial publication and subsequent revisions, this section shall contain the publication date of the initial publication or revision, whichever was published last.

5.2.2.1.2 Reporting of errors <reporting>.

A reporting errors and recommending improvements statement shall be included in all -10 Operator Manuals. The mailing address, email address, and fax number of the responsible proponent shall be inserted in the statement. Additional information may be added as required by the acquiring activity (e.g., how to submit an electronic DA Form 2028 via the Web). The following statement shall be included verbatim:

"REPORTING ERRORS AND RECOMMENDING IMPROVEMENTS

You can help improve this manual. If you find any mistakes or if you know of a way to improve the procedures, please let us know. Mail your letter or DA Form 2028 (Recommended Changes to Publications and Blank Forms), located in the back of this manual directly to: *(name and address of proponent)*. You may also send in your recommended changes via electronic mail or by fax. Our fax number is *(insert DSN and commercial number of proponent)*. Our email address is *(insert address of proponent)*. You may also submit your recommended changes at the following website *(insert appropriate URL)*. A reply will be furnished to you."

A reporting errors and recommending improvements reference shall be included in all -CL and -MTF manuals. The following statement shall be included verbatim: "REPORTING ERRORS AND RECOMMENDING IMPROVEMENTS

Refer to (applicable -10 manual) for instructions on submitting recommendations for improvement to this manual."

5.2.2.1.3 Explanation of warnings, cautions, and notes <explanwcn>.

The below explanation of the use of warnings, cautions, and notes that the operators will find in the TM and the importance of observing these safety alerts shall be provided verbatim in all -10 Operator Manuals. It is prohibited in -CL and -MTF manuals.

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"WARNINGS, CAUTIONS, AND NOTES. Warnings, cautions, and notes are used to emphasize important and critical instructions and are used for the following conditions:

WARNING

Identifies and highlights an essential operating or maintenance procedure, practice, condition, statement, etc. which if not strictly observed, could result in injury to, death of, or long term health hazards to personnel.

CAUTION

Identifies and highlights an essential operating or maintenance procedure, practice, condition, statement, etc., which if not strictly observed, could result in damage to or destruction of equipment, or loss of or degraded mission effectiveness.

NOTE

Identifies and highlights an essential operating or maintenance procedure, condition, or statement."

5.2.2.1.4 Explanation of the use of shall, will, should, and may <explanswsm>.

An explanation of the use of shall, will, should, and may in the TM shall be provided in all -10 Operator Manuals. It is prohibited in -CL and -MTF manuals. For the -10 Operator Manual, the following statement shall be included verbatim in the "About this Manual":

"Within this TM use "shall" whenever a TM expresses a mandatory requirement. "Will" may be used to express a declaration of purpose or procedural result. Use "should" to indicate a non-mandatory but preferred method of accomplishment. The word "may" can be used to indicate an acceptable method of accomplishment."

5.2.2.1.5 Explanation of change symbols <explanchange>.

An explanation of the use of change symbols shall be included in all -10 Operator Manuals. It is prohibited in -CL and -MTF manuals. The following statement shall be included verbatim:

"Changes to the text and tables shall be indicated by a vertical bar in the outer margin extending close to the entire area of the material affected. Pages with emergency markings, which consist of black diagonal lines around three edges, shall have the vertical bar or change symbol placed along the outer margins between the text and the diagonal lines. Changes to an illustration shall be indicated by a vertical bar in the outer margin. Change symbols show current changes only. "

5.2.2.1.6 TM structure <explantm>.

Information to familiarize the user with special or unusual features of the TM shall be prepared. Coverage shall lead the user through the TM and explain important features of the organization and content. For example, the format is explained; equipment data, performance data, normal procedures, emergency procedures, and limitations are explained. For -CL and -MTF manuals, an explanation of tabbed indexes and the requirements detailed in 5.2.2.2 or 5.2.2.3 and their subsections shall be included, depending on publication type. Any peculiarities in the basic arrangement of the TM shall be described. "About this Manual" information shall not repeat instructions given within the modules and/or work packages.

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5.2.2.1.7 Applicability/configurations/symbology <appconfig>.

An explanation of applicability symbols and their corresponding configurations shall be listed in tabular format. A table of all symbols used in the TM, including those used in the amplified and simplified checklists, shall also be provided. The following symbols (TABLE XII) shall be used in the checklist on every step applicable to identify certain conditions or duties:

TABLE XII. Checklist symbols

Symbol	Definition
O	The symbol “O”, which shall precede the step, shall be used to indicate if equipment is installed or available.
④	Those duties that are the responsibility of the pilot/individual (not on the controls) shall be indicated by a circle around the step number or letter.
⋆	The "star" symbol indicates a detailed procedure for the step is included in the Detailed Procedure section and a page reference is provided.
*	The asterisk symbol “*”, which shall precede the step, shall indicate that performance of the step is mandatory for all through-flights. The asterisk steps in this checklist shall be used for combat/tactical operations when authorized by the commander. The asterisk shall apply only to checks performed prior to takeoff.
F	The letter “F”, which shall precede the step, shall indicate a task or step that requires a crewmember, flight engineer, external operator, payload operator, or ground crew personnel function or response.

5.2.2.1.8 Additional introductory information.

Any additional introductory information that may be required such as explanatory information for appendices or indices shall be provided, as applicable.

5.2.2.2 Required content for -CL.

Aviation -CL manuals shall contain a scope paragraph in the TM Structure section **<explantm>** of "About this Manual" that tells the user what is covered in the checklist. The model(s) of equipment covered and applicable software versions shall be included. Specific statements shall be included as detailed in 5.2.2.2.2 through 5.2.2.2.7.

5.2.2.2.1 Precedence statement.

All -CL manuals shall contain the following verbatim statement:

In the event of a conflict between this checklist and the *(insert the relevant TM operator manual)*, the *(insert the relevant TM operator manual)* takes precedence.

5.2.2.2.2 Rotary wing volume 1 scope.

For Rotary Wing aircraft Checklist Volume 1, the following information shall be provided verbatim:

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“Scope.

Volume 1 contains normal procedures, detailed procedures, and performance charts, divided into tabbed sections of like information. Sections may contain one or more work packages of content to allow for easy updates of content. Page numbering is based on the content of the pages; normal procedure pages are numbered N1-1, N1-2, etc. Detailed procedure pages are numbered D1-1, D1-2, etc. Performance chart pages are numbered P1-1, P1-2, etc. Page references to detailed procedures are included at the end of the applicable step. A "RETURN TO:" feature is added at the end of applicable detailed procedures to aid in navigation back to the normal procedures, if required.”

Explanation of references and symbols shall be provided to include explanation of the **<return>** function. A statement explaining that Emergency Procedures are found in Volume 2 and that the CL is a condensed version of the procedures found in the -10 shall be provided.

5.2.2.2.3 Rotary wing volume 2 scope.

For Rotary Wing aircraft Checklist Volume 2, the following information shall be provided verbatim:

“Scope.

These emergency procedure sections list the Warnings, Cautions, and Advisories (W/C/A) that the *(model information)* aircraft displays when emergency conditions exist and outlines the corresponding procedure associated with the emergency. The dividers found in TB 1-1500-1, Supplemental Divider Package for Aviation Checklists, are designed to enhance quicker access to the emergency procedures.”

5.2.2.2.4 Rotary wing volume 2.

For Rotary Wing aircraft, Volume 2, the following information shall be provided verbatim:

“FLIGHT REFERENCE CARD Design.

These cards are designed to differentiate between the Warning, Caution, and Advisory messages displayed to the flight crew. Warnings, along with fires, and emergencies without associated W/C/A messages are located in the RED section; Cautions are in the YELLOW section; Advisories and Annunciators with an associated emergency procedure are in the GREEN section; Mission Equipment emergency procedures without associated W/C/A messages are in the BLUE section. The following examples demonstrate the relationship between indications and construction of the FRC emergency procedure sections: (*An illustration similar to FIGURE 9, representing the applicable indications of the aircraft, shall be provided.*)



FIGURE 9. Indication examples.

Page numbering is based on card sections and tab numbers. Sections are Warnings (W-#), Cautions (C-#) , Advisories (A-#), and Mission Equipment (E-#). The front of the card will be W-1 and the reverse side will be numbered W-1R. Limitations cards are numbered similarly.

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Section Indexes.

Each section's index is arranged alphabetically by the associated *(MPD, MFD, etc as applicable to weapon system)* message. The cards themselves may not be arranged in alphabetical order. The RED section has an additional index for fires and emergencies without W/C/A indications. Emergencies listed in the indexes with BOLD font contain Immediate Action steps.

Emergency Section Format.

In the event of an emergency, the crew shall take action to overcome or alleviate effects of the emergency. The required degree of urgency depends upon the nature of the failure and the prevailing conditions. The FRC emergency section facilitates the U.S. Army Emergency Response Method by enhancing fault diagnosis and recovery actions using the following format:

- **Name of Emergency.** Title of emergency procedure and corresponding failure(s).
- **Indications.** The first indication of many emergency conditions will be activation of the voice warning system, illumination of the *(Enter applicable weaponsystem nomenclature; ie, MASTER WARNING, MASTER CAUTION, MFD, MPD, etc)* messages. The indication section will include these aircraft-displayed W/C/A messages and any associated voice messages. It will also include proprioceptive, audio, and visual cues associated with the malfunction (e.g. fire indications include identification of smoke).
- **Emergency Procedure Steps.** Aircraft control is the most important consideration during any emergency. The urgency of certain emergencies may require immediate and instinctive action by the crew to maintain aircraft control. Therefore, an emergency action step that is underlined shall be committed to memory. The emergency procedure steps detailed in these FRCs are divided into four categories:
  - **Immediate Actions (IA).** Boxed and underlined emergency action steps that shall be carried out immediately after the failure is positively identified to ensure safe flight is not jeopardized. Due to insufficient time to refer to the FRCs, all IA steps will be underlined and committed to memory. Time permitting, during execution of “E” in the Emergency Response Method, IA steps should be confirmed using the FRCs.
  - **Actions.** Boxed emergency action steps that do not justify immediate and subsequent actions. Aircrews will utilize the Emergency Response Method and the FRCs to execute the steps listed under actions.
  - **Subsequent Actions.** Actions to be taken once IA steps are complete and danger to aircraft and aircrew has passed. Aircrews will utilize the Emergency Response Method and the FRCs to execute subsequent action steps.
  - **Considerations.** Considerations offer additional information that is not essential, but which may assist in assessing the longer-term implications of the emergency.
- **Urgency to Land.** The degree of urgency to land the aircraft depends on the nature of the failure and the prevailing conditions. *Use only the applicable definitions.*
  - **Land Without Delay.** A landing in which the primary consideration is continued control of the aircraft and survival of the occupants. It is meant to be more urgent than Land As Soon As Possible. The situation may not permit the crew to continue to maneuver the aircraft to a suitable landing area (e.g., open field). If maneuvering to an open area is not practical, then the crew shall make the decision to land in an area that will have the least amount of negative impact on crew survivability. (e.g. Over dense forest, select an area with the smallest trees, in a mountainous area, choose an area with the least amount of slope).
  - **Land as Soon as Possible.** A landing at the nearest suitable landing area (e.g. open field). The primary consideration is to ensure the survival of occupants.
  - **Land as Soon as Practicable.** A landing at a suitable landing area. The primary consideration is the urgency of the emergency.
- **Cross-Referencing.** Some emergency procedures contain a cross-reference to other emergency procedures or procedures in the Normal Procedures (white) section. In these cases, the cards are referred using the card number (e.g. CARD W-6 or Page N1-2).

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- **Emergency Procedure Completion.** A horizontal black line denotes the end of an emergency procedure and the start of another. Multiple procedures may be found on the same card separated by a horizontal black line.
- **Fire Emergencies.** When dealing with a FIRE emergency, aircrews will check for positive signs of fire during the diagnosis. If unable, crews shall assume there is a fire and execute the appropriate emergency action steps.
- **Limitations.** Standard limitations tables have been added to the end of the emergency procedures and marked with a white tab identified as 'Limitations'."

5.2.2.2.5 Fixed wing.

The following information shall be provided verbatim:

Fixed Wing Verbatim Statement.

"The checklist contains normal procedures, detailed procedures, performance charts, and emergency procedures divided into tabbed sections of like information. Sections may contain one or more work packages of content to allow for easy updates of content. Page numbering is based on the content of the pages; normal procedure pages are numbered N1-1, N1-2, etc. Detailed procedure pages are numbered D1-1, D1-2, etc. Performance chart pages are numbered P1-1, P1-2, etc. Emergency procedure pages are numbered E1-1, E1-2, etc. Page references to detailed procedures are included at the end of the applicable step. A "RETURN TO:" feature is added at the end of applicable detailed procedures to aid in navigation back to the normal procedures, if required."

Explanation of references and symbols shall be provided to include explanation of the **<return>** function. A statement explaining that the CL is a condensed version of the procedures found in the -10 shall be provided.

A complete explanation of how to use the Emergency procedures shall be provided, similar to the following:

"Emergency Procedures.

This Section of the checklist provides procedures for coping with various types of emergencies or critical situations. Procedures for handling malfunctions or other abnormalities that are not of emergency nature, or involve a potential emergency that may be deferred, are included in this Section or in Abnormal Procedures. The emergency procedure section is identified by RED tabbed pages. An index of all emergency procedures can be found on the first page of this section.

IMMEDIATE ACTION EMERGENCY CHECKS.

Steps that shall be performed immediately in an emergency situation are underlined. These steps may be performed from memory without reference to the manual. When the situation permits, confirm steps accomplished and complete subsequent actions from the manual. The urgency of certain emergencies requires immediate action by the pilot. The most important single consideration is aircraft control. All procedures are subordinate to this requirement.

Reset MASTER CAUTION and MASTER WARNING after each malfunction to allow systems to respond to subsequent malfunctions.

DEFINITIONS OF LANDING TERMS.

Terms as defined in *(provide **<xref>** to the table of definitions)* shall be used. *(The definitions table in the corresponding operator's manual emergency procedures introduction work package shall be included in this section.)*

AFTER EMERGENCY ACTION.

After a malfunction has occurred, appropriate emergency actions have been taken, and the aircraft is on the ground, make an entry in the remarks section of DA Form 2408-13-1 describing the malfunction.

EMERGENCY EXITS AND EQUIPMENT.

Refer to *(Insert applicable operator's manual)* for description and locations of emergency exits and equipment.

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Abnormal Procedures.

This section of the checklist provides procedures for handling malfunctions of equipment, or other abnormalities, that are not of an emergency nature or involve a potential emergency that may be deferred.

Emergency and Abnormal Section Format.

Both sections will begin with an index of all the procedures found within the section. The index is found on the first page of the applicable section."

5.2.2.2.6 UAS.

The following information shall be provided verbatim:

UAS Verbatim Statement.

“The checklist contains normal procedures, detailed procedures, performance charts, and emergency procedures divided into tabbed sections of like information. Sections may contain one or more work packages of content to allow for easy updates of content. Page numbering is based on the content of the pages; normal procedure pages are numbered N1-1, N1-2, etc. Detailed procedure pages are numbered D1-1, D1-2, etc. Performance chart pages are numbered P1-1, P1-2, etc. Emergency procedure pages are numbered W1-1, W1-2, etc. or C1-1, C1-2, etc. or A1-1, A1-2, etc. Page references to detailed procedures are included at the end of the applicable step. A "RETURN TO:" feature is added at the end of applicable detailed procedures to aid in navigation back to the normal procedures, if required.”

Explanation of references and symbols shall be provided to include explanation of the **<return>** function. A statement explaining that the CL is a condensed version of the procedures found in the -10 shall be provided.

A complete explanation of how to use the Emergency procedures shall be provided, similar to the following:

“Emergency Procedures.

This Section of the checklist provides procedures for coping with various types of emergencies or critical situations. Procedures for handling malfunctions or other abnormalities that are not of emergency nature, or involve a potential emergency that may be deferred, are included in this Section or in Abnormal Procedures. The emergency procedure section is identified by RED tabbed pages. An index of all emergency procedures can be found on the first page of this section.

IMMEDIATE ACTION EMERGENCY CHECKS.

Steps that shall be performed immediately in an emergency situation are underlined. These steps may be performed from memory without reference to the manual. When the situation permits, confirm steps accomplished and complete subsequent actions from the manual. The urgency of certain emergencies requires immediate action by the pilot. The most important single consideration is aircraft control. All procedures are subordinate to this requirement.

Reset MASTER CAUTION and MASTER WARNING after each malfunction to allow systems to respond to subsequent malfunctions.

DEFINITIONS OF LANDING TERMS.

Terms as defined in *(provide **<xref>** to the table of definitions)* shall be used. *(The definitions table in the corresponding operator's manual emergency procedures introduction work package shall be included in this section.)*

AFTER EMERGENCY ACTION.

After a malfunction has occurred, appropriate emergency actions have been taken, and the aircraft is on the ground, make an entry in the remarks section of DA Form 2408-13-1 describing the malfunction.

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EMERGENCY EXITS AND EQUIPMENT.

Refer to *(Insert applicable operator's manual)* for description and locations of emergency exits and equipment.

Caution and Advisory Procedures.

This section of the checklist provides procedures for handling malfunctions of equipment, or other abnormalities, that are not of an emergency nature or involve a potential emergency that may be deferred.

Emergency, Caution, and Advisory Section Format.

All sections will begin with an index of all the procedures found within the section. The index is found on the first page of the applicable section."

5.2.2.2.7 Emergency response method.

The following shall appear verbatim on the last page of the introduction.

"Emergency Response Method.

The Emergency Response Method (using the mnemonic FADEC-F) increases crew situational awareness and enhances crew coordination during an emergency. FADEC-F is defined as:

F - Fly the aircraft. Gain or maintain a safe rotor speed, attitude, altitude, speed and heading.

A - Alert the crew to the problem. It is important to near simultaneously alert the crew to the emergency condition.

D - Diagnose the emergency. Conduct malfunction analysis and confirm diagnosis with the crew.

E - Execute emergency procedure steps IAW the FRCs. Immediate action (underlined) steps will be performed from memory. Actions and subsequent actions (non-underlined) steps will be performed using the FRCs.

C - Communicate the plan of action to the crew and supporting agencies (Air Traffic Control (ATC), wingman, etc.).

F - Fly the aircraft."

5.2.2.3 Required content for -MTF.

Aviation -MTF manuals shall contain a scope paragraph in the TM Structure section **<explantm>** of "About this Manual" that tells the user information of a general nature including the definition of an MTF, the purpose, and instructions specific to the checklist shall be provided. The purpose of the MTF manual shall be to provide complete instructions for performing an MTF for a specific model, type, and series aircraft. For the specific criteria which require a general or limited MTF, reference shall be made to TM 1-1500-328-23 and the applicable aviation field and sustainment maintenance manuals. Specific statements shall be included as detailed in 5.2.2.3.1 through 5.2.2.3.4.

5.2.2.3.1 Warning statement.

The following warning shall be included verbatim first under the "About this Manual".

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"**WARNING:** A maintenance test flight is an exceptionally demanding operation and requires a thorough flight readiness inspection (pre-flight). The flight readiness inspection is prescribed in *(enter TM number of applicable operator's manual)* (operator's manual) and shall be completed prior to each MTF. Emergency procedures are found in the applicable operator's manual (-10) and checklist (-CL) and are not duplicated in this publication. Prior to each maintenance test flight, the pilot shall contact maintenance/quality control personnel to determine the maintenance that has been performed. This manual should be used only by qualified maintenance test flight pilots as required by AR 95-1."

Maintenance Test Flight Structure

The maintenance test flight manual is divided into the structure detailed in TABLE XIII. These sections are divided by tabbed pages to aid the operator in navigating to specific content.

TABLE XIII. MTF structure

Section	Description
MTF Checklist	Checklist procedures are indicated by pages numbered as N1-1, N1-2, N2-1, etc.
Troubleshooting Guide	Troubleshooting guides follow the MTF Checklist procedures and are indicated by pages numbered as T1-1, T1-2, T2-1, etc.. If found in the 23&P, reference here and include no section in MTF.
Special / Detailed Procedures	Detailed Procedures are indicated by pages numbered as D1-1, D1-2, D2-1, etc.
Performance Charts	Performance Charts are indicated by pages numbered as P1-1, P1-2, etc.
Forms	Forms will be found immediately after charts and page numbering will be continued from the Charts section.
Foldouts	Foldouts appear at the rear of the book and may contain figures, charts, or forms to assist the operator.

Symbols Used in the MTF Manual

The following symbols (TABLE XIV) are used throughout this MTF manual (define all that apply, include any designators or configuration definitions):

TABLE XIV. MTF symbols

Symbol	Description
**	The double astericks symbol "**", which shall precede the step, shall indicate that the performance of the step is mandatory for all maintenance test flights.
T	The letter "T", which shall precede the step, shall indicate a task or step required by the operator's manual.

5.2.2.3.2 Required definitions.

The following statement shall be included verbatim:

"Maintenance test flight. A flight for which the primary mission is to determine airworthiness, i.e., that the airframe, power plant accessories and items of equipment are functioning in accordance with predetermined requirements in the intended operational environment."

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5.2.2.3.3 Required statements.

The following statements shall be included verbatim:

a. "This manual shall cover only MTFs of aircraft *(insert type, model, and series)* and in no way supersedes any information contained in *(enter TM number of applicable operator's manual)* or -CL, but is to be used in conjunction with the -10 or -CL. For the purpose of MTFs only, the -MTF manual shall satisfy all of the requirements of the -CL from Interior Check through Engine Shutdown."

b. "Crew requirements shall be as specified in TM 1-1500-328-23 and *(enter TM number of applicable checklist)*."

c. "The duration of a general or limited test flight shall be in accordance with the requirements of TM 1-1500-328-23."

5.2.2.3.4 Special instructions.

The following special items of interest shall be included:

a. Cargo and passengers shall be prohibited on MTFs.

b. Forms and records shall be checked prior to the MTF to determine what maintenance has been performed and the type of MTF required (i.e., general or limited).

c. The configuration of the aircraft shall be established prior to each MTF in order to determine performance parameters.

d. A thorough post test flight inspection shall be performed to the extent necessary to ensure that deficiencies that may have developed as a result of the MTF are detected.

e. When an MTF is required to ensure proper operation of a specific system(s), references shall be made to the applicable maintenance manuals for the limits of that system.

f. A check sheet shall be developed for recording the results of test flights. When a test flight is performed to determine if specific equipment or systems are operating property, completion of only that portion of the MTF check sheet applicable to the specific equipment or system being tested shall be required. Continuation sheets may be used when necessary. Items that prove to be unsatisfactory during the test flight and require corrective action shall be listed in the remarks block during flight and transferred to DA Form 2408-13-1 immediately after termination of the flight. The sheet shall be attached to the DA Form 2408-13-1 upon completion. After accumulation of two or more sheets, the data shall be reviewed to determine if trends are developing.

5.2.3 Rear matter <rear>.

Refer to MIL-STD-40051 for all requirements unless otherwise specified.

5.2.3.1 Foldout pages in MTF manuals.

Foldout pages for textural data shall not be used for maintenance test flight checklist TMs. However, for ease of use, graphs included in the checklist may be placed on a foldout page following the charts and forms figures.

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6 NOTES

This section contains information of a general or explanatory nature which may be helpful but is not mandatory.

6.1 Intended use.

MIL-STD-3083(AV) prescribes requirements applicable to technical publications for operator's manuals, checklists, and -MTF manuals for Army aircraft.

6.2 Acquisition requirements.

Acquisition documents should specify the following:

a. Title, number, and date of this standard.

b. Filled-out content selection matrix.

6.3 Tailoring guidance.

The acquiring activity in coordination with user representatives should tailor any required options offered herein in accordance with APPENDIX A.

6.4 Subject term (key word) listing.

The following terms are to be used to identify the MIL-STD-3083(AV) document during retrieval searches:

a. Advisories

b. Avionics

c. Cautions

d. Center-of-Gravity (CG)

e. Checklist, alternate

f. Checklist, standard

g. Condition heading

h. Cover

i. Data, graphical

j. Data, performance

k. Diagram, moment

l. Equipment, mission

m. Fires

n. Flight Reference Cards

o. Instruments, flight

p. Limits, operating

q. Number, publication

r. Page, warning

s. Procedures, emergency

t. Procedures, normal

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u. Statement, distribution

v. Symbol, designator

w. System, flight control

x. System, hydraulic

y. System, power train

z. System, utility

aa. Tabs

ab. Test flight, maintenance

ac. Through-flight

ad. Uncrewed

ae. Warnings

af. Weight and balance

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APPENDIX A

REQUIREMENTS MATRICES

A.1 Scope.

This appendix provides the technical content requirements for development and presentation of Aviation Operator's manuals. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

A.2 Applicable publications.

The applicable publications in Section 2 apply to this appendix.

A.3 Definitions.

The definitions in Section 3 apply to this appendix.

A.4 General requirements.

This section is not applicable to this appendix.

A.5 Detailed requirements.

The requirements in A.5.1 through A.5.5 shall be used to tailor the technical contents of an aviation operator manual and properly assign it a title. Intended use of the content selection matrices and acquisition requirements are also detailed in this appendix.

A.5.1 Tailoring requirements for technical manuals.

Tailoring of the technical content requirements contained in APPENDIX B through APPENDIX N is provided in the content matrix tables, TABLE A-II through TABLE A-VII. The tables list all applicable technical content requirements for the development of the following aviation operator TMs. Copies of the applicable tables will be completed and added as an attachment to the Document Summary List of the contract.

A.5.2 Publication titles.

All publications prepared in accordance with this standard shall have a title in accordance with TABLE A-I.

TABLE A-I. Publication type and title with associated context matrix table.

Publication Type	Publication Title	Applicable Matrix
Aircraft Operator’s Manual (-10)	Operator Manual for *Insert System*	TABLE A-II
Aircraft Operator’s Rotary Wing Check-list (-CL)	Operator Checklist Manual for *Insert System*	TABLE A-III
Aircraft Operator’s Fixed Wing Checklist (-CL)	Operator Checklist Manual for *Insert System*	TABLE A-IV
Aircraft Operator’s Uncrewed Aircraft System Checklist (-CL)	Operator Checklist Manual for *Insert System*	TABLE A-V

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TABLE A-I. Publication type and title with associated context matrix table. (continued)

Publication Type	Publication Title	Applicable Matrix
Alternate Operator's Checklist (-CL)	Operator Checklist Manual for *Insert System*	TABLE A-VI
Aircraft Operator's Maintenance Test Flight Manual (-MTF)	Maintenance Test Flight Manual for *Insert System*	TABLE A-VII

A.5.3 Technical content tables.

TABLE A-II through TABLE A-VII shall be used to tailor the content requirements based on the equipment and publication type. They serve as a checklist for the technical content requirements. The tables indicate which content is required by this standard and shall be included as indicated by "R," which content is conditionally required by this standard and shall be included as indicated by "CR," which content is prohibited by this standard and shall not be included as indicated by "P," and content that needs to be assessed by the acquiring activity as indicated with gray shading.

A.5.4 Intended use.

First, determine the types of TMs required for each acquisition and then duplicate the table(s) that contains the content requirements for those types of TMs. Indicate the types of TMs needed by filling in the blank after "Requirements Matrix for" at the top of each matrix.

a. For each type of TM selected, indicate in the empty shaded blocks the "TM" content desired by entering an "R" for "REQUIRED" content, an "NR" for content that is "NOT REQUIRED," or an "AR" for content that is "AS REQUIRED." All blocks for the selected TM type shall be filled in. Further guidance is provided below.

(1) For tables with multiple columns, allowing a choice of TM type (RW, FW, UAS), unused columns shall either be removed during the editing process or blacked out to indicate that manual type is not required. Rows in the matrix tables shall not be removed or added.

(2) Blocks that contain an "R" are content required by this standard and shall not be changed.

(3) Blocks that contain a "P" are content prohibited by this standard and shall not be changed.

(4) Information not included in a specific table is prohibited for that type of TM and shall not be included.

b. The blocks that are shaded are content items where a decision shall be made by the acquiring activity whether they are required to support the equipment. The blocks that are shaded shall be filled in with an "R," "NR," or "AR."

(1) If a decision on a shaded item cannot be made, the shaded item shall be marked with an "AR."

(2) When a decision has been made, the shaded item shall be marked with an "NR" or "R."

(3) Many shaded items identify a parent module or work package (e.g., Revision Summary) that when selected shall contain certain content. Shaded cells with a "CR" identify requirements that shall be followed when the parent item is selected. If a parent module is determined not to be needed and is marked with an "NR," then the children are also not required and should be marked as such. For example, the Revision Summary is optional (shaded) at the parent level. If the revision summary is in a publication, the Safety Summary section is required.

c. An "R" at the module level means the module is required. It does not mean everything below it shall be included. Individual items that can be included with the module carry their own marking ("R," "CR," "P," or "AR").

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A.5.5 Acquisition requirements.

The properly executed Technical Manual Content Selection Matrix table becomes contractually binding when it is made part of the contract, statement of work, or any other contractual instrument. The Content Selection Matrix table should be used to review the CDRL delivery to ensure accuracy of the deliverable. Reference DD Form 1423.

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TABLE A-II. Requirements matrix for aircraft operator's manual (-10)
_____.

TM CONTENT	RW	FW	UAS	MIL-STD-3083 (AV) Reference	Element Name
FRONT MATTER	R	R	R	5.2	**<av_paperfrnt>**
FRONT COVER	R	R	R	5.2	**<frntcover>**
WARNING SUMMARY				5.2	**<warnsum>**
CHANGE TRANSMITTAL PAGE				5.2	**<chgsheet>**
REVISION SUMMARY				5.2.1.1	**<revisionsummary>**
Safety Summary	CR	CR	CR	5.2.1.1.1	**<safesum>**
TABLE OF CONTENTS	R	R	R	5.2.1.2	**<contents>**
ABOUT THIS MANUAL	R	R	R	5.2.2	**<about>**
Change History	R	R	R	5.2.2.1.1	**<changedates>**
Reporting of Errors	R	R	R	5.2.2.1.2	**<reporting>**
Explanation of Warnings, Cautions, and Notes	R	R	R	5.2.2.1.3	**<explanwcn>**
Explanation of the Use of Shall, Will, Should, and May	R	R	R	5.2.2.1.4	**<explanswsm>**
Explanation of Change Symbols	R	R	R	5.2.2.1.5	**<explanchange>**
TM Structure	R	R	R	5.2.2.1.6	**<explantm>**
Applicability/Configurations/Symbology	R	R	R	5.2.2.1.7	**<appconfig>**
MODULE 1 - INTRODUCTION	R	R	R	APPENDIX B	**<avgim>**
AVIATION GENERAL INFORMATION WORK PACKAGE	R	R	R	B.5.1	**<av_ginfowp>**
Scope	R	R	R	B.5.1.2	**<scope>**
Army Aviation Safety Program	R	R	R	B.5.1.3	**<av_safety>**
Forms and Records	R	R	R	B.5.1.4	**<frr>**
Reporting Equipment Improvement Recommendations (EIR)	R	R	R	B.5.1.5	**<eir>**
Destruction of Army Material	R	R	R	B.5.1.6	**<destructmat>**
Nomenclature Cross Reference List				B.5.1.7	**<nomenreflist>**
ABBREVIATIONS, ACRONYMS, AND DEFINITIONS WORK PACKAGE	R	R	R	B.5.2	**<av_acdwp>**
List of Abbreviations/Acronyms	R	R	R	B.5.2.2	**<loa>**
Definitions List	R	R	R	B.5.2.3	**<deflist>**
MODULE 2 - AIRCRAFT AND SYSTEMS DESCRIPTION AND OPERATION	R	R	R	APPENDIX C	**<avdim>**
AIRCRAFT/AIR VEHICLE WORK PACKAGE	R	R	R	C.5.1	**<airvehiclewp>**

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TABLE A-II. Requirements matrix for aircraft operator's manual (-10) _____. (continued)

TM CONTENT	RW	FW	UAS	MIL-STD-3083 (AV) Reference	Element Name
GROUND SYSTEMS WORK PACKAGE			R	C.5.2	**<groundsystemswp>**
DATA LINK WORK PACKAGE			R	C.5.3	**<datalinkwp>**
EMERGENCY RECOVERY SYSTEM WORK PACKAGE				C.5.4	**<uasrecoverwp>**
EMERGENCY EQUIPMENT WORK PACKAGE	R	R	R	C.5.5	**<emersyswp>**
ENGINES AND RELATED SYSTEMS WORK PACKAGE	R	R	R	C.5.6	**<engsyswp>**
FUEL SYSTEM WORK PACKAGE	R	R	R	C.5.7	**<fuelsyswp>**
FLIGHT CONTROL SYSTEM WORK PACKAGE	R	R	R	C.5.8	**<fltcntrlsyswp>**
HYDRAULIC AND PNEUMATIC SYSTEMS WORK PACKAGE	R	R		C.5.9	**<hydrpneusyswp>**
POWER TRAIN SYSTEM WORK PACKAGE	R	R	R	C.5.10	**<pwrtrainwp>**
ROTORS OR PROPELLERS WORK PACKAGE	R	R	R	C.5.11	**<proprotorsyswp>**
UTILITY SYSTEMS WORK PACKAGE	R	R	R	C.5.12	**<utilitysyswp>**
HEATING, VENTILATION, COOLING, AND ENVIRONMENTAL CONTROL SYSTEMS WORK PACKAGE	R	R	R	C.5.13	**<hvacsyswp>**
ELECTRICAL POWER SUPPLY AND DISTRIBUTION SYSTEMS WORK PACKAGE	R	R	R	C.5.14	**<elecpwrsyswp>**
AUXILIARY POWER UNIT WORK PACKAGE	R	R	R	C.5.15	**<apusyswp>**
LIGHTING SYSTEMS WORK PACKAGE	R	R	R	C.5.16	**<lightsyswp>**
FLIGHT INSTRUMENTS WORK PACKAGE	R	R		C.5.17	**<fltinstrumentswp>**
SERVICING, PARKING, AND MOORING WORK PACKAGE	R	R	R	C.5.18	**<servparksyswp>**
APPROVED PRIMARY, ALTERNATE, AND EMERGENCY FUELS WORK PACKAGE	R	R	R	C.5.19	**<paefuelwp>**
GENERAL SYSTEMS WORK PACKAGE				C.5.20	**<sysgenwp>**
MODULE 3 - AVIONICS	R	R	R	APPENDIX D	**<avaim>**
AVIONICS INTRODUCTION WORK PACKAGE	R	R	R	D.5.2	**<avionicintrowp>**
COMMUNICATIONS WORK PACKAGE	R	R	R	D.5.3	**<commwp>**
Description	R	R	R	D.5.1.2.1	**<descrip>**
Controls and Functions	R	R	R	D.5.1.2.2	**<controls>**
Operation	R	R	R	D.5.1.2.3	**<operations>**

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TABLE A-II. Requirements matrix for aircraft operator's manual (-10) _____. (continued)

TM CONTENT	RW	FW	UAS	MIL-STD-3083 (AV) Reference	Element Name
Emergency Operations				D.5.1.2.4	**<emeroper>**
Power Source				D.5.1.2.5	**<power>**
NAVIGATION WORK PACKAGE	R	R	R	D.5.4	**<navwp>**
Description	R	R	R	D.5.1.2.1	**<descrip>**
Controls and Functions	R	R	R	D.5.1.2.2	**<controls>**
Operation	R	R	R	D.5.1.2.3	**<operations>**
Emergency Operations				D.5.1.2.4	**<emeroper>**
Power Source				D.5.1.2.5	**<power>**
TRANSPONDER AND RADAR WORK PACKAGE	R	R	R	D.5.5	**<tran_radarwp>**
Description	R	R	R	D.5.1.2.1	**<descrip>**
Controls and Functions	R	R	R	D.5.1.2.2	**<controls>**
Operation	R	R	R	D.5.1.2.3	**<operations>**
Emergency Operations				D.5.1.2.4	**<emeroper>**
Power Source				D.5.1.2.5	**<power>**
GENERAL AVIONICS EQUIPMENT WORK PACKAGE				D.5.6	**<avionicgenwp>**
Description	CR	CR	CR	D.5.1.2.1	**<descrip>**
Controls and Functions	CR	CR	CR	D.5.1.2.2	**<controls>**
Operation	CR	CR	CR	D.5.1.2.3	**<operations>**
Emergency Operations				D.5.1.2.4	**<emeroper>**
Power Source				D.5.1.2.5	**<power>**
MODULE 4 - MISSION EQUIPMENT	R	R	R	APPENDIX E	**<missn_opim>**
MISSION EQUIPMENT INTRODUCTION WORK PACKAGE	R	R	R	E.5.2	**<missneqp_introwp>**
NON-STANDARD MISSION AVIONICS EQUIPMENT WORK PACKAGE	R	R	R	E.5.3	**<nonstandwp>**
Description	R	R	R	E.5.1.2.1	**<descrip>**
Controls and Functions	R	R	R	E.5.1.2.2	**<controls>**
Operation	R	R	R	E.5.1.2.3	**<operations>**
Emergency Operations				E.5.1.2.4	**<emeroper>**
Power Source				E.5.1.2.5	**<power>**
ARMAMENT WORK PACKAGE				E.5.4	**<armamentwp>**

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TABLE A-II. Requirements matrix for aircraft operator's manual (-10) _____. (continued)

TM CONTENT	RW	FW	UAS	MIL-STD-3083 (AV) Reference	Element Name
Description	CR	CR	CR	E.5.1.2.1	**<descrip>**
Controls and Functions	CR	CR	CR	E.5.1.2.2	**<controls>**
Operation	CR	CR	CR	E.5.1.2.3	**<operations>**
Emergency Operations				E.5.1.2.4	**<emeroper>**
Power Source				E.5.1.2.5	**<power>**
CARGO HANDLING WORK PACKAGE				E.5.5	**<cargowp>**
Description	CR	CR	CR	E.5.1.2.1	**<descrip>**
Controls and Functions	CR	CR	CR	E.5.1.2.2	**<controls>**
Operation	CR	CR	CR	E.5.1.2.3	**<operations>**
Emergency Operations				E.5.1.2.4	**<emeroper>**
Power Source				E.5.1.2.5	**<power>**
PASSIVE DEFENSE WORK PACKAGE	R			E.5.6	**<passdefwp>**
Description	R	CR	CR	E.5.1.2.1	**<descrip>**
Controls and Functions	R	CR	CR	E.5.1.2.2	**<controls>**
Operation	R	CR	CR	E.5.1.2.3	**<operations>**
Emergency Operations				E.5.1.2.4	**<emeroper>**
Power Source				E.5.1.2.5	**<power>**
SYSTEM PAYLOAD WORK PACKAGE				E.5.7	**<syspayloadwp>**
Description	CR	CR	CR	E.5.1.2.1	**<descrip>**
Controls and Functions	CR	CR	CR	E.5.1.2.2	**<controls>**
Operation	CR	CR	CR	E.5.1.2.3	**<operations>**
Emergency Operations				E.5.1.2.4	**<emeroper>**
Power Source				E.5.1.2.5	**<power>**
GENERAL MISSION EQUIPMENT WORK PACKAGE				E.5.8	**<missngenwp>**
Description	CR	CR	CR	E.5.1.2.1	**<descrip>**
Controls and Functions	CR	CR	CR	E.5.1.2.2	**<controls>**
Operation	CR	CR	CR	E.5.1.2.3	**<operations>**
Emergency Operations				E.5.1.2.4	**<emeroper>**
Power Source				E.5.1.2.5	**<power>**
MODULE 5 - OPERATING LIMITS AND RESTRICTIONS	R	R	R	APPENDIX F	**<avlim>**

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TABLE A-II. Requirements matrix for aircraft operator's manual (-10) _____. (continued)

TM CONTENT	RW	FW	UAS	MIL-STD-3083 (AV) Reference	Element Name
LIMITATIONS AND RESTRICTIONS INTRODUCTION WORK PACKAGE	R	R	R	F.5.1	**<limit_introwp>**
SYSTEM LIMITS WORK PACKAGE	R	R	R	F.5.2	**<syslimitwp>**
POWER LIMITS WORK PACKAGE	R	R	R	F.5.3	**<pwrlimitwp>**
LOADING LIMITS WORK PACKAGE	R	R		F.5.4	**<loadlimitwp>**
MAXIMUM AND MINIMUM AIR SPEED LIMITS WORK PACKAGE	R	R	R	F.5.5	**<airspdlimitwp>**
MANEUVERING LIMITS WORK PACKAGE	R	R	R	F.5.6	**<maneuvlimitswp>**
ENVIRONMENTAL RESTRICTIONS WORK PACKAGE	R	R	R	F.5.7	**<envlimitwp>**
GENERAL LIMITS OR RESTRICTIONS WORK PACKAGE				F.5.8	**<genlimitwp>**
MODULE 6 - WEIGHT/BALANCE AND LOADING	R	R	R	APPENDIX G	**<avwim>**
WEIGHT/BALANCE AND LOADING INTRODUCTION WORK PACKAGE	R	R	R	G.5.1	**<wtbal_introwp>**
WEIGHT AND BALANCE WORK PACKAGE	R	R	R	G.5.2	**<wtbalwp>**
FUEL/OIL WORK PACKAGE	R	R	R	G.5.3	**<fuelwtbalwp>**
PERSONNEL WORK PACKAGE	R	R	P	G.5.4	**<perswtbalwp>**
MISSION EQUIPMENT WORK PACKAGE	R	R	R	G.5.5	**<missneqpwtbalwp>**
CARGO LOADING WORK PACKAGE		R		G.5.6	**<cargowtbalwp>**
CENTER-OF-GRAVITY WORK PACKAGE	R	R	R	G.5.7	**<cgwtbalwp>**
GENERAL WEIGHT/BALANCE AND LOADING WORK PACKAGE				G.5.8	**<wtbalgenwp>**
MODULE 7 - PERFORMANCE DATA	R	R	R	APPENDIX H	**<avpim>**
PERFORMANCE DATA INTRODUCTION WORK PACKAGE	R	R	R	H.5.2	**<chart_introwp>**
Aircraft Specific Performance Data Charts Work Package Categories	R	R	R	H.5.1.1	**<rwcharts>** or **<fwcharts>** or **<uascharts>**
FUEL FLOW CHART WORK PACKAGE	R	P	P	H.5.3.1	**<fuelflowchartwp>**
MAXIMUM TORQUE AVAILABLE CHART WORK PACKAGE	R	P	P	H.5.3.2	**<maxtrqchartwp>**
HOVER CHART WORK PACKAGE	R	P	P	H.5.3.3	**<hoverchartwp>**
CRITICAL DATA CHART WORK PACKAGE	R	P	P	H.5.3.4	**<critdatachartwp>**
TAKEOFF CHART WORK PACKAGE	R	P	P	H.5.3.5	**<takeoffchartwp>**

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TABLE A-II. Requirements matrix for aircraft operator's manual (-10) _____. (continued)

TM CONTENT	RW	FW	UAS	MIL-STD-3083 (AV) Reference	Element Name
DRAG CHART WORK PACKAGE	R	P	P	H.5.3.6	**<dragchartwp>**
CRUISE CHART WORK PACKAGE	R	P	P	H.5.3.7	**<cruisechartwp>**
CLIMB-DESCENT CHART WORK PACKAGE	R	P	P	H.5.3.8	**<clmbdeschartwp>**
AIRSPEED CALIBRATION CHART WORK PACKAGE	R	P	P	H.5.3.9	**<airspdcalchartwp>**
OPTIMUM CRUISE CHART WORK PACKAGE		P	P	H.5.3.10	**<opcruisechartwp>**
CROSSWIND CHART WORK PACKAGE	P	R	R	H.5.4.1	**<crswindchartwp>**
AIRSPEED CALIBRATION CHART WORK PACKAGE	P	R		H.5.4.2	**<airspdcalchartwp>**
ALTIMETER CORRECTIONS CHART WORK PACKAGE	P	R		H.5.4.3	**<altimeterchartwp>**
IDLE FUEL FLOW CHART WORK PACKAGE	P		R	H.5.4.4	**<fuelflowchartwp>**
TORQUE AVAILABLE FOR TAKEOFF CHART WORK PACKAGE	P	R		H.5.4.5	**<trqavailchartwp>**
TAKEOFF CHART WORK PACKAGE	P	R	R	H.5.4.6	**<takeoffchartwp>**
TAKEOFF AIRSPEED CHART WORK PACKAGE	P	R		H.5.4.7	**<takeoffairspd-chartwp>**
ACCELERATION CHECK DISTANCE CHART WORK PACKAGE	P			H.5.4.8	**<accelchkdischartwp>**
ACCELERATE-STOP DISTANCE CHART WORK PACKAGE	P	R		H.5.4.9	**<accelstpdistchartwp>**
ACCELERATE AFTER TAKEOFF CHART WORK PACKAGE	P	R		H.5.4.10	**<accelaftertkoff-chartwp>**
MINIMUM SINGLE ENGINE CONTROL AIR-SPEED CHART WORK PACKAGE	P			H.5.4.11	**<minsglengchartwp>**
SINGLE ENGINE CLIMB CHART WORK PACKAGE	P			H.5.4.12	**<singengclmb-chartwp>**
SERVICE CEILING, ONE ENGINE INOPERATIVE CHART WORK PACKAGE	P	R		H.5.4.13	**<servoneinopwp>**
RATE OF CLIMB WORK PACKAGE	P	R		H.5.4.14	**<rateclimbwp>**
CRUISE CLIMB CHART WORK PACKAGE	P	R	R	H.5.4.15	**<cruiseclmbchartwp>**
DRAG CHART WORK PACKAGE	P			H.5.4.16	**<dragchartwp>**
CRUISE CHART WORK PACKAGE	P	R	R	H.5.4.17	**<cruisechartwp>**
CLIMB/DESCENT CHART WORK PACKAGE	P	R	R	H.5.4.18	**<clmbdeschartwp>**
APPROACH SPEED CHART WORK PACKAGE	P	R	R	H.5.4.19	**<appspdchartwp>**

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TABLE A-II. Requirements matrix for aircraft operator's manual (-10) _____. (continued)

TM CONTENT	RW	FW	UAS	MIL-STD-3083 (AV) Reference	Element Name
LANDING CHART WORK PACKAGE	P	R	R	H.5.4.20	**<landchartwp>**
OPTIMUM CRUISE CHART WORK PACKAGE	P	R	R	H.5.4.21	**<optcruisechartwp>**
STALL CHART WORK PACKAGE	P	R		H.5.4.22	**<stallchartwp>**
ADDITIONAL PERFORMANCE CHART WORK PACKAGE				H.5.4.23	**<genchartwp>**
MODULE 8 - NORMAL PROCEDURES	R	R	R	APPENDIX I	**<avnim>**
NORMAL PROCEDURES INTRODUCTION WORK PACKAGE	R	R	R	I.5.2	**<norm_introwp>**
GROUND EQUIPMENT PROCEDURES WORK PACKAGE			R	I.5.3	**<groundnormwp>**
PREFLIGHT PROCEDURES WORK PACKAGE	R	R	R	I.5.4	**<preflightnormwp>**
FLIGHT PROCEDURES WORK PACKAGE	R	R	R	I.5.5	**<flightnormwp>**
POST-FLIGHT PROCEDURES WORK PACKAGE	R	R	R	I.5.6	**<postflightnormwp>**
INSTRUMENT FLIGHT WORK PACKAGE	R	R		I.5.7	**<instrumentwp>**
TRANSFER PROCEDURES WORK PACKAGE			R	I.5.8	**<xfer_normwp>**
MISSION EQUIPMENT PROCEDURES WORK PACKAGE	P	P	R	I.5.9	**<mis_wp>**
COMMUNICATIONS EQUIPMENT WORK PACKAGE	R	R	R	I.5.10	**<commequipwp>**
FLIGHT CHARACTERISTICS WORK PACKAGE	R	R		I.5.11	**<flt_charwp>**
COLD WEATHER OPERATIONS WORK PACKAGE	R	R	R	I.5.12	**<adverse_envwp>**
DESERT AND HOT WEATHER OPERATIONS WORK PACKAGE	R	R	R	I.5.12	**<adverse_envwp>**
TURBULENCE AND THUNDERSTORM OPERATIONS WORK PACKAGE	R	R	R	I.5.12	**<adverse_envwp>**
RAIN OPERATIONS WORK PACKAGE	R	R	R	I.5.12	**<adverse_envwp>**
GENERAL ADVERSE ENVIRONMENT CONDITIONS WORK PACKAGE				I.5.12	**<adverse_envwp>**
GENERAL NORMAL PROCEDURES WORK PACKAGE				I.5.13	**<gennormwp>**
MODULE 9 - EMERGENCY PROCEDURES	R	R	R	APPENDIX J	**<aveim>**
EMERGENCY INTRODUCTION WORK PACKAGE	R	R	R	J.5.2	**<emergency_introwp>**

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TABLE A-II. Requirements matrix for aircraft operator's manual (-10) _____. (continued)

TM CONTENT	RW	FW	UAS	MIL-STD-3083 (AV) Reference	Element Name
ENGINE WORK PACKAGE	R	R	R	J.5.3	**<emenginewp>**
PROPELLER/ROTOR WORK PACKAGE	R	R	R	J.5.4	**<emproprotwp>**
FIRE WORK PACKAGE	R	R	R	J.5.5	**<emfirewp>**
FUEL WORK PACKAGE	R	R	R	J.5.6	**<emfuelwp>**
ELECTRICAL WORK PACKAGE	R	R	R	J.5.7	**<emelectricwp>**
HYDRAULIC WORK PACKAGE	R	R		J.5.8	**<emhydraulicwp>**
LANDING AND DITCHING WORK PACKAGE *(for UAS: Takeoff, Landing and Ditching Work Package)*	R	R	R	J.5.9	**<emlandwp>**
FLIGHT CONTROLS WORK PACKAGE	R	R	R	J.5.10	**<emfligcontwp>**
BAILOUT/EJECTION WORK PACKAGE	R	R	P	J.5.11	**<embailejectwp>**
MISSION EQUIPMENT WORK PACKAGE				J.5.12	**<emmissequipwp>**
GROUND CONTROL SUPPORT EQUIPMENT MALFUNCTIONS WORK PACKAGE	P	P	R	J.5.13	**<emshelterwp>**
OTHER EMERGENCY PROCEDURES WORK PACKAGE				J.5.14	**<emgenwp>**
MODULE 10 - SUPPORTING INFORMATION	R	R	R	APPENDIX K	**<avsim>**
REFERENCE WORK PACKAGE	R	R	R	K.5.2	**<refwp>**
COCKPIT INDEX WORK PACKAGE				K.5.3	**<cockpitwp>**
ADDITIONAL SUPPORTING WORK PACKAGES				K.5.4	**<genwp>**
REAR MATTER	R	R	R	5.2.3	**<rear>**
DA FORM 2028	R	R	R	5.2.3	**<da2028>**
AUTHENTICATION PAGE	R	R	R	5.2.3	**<authent>**
FOLDOUT PAGES				5.2.3	**<foldsect>**
BACK COVER	R	R	R	5.2.3	**<back>**

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TABLE A-III. Requirements matrix for operator's rotary wing checklist (-CL)

TM CONTENT	RW	MIL-STD-3083(AV) Reference	Element Name
VOLUME 1: OPERATOR'S NORMAL AND DETAILED PROCEDURES	R	L.5.1	**<clvolume1>**
VOLUME 1 FRONT MATTER	R	5.2	**<av_paperfrnt>**
FRONT COVER	R	5.2	**<frntcover>**
CHANGE TRANSMITTAL PAGE		5.2	**<chgsheet>**
REVISION SUMMARY		5.2.1.1	**<revisionsummary>**
Safety Summary	CR	5.2.1.1.1	**<safesum>**
ABOUT THIS MANUAL	R	5.2.2	**<about>**
Change History	R	5.2.2.1.1	**<changedates>**
Reporting of Errors	R	5.2.2.1.2	**<reporting>**
TM Structure	R	5.2.2.1.6	**<explantm>**
Applicability/Configurations/Symbology	R	5.2.2.1.7	**<appconfig>**
NORMAL PROCEDURE SECTION *(Normal sections will be developed into Normal work packages as determined by the acquiring activity.)*	R	L.5.3.2.1 L.5.3.2.2	**<normsect>** **<normalwp>**
Normal Procedures	R	L.5.3.2.3	**<normaltsk>**
DETAILED PROCEDURE SECTION *(Detailed sections will be developed into Detailed work packages as determined by the acquiring activity.)*		L.5.3.3.1 L.5.3.3.2	**<detailsect>** **<detailwp>**
Detailed Procedures	CR	L.5.3.3.3	**<detailtsk>**
PERFORMANCE DATA CHARTS SECTION		L.5.3.4.1	**<perfdatasect>**
CHARTS WORK PACKAGE	CR	L.5.3.4.2	**<perfdatawp>**
VOLUME 1 REAR MATTER	R	L.5.3.7	**<rear>**
EMAIL 2028	R	5.2.3	**<da2028>**
AUTHENTICATION	R	5.2.3	**<authent>**
FOLDOUTS		5.2.3	**<foldsect>**
BACK COVER	R	5.2.3	**<back>**
VOLUME 2: OPERATOR'S EMERGENCY PROCEDURES AND LIMITATIONS	R	L.5.1	**<clvolume2>**
VOLUME 2 FRONT MATTER	R	5.2	**<av_paperfrnt>**
FRONT COVER	R	5.2	**<frntcover>**
CHANGE TRANSMITTAL PAGE		5.2	**<chgsheet>**

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TABLE A-III. Requirements matrix for operator's rotary wing checklist (-CL) _____. (continued)

TM CONTENT	RW	MIL-STD-3083(AV) Reference	Element Name
REVISION SUMMARY		5.2.1.1	**<revisionsummary>**
Safety Summary	CR	5.2.1.1.1	**<safesum>**
ABOUT THIS MANUAL	R	5.2.2	**<about>**
Change History	R	5.2.2.1.1	**<changedates>**
Reporting of Errors	R	5.2.2.1.2	**<reporting>**
TM Structure	R	5.2.2.1.6	**<explantm>**
Applicability/Configurations/Symbology	R	5.2.2.1.7	**<appconfig>**
WARNINGS	R	L.5.4.4.1	**<warnsect>**
Introduction	R	L.5.4.1	**<intro>**
Emergency Tasks	R	L.5.4.2	**<emertsk>**
CAUTIONS	R	L.5.5.4.4	**<cautsect>**
Introduction	R	L.5.4.1	**<intro>**
Emergency Tasks	R	L.5.4.2	**<emertsk>**
ADVISORIES		L.5.4.4.3	**<advssect>**
Introduction	CR	L.5.4.1	**<intro>**
Emergency Tasks	CR	L.5.4.2	**<emertsk>**
MISSION EQUIPMENT		L.5.4.4.4	**<missnsect>**
Introduction	CR	L.5.4.1	**<intro>**
Emergency Tasks	CR	L.5.4.2	**<emertsk>**
LIMITATIONS		L.5.3.6	**<limitsect>**
Introduction	CR	L.5.4.1	**<intro>**
Limitations Table	CR	L.5.3.6	**<table>**
VOLUME 2 REAR MATTER	R	L.5.3.7	**<rear>**
EMAIL 2028	R	5.2.3	**<da2028>**
AUTHENTICATION	R	5.2.3	**<authent>**
FOLDOUTS		5.2.3	**<foldsect>**
BACK COVER	R	5.2.3	**<back>**

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TABLE A-IV. Requirements matrix for operator's fixed wing checklist (-CL)

TM CONTENT		MIL-STD-3083(AV) Reference	Element Name
FRONT MATTER	R	L.5.3.1	**<av_paperfrnt>**
FRONT COVER	R	5.2	**<frntcover>**
CHANGE TRANSMITTAL PAGE		5.2	**<chgsheet>**
REVISION SUMMARY		5.2.1.1	**<revisionsummary>**
Safety Summary	CR	5.2.1.1.1	**<safesum>**
ABOUT THIS MANUAL	R	5.2.2	**<about>**
Change History	R	5.2.2.1.1	**<changedates>**
Reporting of Errors	R	5.2.2.1.2	**<reporting>**
TM Structure	R	5.2.2.1.6	**<explantm>**
Applicability/Configurations/Symbology	R	5.2.2.1.7	**<appconfig>**
NORMAL PROCEDURE INTRODUCTION SECTION *(Normal sections will be developed into Normal work packages as determined by the acquiring activity.)*	R	L.5.3.2.1 L.5.3.2.2	**<normsect>** **<normalwp>**
Operating procedures and maneuvers	R	L.5.3.2.3	**<normaltsk>**
Mission Planning	R	L.5.3.2.3	**<normaltsk>**
Crew and Passenger Briefing	R	L.5.3.2.3	**<normaltsk>**
Aviation Life Support Equipment		L.5.3.2.3	**<normaltsk>**
Departure and Descent/arrival briefing		L.5.3.2.3	**<normaltsk>**
Additional Information		L.5.3.2.3	**<normaltsk>**
NORMAL PROCEDURE PREFLIGHT PROCEDURES SECTION *(Normal sections will be developed into Normal work packages as determined by the acquiring activity.)*	R	L.5.3.2.1 L.5.3.2.2	**<normsect>** **<normalwp>**
Preflight Check	R	L.5.3.2.3	**<normaltsk>**
Before exterior check	R	L.5.3.2.3	**<normaltsk>**
Exterior check	R	L.5.3.2.3	**<normaltsk>**
Interior check	R	L.5.3.2.3	**<normaltsk>**
Crew/passenger briefing check	R	L.5.3.2.3	**<normaltsk>**
NORMAL PROCEDURE GROUND PROCEDURES SECTION	R	L.5.3.2.1	**<normsect>**

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TABLE A-IV. Requirements matrix for operator's fixed wing checklist (-CL) _____. (continued)

TM CONTENT		MIL-STD-3083(AV) Reference	Element Name
(Normal sections will be developed into Normal work packages as determined by the acquiring activity.)		L.5.3.2.2	**<normalwp>**
Before Starting Engine(s)	R	L.5.3.2.3	**<normaltsk>**
Starting Engine(s)	R	L.5.3.2.3	**<normaltsk>**
Engine Ground operations	R	L.5.3.2.3	**<normaltsk>**
Before Taxiing	R	L.5.3.2.3	**<normaltsk>**
Taxiing	R	L.5.3.2.3	**<normaltsk>**
Engine Runup	R	L.5.3.2.3	**<normaltsk>**
Before takeoff	R	L.5.3.2.3	**<normaltsk>**
Lineup Check	R	L.5.3.2.3	**<normaltsk>**
NORMAL PROCEDURE FLIGHT PROCEDURES SECTION *(Normal sections will be developed into Normal work packages as determined by the acquiring activity.)*	R	L.5.3.2.1 L.5.3.2.2	**<normsect>** **<normalwp>**
Takeoff	R	L.5.3.2.3	**<normaltsk>**
After Takeoff	R	L.5.3.2.3	**<normaltsk>**
Climb	R	L.5.3.2.3	**<normaltsk>**
Cruise	R	L.5.3.2.3	**<normaltsk>**
Descent/Arrival	R	L.5.3.2.3	**<normaltsk>**
Before Landing	R	L.5.3.2.3	**<normaltsk>**
Landing	R	L.5.3.2.3	**<normaltsk>**
Touch and go landing/go-around	R	L.5.3.2.3	**<normaltsk>**
NORMAL PROCEDURE POST FLIGHT PROCEDURES SECTION *(Normal sections will be developed into Normal work packages as determined by the acquiring activity.)*	R	L.5.3.2.1 L.5.3.2.2	**<normsect>** **<normalwp>**
After Landing	R	L.5.3.2.3	**<normaltsk>**
Engine Shutdown	R	L.5.3.2.3	**<normaltsk>**
Before Leaving Aircraft	R	L.5.3.2.3	**<normaltsk>**
DETAILED PROCEDURES SECTION		L.5.3.3.1	**<detailsect>**

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TABLE A-IV. Requirements matrix for operator's fixed wing checklist (-CL) _____. (continued)

TM CONTENT		MIL-STD-3083(AV) Reference	Element Name
(Detailed sections will be developed into Detailed work packages as determined by the acquiring activity.)		L.5.3.3.2	**<detailwp>**
Detailed Procedures	CR	L.5.3.3.3	**<detailtsk>**
FLIGHT PLANNING/PERFORMANCE DATA SECTION *(Performance data sections will be developed into Performance Data work packages as determined by the acquiring activity.)*		L.5.3.4.1 L.5.3.4.2	**<perfdatasect>** **<perfdatawp>**
Charts	CR	L.5.3.4.2	**<chart>**
EMERGENCY PROCEDURES SECTION: ENGINE *(Emergency sections will be developed into Emergency work packages as determined by the acquiring activity.)*	R	L.5.5.2 L.5.3.3.2	**<warnsect>** **<emergwp>**
Emergency Tasks	R	L.5.5.2	**<fwuas_emertsk>**
EMERGENCY PROCEDURES SECTION: PROPELLER *(Emergency sections will be developed into Emergency work packages as determined by the acquiring activity.)*	R	L.5.5.2 L.5.3.3.2	**<warnsect>** **<emergwp>**
Emergency Tasks	R	L.5.5.2	**<fwuas_emertsk>**
EMERGENCY PROCEDURES SECTION: FIRE *(Emergency sections will be developed into Emergency work packages as determined by the acquiring activity.)*	R	L.5.5.2 L.5.3.3.2	**<warnsect>** **<emergwp>**
Emergency Tasks	R	L.5.5.2	**<fwuas_emertsk>**
EMERGENCY PROCEDURES SECTION: FUEL *(Emergency sections will be developed into Emergency work packages as determined by the acquiring activity.)*	R	L.5.5.2 L.5.3.3.2	**<warnsect>** **<emergwp>**
Emergency Tasks	R	L.5.5.2	**<fwuas_emertsk>**
EMERGENCY PROCEDURES SECTION: ELECTRICAL *(Emergency sections will be developed into Emergency work packages as determined by the acquiring activity.)*	R	L.5.5.2 L.5.3.3.2	**<warnsect>** **<emergwp>**

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TABLE A-IV. Requirements matrix for operator's fixed wing checklist (-CL) _____. (continued)

TM CONTENT		MIL-STD-3083(AV) Reference	Element Name
Emergency Tasks	R	L.5.5.2	**<fwuas_emertsk>**
EMERGENCY PROCEDURES SECTION: HYDRAULIC *(Emergency sections will be developed into Emergency work packages as determined by the acquiring activity.)*	R	L.5.5.2 L.5.3.3.2	**<warnsect>** **<emergwp>**
Emergency Tasks	R	L.5.5.2	**<fwuas_emertsk>**
EMERGENCY PROCEDURES SECTION: LANDING AND DITCHING *(Emergency sections will be developed into Emergency work packages as determined by the acquiring activity.)*	R	L.5.5.2 L.5.3.3.2	**<warnsect>** **<emergwp>**
Emergency Tasks	R	L.5.5.2	**<fwuas_emertsk>**
EMERGENCY PROCEDURES SECTION: FLIGHT CONTROLS *(Emergency sections will be developed into Emergency work packages as determined by the acquiring activity.)*	R	L.5.5.2 L.5.3.3.2	**<warnsect>** **<emergwp>**
Emergency Tasks	R	L.5.5.2	**<fwuas_emertsk>**
EMERGENCY PROCEDURES SECTION: MISSION EQUIPMENT *(Emergency sections will be developed into Emergency work packages as determined by the acquiring activity.)*	R	L.5.5.2 L.5.3.3.2	**<warnsect>** **<emergwp>**
Emergency Tasks	R	L.5.5.2	**<fwuas_emertsk>**
ABNORMAL PROCEDURES SECTION: ENGINE *(Abnormal sections will be developed into Emergency work packages as determined by the acquiring activity.)*	R	L.5.5.2 L.5.3.3.2	**<abnormsect>** **<emergwp>**
Emergency Tasks	R	L.5.5.2	**<fwuas_emertsk>**
ABNORMAL PROCEDURES SECTION: PROPELLER *(Abnormal sections will be developed into Emergency work packages as determined by the acquiring activity.)*	R	L.5.5.2 L.5.3.3.2	**<abnormsect>** **<emergwp>**
Emergency Tasks	R	L.5.5.2	**<fwuas_emertsk>**

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TABLE A-IV. Requirements matrix for operator's fixed wing checklist (-CL) _____. (continued)

TM CONTENT		MIL-STD-3083(AV) Reference	Element Name
ABNORMAL PROCEDURES SECTION: FIRE *(Abnormal sections will be developed into Emergency work packages as determined by the acquiring activity.)*	R	L.5.5.2 L.5.3.3.2	**<abnormsect>** **<emergwp>**
Emergency Tasks	R	L.5.5.2	**<fwuas_emertsk>**
ABNORMAL PROCEDURES SECTION: FUEL *(Abnormal sections will be developed into Emergency work packages as determined by the acquiring activity.)*	R	L.5.5.2 L.5.3.3.2	**<abnormsect>** **<emergwp>**
Emergency Tasks	R	L.5.5.2	**<fwuas_emertsk>**
ABNORMAL PROCEDURES SECTION: ELECTRICAL *(Abnormal sections will be developed into Emergency work packages as determined by the acquiring activity.)*	R	L.5.5.2 L.5.3.3.2	**<abnormsect>** **<emergwp>**
Emergency Tasks	R	L.5.5.2	**<fwuas_emertsk>**
ABNORMAL PROCEDURES SECTION: HYDRAULIC *(Abnormal sections will be developed into Emergency work packages as determined by the acquiring activity.)*	R	L.5.5.2 L.5.3.3.2	**<abnormsect>** **<emergwp>**
Emergency Tasks	R	L.5.5.2	**<fwuas_emertsk>**
ABNORMAL PROCEDURES SECTION: LANDING AND DITCHING *(Abnormal sections will be developed into Emergency work packages as determined by the acquiring activity.)*	R	L.5.5.2 L.5.3.3.2	**<abnormsect>** **<emergwp>**
Emergency Tasks	R	L.5.5.2	**<fwuas_emertsk>**
ABNORMAL PROCEDURES SECTION: FLIGHT CONTROLS *(Abnormal sections will be developed into Emergency work packages as determined by the acquiring activity.)*	R	L.5.5.2 L.5.3.3.2	**<abnormsect>** **<emergwp>**
Emergency Tasks	R	L.5.5.2	**<fwuas_emertsk>**
ABNORMAL PROCEDURES SECTION: MISSION EQUIPMENT	R	L.5.5.2	**<abnormsect>**

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TABLE A-IV. Requirements matrix for operator's fixed wing checklist (-CL) _____. (continued)

TM CONTENT		MIL-STD-3083(AV) Reference	Element Name
(Abnormal sections will be developed into Emergency work packages as determined by the acquiring activity.)		L.5.3.3.2	**<emergwp>**
Emergency Tasks	R	L.5.5.2	**<fwuas_emertsk>**
ADVISORIES *(Emergency sections will be developed into Emergency work packages as determined by the acquiring activity.)*		L.5.5.4.5 L.5.3.3.2	**<advssect>** **<emergwp>**
Emergency Tasks	CR	L.5.5.2	**<fwuas_emertsk>**
MISSION EQUIPMENT *(Emergency sections will be developed into Emergency work packages as determined by the acquiring activity.)*		L.5.5.4.6 L.5.3.3.2	**<missnsect>** **<emergwp>**
Emergency Tasks	CR	L.5.5.2	**<fwuas_emertsk>**
LIMITATIONS		L.5.3.6	**<limitssect>**
REAR MATTER	R	L.5.3.7	**<rear>**
ANNUNCIATOR PANEL	R	L.5.3.7.1	**<annunciator>**
EMAIL 2028	R	L.5.3.7	**<da2028>**
AUTHENTICATION PAGE	R	5.2.3	**<authent>**
FOLDOUTS		L.5.3.7	**<foldsect>**
BACK COVER	R	5.2.3	**<back>**

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TABLE A-V. Requirements matrix for operator's uncrewed aircraft systems checklist (-CL) _____.

TM CONTENT		MIL-STD-3083(AV) Reference	Element Name
FRONT MATTER	R	L.5.3.1	**<av_paperfrnt>**
FRONT COVER	R	5.2	**<frntcover>**
CHANGE TRANSMITTAL PAGE		5.2	**<chgsheet>**
REVISION SUMMARY		5.2.1.1	**<revisionsummary>**
Safety Summary	CR	5.2.1.1.1	**<safesum>**
ABOUT THIS MANUAL	R	5.2.2	**<about>**
Change History	R	5.2.2.1.1	**<changedates>**
Reporting of Errors	R	5.2.2.1.2	**<reporting>**
TM Structure	R	5.2.2.1.6	**<explantm>**
Applicability/Configurations/Symbology	R	5.2.2.1.7	**<appconfig>**
NORMAL PROCEDURE INTRODUCTION SECTION *(Normal sections will be developed into Normal work packages as determined by the acquiring activity.)*	R	L.5.3.2.1 L.5.3.2.2	**<normsect>** **<normalwp>**
Normal Procedures	R	L.5.3.2.3	**<normaltsk>**
DETAILED PROCEDURES SECTION *(Detailed sections will be developed into Detailed work packages as determined by the acquiring activity.)*		L.5.3.3.1 L.5.3.3.2	**<detailsect>** **<detailwp>**
Detailed Procedures	CR	L.5.3.3.3	**<detailtsk>**
PERFORMANCE DATA CHARTS SECTION		L.5.3.4.1	**<perfdatasect>**
CHARTS WORK PACKAGE	CR	L.5.3.4.2	**<perfdatawp>**
WARNINGS *(Warning sections will be developed into Warning work packages as determined by the acquiring activity.)*	R	L.5.5.4.3 L.5.5.1	**<warnsect>** **<emergwp>**
Emergency Tasks	R	L.5.5.2	**<fwuas_emertsk>**
CAUTIONS *(Caution sections will be developed into Caution work packages as determined by the acquiring activity.)*	R	L.5.5.4.4 L.5.5.1	**<cautsect>** **<emergwp>**
Emergency Tasks	R	L.5.5.2	**<fwuas_emertsk>**
ADVISORIES		L.5.5.4.5	**<advssect>**

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TABLE A-V. Requirements matrix for operator's uncrewed aircraft systems checklist (-CL) _____. (continued)

TM CONTENT		MIL-STD-3083(AV) Reference	Element Name
(Advisories sections will be developed into Advisory work packages as determined by the acquiring activity.)		L.5.5.1	**<emergwp>**
Emergency Tasks	CR	L.5.5.2	**<fwuas_emertsk>**
MISSION EQUIPMENT *(Mission Equipment sections will be developed into Mission Equipment work packages as determined by the acquiring activity.)*		L.5.5.4.6 L.5.5.1	**<missnsect>** **<emergwp>**
Emergency Tasks	CR	L.5.5.2	**<fwuas_emertsk>**
LIMITATIONS		L.5.3.6	**<limitssect>**
REAR MATTER	R	L.5.3.7	**<rear>**
EMAIL 2028	R	L.5.3.7	**<da2028>**
AUTHENTICATION PAGE	R	5.2.3	**<authent>**
FOLDOUTS		L.5.3.7	**<foldsect>**
BACK COVER	R	5.2.3	**<back>**

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TABLE A-VI. Requirements matrix for alternate operator's checklist (-CL) _____.

TM CONTENT		MIL-STD-3083(AV) Reference	Element Name
ALTERNATE OPERATOR'S CHECKLIST		L.6	**<alternatecl>**
FRONT MATTER	R	L.6.1	**<alternatefront>**
General Information and Scope	R	L.6.2	**<geninfo>**
NORMAL PROCEDURES	R	L.6.3	**<normal>**
AUTHENTICATION BLOCK	R	L.6	**<authent>**
EMERGENCY PROCEDURES	R	L.6.4	**<emergency>**
OTHER PROCEDURES		L.6.5	**<otherprocs>**
PERFORMANCE CHARTS		L.6.6	**<perfdata>**

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TABLE A-VII. Requirements matrix for operator's maintenance test flight manual (-MTF) _____.

TM CONTENT	RW	FW	MIL-STD-3083(AV) Reference	Element Name
FRONT MATTER	R	R	L.5.3.1 5.2.1	**<av_paperfrnt>**
FRONT COVER	R	R	5.2	**<frntcover>**
CHANGE TRANSMITTAL PAGE			5.2	**<chgsheet>**
REVISION SUMMARY			5.2.1.1	**<revisionsummary>**
Safety Summary	CR	CR	5.2.1.1.1	**<safesum>**
TABLE OF CONTENTS	R	R	5.2.1.2	**<contents>**
ABOUT THIS MANUAL	R	R	5.2.2	**<about>**
Change History	R	R	5.2.2.1.1	**<changedates>**
Reporting of Errors	R	R	5.2.2.1.2	**<reporting>**
TM Structure	R	R	5.2.2.1.6	**<explantm>**
Applicability/Configurations/Symbology	R	R	5.2.2.1.7	**<appconfig>**
MAINTENANCE TEST FLIGHT CHECK-LIST SECTION *(Maintenance test flight checklist sections will be developed into Normal work packages as determined by the acquiring activity.)*	R	R	M.5.3 M.5.3.1	**<normsect>** **<normalwp>**
Normal Procedures	R	R	M.5.3.2	**<normaltsk>**
TROUBLESHOOTING GUIDES SECTION *(Troubleshooting guide sections will be developed into Troubleshooting guide work packages as determined by the acquiring activity.)*			M.5.4 M.5.4.1	**<troublesect>** **<troublewp>**
Troubleshooting Guide Procedures	CR	CR	M.5.4.2	**<troubletsk>**
DETAILED PROCEDURES SECTION *(Detailed procedure sections will be developed into Detailed work packages as determined by the acquiring activity.)*			M.5.5 M.5.5.1	**<detailsect>** **<detailwp>**
Detailed Procedures	CR	CR	M.5.5.1.1	**<detailtsk>**
CHARTS SECTION	R	R	M.5.6.1	**<perfdatasect>**
CHARTS WORK PACKAGE	R	R	M.5.6.1.1	**<perfdatawp>**
FORMS SECTION	R	R	M.5.6.2	**<perfdatasect>**
FORMS WORK PACKAGE	R	R	M.5.6.2.1	**<perfdatawp>**
REAR MATTER	R	R	M.5.7	**<rear>**
ANNUNCIATOR PANEL		R	M.5.8	**<annunciator>**
EMAIL 2028	R	R	5.2.3	**<da2028>**

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TABLE A-VII. Requirements matrix for operator's maintenance test flight manual (-MTF) _____. (continued)

TM CONTENT	RW	FW	MIL-STD-3083(AV) Reference	Element Name
AUTHENTICATION	R	R	5.2.3	**<authent>**
FOLDOUTS			5.2.3	**<foldsect>**
BACK COVER	R	R	5.2.3	**<back>**

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APPENDIX B

AVIATION INTRODUCTION AND GENERAL INFORMATION (MODULE 1)

B.1 Scope.

This appendix establishes the technical content requirements for the preparation of the general information module for aviation operator's manuals (-10). This appendix is a mandatory part of this standard. The information contained herein is intended for compliance. The requirements are applicable for all aviation platform Operator Manuals (-10). This module shall consist, at a minimum, of introductory material that applies to the manual as a whole.

B.2 Applicable publications.

The applicable publications in Section 2 apply to this appendix.

B.3 Definitions.

The definitions in Section 3 apply to this appendix.

B.4 General requirements.

General introductory material that applies to the entire manual as a whole shall be developed.

B.4.1 Preparation of digital data for electronic delivery.

Data prepared and delivered digitally in accordance with this standard shall be XML tagged using the current Army DTD. Data shall be formatted for presentation using the Army stylesheets. Refer to B.4.2 for information on obtaining or accessing the Army DTD and stylesheets. XML tags used in the DTD are noted throughout the text of this standard in bracketed, bold characters (e.g., **<geninfo>**) as a convenience for the author and to ensure that the tags are used correctly when developing a document instance.

B.4.2 Use of the Document Type Definition (DTD)/stylesheet.

The DTD referenced in this appendix interprets the technical content and structure for the functional requirements contained in this standard. Development of the TMs shall be accomplished through the use of this standard, the Army DTD, and the Army stylesheets.

B.4.3 Content structure.

The examples provided herein are an accurate representation of the content structure requirements contained in this appendix and shall be followed to permit the effective use of the DTD.

B.4.4 Style and format.

This standard provides style and format requirements for the technical content requirements described in this appendix. These requirements are considered mandatory and are intended for compliance.

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B.4.5 Work package development.

Data developed in accordance with this appendix shall be divided into work packages. These work packages shall be stand alone and shall contain all information and references required to support the work package type.

B.4.6 Selective application and tailoring of content using Appendix A matrices.

This standard contains some requirements that may not be applicable to the preparation of all TMs. Selective application and tailoring of requirements contained in this standard are the responsibility of the acquiring activity and shall be accomplished using APPENDIX A. The applicability of some requirements is also designated by one of the following statements: unless specified otherwise by the acquiring activity; as specified by the acquiring activity; or when specified by the acquiring activity.

B.5 Detailed requirements.

The introduction and general information module for aviation -10 Operator Manuals shall be developed utilizing work package format as detailed in B.5.1 through B.5.2.

B.5.1 General information work package <av_ginfowp>.

This work package shall contain the requirements in B.5.1.1 through B.5.2.3.

B.5.1.1 Work package identification information <wpidinfo>.

Work package identification information is required for this work package (refer to MIL-STD-40051).

B.5.1.2 Scope <scope>.

A brief summary of the contents of the manual shall be provided. A succinct summary of the aircraft's description and primary mission, omitting any extraneous mission capabilities statements, shall be provided. One of the following statements shall be included verbatim:

For Rotary Wing/Fixed Wing:

"This manual contains the best operating instructions and procedures for the *(insert aircraft designation)*, under most circumstances. The observation of limitations, performance and weight/balance data provided is mandatory. The adherence to procedures is mandatory except when modifications are required because of multiple emergencies, adverse weather, terrain, etc. Basic flight principles are not included. THIS MANUAL SHALL BE ACCESSIBLE BY THE OPERATOR IN THE AIRCRAFT DURING ALL FLIGHTS IN ACCORDANCE WITH AR 95-1."

For UAS:

"This manual contains the best operating instructions and procedures for the (insert aircraft designation), under most circumstances. The observation of limitations, performance and weight/balance data provided is mandatory. The adherence to procedures is mandatory except when modifications are required because of multiple emergencies, adverse weather, terrain, etc. Basic flight principles are not included. THIS MANUAL SHALL BE ACCESSIBLE TO THE OPERATOR DURING ALL FLIGHTS IN ACCORDANCE WITH AR 95-1."

B.5.1.3 Aviation safety program <av_safety>.

In accordance with AR 385-10, the following statement shall be included verbatim:

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“Reports necessary to comply with the Army Aviation Safety Program are prescribed in DA PAM 385-10.”

B.5.1.4 Forms and records <frr>.

Flight records and aircraft maintenance records which are used by the operators and crewmembers shall be described. References shall be made to DA Pam 738-751 and TM 1-1500-328-23.

B.5.1.5 Reporting Equipment Improvement Recommendations (EIR) <eir>.

The following statement shall be included verbatim (italicized text within parentheses shall be replaced with the appropriate information):

"REPORTING EQUIPMENT IMPROVEMENT RECOMMENDATIONS (EIR) If your *(insert equipment short item name)* needs improvement, let us know. Send us an EIR. You, the user, are the only one who can tell us what you do not like about your equipment. Let us know why you do not like the design or performance. All AMCOM (Aviation and Missile Command) Deficiency Reports (DRs), (Warranty, EIR, and PQDRs) shall be submitted through the Joint Deficiency Reporting System (JDRS) at https://jdrs.mil/."

B.5.1.6 Destruction of army material <destructmat>.

Information on procedures for destroying Army materiel to prevent enemy use shall be included. Reference shall be made to TM 750-244-1-5.

B.5.1.7 Nomenclature cross-reference list <nomenreflist>.

A cross-reference list shall be prepared when unofficial nomenclature (common name) is approved by the proponent activity.

B.5.2 Abbreviation, acronyms, and definitions work package <av_acdwp>.

A list of all abbreviations and acronyms used in the manual shall be prepared and included in this work package. As determined by the acquiring activity, a list of terms and definitions may be provided.

B.5.2.1 Work package identification information <wpidinfo>.

Work package identification information is required for this work package (refer to MIL-STD-40051).

B.5.2.2 List of abbreviations/acronyms <loa>.

A list of all abbreviations, acronyms, signs, or symbols used in the manual shall be prepared. Warning icons are defined in the Warning Summary. Designator symbols are defined in the general information work package. A statement shall be prepared that abbreviations are in accordance with abbreviations contained in the Abbreviations, Brevity Codes, and Acronyms (ABCA) tool at https://armypubs.army.mil, except when the abbreviation stands for a marking actually found in the aircraft.

B.5.2.3 Definitions list <deflist>.

A definitions list shall be prepared for TMs for all terms that are uncommon and are not adequately defined in the text or in the Army, DoD, or standard dictionary. The definitions list shall include a list of terms **<term>** followed by

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definitions **<def>**. The terms shall be listed in alphabetical order. At a minimum, the definitions required by H.5.1.10 shall be included.

B.6 Notes.

The notes in Section 6 apply to this appendix.

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APPENDIX C

AIRCRAFT AND SYSTEMS DESCRIPTION AND OPERATION (MODULE 2)

C.1 Scope.

This appendix establishes the technical content requirements for the preparation of aircraft and systems description information and operation for aviation weapon systems. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

C.2 Applicable publications.

The applicable publications in Section 2 apply to this appendix.

C.3 Definitions.

The definitions in Section 3 apply to this appendix.

C.4 General requirements.

Descriptive information and operational procedures for the airframe and all aircraft systems and controls shall be developed. Information that is required to provide the operator with a physical description and to functionally explain how the aircraft system and controls operate shall be included.

C.4.1 Preparation of digital data for electronic delivery.

Data prepared and delivered digitally in accordance with this standard shall be XML tagged using the current Army DTD. Data shall be formatted for presentation using the Army stylesheets. Refer to C.4.2 for information on obtaining or accessing the Army DTD and stylesheets. XML tags used in the DTD are noted throughout the text of this standard in bracketed, bold characters (e.g., **<geninfo>**) as a convenience for the author and to ensure that the tags are used correctly when developing a document instance.

C.4.2 Use of the Document Type Definition (DTD)/stylesheet.

The DTD referenced in this appendix interprets the technical content and structure for the functional requirements contained in this standard. Development of the TMs shall be accomplished through the use of this standard, the Army DTD, and the Army stylesheets.

C.4.3 Content structure.

The examples provided herein are an accurate representation of the content structure requirements contained in this appendix and shall be followed to permit the effective use of the DTD.

C.4.4 Style and format.

This standard provides style and format requirements for the technical content requirements described in this appendix. These requirements are considered mandatory and are intended for compliance.

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C.4.5 Work package development.

Data developed in accordance with this appendix shall be divided into work packages. These work packages shall be stand alone and shall contain all information and references required to support the work package type.

C.4.6 Selective application and tailoring of content using Appendix A matrices.

This standard contains some requirements that may not be applicable to the preparation of all TMs. Selective application and tailoring of requirements contained in this standard are the responsibility of the acquiring activity and shall be accomplished using APPENDIX A. The applicability of some requirements is also designated by one of the following statements: unless specified otherwise by the acquiring activity; as specified by the acquiring activity; or when specified by the acquiring activity.

C.5 Detailed requirements.

Aircraft and systems description work packages shall be developed as detailed in C.5.1 through C.5.18.

C.5.1 Aircraft/air vehicle work package <airvehiclewp>.

The Aircraft/Air Vehicle WP shall provide a complete but concise description of the aircraft. At a minimum, the subjects and illustrations described in C.5.1.1 through C.5.1.7 shall be included. To better organize and manage the data required in the WP, it may be separated into multiple Aircraft/Air Vehicle WPs or combined into one WP so long as the WP remains under 50 pages.

C.5.1.1 Airframe and aircraft systems and controls general information.

A description of the airframe and all aircraft systems and controls shall be provided in this WP. Major assemblies such as fuselage, wings, and tail boom shall be described. Each compartment of the aircraft such as cockpits and cabins shall be described and illustrated as required. Individual sections shall be developed for the description and operation of the aircraft and each aircraft system as applicable.

C.5.1.1.1 UAS specific requirements.

UAS manuals shall provide a complete description and shall include the entire system by major components such as Air Vehicle, Ground Control Shelter, Ground Control Station/Portable Control Station, Data Link, and Launcher. The description of each system and associated controls and equipment shall be brief and concise. Major UAS assemblies such as shelter, portable cases, and workstations shall be described. Each workstation of the ground control station shall be described and illustrated as required. Illustrations shall be used to identify and locate the systems, controls and equipment.

C.5.1.1.2 Emergency systems.

Emergency systems may be located in Module 9 at the discretion of the acquiring activity. When this is done, include the following statement in the WP: “Emergency equipment information is located in Module 9.”

C.5.1.1.3 Interactive displays.

For aircraft equipped with interactive displays, (i.e. Multifunction Display (MFD), Primary Function/Flight Display (PFD), etc.), the data management system, including the interactive display, shall be described in this module. For each section within this module that describes a subsystem, the appropriate top page of the interactive display,

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including each button, shall be fully described and explained. When needed for clarity, illustrations or representative items displayed on the interactive display shall be included throughout the TM.

C.5.1.1.4 Controls.

Each control contributing to the operation of a system shall be described and its location given. The function of the control and the end result produced when the control is moved to each of its possible positions shall be included in the description. Any effect which this control may have on other systems, or which the other systems may have on the control shall be stated. If movement of the control requires any special action because of locks, gates, etc., it shall be so stated. When feasible, a separate paragraph and illustration shall be devoted to each control. It shall be preferable to divide the control description into two portions, normal controls and emergency controls, if emergency capabilities exist.

C.5.1.1.5 Indicators.

All indicators, instruments, and warning devices that are a part of the aircraft system shall be described and illustrated. This shall include location, function, power source, and interpretation of the indications.

C.5.1.2 Illustrations and tables.

The following illustrations and tables shall be included in the Aircraft/Air Vehicle WP.

C.5.1.2.1 General arrangement illustration.

The aircraft's general arrangement shall depict all access openings that will be checked during preflight of the aircraft (FIGURE C-1). The general arrangement shall be placed as near to the beginning of the Aircraft/Air Vehicle WP as practicable. These diagrams shall not include individual controls or aircraft systems. Diagrams that are needed for clarity shall be used. Two or more of these illustrations, such as crew movement diagrams and compartment diagrams, may be combined into one.

C.5.1.2.2 Turning radius, ground clearance, dimensions, and danger areas.

Illustrations showing minimum turning radius, ground clearance, dimensions and danger areas shall be included (FIGURE C-2 and FIGURE C-3).

a. The minimum turn shall be based on a turn permitted on one wheel (tire hub), with and without power steering assist.

b. Minimum ground clearance shall also be shown. The turning radius for skid equipped aircraft shall be based on turning the aircraft on an identifiable reference point on the aircraft or an identifiable reference point on the ground.

c. An illustration shall be included showing danger areas around the aircraft for all modes of operations on or near the ground. Areas to be avoided to prevent damage to equipment or injury to personnel shall be depicted or described. These figures shall be provided for idle and maximum power.

d. For rotary wing aircraft, illustrations shall be based on hover power required at maximum gross weight. Danger areas of main rotors, tail rotors, or propellers shall also be depicted.

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C.5.1.2.3 Significant differences.

Significant differences in design and operation between each aircraft series included in the manual shall be provided. Special emphasis shall be placed on features that will affect recognition and operation of the various series. This information shall be contained in a table.

C.5.1.2.4 Major compartments.

Each major compartment, such as cockpit or cabin (FIGURE C-4), that can carry payload or that can be entered by personnel shall be illustrated and identified.

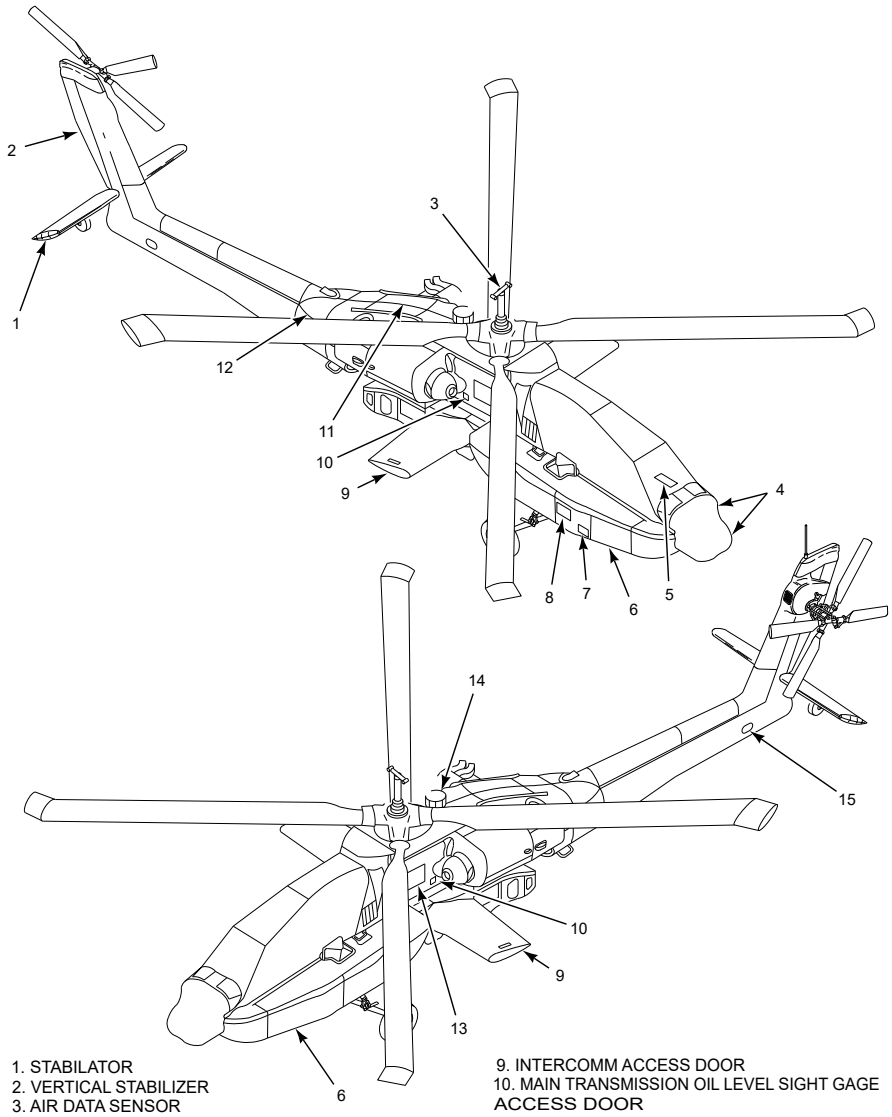


FIGURE C-1. Aircraft general arrangement diagram.

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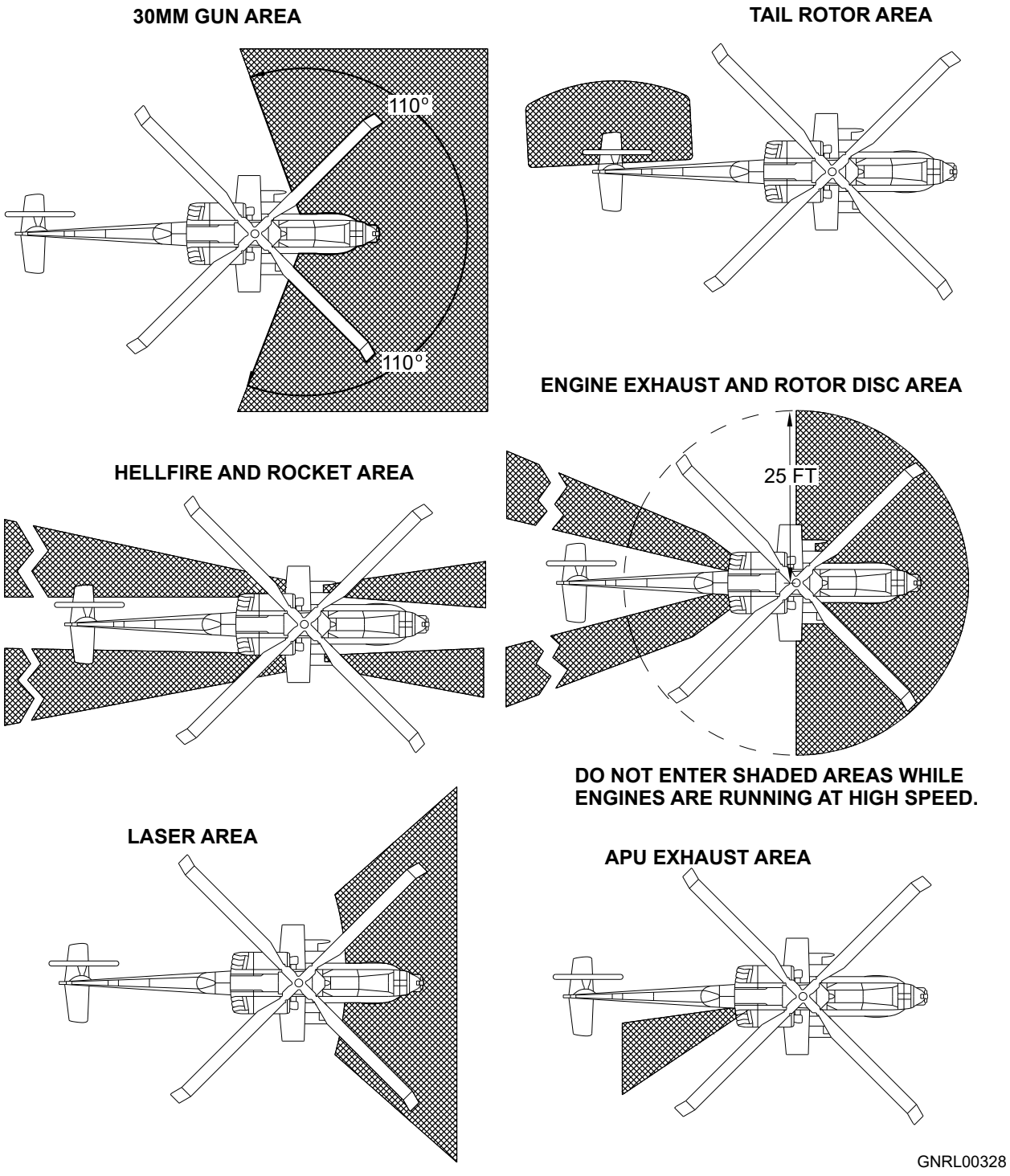


FIGURE C-2. Example of danger area diagram.

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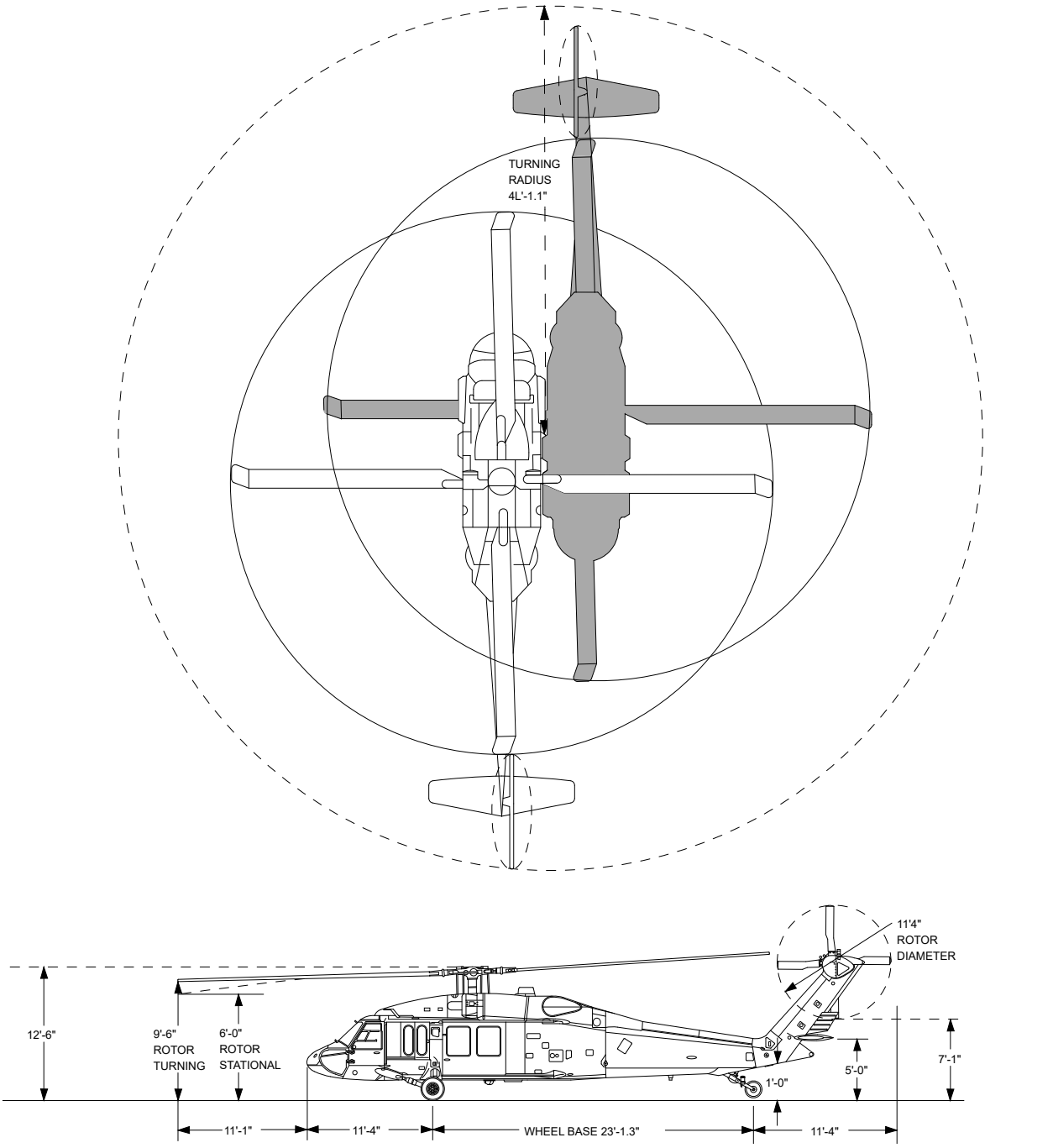
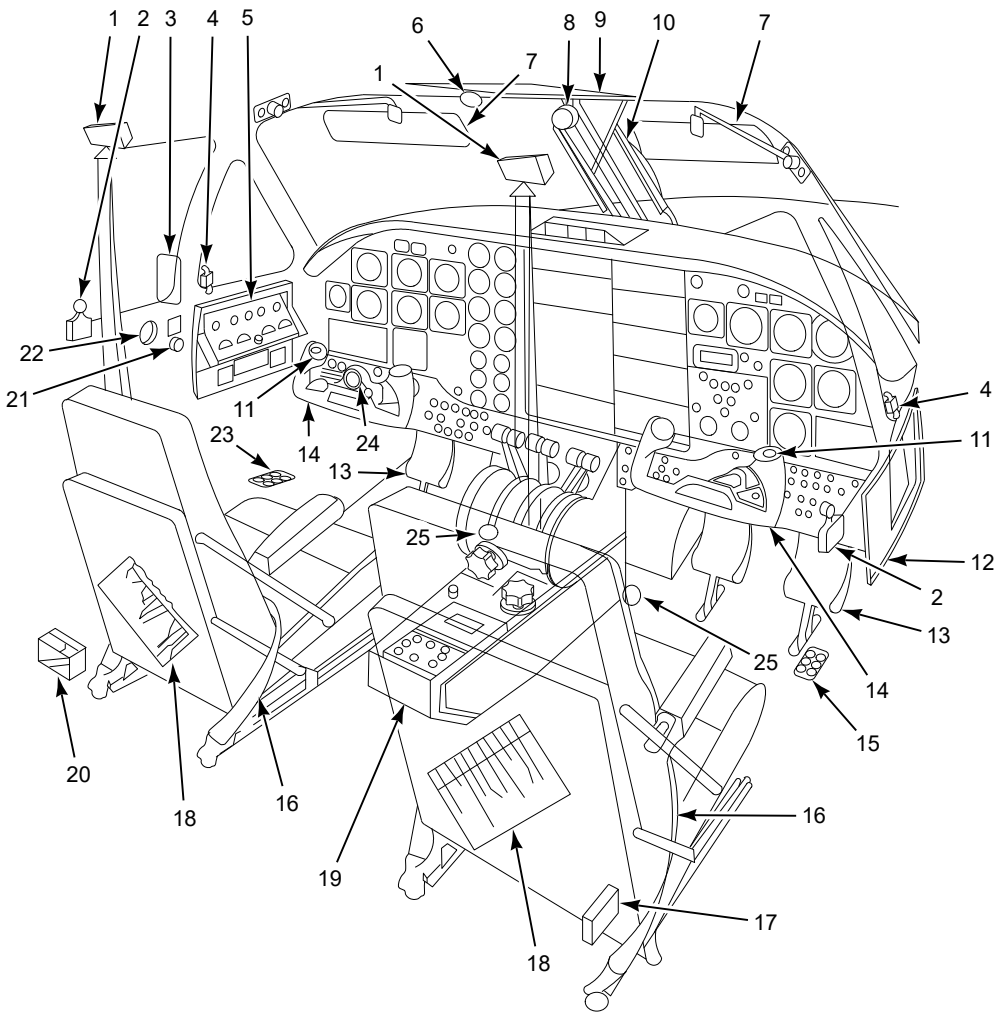


FIGURE C-3. Example of turning radius and dimensions diagram.

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1. SHOULDER HARNESS INERTIA REEL
2. SHOULDER HARNESS LOCK LEVER
3. EXTERNAL REAR VIEW MIRROR
4. STORM WINDOW LOCK
5. FUEL MANAGEMENT PANEL
6. FREE AIR TEMPERATURE GAUGE
7. SUN VISOR
8. MAGNETIC COMPASS
9. OVERHEAD CONTROL PANEL
10. WINDSHIELD WIPERS
11. MICROPHONE SWITCH
12. CO-PILOT'S CIRCUIT BREAKER PANEL
13. RUDDER PEDALS
14. CONTROL WHEEL
15. OXYGEN REGULATOR CONTROL PANEL
16. SEAT BELT
17. VERTICAL GYRO CIRCUIT BREAKER BOX
18. UTILITY POCKET
19. CONTROL PEDESTAL
20. AUDIO CONTROL PANEL
21. EXTERNAL MIRROR ADJUSTMENT KNOB
22. OXYGEN SYSTEM GAUGE
23. OXYGEN SYSTEM CONTROLS AND REGULATOR CONTROL PANEL
24. EIGHT-DAY CLOCK
25. FOOT MICROPHONE SWITCH

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FIGURE C-4. Example of compartments illustration.

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C.5.1.2.5 UAS specific illustrations and tables.

As specified by the acquiring activity, the following illustrations and tables shall be provided for UAS manuals:

a. The ground control station or portable ground control stations general arrangement shall depict all access openings. These diagrams shall not include individual controls or systems. Diagrams that are needed for clarity shall be used. Two or more of these illustrations, such as crew movement diagrams and compartment diagrams, may be combined into one.

b. Danger areas around the ground control station for all modes of operations shall be illustrated. Areas to be avoided to prevent damage to equipment or injury to personnel shall be depicted or described.

c. Significant differences in design and operation between each ground control station series included in the manual shall be provided. Special emphasis shall be placed on features that will affect recognition and operation of the various series. This information shall be contained in a table.

d. The Ground Data Terminal/Antenna System general arrangement shall depict all areas that will be checked during preflight of the system. The general arrangement shall be placed as near to the beginning of the Aircraft/Air Vehicle WP as practicable. These diagrams shall include all components of the Ground Data Terminal/Antenna System. Diagrams that are needed for clarity shall be used.

e. Illustrations showing setup distance, dimensions and date areas shall be included. Minimum setup distance from UAS system or other structures shall also be shown. An illustration shall be included showing danger areas around the Ground Data Terminal/Antenna System. Areas to be avoided to prevent damage to equipment or injury to personnel shall be depicted or described.

f. Significant differences in design and operation between each Ground Data Terminal/Antenna System series included in the manual shall be provided. Special emphasis shall be placed on features that will affect recognition and operation of the various series. This information shall be contained in a table.

C.5.1.3 Landing gear system.

Information describing the landing gear system shall be presented in detail for the operator's use. The following shall also be included.

a. The steering system including any special or unusual features shall be described.

b. The brake system, including all emergency provisions, shall be described. Brake provisions for aircraft equipment with floats shall also be described.

C.5.1.4 Instruments, panels and consoles.

All instruments, panels, and consoles shall be described and illustrated. Several configurations may be covered by one illustration labeled typical. Minor variations in number or type of controls and instruments shall be indicated by detailed views to the illustration and by notations in the key. The panels or console shall be shown more than once when major changes in configuration are involved. UAS instruments, panels and consoles include those on or within the ground control station and on or within the Data Link System.

C.5.1.5 Canopies.

The canopies shall be described and illustrated. Several configurations may be covered by one illustration labeled typical. All normal and emergency canopy controls, both external and internal, shall be described and illustrated.

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C.5.1.6 Doors.

All doors to include ramps, hatches, etc., controls for normal and emergency operations, and their sources of power shall be described.

C.5.1.7 Seats.

As applicable, pilot and other flight compartment seat controls shall be described and illustrated. Emergency and ejection seat controls, inertia reels, harnesses, and seat belts shall be described and illustrated in detail, emphasizing how they are affected by other systems.

C.5.2 Ground systems work package <groundsystemswp>.

This section shall provide a complete but concise description of the UAS System specific Data Link Systems used in transmitting flight control and payload information between the Ground Control Station and the Air Vehicle. These may include systems such as Ground Data Terminals, Portable Ground Data Terminals, Tactical Common Data Link System, or Tactical Automated Landing Systems.

C.5.3 Data link system work package <datalinkwp>.

An overall description and general arrangement of the Data Link System shall be included. Major assemblies such as shelters, portable cases and workstations shall be described. Each workstation of the ground control station shall be described and illustrated as required. Diagrams needed for clarity shall be used.

C.5.4 Emergency recovery system work package <uasrecoverwp>.

As required by the acquiring activity, all emergency equipment used in the recovery of Uncrewed aircraft shall be described (i.e., parachute systems, deadstick landing systems).

C.5.5 Emergency equipment work package <emersyswp>.

All emergency equipment, except that which forms part of a complete system, shall be described. For example, emergency landing gear controls shall be treated under the landing gear system and emergency fuel pumps under the fuel system. Emergency equipment in this section shall include, but shall not be limited to, hand fire extinguishers, engine fire extinguishers, emergency alarms, pyrotechnic equipment, axes, emergency hatches, signal lamps, ditching jackets, first aid kits, and survival kits. Emergency procedures shall be described only in Module 9. Illustrations showing locations of emergency equipment or systems shall be shown as needed but only in Module 9.

C.5.6 Engines and related systems work package <engsyswp>.

The most important characteristics and special features of the engine shall be described. Model designation shall be included for all engines used in the subject aircraft. The engine and its related controls shall be described as detailed in the C.5.6.1 through C.5.6.10.

C.5.6.1 Cooling system.

Cooling system and controls such as cowl flaps and engine cooling fans shall be described.

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C.5.6.2 Anti-icing/deicing system.

Engine/engine inlet anti-icing/deicing system shall be described.

C.5.6.3 Fuel control system.

The engine fuel control system, which applies to jet and turbine powered aircraft and extends from the engine fuel control unit through the burner ring or combustor section. When applicable, special emphasis shall be placed on the emergency fuel control systems. Any special or unusual characteristics of the system shall also be described. Theory of operation shall not be included. Discussion of the throttle/power level shall be included, as well as all systems affected by throttle/power lever operation.

C.5.6.4 Oil system.

Information on all controls affecting the oil system shall be described.

C.5.6.5 Ignition system.

Ignition system controls shall be described.

C.5.6.6 Starter controls.

Starter controls shall be described.

C.5.6.7 Infrared (IR) suppression system.

IR suppression system shall be described.

C.5.6.8 Engine instruments and indicators.

Engine instruments and indicators shall be described. For the purpose of the operator's manual, the fuel and oil supply systems shall be treated as ending at the point where they deliver the fluid to the carburetor, fuel control unit, or the engine-driven oil pump.

C.5.6.9 Engine management system.

Engine management system shall be described.

C.5.6.10 Fire detection/extinguishing system.

The engine and APU fire detection/extinguishing system shall be described. The description, normal operation, and emergency operation for each of the systems shall be presented in the appropriate locations (emergency operation under emergency procedures, etc.), as applicable. The fire detection/ extinguishing controls and displays shall be described.

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C.5.7 Fuel system work package <fuelsyswp>.

A full description of the fuel system shall be given. Coverage of drop tank release controls shall be included. Reference shall be made to fuel grades and specifications in the Servicing, Parking, and Mooring WP. Diagrams of the typical courses of fuel flow, including fuel system control positions for takeoff, cruising, landing, and emergency operation shall be included.

C.5.7.1 Controls and indicators.

Fuel system controls and indicators shall be described.

C.5.7.2 Fuel system management.

The fuel system management process shall be described, including auxiliary fuel, booster pump use, fuel transfer procedures, tank selection procedures, and courses of fuel flow. All possible courses of fuel flow, such as inoperative engines and failed boost pump, shall be included. The sequence in which fuel tanks shall be used shall be stated with corresponding reasons (strength or balance). When applicable, reference shall be made to the pertinent portion of Module 6 when weight distribution becomes a problem. The required sequence of use of tanks to maintain a favorable Center-of-Gravity (CG) shall be described in detail. Remarks shall also be included regarding control of the aircraft if the transfer system fails and results in an unbalanced condition because of improper fuel distribution.

C.5.8 Flight control system work package <fltcntrlsyswp>.

The flight control system and its location in the aircraft or UAS shall be described in its entirety. Flight controls, indicators, trim tabs, force trim, control locks, UAS data links, etc. shall be discussed as stated in paragraph C.5.8.1. In addition, all other controls located on the control sticks, wheels, yokes, pedals, cyclic and collective, shall be discussed. Illustrations shall be provided for buttons, friction devices, locks, etc. Variations in controls between aircraft series or serial numbers, or both, shall also be shown.

C.5.8.1 Automatic flight control system.

Detailed coverage of automatic stabilization equipment, stability augmentation control system, autopilot, and UAS automatic flight modes shall be provided. All modes of operation shall be described. If any additional systems are required to operate in conjunction with the stabilization equipment, a statement shall be included to that effect. Applicable precautionary data shall be included for conditions of partial or temporary electrical power failure, manual override, etc. When applicable, reference shall be made to navigation equipment descriptions and operations contained in Module 3.

C.5.9 Hydraulic and pneumatic systems work package <hydrpneusyswp>.

A description of all hydraulic and pneumatic systems shall be provided. At a minimum, test switches, indicators and gauges, caution/warning lights, and controls shall be discussed.

C.5.10 Power train system work package <pwrtrainwp>.

The power train system shall be described in detail to include the transmission and gearbox systems, drive shafting, system controls, and indicators.

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C.5.11 Rotors or propellers work package <proprotorsyswp>.

The propellers or rotors, as applicable, and their functions shall be described, including a detailed description of operation.

C.5.12 Utility systems work package <utilitysyswp>.

A description of the defrosting, anti-icing/deicing, pressurization, oxygen, and rain removal systems and miscellaneous equipment shall be provided. Coverage shall be brief and shall focus on the location of the equipment and its controls, source of power, illustration of controls (if not covered previously), and a brief discussion of function and operation. Control/switch panels that control several different utility systems shall only be illustrated once, if feasible. Information shall be included on all non-emergency equipment which is not part of a system. All miscellaneous equipment and normal and emergency operation procedures shall be included. Miscellaneous equipment shall include, but shall not be limited to, seats (other than pilot and flight engineer), hatches, heated blanket provisions, data case, benching gear, night flying curtains, ladders, relief equipment, food warmers, water containers, and tools kits. Items covered as aircraft loading equipment in Module 6 shall not be included here. Items dealing with aircraft servicing and ground handling shall be contained in the Servicing, Parking, and Mooring WP.

C.5.13 Heating, ventilation, cooling, and environmental control systems work package <hvacsyswp>.

The heating, ventilation, cooling, and environmental control systems and their displays and controls shall be described. The description, normal operation, and emergency operation for each of these systems shall be discussed under separate paragraphs, as applicable.

C.5.14 Electrical power supply and distribution systems work package <elecpwrsyswp>.

The electrical power supply and distribution systems and controls shall each be described and illustrated (FIGURE C-5 and FIGURE C-6). Where pertinent, reference shall be made to auxiliary power systems that are described elsewhere. The external power source and the interaction between the auxiliary power plant and the electrical system shall be described. General arrangement and order of the primary system shall be covered first, followed by the secondary system.

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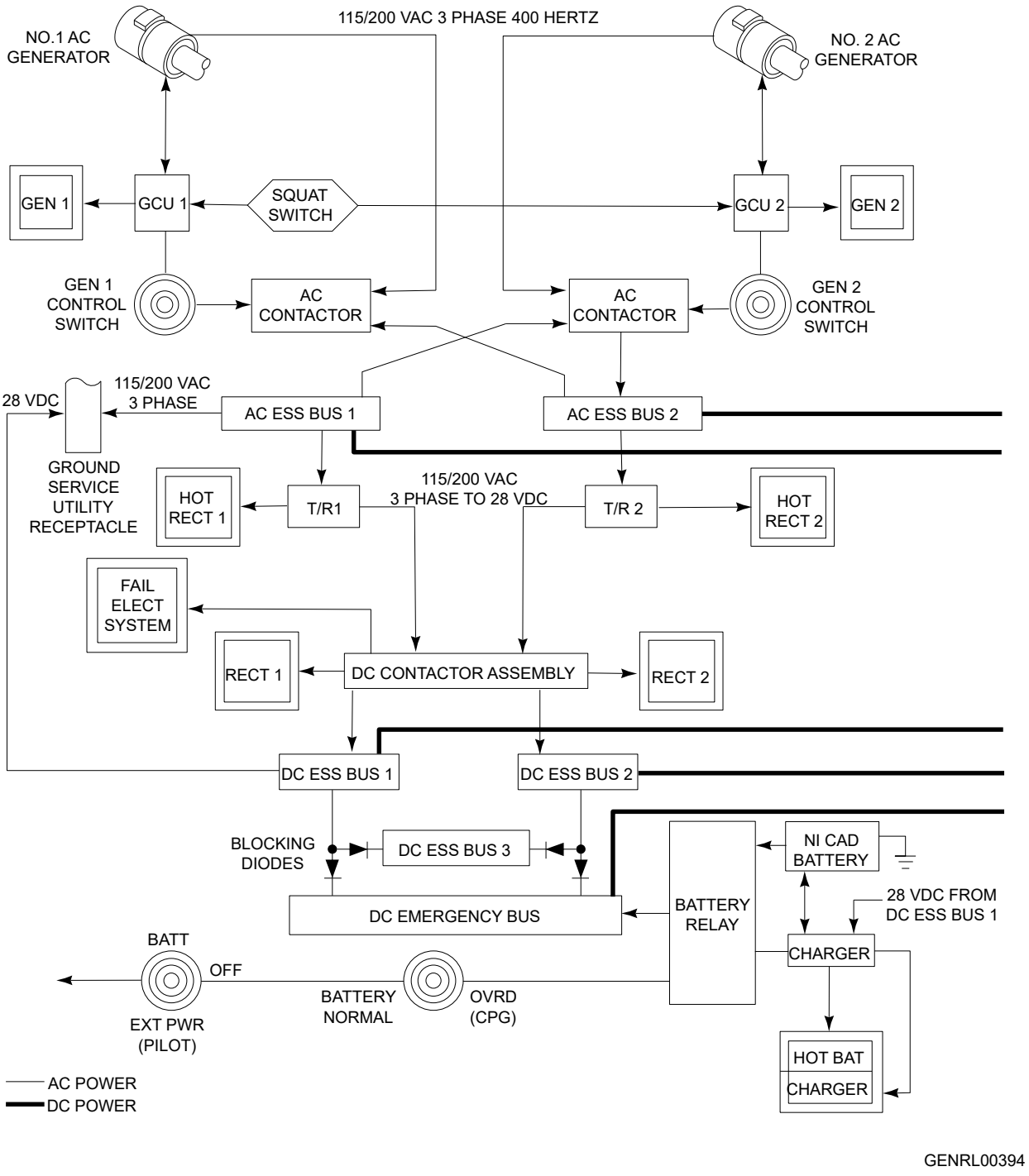


FIGURE C-5. Example 1 of electrical power supply and distribution system.

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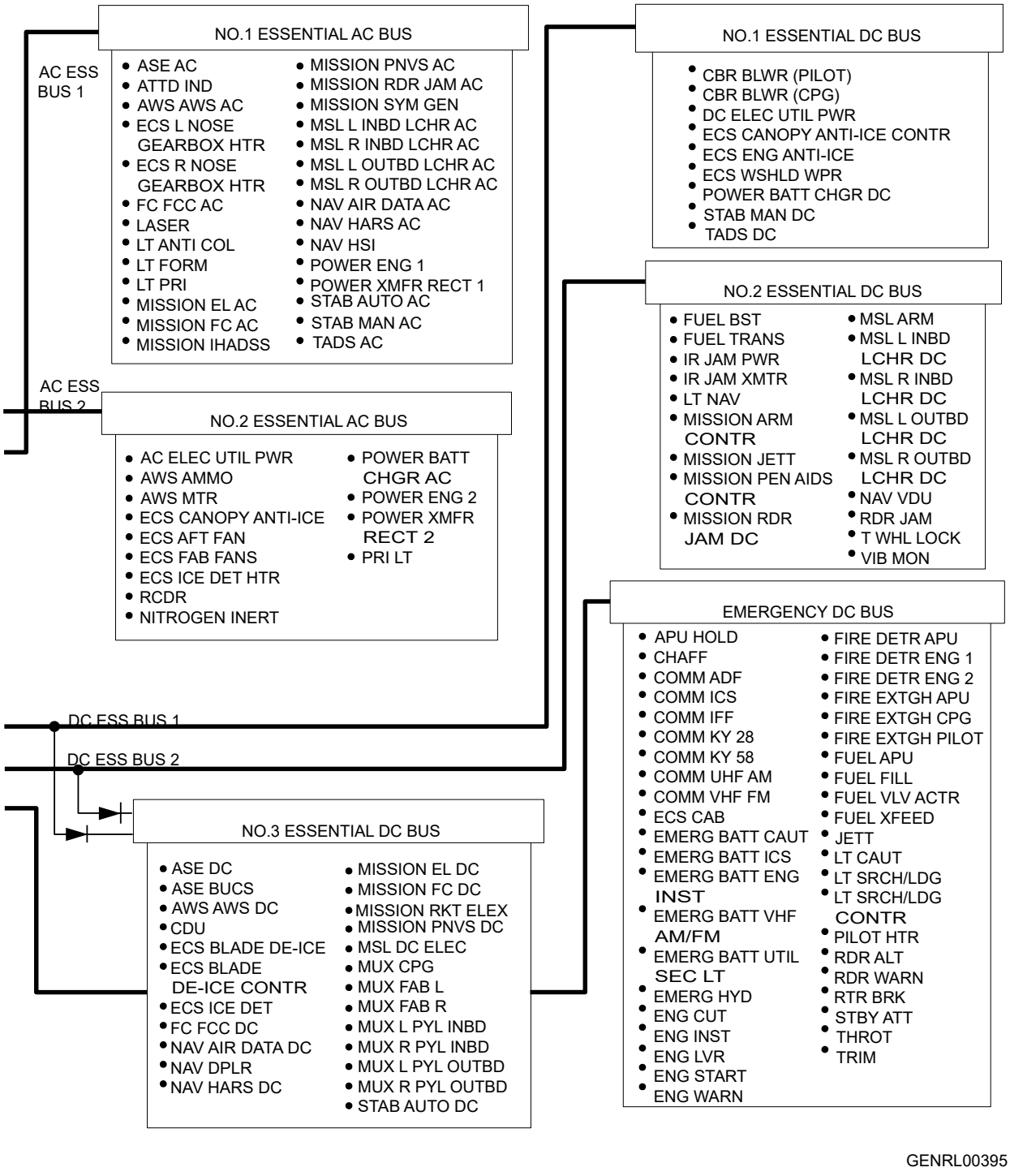


FIGURE C-6. Example 2 of electrical power supply and distribution system.

C.5.14.1 Direct Current (DC) power supply system.

DC Power supply system's shall include battery; starter-generators, generators, alternators, and converters; indicators, gauges, and controls; circuit breaker and junction boxes; auxiliary power; and ground power. Refer to FIGURE C-5.

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C.5.14.2 Alternating Current (AC) power supply system.

These systems shall include inverters and alternators; indicators, gauges, and controls; AC circuit breaker and junction box diagram; auxiliary power; and ground power. Refer to FIGURE C-5.

C.5.14.3 Breakers illustration.

The location of each circuit breaker panel shall be shown, and each circuit breaker in the panels shall be identified on standardized installation. The illustration shall depict a typical installation of both systems (AC/DC) and may be combined on one illustration. In those instances where a standardized circuit breaker location does not exist, the location of circuit breakers or fuses shall be given. Refer to FIGURE C-5.

C.5.15 Auxiliary power unit work package <apusyswp>.

A description of the auxiliary power unit, controls, and its interaction with other systems shall be provided. Starting, stopping, and in-flight operating procedures shall be contained in Module 8 and emergency procedures in Module 9.

C.5.16 Lighting work package <lightsyswp>.

Lighting information shall be developed. Coverage shall concern itself largely with locations, controls, power sources, and a discussion of functions. Illustrations may be used if equipment is not depicted elsewhere in the manual. Information shall be provided for, but shall not be limited to the following:

a. Formation lights

b. Landing lights

c. Fuselage lights

d. Cabin lights

e. Instruments lights

f. Wheel well lights

g. Taxi lights

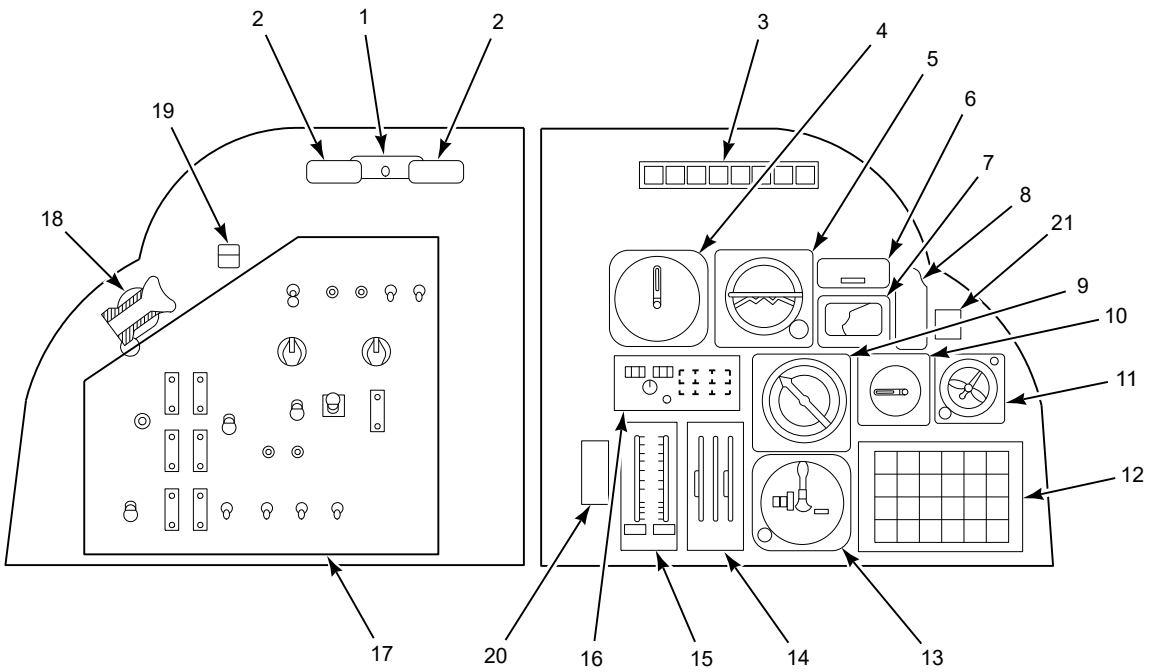
h. Navigation lights

i. Anti-collision lights

C.5.17 Flight instruments work package <fltinstrumentswp>.

All flight instruments, indicators, gauges, and miscellaneous instruments and systems shall be described. Miscellaneous instruments and systems shall include such items as master caution systems, Revolutions Per Minute (RPM) high/low warning systems, trainer instrument panel, and clocks. Special problems, such as erroneous readings of the airspeed indicating system resulting from installation error or hovering, shall be included with references to correction charts, when applicable. Complex display systems shall be included under a separate primary heading. Line drawings shall be provided for all instruments. Each indicator, gauge, and control shall be shown (FIGURE C-7). Each item shall be indexed or posted and references or links shall be used within the text as appropriate. For UAS, this may be included in the ground systems WP per the acquiring activity.

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1. FIRE EXTINGUISHER BOTTLE SELECT SWITCH
2. ENGINE FIRE PULL HANDLES
3. MASTER CONTROL WARNING PANEL
4. AIRSPEED INDICATOR
5. REMOTE ALTITUDE INDICATOR
6. RADIO CALL PLACARD
7. STABILATOR POSITION INDICATOR
8. STABILATOR/AIRSPEED PLACARD
9. RADIO MAGNET INDICATOR (RMI)
10. VERTICAL SPEED INDICATOR (VSI)
11. CLOCK
12. CAUTION/WARNING PANEL
13. BAROMETRIC ALTIMETER
14. ENGINE ROTOR INDICATOR
15. ENGINE TORQUE INDICATOR
16. SELECTABLE DIGITAL DISPLAY PANEL
17. FIRE CONTROL PANEL
18. CANOPY JETTISON HANDLE
19. ARM SAFE INDICATOR
20. ENGINE INSTRUMENT DIM/TEST PANEL
21. FUEL TRANSFER INDICATOR
    (UNMODIFIED CAUTION/WARNING PANEL)

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FIGURE C-7. Example instrument panel.

C.5.18 Servicing, parking, and mooring work package <servparksyswp>.

Servicing shall include, but shall not be limited to, flight crew oriented instruction for normal and closed circuit refueling and for replenishment of fuel, oil, hydraulic fluid, other fluids, and air in tires. Servicing shall also include all other such items involved in servicing the aircraft that a crew could be expected to perform while away from military maintenance support. Safety precautions to observe in servicing a particular tank or reservoir, such as

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grounding and supplemented with a diagram showing locations of regular and alternate servicing points. "NO STEP" areas on walkways leading to tanks shall be indicated, with necessary precautions. Reference shall be made to graphs or data in other parts of the manual pertinent to servicing, such as tire pressure versus gross takeoff weight.

C.5.18.1 Servicing information.

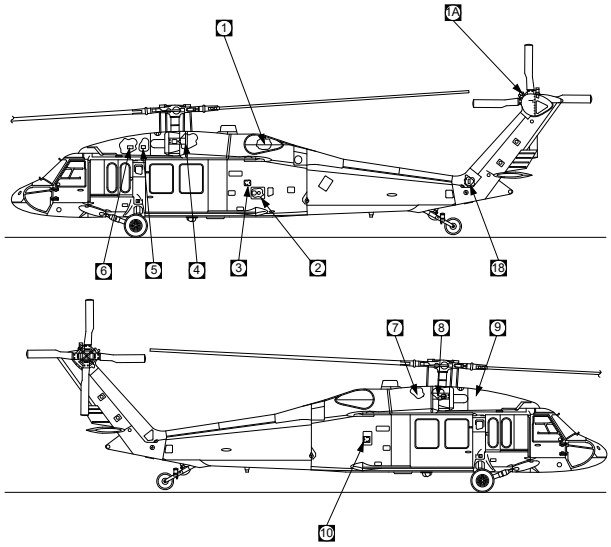
Servicing data shall be in tabular form as show in FIGURE C-9. Each item of equipment including, but not limited to, engine, transmission, gearboxes, reservoirs (hydraulic, anti-icing), auxiliary power unit, and oxygen systems shall be listed under “System.” Under the heading of “Specification,” the military specification for the fuel, oil, fluid, or lubricant shall be listed, including references to include total, servicing capacity, and usable capacity in U.S. measurements to the nearest tenth of a gallon, and metric equivalent.

C.5.18.1.1 Servicing diagram.

The servicing diagram shall depict each servicing point, including, but not limited to, tanks, reservoirs, filler caps, receptacles, oxygen bottles, and accumulators and shall be show as viewed (FIGURE C-8). Illustrations of site gauges and other indicators shall clearly depict proper servicing levels.

C.5.18.1.2 Additional servicing instructions.

Information shall include a listing of acceptable commercial engine oils as indicated in TB 1-9150-200-24 and as authorized for use in the aircraft (FIGURE C-9).



1. AUXILIARY POWER UNIT
1A. TAIL ROTOR GEAR BOX OIL LEVEL SIGHT GAGE
1B. INTERMEDIATE GEAR BOX OIL LEVEL SIGHT GAGE
2. CLOSED CIRCUIT AND PRESSURE REFUELING PORTS
3. NO.1 (LEFT) FUEL TANK GRAVITY REFUEL PORT
4. NO. 1 ENGINE OIL LEVEL SIGHT GAGE
5. NO. 1 HYDRAULIC PUMP MODULE
6. BACKUP HYDRAULIC PUMP MODULE
7. MAIN TRANSMISSION OIL FILLER PORT AND DIP STICK
8. NO. 2 ENGINE OIL FILLER PORT AND SIGHT GAGE
9. NO. 2 HYDRAULIC PUMP MODULE AND PUMP MODULE FLUID FILLER PUMP
10. NO. 2 (RIGHT) FUEL TANK GRAVITY REFUEL PORT

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FIGURE C-8. Example servicing diagram.

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C.5.18.2 Ground handling.

Instructions and necessary precautions for ground handling of the aircraft shall be provided, including any information needed in extreme cold, heat, humidity, and dust. A description and instructions for operating any ground handling equipment involved shall also be provided. Left and right turning limits while towing (with or without external stores) shall be listed. Aircraft ground handling procedures relating to electronics equipment shall be stated when applicable.

C.5.18.3 Parking and mooring.

Instructions for parking and mooring and the installation and stowage of aircraft covers, control locks, chocks, and tie down devices shall be described and illustrated. If feasible, ground handling, parking, and mooring may be shown on a single page illustration.

C.5.19 Approved primary, alternate, and emergency fuels work package <paefuelwp>.

A tabular listing of primary, alternate, and emergency fuels shall be included, to include North Atlantic Treaty Organization (NATO) and commercial brand names authorized for use in the aircraft for which this manual applies. Warnings and cautions regarding additives shall be presented in the table. Also, restrictions on the use of any fuels shall be stated. The fuels contained in this listing shall only be those authorized for use by TB 1-9150-200-24 and by the acquiring activity. This information shall not be repeated in the manual. (FIGURE C-9)

C.5.20 General systems work package <sysgenwp>.

As determined by the acquiring activity, additional work packages necessary for describing aircraft and associated systems may be included when the specific type of aircraft/system is not covered as described in C.1 through C.5. Each additional aircraft/system shall be a separate work package and shall comply with all of the requirements of C.1 through C.5. If additional work packages are used, the proponent shall submit to AMCOM the requirements for this aircraft/system type for possible incorporation within future revisions to this standard.

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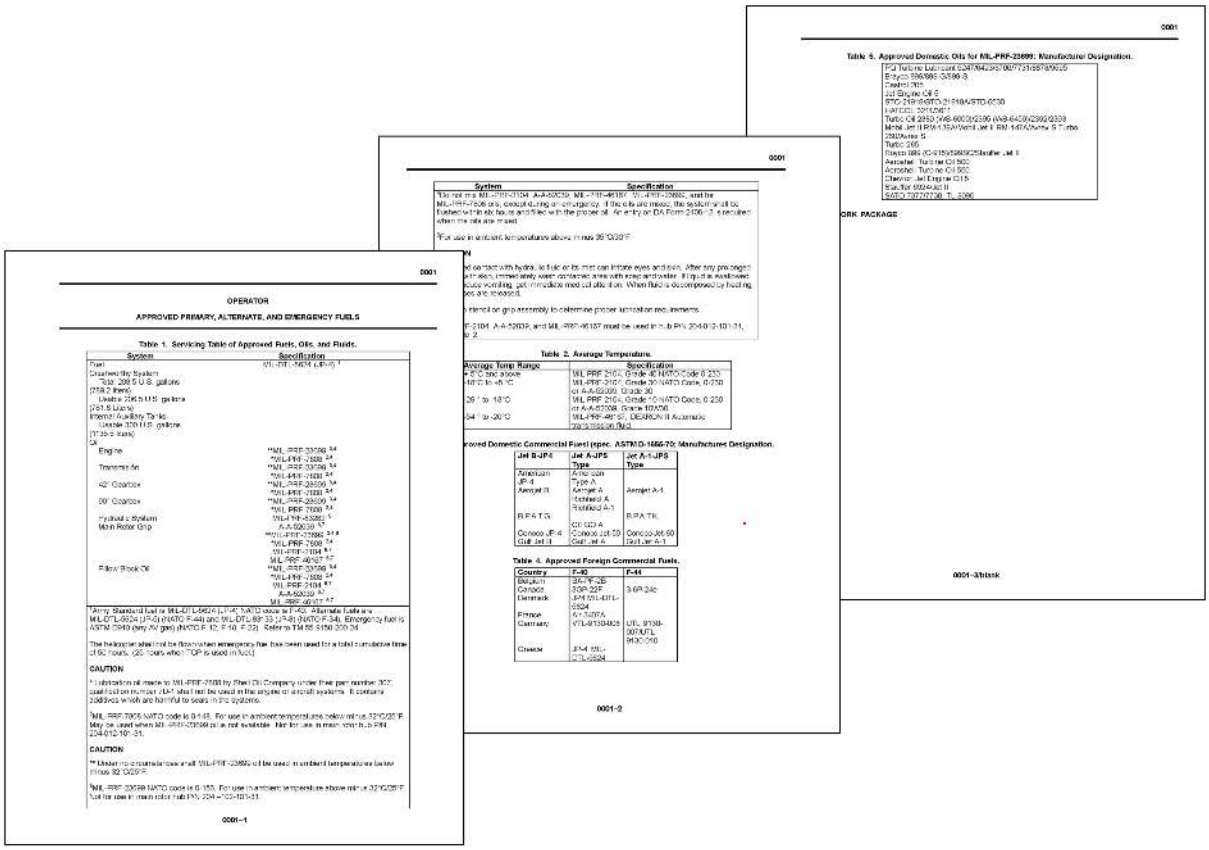


FIGURE C-9. Example of table of approved fuels, oils, and fluids.

C.6 Notes.

The notes in Section 6 apply to this appendix.

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APPENDIX D

AVIONICS (MODULE 3)

D.1 Scope.

This appendix establishes the technical content requirements for the preparation of avionics information and operation for aviation weapon systems. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

D.2 Applicable publications.

The applicable publications in Section 2 apply to this appendix.

D.3 Definitions.

The definitions in Section 3 apply to this appendix.

D.4 General requirements.

Avionics equipment shall be described in detail, including controls, indicators, instruments (if applicable), jacks, switches, and control panels, etc. The proper techniques and procedures to be employed when operating the equipment shall also be described.

D.4.1 Preparation of digital data for electronic delivery.

Data prepared and delivered digitally in accordance with this standard shall be XML tagged using the current Army DTD. Data shall be formatted for presentation using the Army stylesheets. Refer to D.4.2 for information on obtaining or accessing the Army DTD and stylesheets. XML tags used in the DTD are noted throughout the text of this standard in bracketed, bold characters (e.g., **<geninfo>**) as a convenience for the author and to ensure that the tags are used correctly when developing a document instance.

D.4.2 Use of the Document Type Definition (DTD)/stylesheet.

The DTD referenced in this appendix interprets the technical content and structure for the functional requirements contained in this standard. Development of the TMs shall be accomplished through the use of this standard, the Army DTD, and the Army stylesheets.

D.4.3 Content structure.

The examples provided herein are an accurate representation of the content structure requirements contained in this appendix and shall be followed to permit the effective use of the DTD.

D.4.4 Style and format.

This standard provides style and format requirements for the technical content requirements described in this appendix. These requirements are considered mandatory and are intended for compliance.

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D.4.5 Work package development.

Data developed in accordance with this appendix shall be divided into work packages. These work packages shall be stand alone and shall contain all information and references required to support the work package type.

D.4.6 Selective application and tailoring of content using Appendix A matrices.

This standard contains some requirements that may not be applicable to the preparation of all TMs. Selective application and tailoring of requirements contained in this standard are the responsibility of the acquiring activity and shall be accomplished using APPENDIX A. The applicability of some requirements is also designated by one of the following statements: unless specified otherwise by the acquiring activity; as specified by the acquiring activity; or when specified by the acquiring activity.

D.5 Detailed requirements.

Avionics equipment shall be described in detail, including controls, indicators, instruments (if applicable), jacks, switches, and control panels, etc. The proper techniques and procedures to be employed when operating the equipment shall also be described.

D.5.1 Preparation of avionics work packages.

Avionics work packages shall be developed as detailed in D.5.1.1 through D.5.1.2.

D.5.1.1 Avionics work packages.

The avionics module shall be divided into the following work packages:

a. Avionics Introduction work package

b. Communications work package

c. Navigation work package

d. Transponder and Radar work package

e. Additional Avionics work package

Each work package type, with the exception of the introduction work package, is repeatable and may be used multiple times to better organize and structure the content for the operator.

D.5.1.2 Work package contents.

For each item of avionics equipment contained within the work packages detailed in D.5.1.2.1 through D.5.1.2.5, the following information shall be included:

a. Description.

b. Controls and functions.

c. Operation.

d. Emergency operation (if applicable).

e. Power source (if applicable).

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D.5.1.2.1 Description <descrip>.

Avionics equipment shall be described in detail, including controls, indicators, instruments (if applicable), jacks, switches, and control panels, etc. Antenna locations shall be shown on appropriate illustrations. Antenna arrangement illustrations shall be included in the Avionics Introduction WP and referenced or linked when required. The proper techniques and procedures to be employed when operating the equipment shall also be described.

D.5.1.2.2 Controls and functions <controls>.

Controls and functions information shall be developed as described below:

a. The location and function of each control, including built-in test capability, contributing to the operation of the avionics equipment shall be listed. Each control panel shall be discussed separately. Reference or links shall be made to illustrations in Module 2 regarding controls and control panels.

b. A tabular listing may be included for each control panel (see TABLE D-I for formatting example). each control or indicator shall be listed and its function defined in terms of what the operator of the control shall see, hear, or do as a result of the control setting. Terms of simple, immediate, and observable results shall be used. No attempt shall be made to give the operator the exact technical details about what happens when the control is used.

c. A tabular listing may be included for each control display unit (see TABLE D-I for formatting example). Each key that shall be pressed shall be listed and a description of the function shall be included in the table.

TABLE D-I. Formatting example of control and indicator table or control display unit table.

Control/Indicator	Function
LO SET knob	Either pilot's LO SET knob applies power to the altimeter system. LO set indicated on the altimeters system. LO set indicated on the altimeters can be set independently. Both LO set indices shall be masked to turn the set off.
HI SET knob	Sets position of HI set index and tests altimeter system when pressed.
LO set index	Indicates altitude trip point for LO caution light.
HI set index	Indicates altitude trip point for HI caution light.
Indicator pointer	Indicates absolute altitude from 0 to 1,500 feet.
Digital indicator	Provides direct reading four digit indication of absolute altitude from 0 to 1,500 feet.
LO caution light	Comes on when helicopter descends below altitude on LO set index.
HI caution light	Light comes on when helicopter rises above altitude on HI set index.

D.5.1.2.3 Operation <operations>.

A description of the operating details for each item of avionics equipment shall be provided. Whenever standard operational avionics data exist within the government, such data shall be furnished by the acquiring activity. Complete operating procedures shall be included as follows:

a. When separate modes of operation are available (e.g., when the equipment serves two or more systems), each mode shall be described. These shall be listed as modes of operation and each shall be briefly described.

b. The sequence of settings and the position to which the controls should be set to ensure proper results each time the equipment is energized shall be explained. Instructions shall be provided to prevent the possibility of damage through improper settings or sequence of operations. When appropriate, operating tolerances shall be called out. When operation of a unit is related to or dependent on the operation of a similar or independent control unit, this information shall be included in the operating procedure. Only those controls normally used

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by the operator shall be included; control adjustments that are the responsibility of maintenance personnel shall not be included.

c. If the configuration provides for a parallel operation from various positions in the aircraft, similar, separate, and complete coverage for each position shall be provided. When the procedure is identical to a position previously covered, it shall be covered by a reference to the previous procedure.

D.5.1.2.4 Emergency operations <emeroper>.

When applicable, settings and operations of avionics equipment during emergency operations shall be described.

D.5.1.2.5 Power source <power>.

When applicable, a brief description of the power sources for avionics equipment shall be provided, including any special procedures or limitations using, but not limited to, external power and battery power.

D.5.2 Avionics introduction work package <avionicintrowp>.

Except for mission avionics, general overall description covering the avionics equipment configurations installed on a specific aircraft shall be provided. It shall include a brief description of avionics equipment, its technical characteristics, capabilities, and locations. Mission avionics equipment shall be covered in Module 4.

D.5.3 Communications work package <commwp>.

Information for communications equipment installed in the aircraft shall be developed.

D.5.4 Navigation work packages <navwp>.

A description of all navigation systems and indicators shall be provided. When there is doubt as to whether the system should be covered under communications or navigation, the primary use of the system shall be the deciding factor. A suitable reference shall be made in the manual to aid in locating the material.

a. Non-directional Radio Beacon (NDB)

b. Very High Frequency (VHF) Omni-directional Range (VOR)

c. VOR receiver check

d. Tactical Air Navigation (TACAN)

e. VOR/TACAN (VORTAC)

f. Distance Measuring Equipment (DME)

g. Navigational Aid (NAVAID) service volumes

h. Instrument Landing System (ILS)

i. Simplified Directional Facility (SDF)

j. Microwave Landing System (MLS)

k. Long Range Navigation (LORAN)

l. Inertial Reference Unit (IRU), Inertial Navigation System (INS), and Attitude Heading Reference System (AHRS)

m. Doppler radar

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n. Global Positioning Radar (GPS)

o. Wide Area Augmentation System (WAAS)

p. Ground Based Augmentation System (GBAS) Landing System (GLS)

q. Precision approach systems other than ILS, GLS, and MLS

r. Automatic Direction Finder (ADF)

s. Gyro compass and magnetic indicators

t. Marker beacon

u. Flight director

v. Autopilot

w. Other

D.5.5 Transponder and radar work package <tran_radarwp>.

All information for transponders, collision warning systems, and radar systems and indicators, as applicable, shall be provided.

D.5.6 General avionics work packages <avionicgenwp>.

When specified by the acquiring activity, additional work packages may be developed when the specific type of avionics information and operation is not covered as described in D.1 through D.5. Each additional avionics equipment/system shall be a separate work package and shall comply with all of the requirements of D.1 through D.5. If additional avionics work packages are used, the proponent shall submit to AMCOM the requirements for this avionics work package type for possible incorporation within future revisions to this standard.

D.6 Notes.

The notes in Section 6 apply to this appendix.

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APPENDIX E

MISSION EQUIPMENT (MODULE 4)

E.1 Scope.

This appendix establishes the description of all standard mission equipment that may be utilized with the aircraft. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

E.2 Applicable publications.

The applicable publications in Section 2 apply to this appendix.

E.3 Definitions.

The definitions in Section 3 apply to this appendix.

E.4 General requirements.

A description of all standard mission equipment that may be utilized with the aircraft shall be provided. Coverage shall include description, controls and function, operating procedures, power sources, and illustrations.

E.4.1 Preparation of digital data for electronic delivery.

Data prepared and delivered digitally in accordance with this standard shall be XML tagged using the current Army DTD. Data shall be formatted for presentation using the Army stylesheets. Refer to E.4.2 for information on obtaining or accessing the Army DTD and stylesheets. XML tags used in the DTD are noted throughout the text of this standard in bracketed, bold characters (e.g., **<geninfo>**) as a convenience for the author and to ensure that the tags are used correctly when developing a document instance.

E.4.2 Use of the Document Type Definition (DTD)/stylesheet.

The DTD referenced in this appendix interprets the technical content and structure for the functional requirements contained in this standard. Development of the TMs shall be accomplished through the use of this standard, the Army DTD, and the Army stylesheets.

E.4.3 Content structure.

The examples provided herein are an accurate representation of the content structure requirements contained in this appendix and shall be followed to permit the effective use of the DTD.

E.4.4 Style and format.

This standard provides style and format requirements for the technical content requirements described in this appendix. These requirements are considered mandatory and are intended for compliance.

E.4.5 Work package development.

Data developed in accordance with this appendix shall be divided into work packages. These work packages shall be stand alone and shall contain all information and references required to support the work package type.

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E.4.6 Selective application and tailoring of content using Appendix A matrices.

This standard contains some requirements that may not be applicable to the preparation of all TMs. Selective application and tailoring of requirements contained in this standard are the responsibility of the acquiring activity and shall be accomplished using APPENDIX A. The applicability of some requirements is also designated by one of the following statements: unless specified otherwise by the acquiring activity; as specified by the acquiring activity; or when specified by the acquiring activity.

E.5 Detailed requirements.

All standard mission equipment that may be utilized with the aircraft shall be described in detail. Coverage shall include description, controls and function, operating procedures, power sources, and illustrations.

E.5.1 Preparation of mission equipment work packages.

Mission Equipment work packages shall be developed as detailed in E.5.1.1 through E.5.1.2.

E.5.1.1 Mission equipment work packages.

The avionics module shall be divided into the following work packages:

- **a.** Mission Equipment Introduction work package
- **b.** Non-Standard Mission Avionics Equipment work package
- **c.** Armament work package (if applicable)
- **d.** Cargo Handling work package (if applicable)
- **e.** Passive Defense work package (if applicable)
- **f.** System Payload work package (if applicable)
- **g.** General Mission Equipment work packages (if applicable)

Each work package type, with the exception of the introduction work package, is repeatable and may be used multiple times to better organize and structure the content for the operator.

E.5.1.2 Work package contents.

For each item of mission equipment contained within the work packages detailed in E.5.1.2.1 through E.5.1.2.5, the following information shall be included:

- **a.** Description.
- **b.** Controls and functions.
- **c.** Operation.
- **d.** Emergency operation (if applicable).
- **e.** Power source (if applicable).

E.5.1.2.1 Description <descrip>.

Mission equipment shall be described in detail, including controls, indicators, instruments (if applicable), jacks, switches, and control panels, etc. The proper techniques and procedures to be employed when operating the equipment shall also be described.

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E.5.1.2.2 Controls and functions <controls>.

Controls and functions information shall be developed as described below:

a. The location and function of each control, including built-in test capability, contributing to the operation of the mission equipment shall be listed. Each control panel shall be discussed separately. Reference or links shall be made to illustrations in Module 2 regarding controls and control panels.

b. A tabular listing may be included for each control panel (see TABLE D-I for formatting example). Each control or indicator shall be listed and its function defined in terms of what the operator of the control shall see, hear, or do as a result of the control setting. Terms of simple, immediate, and observable results shall be used. No attempt shall be made to give the operator the exact technical details about what happens when the control is used.

c. A tabular listing may be included for each control display unit (see TABLE D-I for formatting example). Each key that shall be pressed shall be listed and a description of the function shall be included in the table.

E.5.1.2.3 Operation <operations>.

A description of the operating details for each item of mission equipment shall be provided. Whenever standard operational mission equipment data exists within the government, such data shall be furnished by the acquiring activity. Complete operating procedures shall be included as follows:

a. When separate modes of operation are available, i.e., when the equipment may serve two or more systems, each mode shall be described. These shall be listed as modes of operation and each shall be briefly described.

b. The sequence of settings and the position to which the controls should be set to ensure proper results each time the equipment is energized shall be explained. Instructions shall be provided to prevent the possibility of damage through improper settings or sequence of operations. When appropriate, operating tolerances shall be called out. When operation of a unit is related to or dependent on the operation of a similar or independent control unit, this information shall be included in the operating procedure. Only those controls normally used by the operator shall be included; control adjustments that are the responsibility of maintenance personnel shall not be included.

c. If the configuration provides for a parallel operation from various positions in the aircraft, similar, separate, and complete coverage for each position shall be provided. When the procedure is identical to a position previously covered, it shall be covered by a reference to the previous procedure.

E.5.1.2.4 Emergency operations <emeroper>.

When applicable, settings and operations of mission equipment during emergency operations shall be described.

E.5.1.2.5 Power source <power>.

When applicable, a brief description of the power sources for mission equipment shall be provided, including any special procedures or limitations using, but not limited to, external power and battery power.

E.5.2 Mission equipment introduction work package <missneqp_introwp>.

An overall description of all standard mission equipment that may be utilized with the aircraft shall be provided. It shall include a brief description of mission equipment, its technical characteristics, capabilities, and locations.

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E.5.3 Non-standard mission avionics work package <nonstandwp>.

Note

Classified information on mission avionics equipment shall be covered in a separate classified publication.

Unclassified information regarding mission avionics equipment that is not a part of the standard flight communication, navigation, transponder, or radar equipment shall be included.

a. Electronic equipment such as radio monitoring systems, Side Looking Airborne Radar (SLAR), IR devices, and photographic equipment shall be covered.

b. Detailed information shall be given regarding the photographic equipment including, but not limited to, types of cameras, control stations, camera doors, and capabilities of the equipment.

c. Gun camera equipment shall be covered.

d. Mission avionics equipment that requires extensive explanation of operating procedures shall be covered in this work package or separately in another Non-Standard Mission Avionics Work Package.

E.5.4 Armament work package <armamentwp>.

The description of gunnery, rocket, tow target, control, and computer equipment and their interrelations when installed shall be provided. Armor protection shall be discussed along with the individual item that is being protected. Precautions and safety considerations shall also be included. All information for armament control systems, gunnery equipment, rocket equipment, missiles, and laser control systems, as applicable, shall be provided. To better organize and manage the data required in the WP, it may be separated into multiple Armament WPs or combined into one WP so long as the WP remains under 50 pages.

E.5.4.1 Armament control system.

Description and operating instructions for the armament control system shall be provided, if applicable. Also, information such as presentation on the scope or sight, when applicable, shall be included. Warm-up time and preflight, in-flight before landing and after landing checks shall be listed.

E.5.4.2 Gunnery equipment.

Information shall be included on all guns and turrets, including quantity of ammunition that can be carried for each gun, if applicable. When describing remote controlled turrets, the manual shall include, but shall not be limited to, the station from which the turret is operated, method of gaining control of the turret, and method of transferring control. All gunnery controls shall be covered, including gun sight and gun heater.

E.5.4.3 Rocket equipment.

Information shall be provided regarding the firing procedures, description and capability, controls, and types and number of rockets that can be carried, if applicable. Typical combinations of rockets and firing order shall be covered. Special precautions, if any, shall be listed.

E.5.4.4 Missiles.

Information shall be provided regarding the firing procedures, description and capability, controls, and types and numbers of missiles that can be carried, if applicable. Special precautions, if any, shall be listed.

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E.5.4.5 Laser control system.

Description and operating instructions for the laser control system shall be provided, if applicable. Also, information such as presentation on the scope or sight, when applicable, shall be included. Warm-up time and preflight, in-flight before landing and after landing checks shall be listed.

E.5.5 Cargo handling work package <cargowp>.

Descriptions and procedures for cargo handling systems and equipment to include hoists, winches, and cargo hooks shall be provided.

E.5.6 Passive defense work package <passdefwp>.

Passive defense equipment shall be described, procedures outlines, and controls and precautions listed. Employment methods shall also be discussed.

E.5.7 System payload work package <syspayloadwp>.

System payload shall be described, as applicable.

E.5.8 General mission equipment work package <missngenwp>.

At the acquiring activity's discretion, additional work packages shall be used to describe systems not covered in other work packages. Each additional system shall be a separate work package and shall comply with all of the requirements of E.1 through E.5. If additional mission equipment work packages are used, the proponent shall submit to AMCOM the requirements for this mission equipment work package type for possible incorporation within future revisions to this standard.

E.6 Notes.

The notes in Section 6 apply to this appendix.

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APPENDIX F

OPERATING LIMITS AND RESTRICTIONS (MODULE 5)

F.1 Scope.

This appendix establishes the technical content requirements for the preparation of operating limits and restrictions information for aviation weapon systems. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

F.2 Applicable publications.

The applicable publications in Section 2 apply to this appendix.

F.3 Definitions.

The definitions in Section 3 apply to this appendix.

F.4 General requirements.

All important operating limits and restrictions that shall be observed during ground and flight operations shall be provided. Special emphasis shall be placed on any unusual restrictions which are particularly characteristic of the aircraft. All time limited operations shall include a time limit and the upper and lower boundaries.

F.4.1 Preparation of digital data for electronic delivery.

Data prepared and delivered digitally in accordance with this standard shall be XML tagged using the current Army DTD. Data shall be formatted for presentation using the Army stylesheets. Refer to F.4.2 for information on obtaining or accessing the Army DTD and stylesheets. XML tags used in the DTD are noted throughout the text of this standard in bracketed, bold characters (e.g., **<geninfo>**) as a convenience for the author and to ensure that the tags are used correctly when developing a document instance.

F.4.2 Use of the Document Type Definition (DTD)/stylesheet.

The DTD referenced in this appendix interprets the technical content and structure for the functional requirements contained in this standard. Development of the TMs shall be accomplished through the use of this standard, the Army DTD, and the Army stylesheets.

F.4.3 Content structure.

The examples provided herein are an accurate representation of the content structure requirements contained in this appendix and shall be followed to permit the effective use of the DTD.

F.4.4 Style and format.

This standard provides style and format requirements for the technical content requirements described in this appendix. These requirements are considered mandatory and are intended for compliance.

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F.4.5 Work package development.

Data developed in accordance with this appendix shall be divided into work packages. These work packages shall be stand alone and shall contain all information and references required to support the work package type.

F.4.6 Selective application and tailoring of content using Appendix A matrices.

This standard contains some requirements that may not be applicable to the preparation of all TMs. Selective application and tailoring of requirements contained in this standard are the responsibility of the acquiring activity and shall be accomplished using APPENDIX A. The applicability of some requirements is also designated by one of the following statements: unless specified otherwise by the acquiring activity; as specified by the acquiring activity; or when specified by the acquiring activity.

F.5 Detailed requirements.

All important operating limits and restrictions that shall be observed during ground and flight operations shall be provided. Special emphasis shall be placed on any unusual restrictions which are particularly characteristic of the aircraft. All time limited operations shall include a time limit and the upper and lower boundaries.

F.5.1 Limitations and restrictions introduction work package <limit_introwp>.

General information on aircraft limits and restrictions, including decals and placards, shall be provided within this work package. The following statements shall be included as applicable:

a. "Purpose. This module identifies or refers to all important operating limits and restrictions that shall be observed during ground and flight operations.

b. General. The operating limitations set forth in this module are the direct result of design analysis, tests, and operating experiences. Compliance with these limits shall allow the pilot to safely perform the assigned missions and to derive maximum utility from the aircraft.

c. Exceeding operational limits. Any time an operational limit is exceeded, an appropriate entry shall be made on DA Form 2408-13-1. The entry shall state what limit or limits were exceeded, range, time beyond limits, and any additional data that would aid maintenance personnel in the maintenance action that may be required.

d. Minimum crew requirements. The minimum crew required for flight is *(fill in proper number)*. Additional crew members, as required, will be added at the discretion of the commander in accordance with pertinent Department of the Army (DA) regulations."

F.5.2 Aircraft system limits work package <syslimitwp>.

All aircraft system limits not covered elsewhere in this module that may restrict operation shall be provided using the System Limits WP.

F.5.2.1 Instrument, interactive display, or display operation ranges and markings.

Each instrument, interactive display, or display that indicates an operating limit(s) shall be illustrated and accurately reflect the actual markings/displays on the instrument, interactive display, or display (FIGURE F-1 and FIGURE F-2).

a. The information appearing on the illustration depicting markings or displays shall not be repeated in the text or table.

b. The color coded markings/displays or interactive display graphic symbols shall be fully explained.

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c. If the instrument, interactive display, or display limits cannot be adequately explained in the space provided for the captions, explanations shall be included under the appropriate paragraph heading.

d. The text shall state or describe all limit ranges, including gaps that may be shown in range markings.

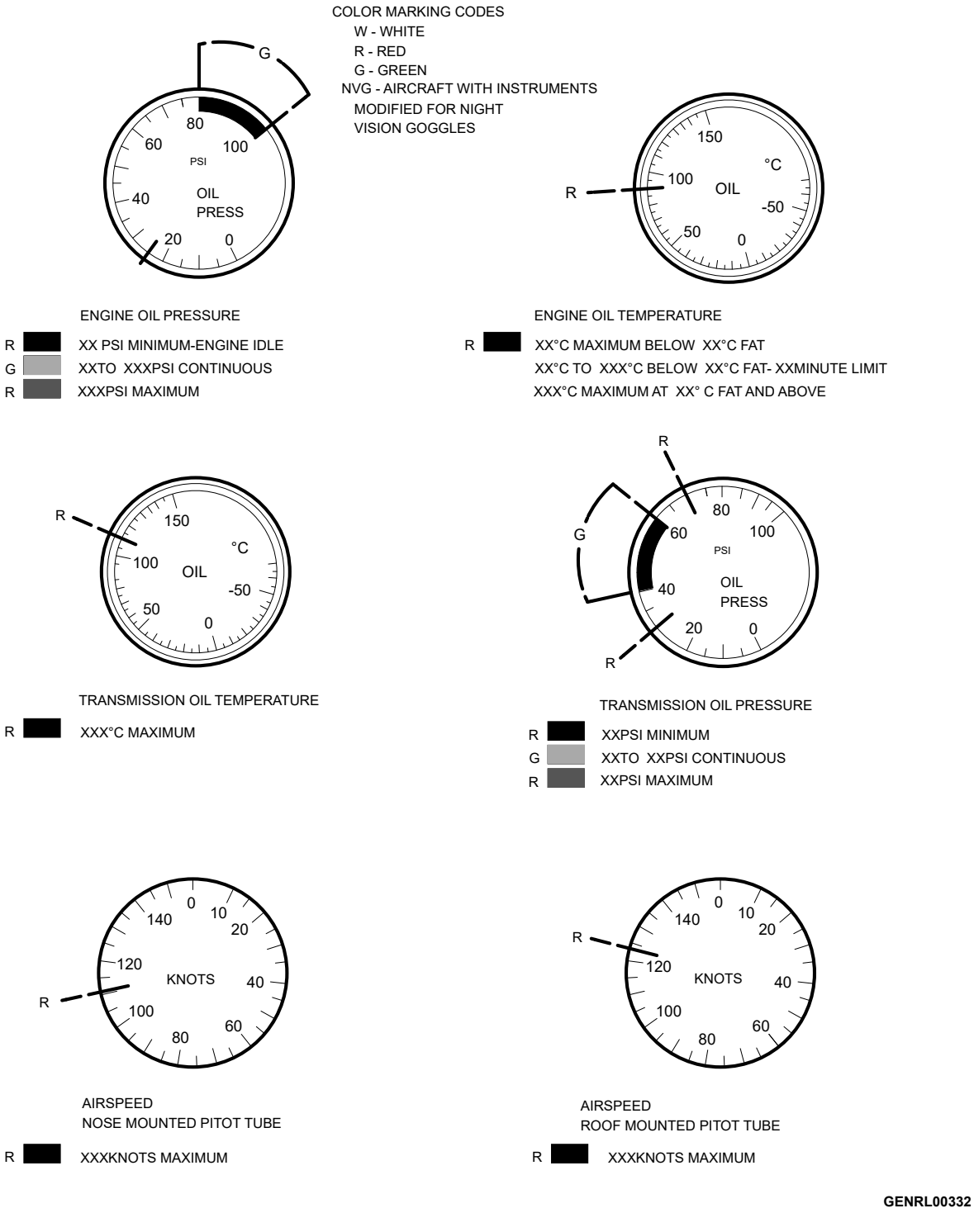
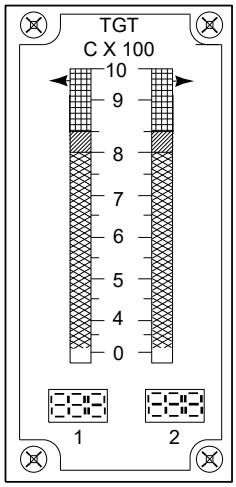
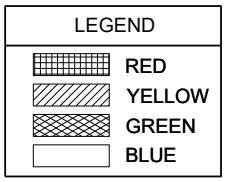


FIGURE F-1. Example 1 of instrument/display operating ranges and markings.

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NOTE:
LIMITS BASED ON INDICATED TGT

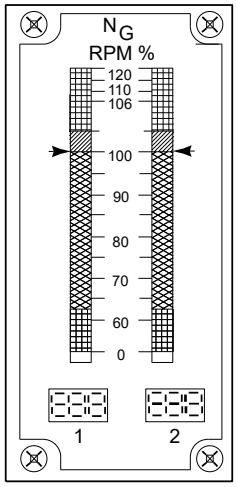


701 ENGINE TURBINE GAS TEMPERATURE (TGT°C)

	LIMITS	
RED	XXX	MAXIMUM
RED	XXX-XXX	TRANSIENT (XXSECONDS)
RED	XXX-XXX	SINGLE ENGINE CONTINGENCY (XXMINUTE UNIT)
RED	XXX	AUTOMATIC DUAL ENGINE TGT UNITING
YELLOW	XXX	MAXIMUM DURING START
YELLOW	XXX-XXX	IRP (XXMINUTES)
GREEN	XXX	NCP
GREEN	X-XXX	NORMAL OPERATION
BLUE		INSTRUMENT POWER ON

701C ENGINE TURBINE GAS TEMPERATURE (TGT°C)

	LIMITS	
	XXX	MAXIMUM
RED	XXX-XXX	TRANSIENT (XXSECONDS)
RED	XXX-XXX	SINGLE ENGINE CONTINGENCY (XXMINUTE UNIT)
RED	XXX	AUTOMATIC DUAL ENGINE TGT UNITING
YELLOW	XXX	MAXIMUM DURING START
YELLOW	XXX-XXX	IRP (XXMINUTES)
YELLOW	XXX-XXX	IRP (XXMINUTES)
GREEN	XXX	NCP
GREEN	X-XXX	NORMAL OPERATION
BLUE		INSTRUMENT POWER ON



ENGINE GAS GENERATOR SPEED (N_G) (RPM%)

		LIMITS	
RED	(UPPER)	XXX-XXX	TRANSIENT XXSECOND MAXIMUM
YELLOW		XX-XXX	XXMINUTE LIMIT
GREEN		XX-XX	NORMAL OPERATION
RED	(LOWER)	XX	MINIMUM-ENGINE OUT WARNING LIGHT SET AT THIS VALUE
BLUE			INSTRUMENT POWER ON
RED			BEGINNING OF A RED RANGE (FROM A NORMAL OPERATING REFERENCE)
YELLOW			BEGINNING OF A YELLOW RANGE (FROM A NORMAL OPERATING RANGE)

GENRL00397

FIGURE F-2. Example 2 of instrument/display operating ranges and markings.

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F.5.2.2 Propeller limitations.

Propeller limitations shall be discussed including, but not limited to, reverse pitch and restricted revolutions per minute (rpm).

F.5.2.3 Rotor limitations.

For rotary wing aircraft, rotor limitations during both flight and ground operation shall be discussed, covering such points as restricted rpm, auto rotational rpm, limitations for startup and shutdown during high winds, and wind gust spread.

F.5.2.4 Additional limitations.

All system limits and restrictions not described by instrument markings shall be included. Limits and restrictions should be observed when operating utility, heating, ventilation, cooling, or rain removal systems shall also be included.

F.5.3 Power limits work package <pwrlimitwp>.

Power limits shall include engine and drive train and idle limitations. Limitations that shall be observed when alternate fuel grades are used shall be included. Acceleration limits and restrictions that apply to the engine shall be covered. Limits shall be expressed in terms of observable indications that are available to the flight crew; e.g., 360 ° C, 46 lbs, 10 pounds per square inch (psi). Terms such as military power or takeoff power shall not be used.

F.5.4 Loading limits work package <loadlimitwp>.

Loading limits pertaining to the aircraft shall be discussed in detail.

F.5.4.1 Center-of-Gravity (CG) limitations.

Longitudinal limitations shall be described. Lateral limitations shall be described when specified by the acquiring activity. Also, the following statement shall be included:

"CG limits for the aircraft to which this manual applies and appropriate charts for computation of the CG are contained in Module 6."

F.5.4.2 Weight limitations.

All minimum/maximum aircraft weight limitations including parking, towing, taxiing, takeoff, and landing from prepared/unprepared fields shall be provided. For aircraft in which weight distribution is a problem (such as minimum fuel to be carried in the wings at various gross weights), coverage of the limitations involved shall be included. References or links shall be made to fuel management in Module 2, as necessary.

F.5.4.3 Turbulence.

Restrictions regarding flying in all levels of turbulence shall be discussed and limitations shall be covered.

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F.5.4.4 Other loading limitations.

Other types of loading limitations that affect operations shall be covered, including the following:

a. Additional restrictions to be observed when carrying stores. For aircraft equipped to carry a variety of external stores, information concerning the stores carried at each station and the maximum lateral unbalanced load that can be carried shall be included.

b. Limitations as to the weight for external sling loads on rotary wing aircraft and speed restrictions, if any.

c. Floor loading limits that are to be observed when carrying internal cargo.

d. Restrictions on jettisoning external stores and sling loads.

F.5.5 Maximum and minimum airspeed limits work package <airspdlimitwp>.

Airspeed limitations shall be discussed, including level flight airspeed, diving airspeed, airspeed for various degrees of flap extension, airspeed for various stabilator positions, airspeed for door opening, and airspeeds under various conditions of weight and configuration.

a. For rotary wing aircraft, sideward and rearward airspeed limits and restrictions shall be discussed.

b. Airspeeds shall be expressed as Knots Indicated Airspeed (KIAS), unless otherwise specified by the acquiring activity.

c. An airspeed operating limits chart shall be included. This chart shall present operating limits for forward flight at various gross weights, pressure altitudes, Free Air Temperature (FAT), and KIAS (FIGURE F-3).

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AIRSPEED OPERATING LIMITS

EXAMPLE

WANTED

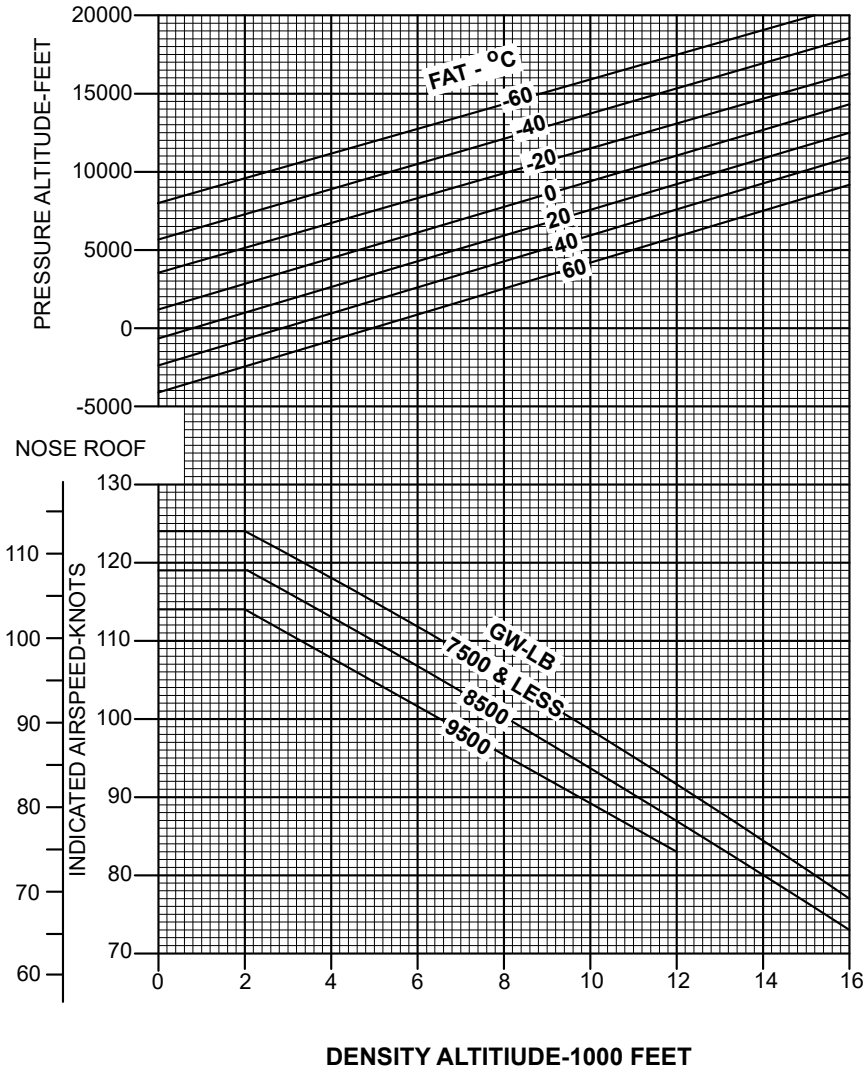
INDICATED AIRSPEED
AND DENSITY ALTITUDE

KNOWN

GROSS WEIGHT = XXXXXLB
PRESSURE ALTITUDE = XXXXFEET
FAT = XX°C
ROOF MOUNTED SYSTEM

METHOD

ENTER PRESSURE ALTITUDE
MOVE RIGHT TO FAT
MOVE DOWN TO GROSS WEIGHT
MOVE LEFT, READ INDICATED
AIRSPEED = XXXKNOTS
REENTER PRESSURE ALTITUDE
MOVE DOWN, READ DENSITY
ALTITUDE = XXXXFEET



DATA BASIS: DERIVED FROM FLIGHT TEST

GENRL00333

FIGURE F-3. Airspeed operating limits chart.

F.5.6 Maneuvering limits work package <maneuvlimitswp>.

Maneuvering flight limitations to include acrobatic flight, if applicable, shall be described.

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a. Acceleration limitations shall also be covered, including maximum acceleration with tip tanks and maximum bank angle at high gross weight.

b. Maximum permissible accelerations under various flight conditions at specific gross weights and fuel weights shall be detailed.

c. For aircraft not equipped with G meters, G forces shall be expressed in terms that are recognizable by the pilot, such as airspeed and bank angle.

d. Restrictions on control movements shall be listed.

e. Material shall be presented on permissible bank angles, side slip, flight loads and slope operations.

f. Prohibited maneuvers shall be listed as appropriate.

g. A flight envelope chart shall be included.

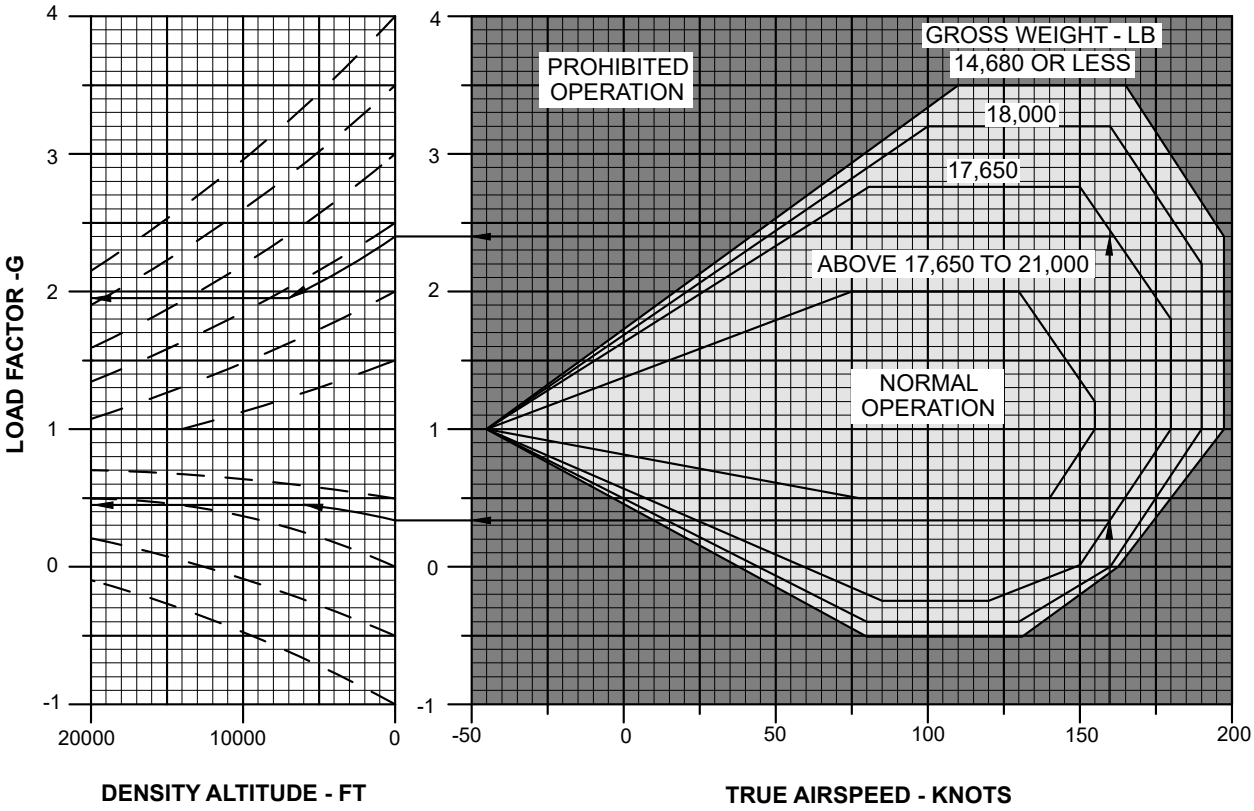
(1) For aircraft with G meters, plots of load factor versus speed for the full range of gross weight shall be shown. The speeds at which maneuvers are restricted and unrestricted, as a function of load limit factors, shall be presented (FIGURE F-4).

(2) When changes in configuration result in variations in airspeed position error, separate airspeed scales shall be shown.

(3) Where direct reading Mach meters are provided, charts for both Indicated Airspeed (IAS) and Indicated Mach Number (IMN) shall be provided.

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FLIGHT ENVELOPE



EXAMPLE

WANTED

MAXIMUM AND MINIMUM LOAD FACTOR

KNOWN

GROSS WEIGHT = XXXXXPOUNDS
DENSITY ALTITUDE = XXXXFEET
AIRSPEED = XXXKTAS

METHOD

ENTER AT V = XXXKT. MOVE UP TO UPPER AND LOWER ENVELOPE BOUNDARIES FOR
GROSS WEIGHT = XXXXXPOUNDS
MOVE LEFT TO DENISTY ALTITUDE OF ZERO FEET
SLIDE TO LEFT ALONG DASHED LINES TO XXXXFEET DENSITY ALTITUDE
MOVE LEFT TO LOAD FACTOR SCALE, READ MAX G = XXX, MIN G = XXX

GENRL00334

FIGURE F-4. Flight envelope chart.

F.5.7 Environmental restrictions work package <envlimitwp>.

As applicable, altitude, temperature, rain, snow, ice, hail and oxygen limits shall be provided.

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a. Material on maximum wind velocity and gust spread, maximum wind velocity for crosswind operations, wind from the critical azimuth, turbulence, and normal operation shall be included.

b. Operations under wind azimuth direction and wind velocity conditions that should be avoided shall be discussed.

c. Where appropriate, charts shall be used to depict the preceding conditions.

d. The definition of Instrument Meteorological Conditions (IMC) and the criteria for such flights shall be provided. In addition, when applicable, information on when a particular aircraft is qualified for operation in instrument meteorological conditions, and when a certain aircraft is restricted to visual flight conditions shall be included.

F.5.8 General operating limits and restrictions work package <genlimitwp>.

When specified by the acquiring activity, additional work packages may be developed when the specific type of operating limit or restriction is not covered as described in F.5.1 through F.5.7. Each additional operating limit or restriction shall be a separate work package and shall comply with all of the requirements of F.1 through F.5. If additional operating limits and restriction work packages are used, the proponent shall submit to AMCOM the requirements for this operating limits and restriction work package type for possible incorporation within future revisions to this standard.

F.6 Notes.

The notes in Section 6 apply to this appendix.

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APPENDIX G

WEIGHT/BALANCE AND LOADING (MODULE 6)

G.1 Scope.

This appendix establishes the technical content requirements for the preparation of weight, balance, and loading information for aviation weapon systems. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

G.2 Applicable publications.

The applicable publications in Section 2 apply to this appendix.

G.3 Definitions.

The definitions in Section 3 apply to this appendix.

G.4 General requirements.

Sufficient instructions and information so that an operator, given the proper data, can compute any combination of weight and balance shall be provided.

G.4.1 Preparation of digital data for electronic delivery.

Data prepared and delivered digitally in accordance with this standard shall be XML tagged using the current Army DTD. Data shall be formatted for presentation using the Army stylesheets. Refer to G.4.2 for information on obtaining or accessing the Army DTD and stylesheets. XML tags used in the DTD are noted throughout the text of this standard in bracketed, bold characters (e.g., **<geninfo>**) as a convenience for the author and to ensure that the tags are used correctly when developing a document instance.

G.4.2 Use of the Document Type Definition (DTD)/stylesheet.

The DTD referenced in this appendix interprets the technical content and structure for the functional requirements contained in this standard. Development of the TMs shall be accomplished through the use of this standard, the Army DTD, and the Army stylesheets.

G.4.3 Content structure.

The examples provided herein are an accurate representation of the content structure requirements contained in this appendix and shall be followed to permit the effective use of the DTD.

G.4.4 Style and format.

This standard provides style and format requirements for the technical content requirements described in this appendix. These requirements are considered mandatory and are intended for compliance.

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G.4.5 Work package development.

Data developed in accordance with this appendix shall be divided into work packages. These work packages shall be stand alone and shall contain all information and references required to support the work package type.

G.4.6 Selective application and tailoring of content using Appendix A matrices.

This standard contains some requirements that may not be applicable to the preparation of all TMs. Selective application and tailoring of requirements contained in this standard are the responsibility of the acquiring activity and shall be accomplished using APPENDIX A. The applicability of some requirements is also designated by one of the following statements: unless specified otherwise by the acquiring activity; as specified by the acquiring activity; or when specified by the acquiring activity.

G.5 Detailed requirements.

Sufficient instructions and information so that an operator, given the proper data, can compute any combination of weight and balance shall be provided. Work packages shall be developed as detailed in G.5.1 through G.5.8.

G.5.1 Weight/balance and loading introduction work package <wtbal_introwp>.

General statements about the importance of weight and balance calculations shall be provided. In addition, when weight and balance computer/calculators are provided for the aircraft, instructions and examples of their use shall be based on gear down configurations, with supplementary data for gear up conditions (when required).

G.5.1.1 Required statements.

The notes and statements detailed in G.5.1.1.1 through G.5.1.1.2 shall be included.

G.5.1.1.1 Note.

A note that Module 6 contains sufficient instructions and data so that an aviator, given the proper data, can compute any combination of weight and balance shall be included.

G.5.1.1.2 Weight and balance class statement.

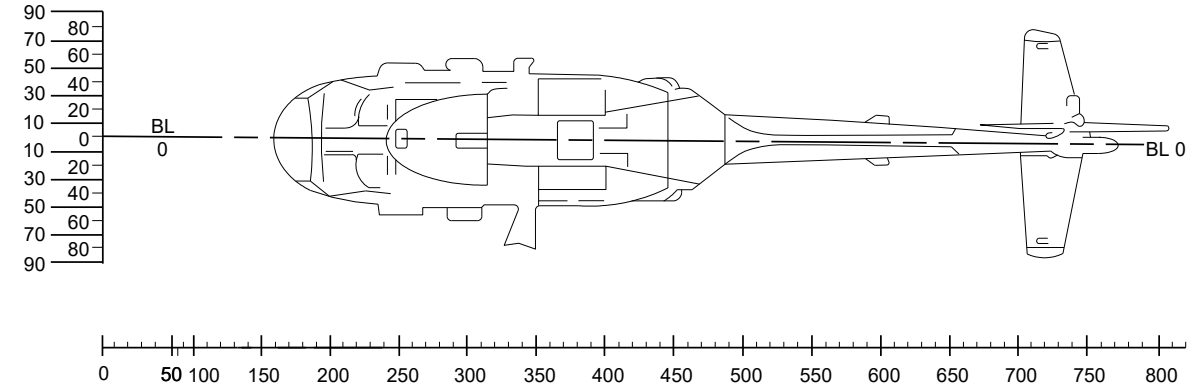
The following statement shall be included verbatim:

"Army *(insert assigned aircraft designation)* is in class *(insert class)*. Additional directives governing weight and balance of class *(insert class)* aircraft forms and records are contained in AR 95-1."

G.5.1.2 Aircraft compartment and station diagram.

A general description of the aircraft compartments and a supporting diagram shall be provided. The diagram shall show the reference datum line, stations, butt lines, and water lines in inches (FIGURE G-1 and FIGURE G-2).

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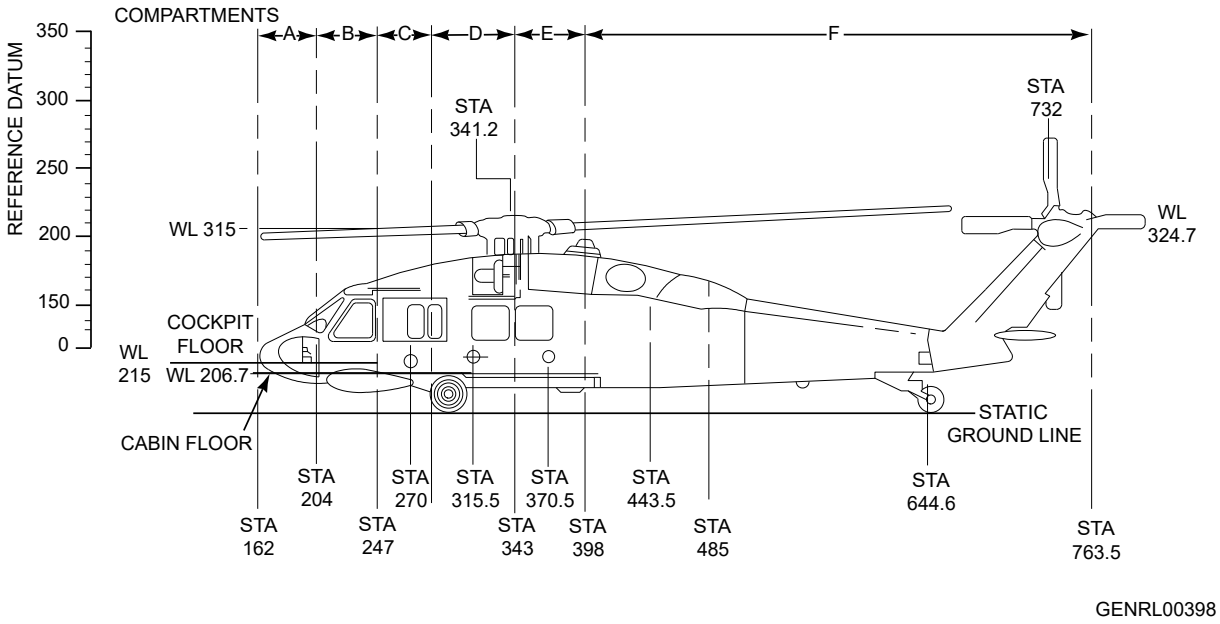


FIGURE G-1. Example aircraft compartments and stations diagram 1.

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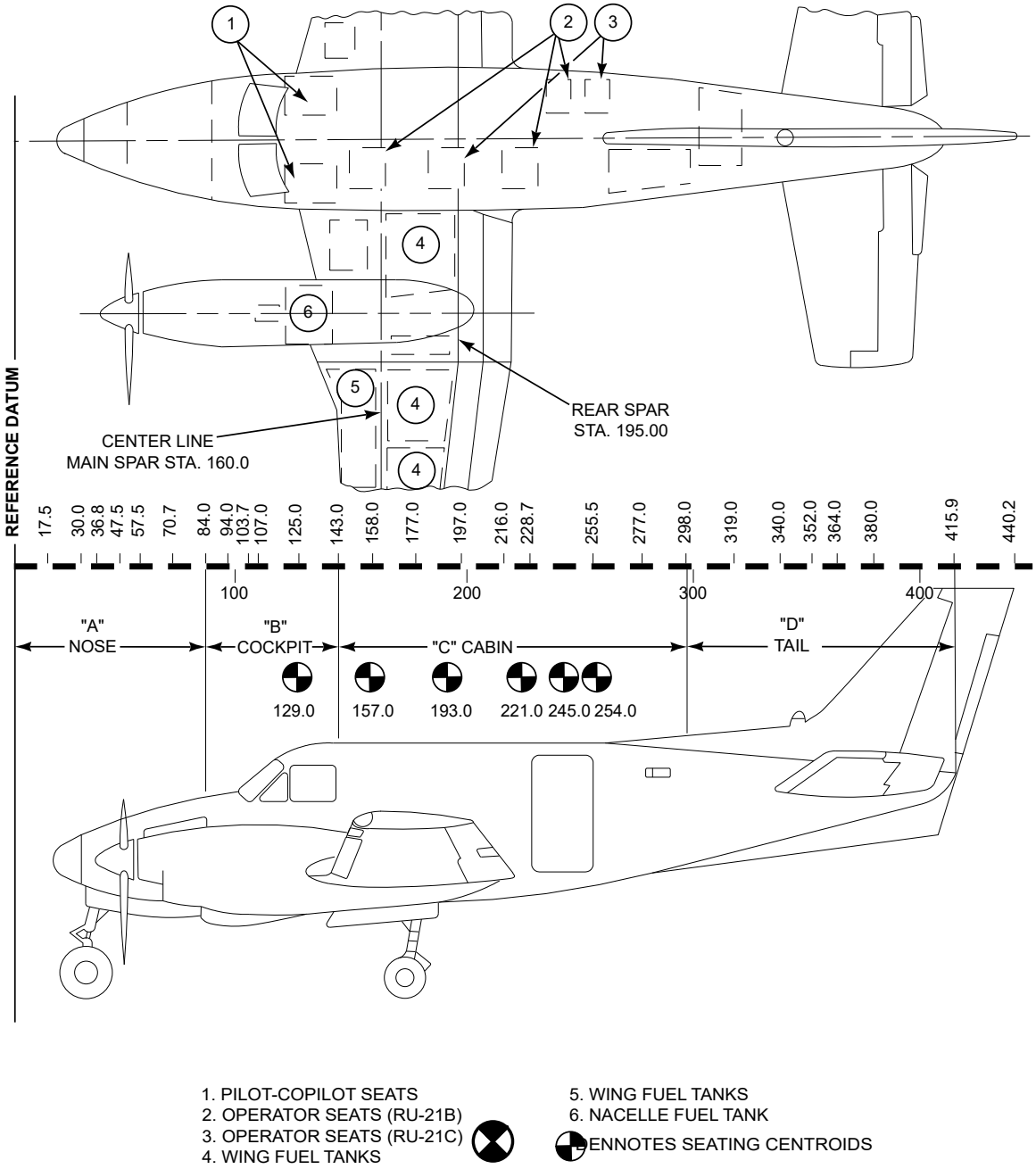


FIGURE G-2. Example aircraft compartments and stations diagram 2.

G.5.2 Weight and balance work package <wtbalwp>.

Information necessary for the computation of weight and balance for loading of specific aircraft shall be provided.

a. Instructions for completion of weight and balance forms (DD Form 365 series) shall not be provided in the manual; however, TM 55-1500-342-23 which provides these instructions shall be referenced.

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b. Sufficient information shall be provided to permit the flight crew to readily use the data presented in the other sections of this module to determine loading arrangements, fuel burn or transfer sequences, and other weight and balance procedures to ensure the aircraft remains within weight and balance limits for the entire flight.

G.5.3 Fuel/oil work package <fuelwtbalwp>.

A fuel and oil work package shall be developed as detailed in G.5.3.1 through G.5.3.2.

G.5.3.1 Fuel quantity data.

Fuel quantity data shall be in chart form (FIGURE G-3).

a. The names of the tanks on the charts shall be identical to the name appearing on the tank selector (a more explanatory title may be carried in parentheses if desired).

b. Any group of tanks or cells that are interconnected to fill and drain shall be treated as a single tank.

c. The chart shall include data on each tank (including droppable and ferry) that is designed for use with the aircraft.

d. Tank volume shall be given in terms of usable fuel rather than total tank volume.

e. Fuel quantities shall be given in gallon regardless of the type of instrumentation.

f. All gallon figures shall be followed by the conversion to pounds.

g. The grid lines within the chart shall be based on fuel weight in pounds of fuel.

h. It shall be stated that the weights are based on a given specific gravity at standard day temperatures.

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EXAMPLE

WANTED

FUEL MOMENT

KNOWN

FUEL QUANTITY
MAIN XXXXPOUNDS

METHOD

FOR MAIN TANK ENTER
AT XXXXPOUNDS AND
MOVE RIGHT TO MAIN LINE.
MOVE DOWN READ
MOMENT/XXXX= XX

FUEL MOMENTS

ITEM	STA	WEIGHT LBS	MOM/XXXX
XXX- GALLON TANK (IB OR OB)	XXX	XXX	XX
XXX- GALLON TANK (IB)	XXX	XXX	XX

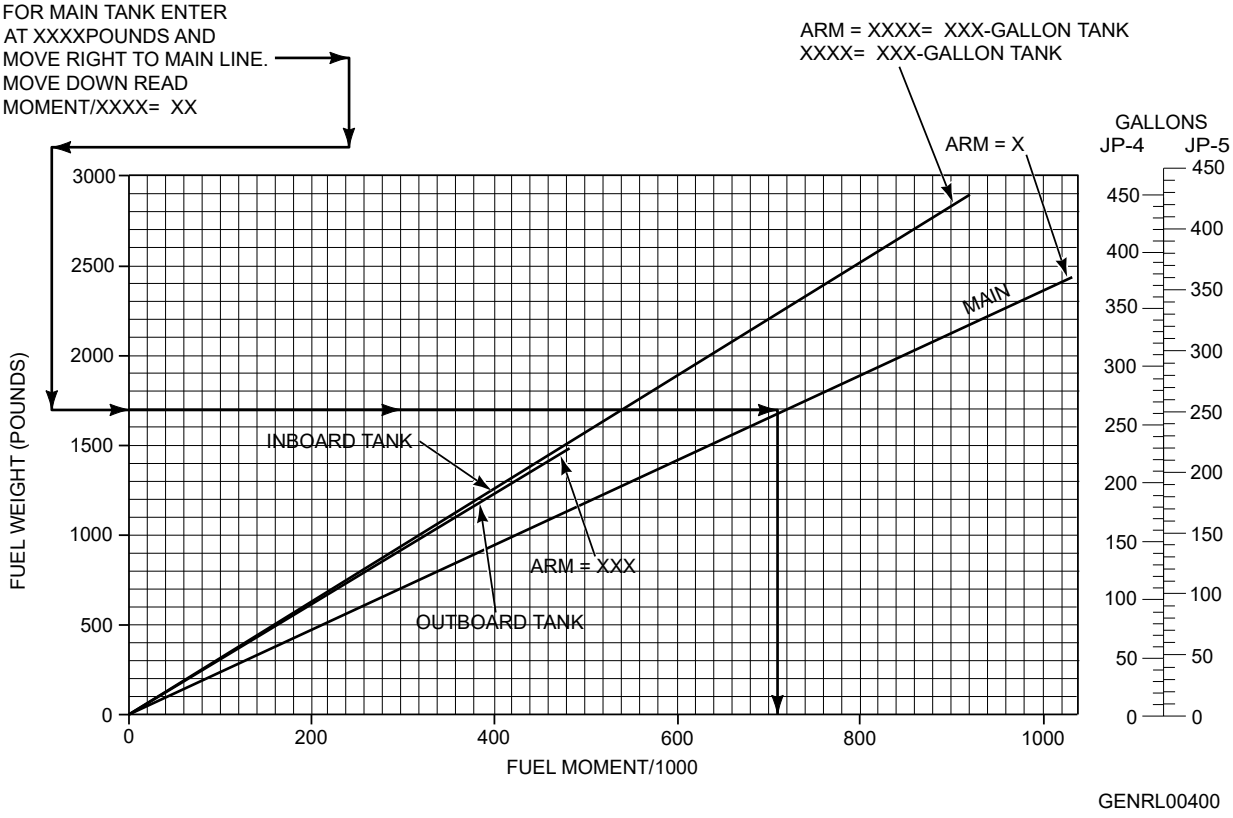


FIGURE G-3. Fuel moment chart.

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G.5.3.2 Oil data.

When specified by the acquiring activity, a statement of usable oil capacity, equivalent in pounds, total moments, and fuselage station number shall be provided.

a. Aircraft that have a large usable oil capacity shall have a tabular listing if oil loading computation is critical.

b. It shall be noted that the weights are based on a given specific gravity at standard day temperature.

G.5.4 Personnel work package <perswtbalwp>.

All essential information and instructions for preparation, loading, and unloading of personnel, including airborne troops shall be provided.

G.5.4.1 Personnel compartment and entrances.

A general description of the personnel compartment and entrances, including profile and cross-section drawings showing all dimensions, in inches, shall be provided. In addition, a description shall be provided of any critical dimensions which limit full use of the personnel compartment.

G.5.4.2 Personnel loading and unloading.

Personnel loading and unloading shall include, but shall not be limited to, a checklist and description of steps necessary for loading and unloading troops as follows:

a. Troop seat installation.

b. A description and operation of safety belts and harness.

c. A check of comfort and emergency provisions.

d. Instructions for troop loading and unloading procedure.

G.5.4.3 Personnel weight.

When aircraft are operated at critical gross weights, the exact weight of each individual occupant, including the weight of the equipment shall be provided. In addition, if weighting facilities are not available, or if the tactical situation dictates otherwise, loads shall be included and computed as follows:

a. Combat equipped soldiers - 240 lbs per individual.

b. Combat equipped paratroopers - 260 lbs per individual.

c. Litter and patient’s weight - 265 lbs per patient.

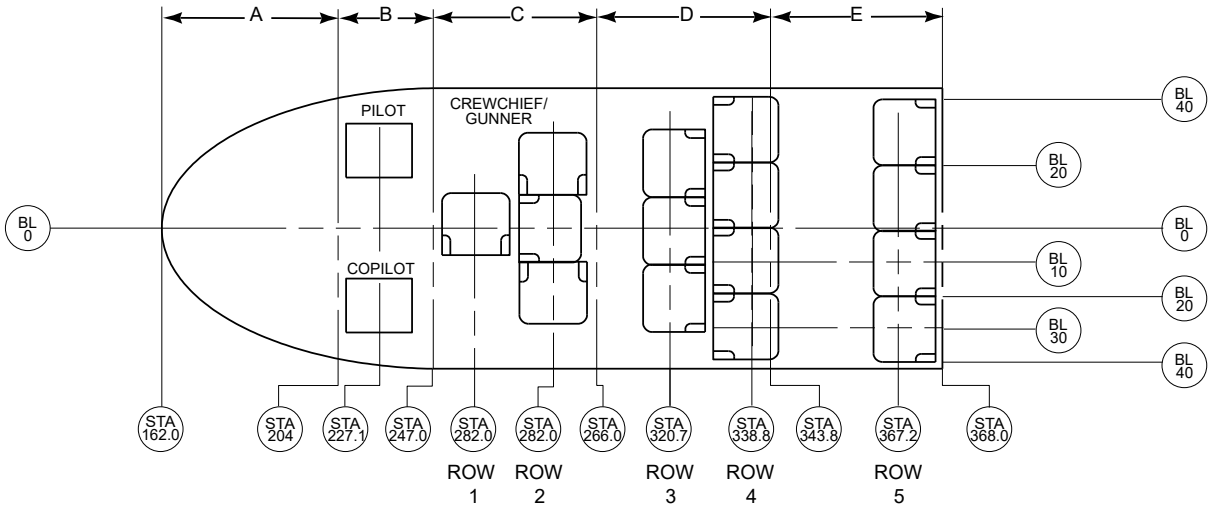
d. Crew and passengers with no equipment - compute weight according to each individual’s estimate.

G.5.4.4 Personnel moments.

Personnel moments charts for personnel in any position shall be provided (FIGURE G-4, FIGURE G-5, and FIGURE G-6).

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PERSONNEL MOMENTS



SEAT WEIGHT - AND MOMENT TABLE *

ITEM	ROW	WEIGHT	MOM / XXX
CREWCHIEF / GUNNER (X) TROOPS (X) TROOPS (X) TROOPS (X)	X X X X	XX XX XX XX	XX XX XX XX
TOTAL - XX SEATS		XXX	XX
ALTERNATE SEATING (BROKEN LINES)			
FORWARD TROOP SEAT (X)	X	XX	X
REAR FACING TROOP SEAT (X)	X	XX	X
REAR FACING TROOP SEAT (X)	X	XX	X
TOTAL - XX SEATS		XXX	XX

* SEAT WEIGHT AND MOMENTS SHOULD BE INCLUDED ON CHART C

EXAMPLE

WANTED:

PERSONNEL MOMENTS

KNOWN:

X PERSONNEL IN ROW X
TOTAL WEIGHT XXX POUNDS

METHOD:

ENTER WEIGHT AT XXX POUNDS - MOVE RIGHT TO ROW X.
MOVE DOWN. READ MOMENT / XXXX = XXX

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FIGURE G-4. Example personnel moments chart 1.

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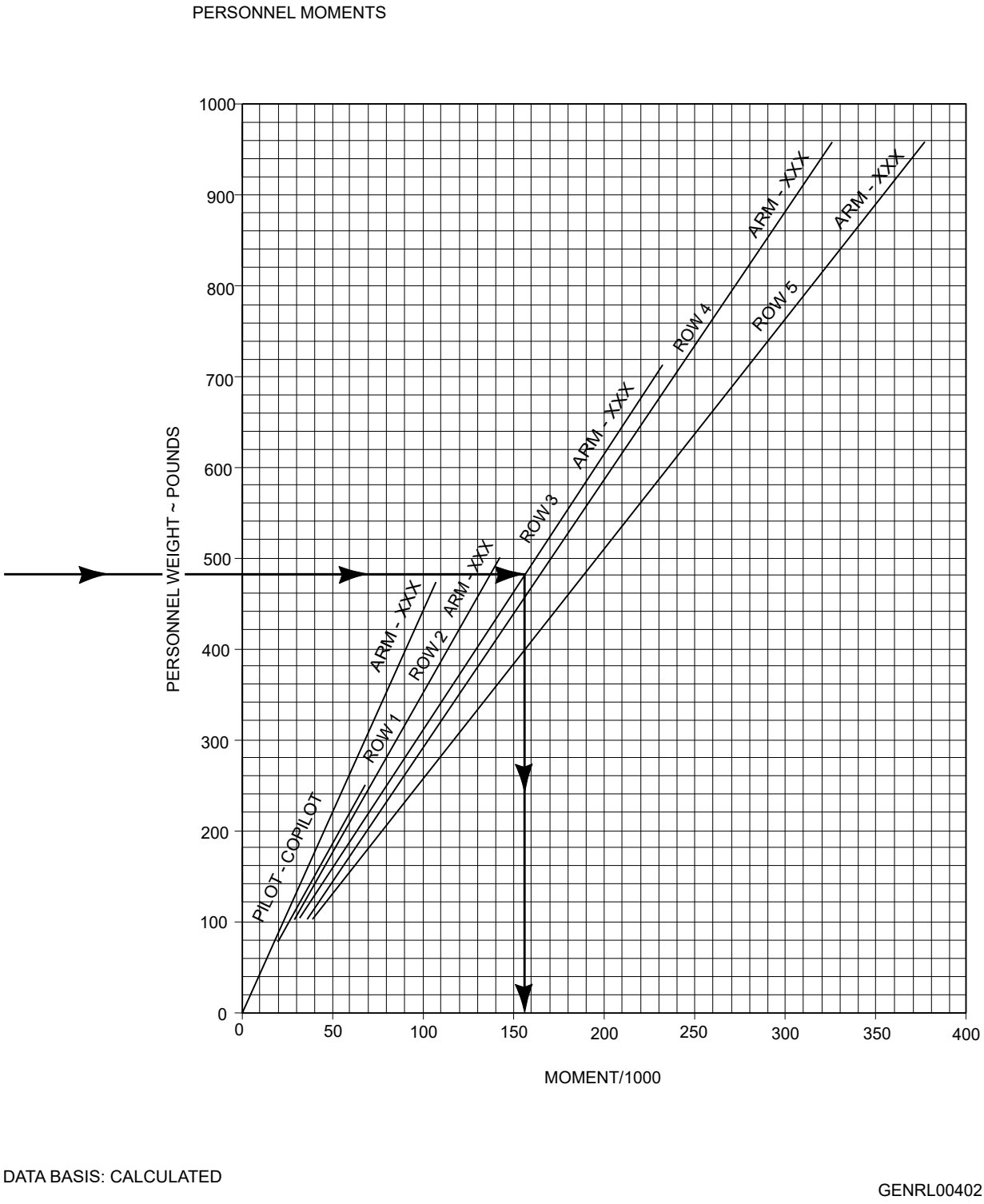
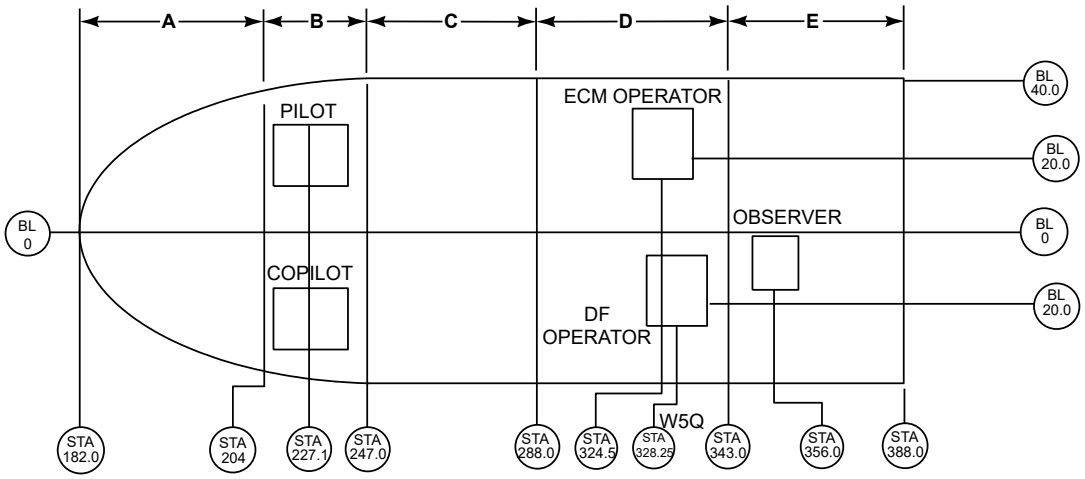


FIGURE G-5. Example personnel moments chart 2.

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PERSONNEL MOMENTS



*ITEM	STA	WEIGHT	MOM/XXXX
OBSERVER SEAT	XXX	XX	X
TOTAL - 1 SEAT	–	XX	X

*SEAT WEIGHT AND MOMENTS SHOULD BE INCLUDED ON CHART C.

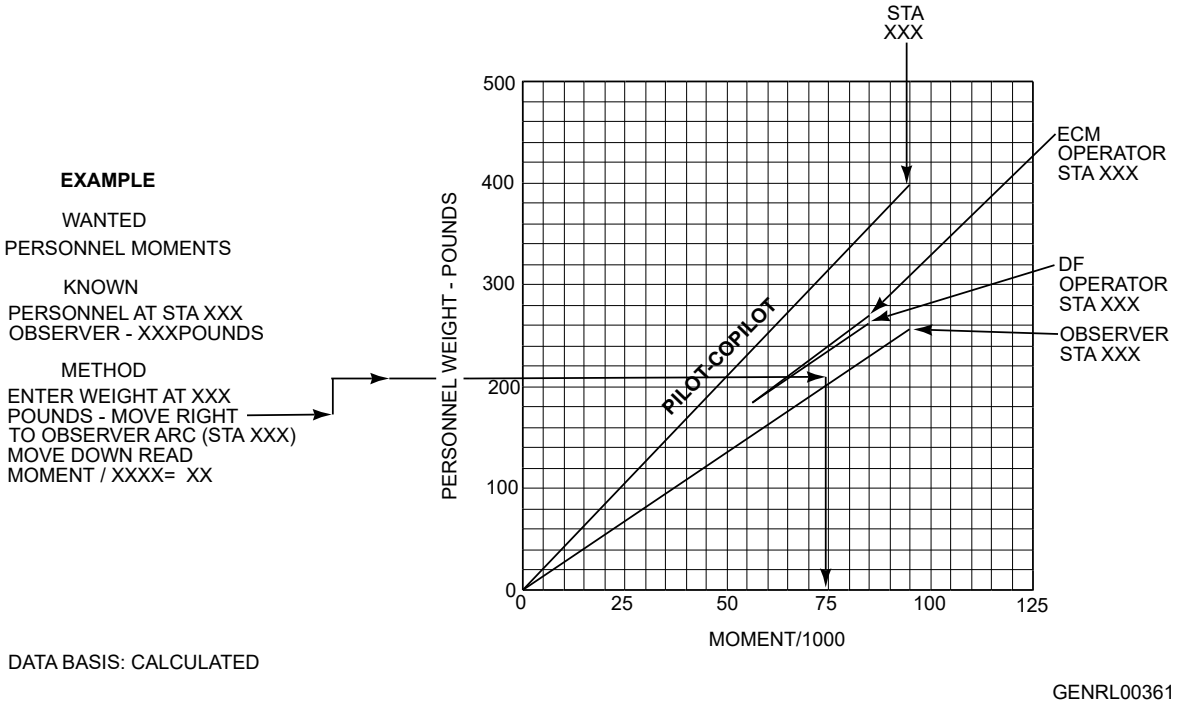


FIGURE G-6. Example personnel moments chart 3.

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G.5.5 Mission equipment work package <missneqpwtbalwp>.

Loading data charts for mission equipment shall provide a tabular listing containing the quantity, weight and moment of each load item up to the maximum quantity for which provisions are available.

a. Only items of load shall be listed.

b. Items that are part of the basic weight shall not be part of this listing.

c. Data shall be provided for all applicable mission systems loads including, but not limited to, armament, avionics, sling, hoist, and litters. Listings shall provide weights and moments of required pylons and launchers.

d. Tabular listing of rockets shall be inclusive for maximum capacity of launchers. Since rockets vary in weight by type, separate listings shall be required (FIGURE G-7).

ITEM	QTY	INBOARD STATION 2, 3 OR OUTBOARD STATION 1, 4	
		ACCUM WEIGHT (LB)	MOMENT (IN.-LB/100)
MISSILE	1	98.5	188
MISSILE	2	197.0	376
MISSILE	3	295.5	564
MISSILE	4	394.0	751

ITEM	QTY	INBOARD STATION 2, 3 OR OUTBOARD STATION 1, 4	
		ACCUM WEIGHT (LB)	MOMENT (IN.-LB/100)
ROCKET	1	20.6	41
ROCKET	2	41.2	81
ROCKET	3	61.8	122
ROCKET	4	82.4	162

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FIGURE G-7. Example armament loading data chart.

G.5.6 Cargo loading work package <cargowtbalwp>.

Detailed information on cargo loading shall be provided, as applicable.

G.5.6.1 Description and illustrations.

A general description of cargo compartments and entrances, including profile and cross-section drawings showing all dimensions (in inches) shall be provided. Also, descriptions of critical dimensions that limit full use of cargo compartments shall be included.

a. A plan view showing dimensions of cargo floor, designation, location, and strength of tie-down fittings, and diagram and limitations on use of fittings, including the desirable cone of action when using fittings, shall be provided. Also, a plan view of cargo floor showing variations in floor strength and weight concentration limitations in various areas shall be included, as applicable.

b. A suitable view of litter provisions showing location shall be presented.

c. A general description of, and operating instructions for, aerial delivery systems shall be included, when applicable.

d. A list and description of all cargo loading aids, unloading aids, cargo securing equipment (including, but not limited to, ramps, hoists, winching provisions, and tie-downs), and stowage provisions shall be provided.

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G.5.6.2 Equipment loading and unloading.

Procedures and a checklist for loading and unloading vehicles and equipment shall be provided, as follows:

a. Assembly of equipment needed for loading.

b. Preparation of cargo compartment and floor and installation of fittings.

c. Preparation of the aerial delivery system.

d. Including, but not limited to, operation of cargo doors, ramps, load assist devices, and aircraft support jacks, including installation and operation, as applicable. Instructions for checking landing gear shall be included, when appropriate.

e. Assembly and checking of unloading aids and releasing of cargo tie-down devices.

G.5.6.3 Preparation of general cargo.

Pre-loading information shall be presented as follows:

a. Instructions that loading personnel should assemble prior to loading data, such as weights, dimensions, CG locations, and contact areas of equipment for use in positioning the load shall be included.

b. Reference or a link shall be made to the weight and balance computations in Section II, and the balance computer, if furnished, for the computation of final load positions in the aircraft.

G.5.6.4 Loading, securing, and unloading cargo.

General methods of loading, save lashing, and unloading of cargo, vehicles, and equipment shall be provided. Rigging of cargo for aerial delivery shall be included, when applicable. The information shall be detailed enough to acquaint service personnel with the factors involved in properly loading, securing loads, and unloading the aircraft.

G.5.6.5 Cargo Center-of-Gravity (CG).

A chart shall be provided (FIGURE G-8) showing approximate allowable cargo CGs versus known weights which may be used for planning purposes for various cargo loads. The chart shall be based on a range of aircraft basic weights and center-of-gravity locations to allow for anticipated variations in these values. The chart shall state that these data are for planning purposes only, that the results are approximate, and final loading shall be checked for the particular aircraft using weight and balance computations and the balance computer, if furnished.

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EXAMPLE

WANTED

CARGO MOMENT FOR A GIVEN CARGO WEIGHT AND FUSELAGE STATION

KNOWN

CARGO WEIGHT XXXXLBS
LOCATION FSXXX

METHOD

ENTER INTERNAL CARGO WEIGHT
MOVE RIGHT TO FSXXX
MOVE DOWN TO BASE-LINE AND
READ XXXXINCH POUNDS/ XXX

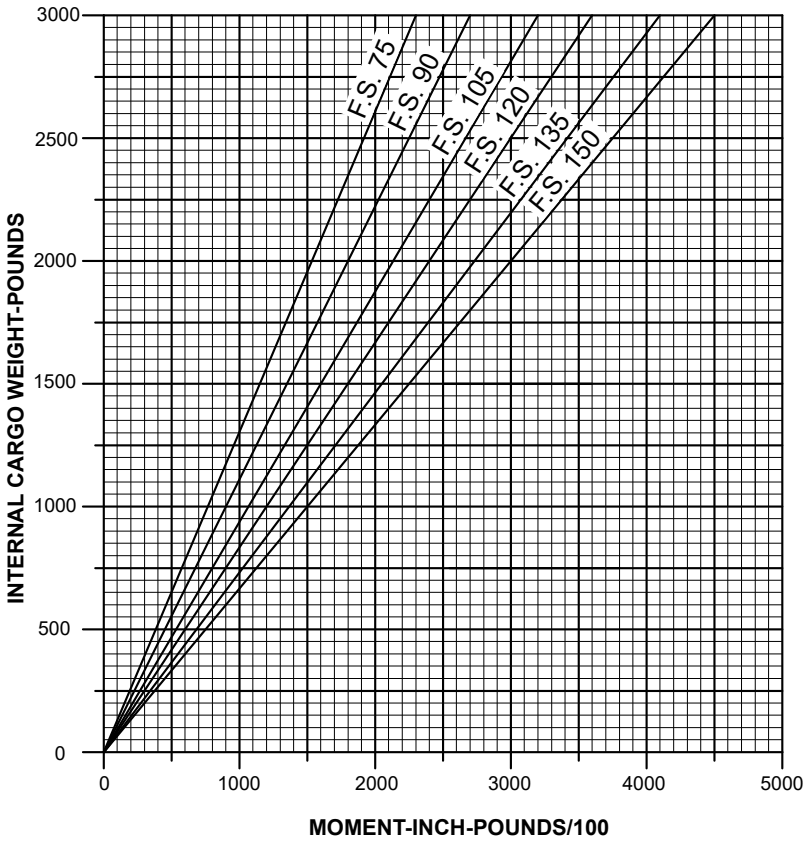


FIGURE G-8. Example cargo moments chart.

G.5.6.6 Loading procedure.

A checklist of the actions required from the time the aircraft is prepared for loading until it is ready for flight shall be provided. It shall include instructions and notes on loading equipment into the aircraft, checking items with CG markings and items 10 feet or longer and placing them in position, determining the amount of shoring required for flight conditions, and general instructions for loading and lashing miscellaneous cargo. Reference shall be made to the appropriate regulations regarding handling of hazardous equipment.

G.5.6.7 Securing loads.

The following items shall be described:

a. Approved restraint criteria including fore, aft, sideward, and vertical restraints.

b. Detailed tie-down instructions shall be provided only for equipment or cargo that is unique to a specific aircraft.

G.5.6.8 Unloading procedures.

Procedures for unloading the aircraft and stowing associated equipment shall be provided.

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G.5.7 Center-of-Gravity work package <cgwtbalwp>.

Longitudinal CG limitations shall be included, and lateral CG limits shall be shown as specified by the acquiring activity.

a. Where possible, the gross weight and CG limitations of the aircraft shall appear on a single chart. However, additional charts may be used if necessary to adequately portray the various configurations of the aircraft. All charts shall be in the style and format as show in FIGURE G-9 and FIGURE G-10.

b. Explanatory text shall explain the purpose and components of the charts; illustrate the use of the charts; emphasize that charts are designed to illustrate degree of risk involved at various weights and CGs; and establish limitations.

c. The chart shall be based on gross weight that is defined as the total weight of the aircraft and its contents. It shall include, but not limited to, operating weight plus fuel, cargo, ammunition, missiles, and external auxiliary fuel tanks. The gross weight in pounds shall be shown on the left side of the chart, and shall range from the aircraft's minimum operating weight to maximum gross weight allowable.

d. At least one example to illustrate the application of the chart shall be included.

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EXAMPLE

WANTED

DETERMINE CENTER OF GRAVITY FOR KNOWN WEIGHT AND MOMENT

KNOWN

GROSS WEIGHT EQUALS XXXX POUNDS,MOMENT/XXXEQUALS XXXXXINCH-POUNDS

METHOD

MOVE RIGHT FROM XXXXPOUNDS TO A POINT APPROXIMATELY 1/2 OF THE DISTANCE BETWEEN XXXXXAND XXXXXINCH-POUND DIAGONAL LINES. FROM THIS POINT PROJECT DOWN TO READ XXXON THE CENTER OF GRAVITY SCALE (FUSELAGE STATION IN INCHES).

NOTE

WHEN CG IS WITHIN SHADED AREA AFT OF STATION XXX, APPROACHES SHOULD BE TERMINATED TO A X-FOOT HOVER FOR ADEQUATE TAIL ROTOR CLEARANCE.

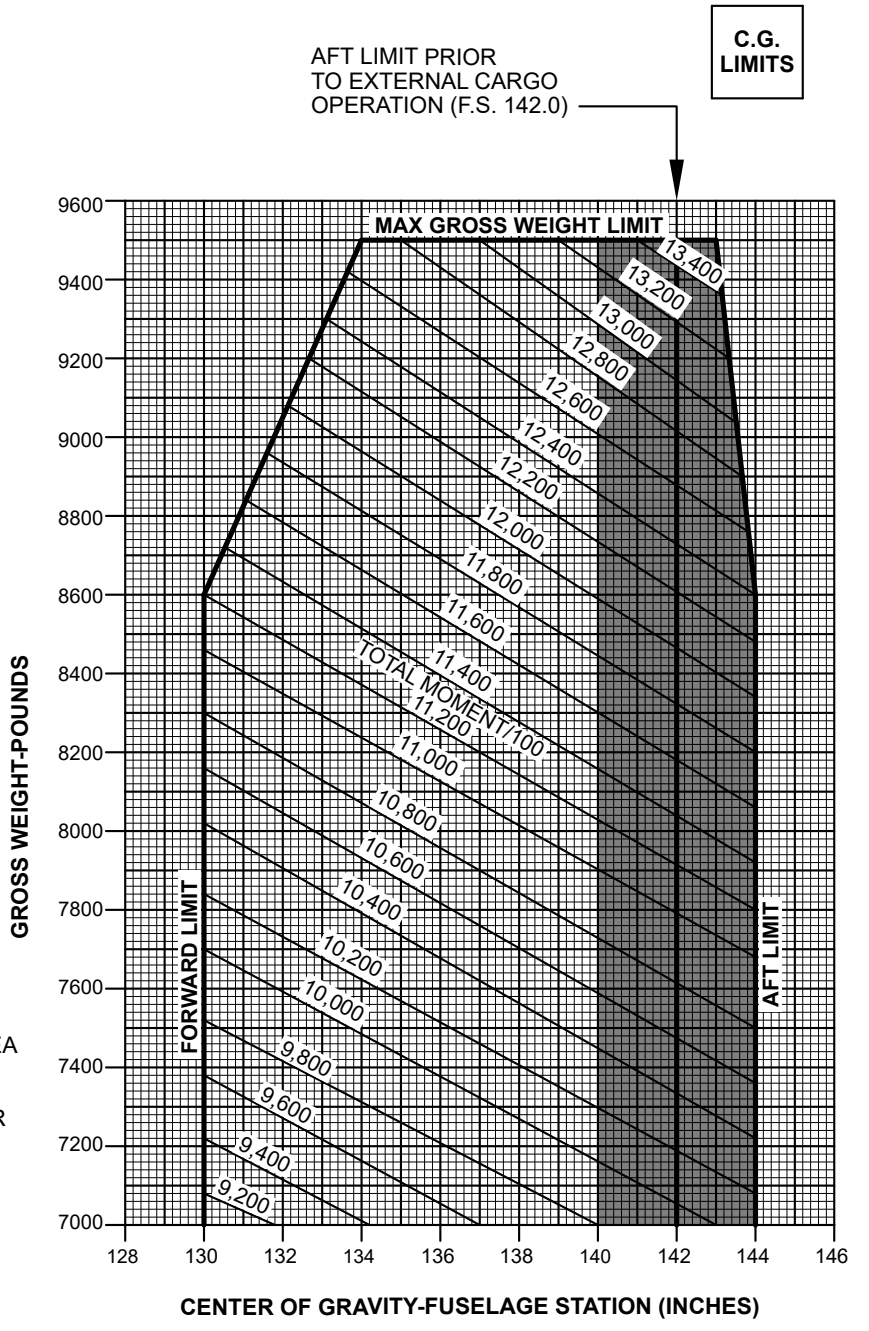


FIGURE G-9. Example center-of-gravity limits 1.

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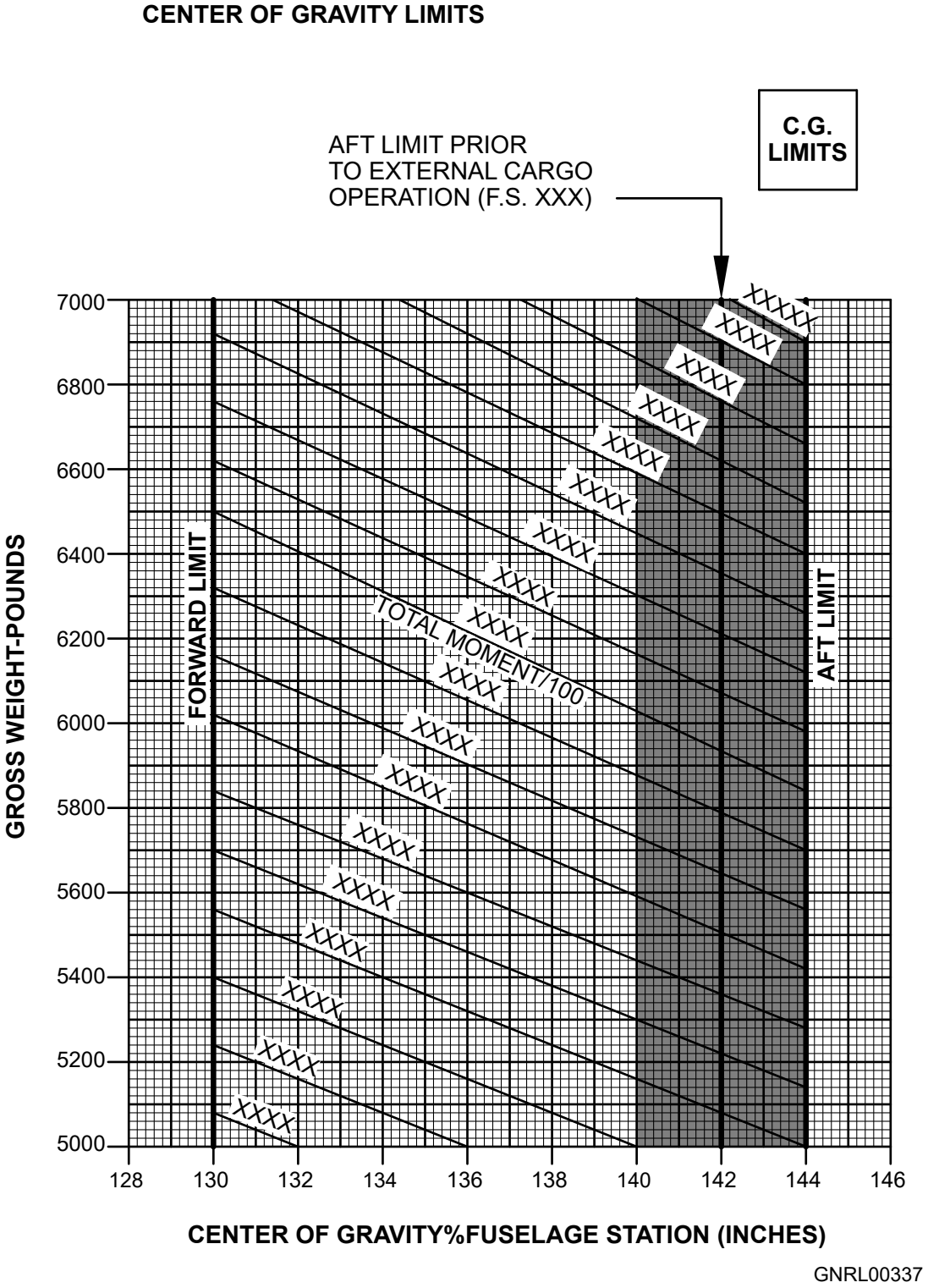


FIGURE G-10. Example center-of-gravity limits 2.

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G.5.8 General weight balance and loading work packages <wtbalgenwp>.

When specified by the acquiring activity, additional work packages may be developed when the specific type of weight/balance and loading information is not covered as described in G.5.1 through G.5.7. Each additional weight/balance and loading requirement shall be a separate work package and shall comply with all of the requirements of G.1 through G.5. If additional weight/balance and loading work packages are used, the proponent shall submit to AMCOM the requirements for this weight/balance and loading work package type for possible incorporation within future revisions to this standard.

G.6 Notes.

The notes in Section 6 apply to this appendix.

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APPENDIX H

PERFORMANCE DATA (MODULE 7)

H.1 Scope.

This appendix establishes the technical content requirements for the preparation of performance data charts for aviation weapon systems. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

H.2 Applicable publications.

The applicable publications in Section 2 apply to this appendix.

H.3 Definitions.

The definitions in Section 3 apply to this appendix.

H.4 General requirements.

All the performance data charts required for the completion of preflight and in-flight mission planning shall be provided. The data presented shall cover the maximum range of conditions and performances for which the aircraft is qualified. Explanatory text applicable to the use of the data presented shall be included for each model of aircraft. Performance data charts shall appear in the initial issue of the manual.

H.4.1 Preparation of digital data for electronic delivery.

Data prepared and delivered digitally in accordance with this standard shall be XML tagged using the current Army DTD. Data shall be formatted for presentation using the Army stylesheets. Refer to H.4.2 for information on obtaining or accessing the Army DTD and stylesheets. XML tags used in the DTD are noted throughout the text of this standard in bracketed, bold characters (e.g., **<geninfo>**) as a convenience for the author and to ensure that the tags are used correctly when developing a document instance.

H.4.2 Use of the Document Type Definition (DTD)/stylesheet.

The DTD referenced in this appendix interprets the technical content and structure for the functional requirements contained in this standard. Development of the TMs shall be accomplished through the use of this standard, the Army DTD, and the Army stylesheets.

H.4.3 Content structure.

The examples provided herein are an accurate representation of the content structure requirements contained in this appendix and shall be followed to permit the effective use of the DTD.

H.4.4 Style and format.

This standard provides style and format requirements for the technical content requirements described in this appendix. These requirements are considered mandatory and are intended for compliance.

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H.4.5 Work package development.

Data developed in accordance with this appendix shall be divided into work packages. These work packages shall be stand alone and shall contain all information and references required to support the work package type.

H.4.6 Selective application and tailoring of content using Appendix A matrices.

This standard contains some requirements that may not be applicable to the preparation of all TMs. Selective application and tailoring of requirements contained in this standard are the responsibility of the acquiring activity and shall be accomplished using APPENDIX A. The applicability of some requirements is also designated by one of the following statements: unless specified otherwise by the acquiring activity; as specified by the acquiring activity; or when specified by the acquiring activity.

H.5 Detailed requirements.

Performance data charts shall be prepared and subdivided into individual work packages that provide the operator of the weapon system/equipment with all of the performance data required for the completion of preflight and in-flight mission planning. The data presented shall cover the maximum range of conditions and performances for which the aircraft is qualified. Explanatory text applicable to the use of the data presented shall be included for each model of aircraft.

a. Information contained on the charts shall be based on, and shall be consistent with, the recommended operating procedures and techniques set forth elsewhere in the manual. Each section shall include an explanation of all applicable charts and a synopsis of pertinent terms used with each chart. The Contractor shall provide a substantiating data report describing all data and methodology used to produce each performance/operating limitation chart published in the Operator's Manual. This report shall in general follow the format of DI-SESS-82321, Air Vehicle Flight Performance Description Report for Rotorcraft and Propeller-Driven Fixed-Wing Aircraft and MIL-STD-3013 Glossary of Definitions, Ground Rules and Mission Profiles to Define Air Vehicle Performance Capability for Fixed Wing Air Vehicles. Each report shall include a description of the static flight performance model, input description, and input database developed to generate the charts. Wherever possible, the performance model shall be correlated with flight test data. Additionally, the analytical methods and supporting data shall be described for each type of performance chart presented in the TM. The substantiating data report shall be reviewed by the airworthiness authority during the TM approval process.

b. In addition to the draft TM, the acquiring activity may require submission of an aerodynamic report illustrating the derivation of the data entered on the charts included in the TM. The report should include an analysis leading to the establishment of lift and drag values used in the calculations, aircraft efficiency and compressibility correction factors, methods of computing power or thrust required and available, a discussion of engine installation losses to include inlet and exhaust loss, power extraction, bleed air offtakes, duct loss, propeller efficiency, transmission efficiency, and adequate references to appropriate wind tunnel or flight test data.

H.5.1 Preparation of performance data chart work packages.

Performance Data Charts shall be developed into separate work packages for each chart listed in H.5.1.1 or H.5.1.2, as applicable. The work packages shall be titled using the applicable performance data chart title. The sample charts provided in this standard are examples only. In addition to the chart itself, each work package shall contain, as a minimum, the following:

a. Description **<descrip>**. A description of the performance data including those parameters obtainable from the chart and information relative to any peculiarity of data presented shall be provided.

b. Use of charts **<useofcharts>**. Reference shall be made to examples used on each chart. Additional use of charts may be included when approved by the acquiring activity. Reference shall be made to related charts that may be used in conjunction with the chart and all information relative to peculiarities of data presented on the chart.

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c. Conditions **<condition_effect>**. Each condition that has a direct or indirect effect on the chart data presented shall be discussed, explaining the effect it may have on the aircraft.

d. Chart **<chart>**. All charts shall be developed in accordance with 4.5.15.2 through 4.5.15.2.5.16.

e. Data Basis **<databasis>**. If the Data Basis is not on the chart itself, then it may be included directly below the chart. When this method is used, the Data Basis shall display on the same page directly below the chart, and shall not be separated from the chart by a page break.

H.5.1.1 Performance data chart work packages.

Required performance data charts are dependent upon the aircraft type as detailed in H.5.1.1.1 through H.5.1.1.3.

H.5.1.1.1 Rotary wing performance data work packages <rwcharts>.

A list of required performance data charts and tables shall be provided by the developer to the acquiring activity before the charts are made so that there is concurrence on what will be included. At a minimum, the following performance data charts shall be required:

a. Fuel flow (FIGURE H-1).

b. Maximum torque available (FIGURE H-2).

c. Hover (FIGURE H-3).

d. Critical Data Chart.

e. Takeoff (FIGURE H-4).

f. Drag (FIGURE H-5).

g. Cruise (FIGURE H-8).

h. Climb-descent (FIGURE H-9).

i. Airspeed calibration (FIGURE H-10).

j. Optimum Cruise.

As specified by the acquiring activity, additional charts peculiar to certain aircraft, such as multi-engine, shall be included. These charts, if required, shall completely define the operation or restrictions of the aircraft.

H.5.1.1.2 Fixed wing performance data work packages <fwcharts>.

Unless otherwise specified by the acquiring activity and Airworthiness Authority, the following performance data shall be presented for fixed wing aircraft:

a. Crosswinds - takeoff and landing (FIGURE H-15).

b. Airspeed calibration.

c. Altimeter corrections.

d. Idle fuel flow (FIGURE H-1).

e. Torque available for takeoff (FIGURE H-16).

f. Takeoff (FIGURE H-17).

g. Takeoff airspeed (FIGURE H-18).

h. Acceleration check distance (FIGURE H-19).

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i. Accelerate-stop distance (FIGURE H-20).

j. Accelerate after lift-off (FIGURE H-21).

k. Minimum single engine control airspeed (FIGURE H-22) (flaps down and up, if applicable).

l. Single engine climb (FIGURE H-23).

m. Service ceiling, one engine inoperative.

n. Rate of climb.

o. Cruise climb (FIGURE H-25).

p. Drag (FIGURE H-5).

q. Cruise (FIGURE H-26).

r. Climb/descent (FIGURE H-27).

s. Approach speed (FIGURE H-28).

t. Landing (FIGURE H-29).

u. Optimum Cruise.

v. Stall.

Additional charts peculiar to certain aircraft, such as multi-engine, shall be included as specified by the acquiring activity and/or Airworthiness Authority. These charts, if required, shall completely define the operation or restrictions of the aircraft.

H.5.1.1.3 Uncrewed aircraft system performance data work packages <uascharts>.

Unless otherwise specified by the acquiring activity and Airworthiness Authority, the following performance data shall be presented for UAS aircraft. UAS is a rapidly evolving aircraft type, and additional charts not listed below may be required for operator planning. Any additional needed charts shall be included in the publication using either an applicable optional UAS chart or the general performance chart work package.

a. Crosswinds - takeoff and landing (FIGURE H-15).

b. Idle fuel flow (FIGURE H-1).

c. Takeoff (FIGURE H-17).

d. Cruise climb (FIGURE H-25).

e. Cruise (FIGURE H-26).

f. Climb/descent (FIGURE H-27).

g. Approach speed (FIGURE H-28).

h. Landing (FIGURE H-29).

i. Optimum Cruise.

H.5.1.2 Data basis.

Unless otherwise specified by the acquiring activity , the preparation of performance data charts shall be derived from flight test reports, when available. Exceptions to this may be authorized by the acquiring activity for new aircraft, provided adequate flight tests have been completed for the prototype. However, for these exceptions, an evaluation of all changes that affect performance shall be obtained by additional flight tests. The basis for data presented shall be clearly defined at the bottom of each chart or in a table below the chart, as determined by the

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acquiring activity, to include data type and source data document. Army test reports shall be used when available. When flight test reports are not available, referenced estimates shall be clearly identified as such. Conservative estimates shall be used until verified by flight test data. Data that are not based strictly on the particular aircraft shall be explained in detail.

H.5.1.3 Identification.

Each chart shall be marked in the following manner:

a. Titles shall be centered above the chart. The name of each chart shall define the type of information to be obtained from that particular chart.

b. Condition headings (4.5.15.2.5.2) shall be centered below the title and, when required, shall contain the following types of information, when applicable:

(1) Pressure altitude.

(2) Situation to which chart applies (takeoff, landing, sling load takeoff, etc.).

(3) Conditions of auxiliary equipment.

(4) Configuration.

(5) Wing flap position.

(6) Rotor or propeller rpm.

(7) Engine rpm.

(8) Fuel type.

(9) Hovering condition (In-Ground Effect (IGE) or Out-of-Ground Effect (OGE)).

(10) Power requirements.

(11) Runway conditions.

(12) Wind conditions.

(13) Gear up/down.

(14) Power required.

(15) Ambient temperature.

(16) Drag coefficient.

c. Titles of figures shall match the title shown at the top of each chart.

H.5.1.4 Factors affecting data.

Conditions that affect the data but are not presented as variables on any specific chart shall be listed as “Conditions” under the title of the chart. An explanation of these factors shall be included in the text that describes that chart.

H.5.1.5 Configuration.

Unless otherwise specified by the acquiring activity, the baseline configuration for all presented data shall be the most probably combat configuration. The baseline configuration shall be labeled and presented as a condition on applicable charts. The baseline configuration shall be completely defined in the “Drag” section. Where inherent configuration variations exist (including, but not limited to, antenna variations, IR suppressors, and engine inlet configurations), the data shall be based on the most conservative configuration combination (highest drag, lowest

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power/thrust available, highest fuel consumption, etc.). The effects of altering these items shall be discussed in each section as applicable.

H.5.1.6 Fuel.

All charts shall be based on the primary fuel for the engine/engines installed unless additional charts are required by the acquiring activity for alternate fuels.

H.5.1.7 Atmospheric conditions.

Where data are presented incrementally, they shall be presented to the next increment beyond the range of probable operating atmospheric conditions as found in MIL-HDBK-310, for guidance only, to permit interpolation. Unless otherwise specified by the acquiring activity, standard day, standard conditions, standard temperature, or density altitude shall not be mentioned or presented. All atmospheric data and formulas including pressure (P) and pressure altitude (H_p) shall be based on the USAF 1976 US Standard Atmosphere as adapted by the International Civil Aviation Organization (ICAO). The following formulas for converting pressure altitude (H_p) to static air pressure (P), and vice versa, shall be used:

$$P \text{ (in. Hg)} = 29.92125(1 - H_p/145{,}442.1)^{5.255876}$$

$$H_p\text{(ft.)} = 145{,}442.1(1 - P/29.92125)^{.1902632}$$

H.5.1.8 Allowances.

Allowances shall be made for all installation losses and a complete analysis of such allowances shall be included in the performance data substantiation report. The following allowances shall be included. An increased allowance of five percent shall be made for fuel consumption data only when data are based on estimates; however, this shall not be stated in the TM.

H.5.1.9 Limitations and restrictions.

Applicable operating limits shall be shown. Restricted operating regions shall be depicted by shaded areas. Data shall be extended to the next normal increment beyond operating limits to aid interpolation. Such data shall be represented by dotted lines. Note: maximum gross weight is an operating limit.

H.5.1.10 Definitions.

Definition of terms used including, but not limited to, takeoff speed, takeoff distance, and rotation speed shall be included in the Abbreviation, Acronyms, and Definitions WP.

H.5.2 Performance data chart introduction work package <chart_introwp>.

An explanation of the performance data including the purpose, scope, limits, uses, and conditions shall be provided.

H.5.2.1 Purpose <purpose>.

The following paragraphs shall be included verbatim:

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"The purpose of this module is to provide the best available performance data for the *(insert assigned aircraft designation)*. Regular use of this information will allow you to receive maximum safe use of the aircraft. Although maximum performance is not always required, regular use of this module is recommended for the following reasons:

a. Knowledge of performance margins will allow you to make better decisions when unexpected conditions or alternate missions are encountered.

b. Situations requiring maximum performance will be more readily recognized.

c. Familiarity with the data will allow performance to be computed more easily and quickly.

d. Experience will be gained in accurately estimating the effects of conditions for which data are not presented.

The information is primarily intended for mission planning and is most useful when planning operations in unfamiliar areas or at extreme conditions. The data may also be used in flight, to establish unit or area Standard Operating Procedures, including pilot aid cards, and to inform ground commanders of performance/risk tradeoffs."

H.5.2.2 General <geninfo>.

This paragraph shall contain the following verbatim statement:

"The data presented cover the maximum range of conditions and performance that can reasonably be expected. In each area of performance, the effects of altitude, temperature, gross weight, and other parameters relating to that phase of flight are presented. In addition to the presented data, judgment and experience will be necessary to accurately determine performance under a given set of circumstances. The conditions for the data are listed under the title of each chart. The effects of different conditions are discussed in the text accompanying each phase of performance. Where practical, data are presented at conservative conditions. However, no general conservatism has been applied.

WARNING

Exceeding operating limits can cause permanent damage to critical components. Over limit operation can decrease performance, cause immediate failure, or failure on a subsequent flight, which may result in injury or death to personnel.

Applicable limits are shown on the charts. Performance generally deteriorates rapidly beyond limits. If limits are exceeded, minimize the amount and time. Enter the maximum value and time beyond limits on DA Form 2408-13-1 so proper maintenance action can be taken. Exceeding operating limits can cause permanent damage to critical components. Over limit operations can decrease performance, cause immediate failure, or failure on a subsequent flight."

H.5.2.3 Use of charts <useofcharts>.

This paragraph shall contain a sample problem typical of a normal mission accomplished by the aircraft. The sample shall be included on or precede the first chart. Additional examples shall be prepared as required for other charts within a section. When possible, actual chart values shall be used throughout the problem. Data for the problem in which derivation may not be entirely clear shall be explained. Additional discussion, sample problems, or illustrations may be used throughout the module to clarify the usage charts.

The TM shall point out that the use of a straight edge (ruler or paper edge) and a hard fine point pencil is recommended to avoid cumulative errors. In addition the primary use, other uses of each chart are explained in the text accompanying each set of performance charts. An example of an auxiliary use of the charts shall be shown by noting that although the hover chart is primarily arranged to find torque required, maximum skid/wheel height or maximum gross weight can also be found. The TM shall note that in general, any single variable can be found if all other variables are know. Also, the tradeoffs between two variables can be found.

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H.5.2.4 Data basis <databasis>.

This paragraph shall contain a statement similar to the following statements and definitions:

"a. The source of data used is indicated at the bottom of each performance chart under "Data Basis". The applicable report and date of the data are also given. The data provided generally are based on one of the following categories.

b. Flight test data are obtained by flight tests of the aircraft at precisely known conditions using sensitive calibrated instruments.

c. Calculated data are data based on tests, but not on flight tests of the complete aircraft.

d. Estimated data are data based on estimates using aerodynamic theory or other means not verified by flight testing."

H.5.2.5 Specific conditions <speccond>.

This paragraph shall contain the following verbatim:

"The data presented are accurate only for specific conditions listed under the title of each chart. Variables for which data are not presented, but which may affect that phase of performance, are discussed in the text. Where data are available or reasonable estimates can be made, the amount that each variable affects performance shall be given."

H.5.2.6 General conditions <gencond>.

General conditions, in addition to specific conditions listed on each chart, shall be included. Examples of general conditions which might affect performance of the aircraft shall include, but shall not be limited to, rigging, pilot technique, sideslip, aircraft variation, engine variation, and instrument variation. Information shall be included which defines what effect the general conditions listed shall have on the performance data of the aircraft.

H.5.2.7 Performance discrepancies <perfdisc>.

The following statement shall be included verbatim in the TM:

"Regular use of this module will also allow monitoring of instruments and other aircraft systems for malfunctions, by comparing actual performance with planned performance. Knowledge will also be gained concerning the effects of variables for which data are not provided, thereby increasing the accuracy of performance predictions."

H.5.2.8 Conversion charts <convchart>.

As specified by the acquiring activity, standard conversion charts shall be developed and included in the performance data introduction work package.

H.5.2.9 International Standard Atmosphere (ISA) conversion chart.

An ISA chart shall be included. The ISA chart shall depict ISA values in relation to variations of pressure altitude in feet versus outside air temperature (OAT) in degrees Celsius.

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H.5.3 Rotary wing performance charts <rwcharts>.

Performance data charts for rotary wing aircraft shall conform to the requirements detailed in H.5.3.1 through H.5.3.11.

H.5.3.1 Fuel flow chart work package <fuelflowchartwp>.

The fuel flow chart (FIGURE H-1) shall show fuel flow at both the engine idle position and at normal rotor speed with flat pitch. The chart shall also present fuel flow conditions when the engine is operational at different configurations, e.g. bleed air On/Off. Pressure altitude and Free Air Temperature (FAT) shall be used as the criteria for fuel flow computations. Reference shall be made to other charts that present fuel flow data at cruise conditions. Fuel flow data shall be based on the primary fuel type. Information shall be included in the supporting text to define additional pertinent information which may affect fuel flow. All data shall be based on normal operating engine rpm.

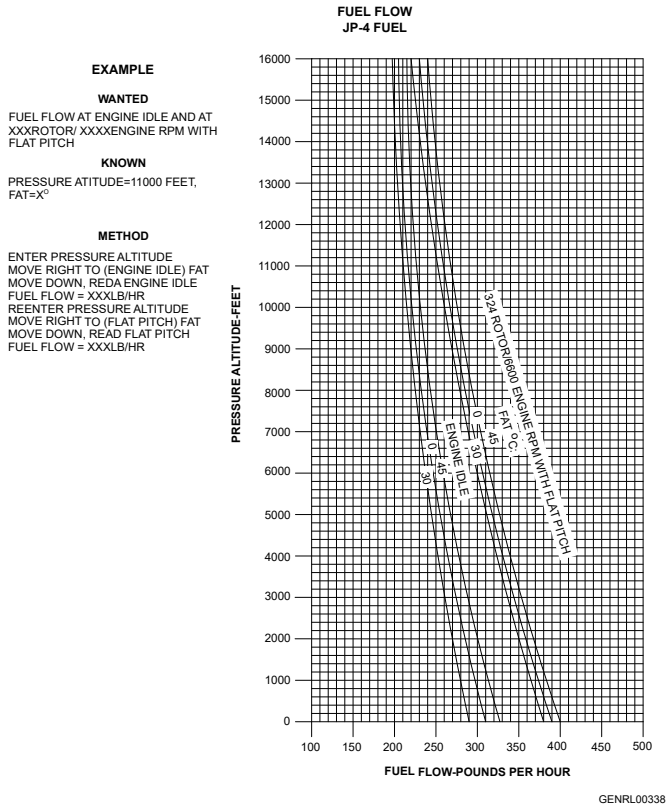


FIGURE H-1. Example fuel flow chart.

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H.5.3.2 Maximum torque available chart work package <maxtrqchartwp>.

The charts for maximum torque available (FIGURE H-2) shall show the effects of altitude and temperature on the maximum torque available and shall take into consideration calibration factors used to correct for known errors in torque indicating systems. Separate charts shall be provided for each applicable set of time limited torque available data. For example, separate charts shall be provided for intermediate (30 minute) and one engine inoperative contingency (10 minute) torque available data. Information shall be provided to allow the operator to correct the data presented on the charts to account for variations in torque available due to operation of IR suppressors, systems requiring bleed air, or other similar operating conditions. Information shall also be provided to allow the operator to correct the data presented to account for known variations in the torque available of the individual engines installed in the aircraft compared to the standard or specification engines depicted by the charts.

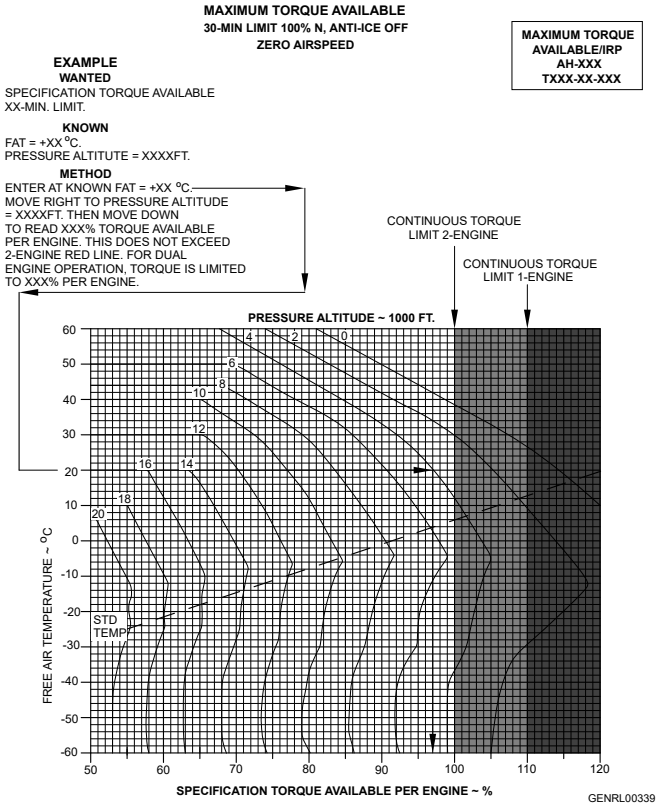


FIGURE H-2. Example maximum torque available (insert time) (RW).

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H.5.3.3 Hover chart work package <hoverchartwp>.

The hover chart (FIGURE H-3) shall present the torque required to hover at given conditions of skid/wheel height, gross weight, temperature and altitude. Basic IGE hover data shall be based on hovering over a level surface. If IGE hover data are presented for other than level surfaces, information shall be included in the supporting text or on the charts. Compressibility effects on hover power required should be included and one method of adding a temperature versus delta torque required table is shown on FIGURE H-3. Also, aircraft limitations may be presented on the Hover chart or additional charts in the Hover section to include marginal areas of performance. When unsafe performance areas could be encountered, the full range of precautionary data shall be presented and safe limits presented to better clarify the use of the data.

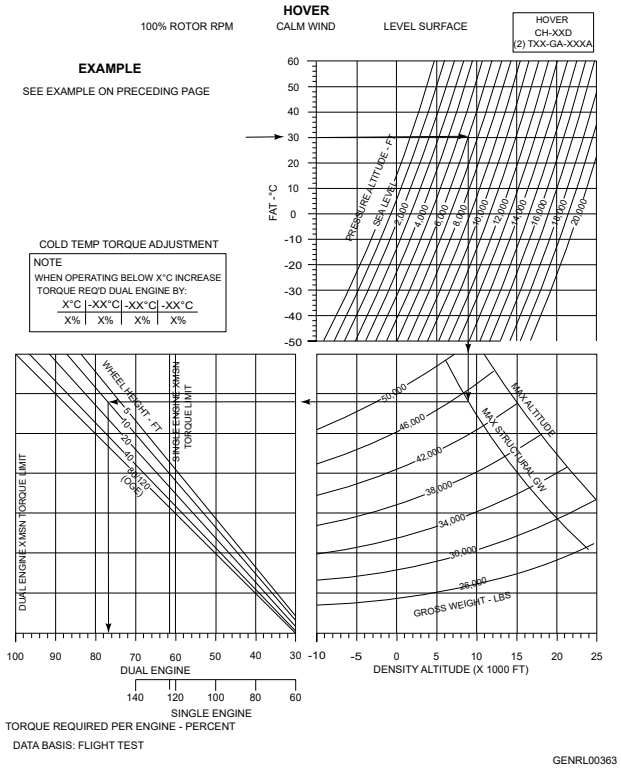


FIGURE H-3. Example hover chart.

H.5.3.4 Critical data chart work package <critdatachartwp>.

Critical wind azimuth and velocities at varying gross weights, pressures altitudes, and FAT during hover and low speed flight shall be presented as required. A separate chart may be used.

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H.5.3.5 Takeoff chart work package <takeoffchartwp>.

The takeoff chart shall consist of all takeoff data required to clear various obstacle heights and shall be based on all the parameters shown on FIGURE H-4. All approved techniques such as level acceleration, coordinated climb, and sling load techniques (as applicable) shall be covered on additional charts as required by the acquiring activity. The primary parameters used for takeoff performance shall be maximum hovering height capability, FAT, gross weight, and maximum torque available. Additional performance charts shall be referenced when required. Takeoff limits shall be stated and indicated on all charts. All takeoff conditions shall be based on calm winds, level hard surfaces, normal rotor/engine speeds, and optimum torque available.

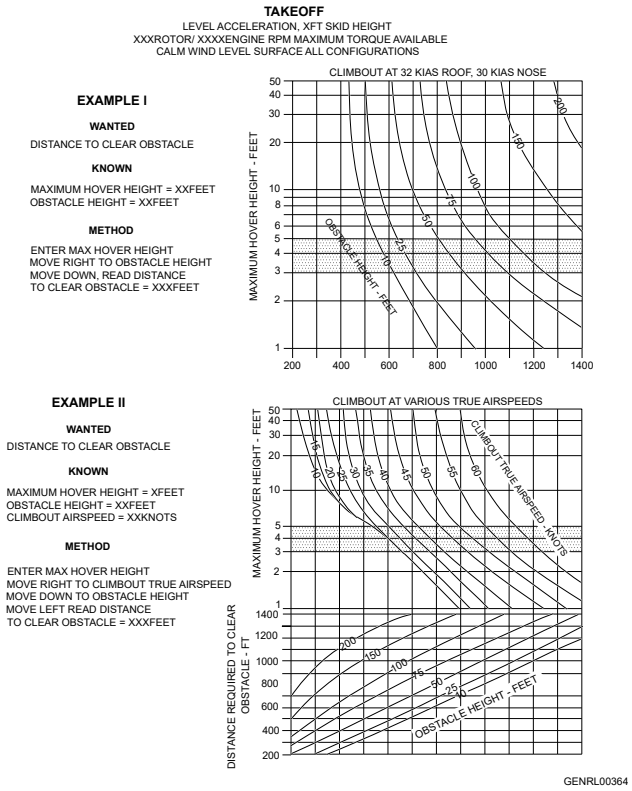


FIGURE H-4. Example takeoff chart.

H.5.3.6 Drag chart work package <dragchartwp>.

The baseline configuration for drag (FIGURE H-5, FIGURE H-6, and FIGURE H-7) shall be completely defined. Inherent or basic equipment variations, existing or anticipated, and any external stores included in the baseline configuration shall be provided. Data shall be prepared to show each drag item and the drag area change in square feet based on additional engine torque or horsepower required. These data shall be prepared in tabular form or shall be conveyed in a manner more suitable for interactive viewing. Negative drag increments from baseline configurations shall be permissible. The drag data shall fall into one of these major categories: (1) inherent or basic aircraft modifications or basic equipment changes; (2) external stores and store combinations; (3) crew alterable configurations; and (4) for helicopters with sling capability, drag of various standard sling loads. A procedure shall be provided for estimating drag of sling loads for which data are not provided. Information to determine the change in maximum range or long range cruise to chart the airspeed with drag variations shall be provided. A supplementary graph on the cruise chart depicting torque/horsepower change for drag change shall be provided. It shall cover the airspeed range from minimum power to limit airspeed. It shall also cover a drag range to one-half the basic aircraft drag or the largest drag increment combination, whichever is larger. One or two alternate total configurations shall be

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depicted on these sub-graphs using special line coding with approval of the acquiring activity. If alternate configurations are depicted, they shall be completely defined using separate charts, as applicable.

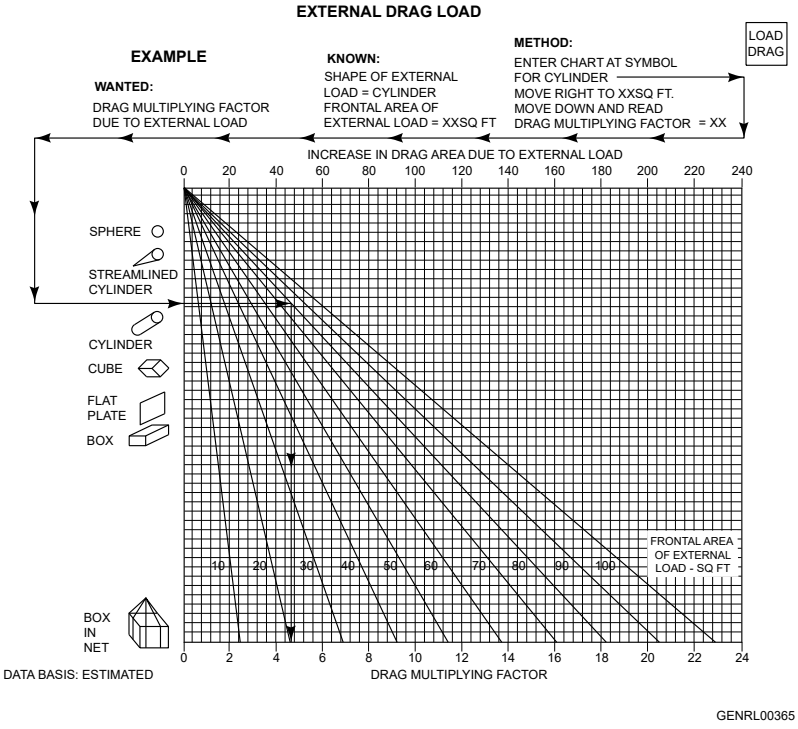


FIGURE H-5. Example drag chart 1.

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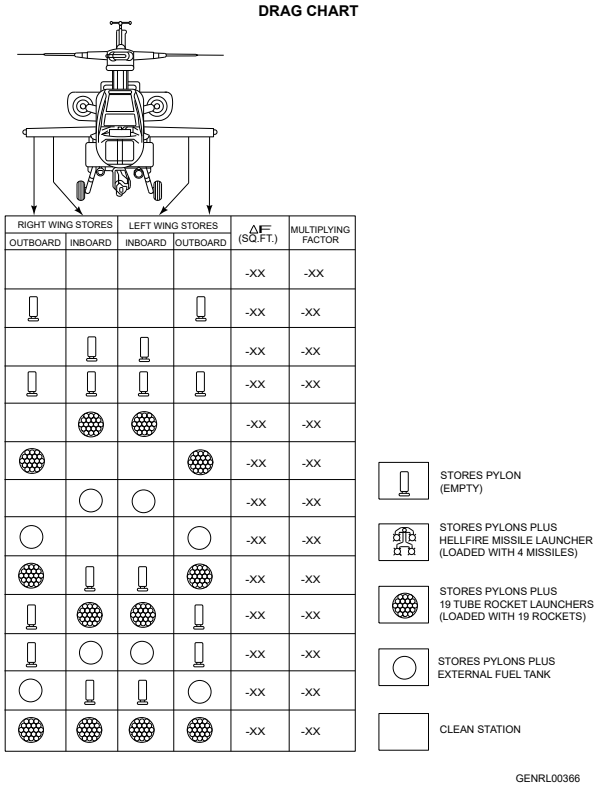


FIGURE H-6. Example drag chart 2.

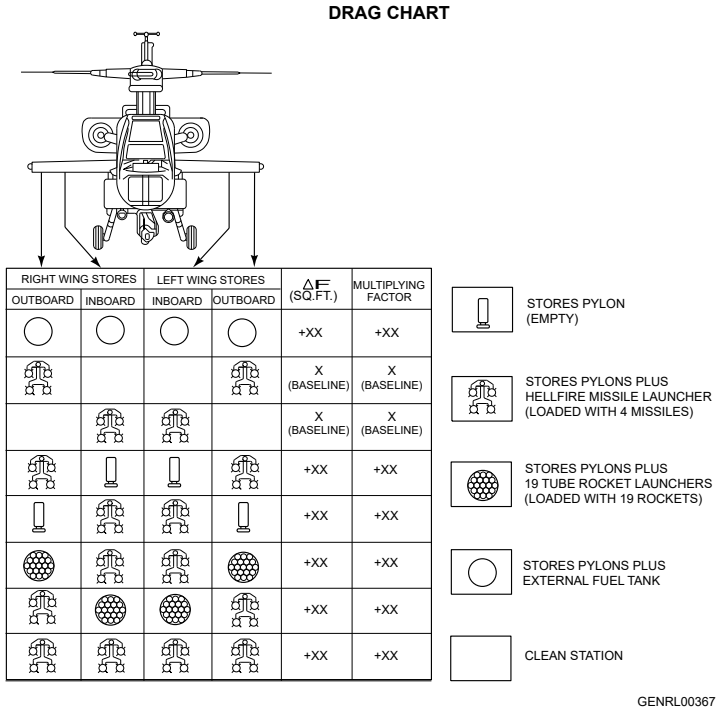


FIGURE H-7. Example drag chart 3.

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H.5.3.7 Cruise chart work package <cruisechartwp>.

Cruise charts (FIGURE H-8) shall present torque requirements for level flight at various airspeeds, gross weights, pressure altitudes, and FAT. The particular altitudes and temperatures at which cruise data are to be presented shall be specified by the acquiring activity. Indicated airspeeds for all airspeed systems used on the aircraft referenced shall be shown on the charts with the exception of aircraft with cockpit displays which provide calibrated airspeed as the primary airspeed display. Fuel flow shall be shown in such a manner as to allow determination of fuel flow as a function of torque required. Torque available may be shown for maximum torque and continuous bleed air On/Off. When torque available is greater than the torque limit only, the torque limit shall be shown. Velocity never exceeded (V_{ne}) shall be shown on each chart, as appropriate. Airspeeds for maximum range, endurance, and rate of climb shall be included on each chart. This information shall be presented for each engine when performance data pertain to multi-engine aircraft. Maximum performance, precautionary, and limits data shall be shown on each chart and explained in the text. Other performance data charts related to the cruise charts shall be referenced. All cruise data shall be based on normal operational rotor and engine speed, on drag area changes, true airspeed, pressure altitude, and FAT. A drag area change table showing the change due to each possible configuration change shall be included. When required by the acquiring activity, a separate chart for one engine inoperative shall be prepared.

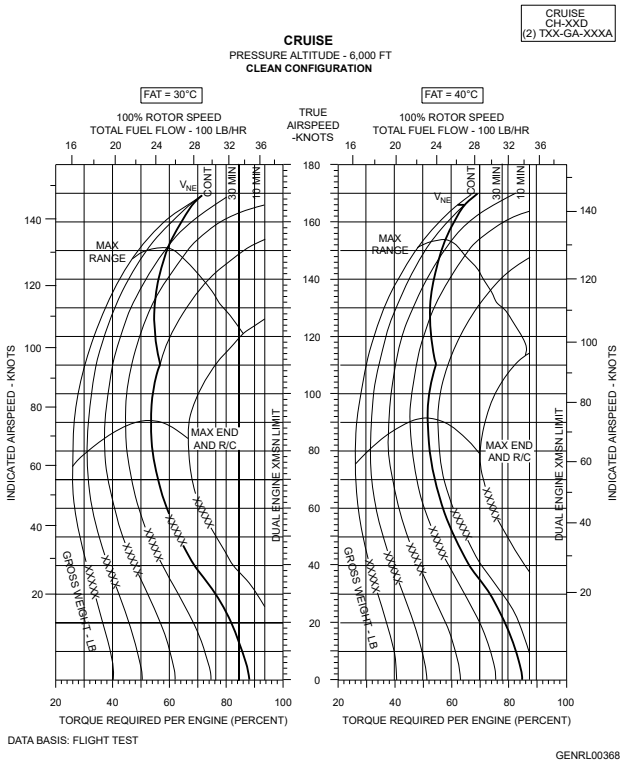


FIGURE H-8. Example cruise chart (RW).

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H.5.3.8 Climb-descent chart work package <clmbdeschartwp>.

The climb-descent chart (FIGURE H-9) shall show the torque required in excess of that needed for level flight to obtain the desired rate of climb. The torque decrease for a desired rate of descent shall also be shown. Desired rate of climb or descent and gross weight shall be used to compute the torque change required.

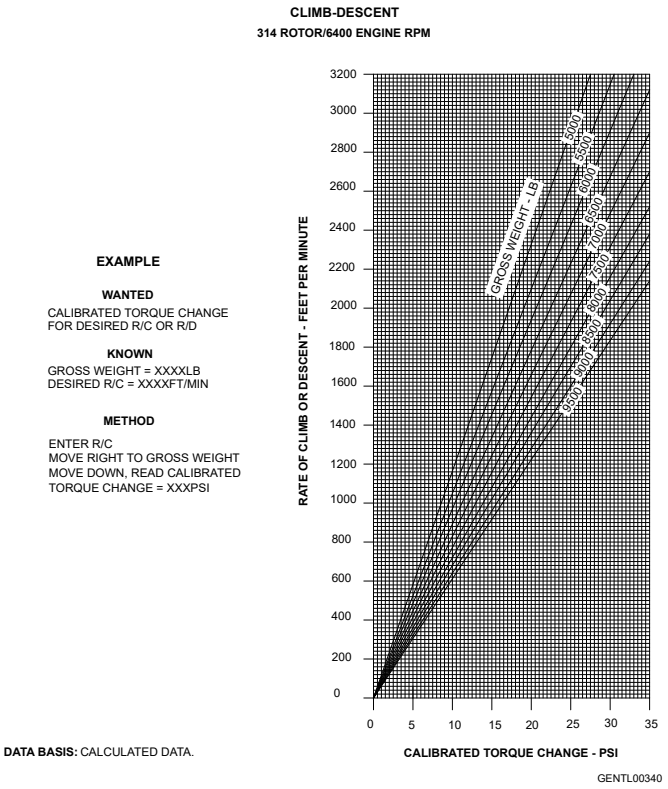


FIGURE H-9. Example climb-descent chart.

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H.5.3.9 Airspeed calibration chart work package <airspdcalchartwp>.

An airspeed calibration chart (FIGURE H-10, FIGURE H-11, FIGURE H-12, FIGURE H-13, and FIGURE H-14), which defines the relationship between the pilot's indicated and calibrated airspeed for level flight, climb, and descent, shall be provided. Instructions and examples shall be provided to show the operator how to determine the level flight indicated airspeed value which corresponds to known indicated airspeeds in climb and descent. Instructions and examples for determining calibrated airspeeds corresponding to known indicated airspeed shall also be provided. Altimeter correction charts that provide position error correction versus indicated airspeed shall be provided for all normal and emergency altimeter systems. Data shall be provided for all applicable flap settings or other variations in configuration. A temperature conversion/correction chart that provides true FAT as a function of true airspeed and indicated temperature shall also be provided for aircraft capable of significant airspeeds. For those aircraft whose air data system position errors are insignificant, calibration data for airspeed, altitude, and temperature shall be omitted, with approval of the acquiring activity.

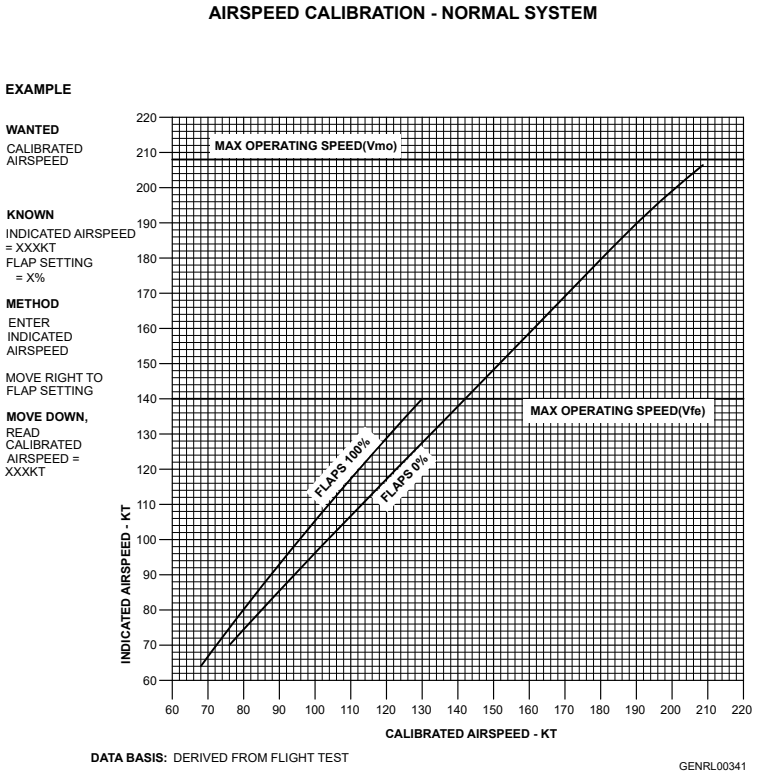


FIGURE H-10. Example airspeed calibration chart.

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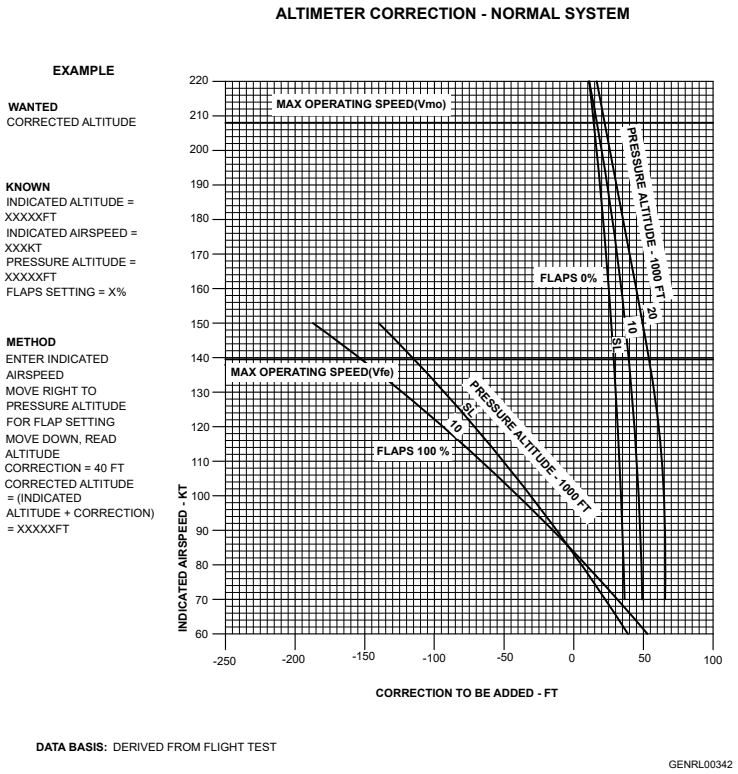


FIGURE H-11. Example airspeed-altimeter correction chart.

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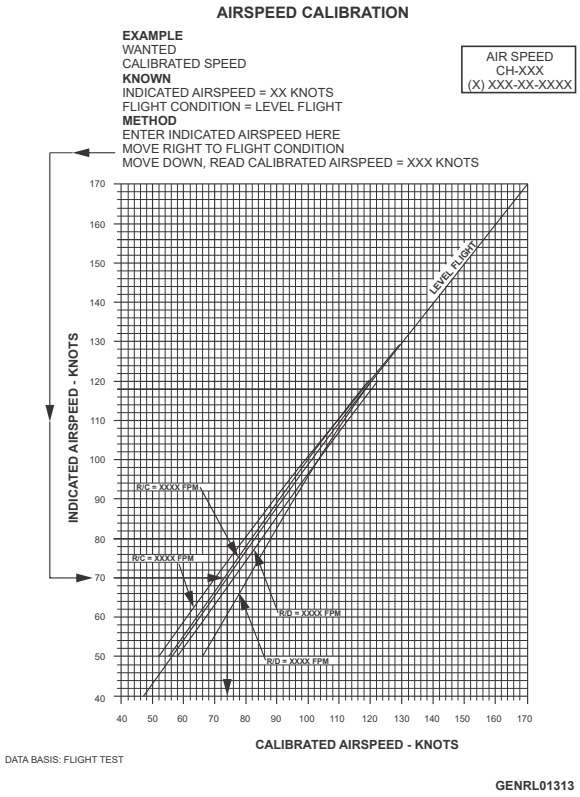


FIGURE H-12. Example airspeed calibration chart.

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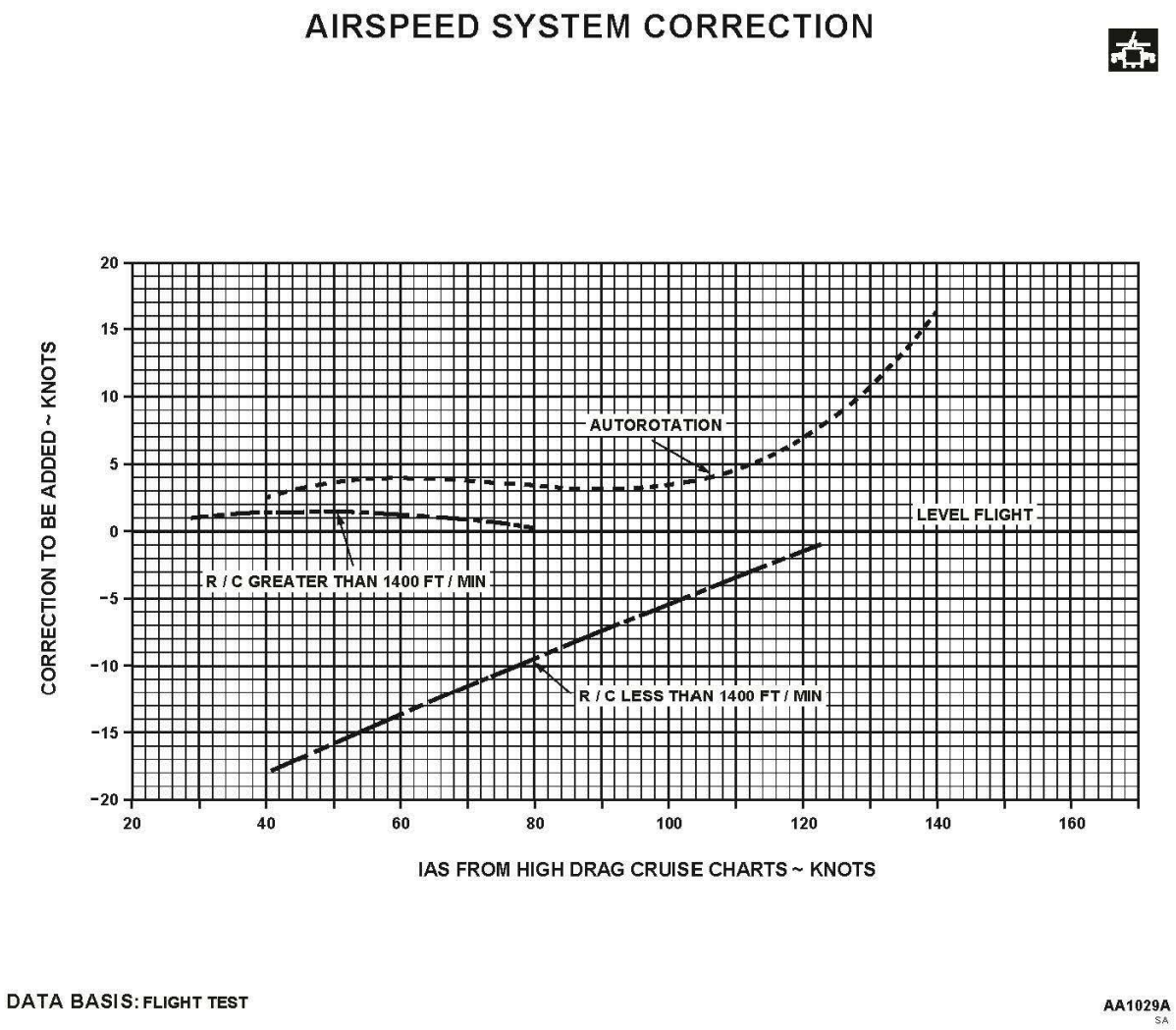


FIGURE H-13. Example airspeed correction chart 1 (Rotary Wing).

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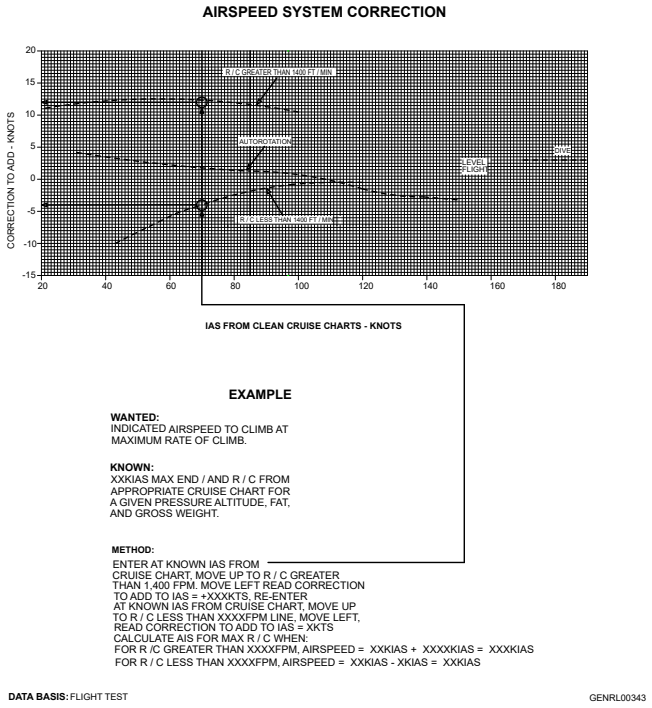


FIGURE H-14. Example airspeed correction chart 2 (Rotary Wing).

H.5.3.10 Optimum cruise work package <opcruisechartwp>.

When requested by the acquiring activity, data shall be provided to determine the altitude for maximum range and maximum endurance as a function of gross weight and ambient temperature. Information shall also be provided for optimum rotor/propeller rpm for maximum range and endurance. Where optimum rpm is different from that presented for the (normal) cruise data, information shall be provided to correct fuel flow for the different RPMs. Optimum cruise speed (maximum range or endurance) presented on the cruise chart shall be referenced and used. Airspeed and power schedules for climb and descent to maximize total range or endurance shall be described. A means shall be provided for estimating ambient temperature at optimum altitude. Also, a means shall be provided for comparing the effects of varying winds with altitude with the change in aircraft performance with altitude. Data shall cover the range of gross weights and ambient temperatures presented on the cruise charts, and the limits of altitude on the cruise charts, if required. If corrections to optimum altitude for configuration variations are significant and capable of being done, this information shall be provided.

H.5.3.11 Additional performance charts work package <genchartwp>.

As determined by the acquiring activity, additional charts necessary for the completion of preflight and in-flight mission planning may be included when the specific type of performance chart is not covered as described in H.5.3 through H.5.3.10. Each additional chart shall be a separate work package and shall comply with all of the requirements of H.5.1 through H.5.1.10. If additional performance charts are used, the proponent shall submit to AMCOM the requirements for this performance chart type for possible incorporation within future revisions to this standard.

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H.5.4 Fixed wing and UAS performance chart content <fwcharts> or <uascharts>.

Performance data charts for fixed wing and UAS aircraft shall conform to the requirements of H.5.4.1 through H.5.4.22 as applicable per APPENDIX A.

H.5.4.1 Crosswind chart work package <crswindchartwp>.

The crosswind chart (FIGURE H-15) shall show the takeoff or landing conditions under which a takeoff or landing is or is not recommended. Various wind velocities, runway wind angle, and rotation or touchdown airspeeds shall be shown.

Additional charts to obtain required information shall be referenced. When more than one configuration is possible for the applicable aircraft, the differences shall be indicated, and the charts adjusted appropriately, or separate charts may be provided for each configuration.

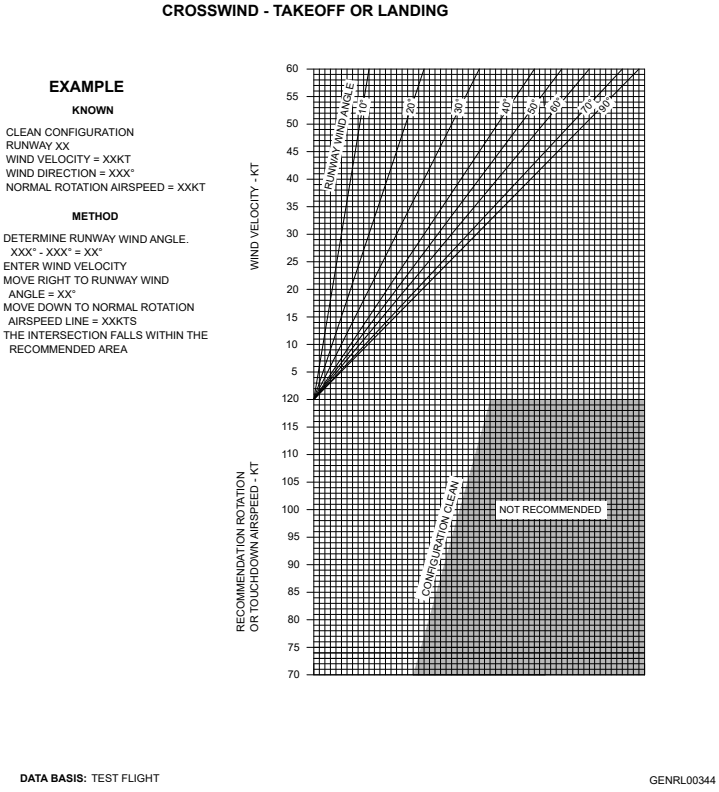


FIGURE H-15. Example crosswind takeoff or landing chart (FW).

H.5.4.2 Airspeed calibration chart work package <airspdcalchartwp>.

An airspeed calibration chart that defines the relationship between the pilot's indicated and calibrated airspeed for level flight, climb, and descent shall be provided. Instructions and examples shall be provided to show the operator how to determine the level flight indicated airspeed value that corresponds to known indicated airspeeds in climb and descent. Instructions and examples for determining calibrated airspeeds corresponding to known indicated airspeed shall also be provided. Altimeter correction charts that provide position error correction versus indicated airspeed shall be provided for all normal and emergency altimeter systems. Data shall be provided for all applicable flap settings or other variations in configuration. A temperature conversion/correction chart that provides true free air temperature (FAT) as a function of true airspeed and indicated temperature shall also be provided. For those aircraft

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whose air data system position errors are insignificant, airspeed, altitude, and temperature calibration data shall be omitted, with approval of the acquiring activity.

H.5.4.3 Altimeter corrections chart work package <altimeterchartwp>.

Data shall be presented as altimeter correction versus indicated airspeed at the option of the manufacturer. The presentation should be included for those configurations and airspeed systems for which airspeed calibration data are presented. As a minimum. data should be presented at 5000 feet. A second table should be added if tabular data are presented at more than one altitude.

H.5.4.4 Idle fuel flow chart work package <fuelflowchartwp>.

The idle fuel flow chart shall show idle fuel flow pounds per hour at the airframe idle throttle position at various altitudes and ambient air temperatures. Additional charts, when applicable, depicting idle fuel flow at various idle conditions shall be included. Differences between idle fuel flow with bleed air On or Off and similar conditions shall also be shown when applicable. The type of fuel used in computation shall be shown in the subheading of this chart.

H.5.4.5 Torque available for takeoff chart work package <trqavailchartwp>.

This chart (FIGURE H-16) shall show the torque/power setting/rpm available for takeoff, per engine for multi-engine aircraft, at various ambient air temperatures and altitudes. Maximum torque limits shall be shown when applicable. The standards for which the chart was compiled shall be shown in the heading and defined in the supporting text. Allowable tolerances for available torque shall be stated when applicable.

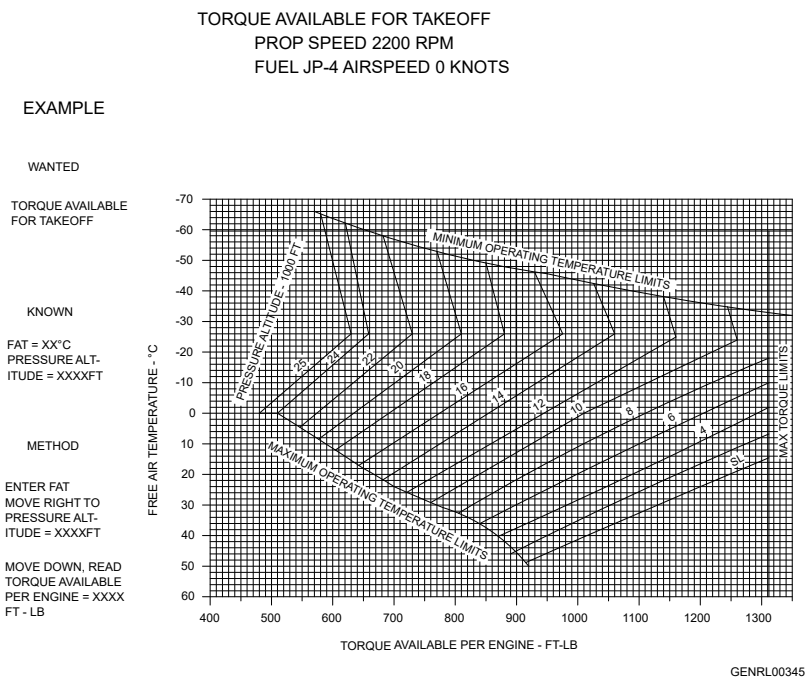


FIGURE H-16. Example torque available for takeoff chart (FW).

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H.5.4.6 Takeoff chart work package <takeoffchartwp>.

The takeoff chart (FIGURE H-17) shall show the ground roll distance and total takeoff distance required to clear different obstacle heights at various temperatures, altitudes, and aircraft gross weights. Wind conditions, aircraft configuration, power requirements, runway surface conditions, and other applicable information shall be given in the subheading and explained in the text. Additional charts required to obtain information shall be referenced. Each approved takeoff technique shall be covered on separate charts.

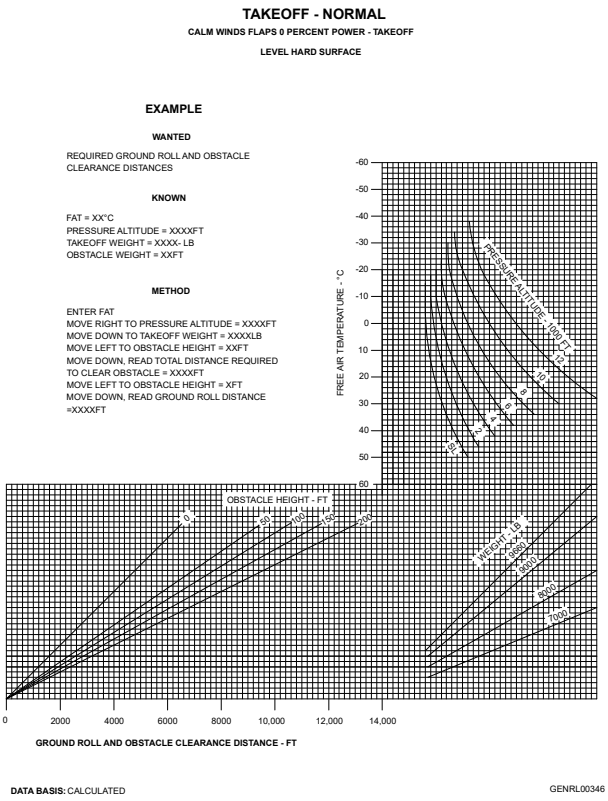


FIGURE H-17. Example takeoff - normal chart (FW).

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H.5.4.7 Takeoff airspeed chart work package <takeoffairspdchartwp>.

The chart (FIGURE H-18) shall show the recommended takeoff airspeeds for the aircraft at various gross weights. Flap settings and other applicable information, as required by the acquiring activity, shall be given in the subheading, or explained in the text. Each approved takeoff technique shall be covered on separate charts.

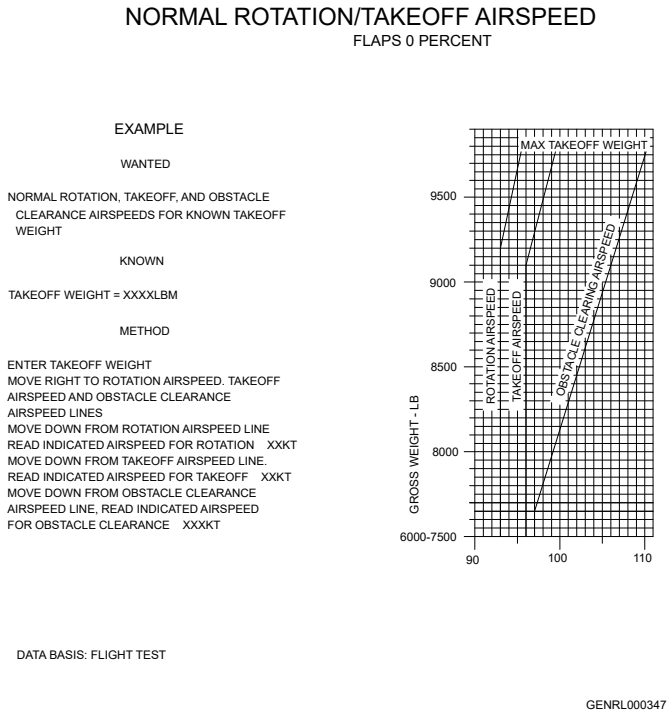


FIGURE H-18. Example normal rotation/takeoff airspeed chart (FW).

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H.5.4.8 Acceleration check distance chart work package <accelchkdischartwp>.

This chart (FIGURE H-19) shall show the relationship between indicated airspeed and ground roll distance during takeoff. The actual indicated airspeed required at any distance traveled along the takeoff airspeeds for various aircraft gross weights and the required ground roll distances for the aircraft shall be provided.

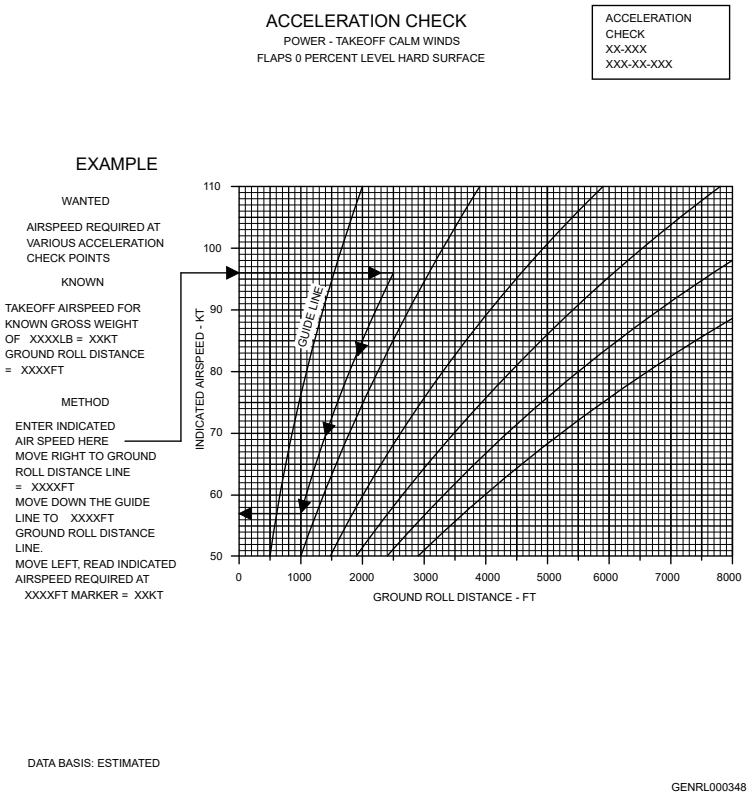


FIGURE H-19. Example acceleration check distance chart (FW).

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H.5.4.9 Accelerate-stop distance chart work package <accelstpdistchartwp>.

The accelerate-stop distance chart (FIGURE H-20) shall show the actual distance required to begin takeoff, accelerate to rotation speed, abort the takeoff, and bring the aircraft to a stop. Variables shall include ambient air temperature, pressure altitude, runway conditions, and gross weight.

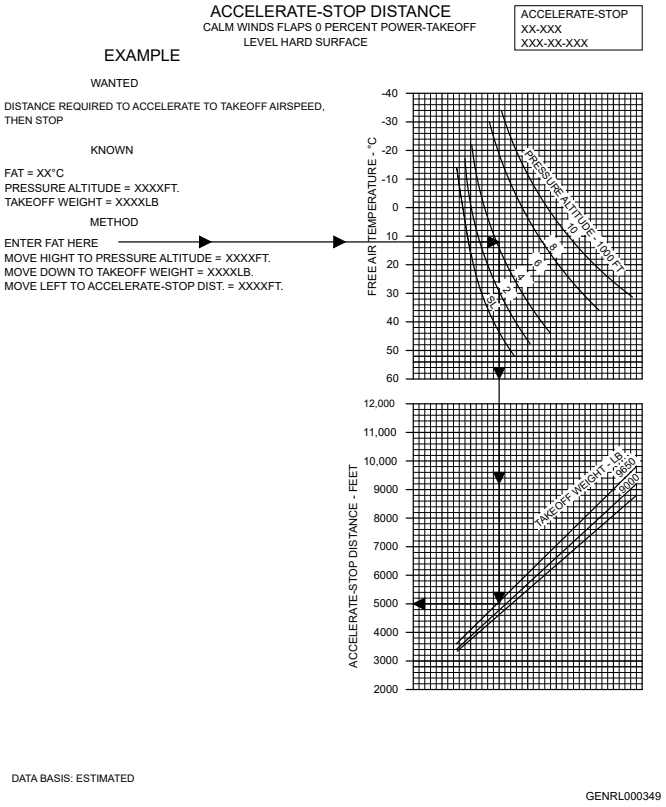


FIGURE H-20. Example accelerate - stop distance chart (FW).

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H.5.4.10 Accelerate after takeoff chart work package <accelaftertkoffchartwp>.

The accelerate after takeoff chart (FIGURE H-21) shall show the actual distance required to clear an obstacle after takeoff. Parameters shall include OAT, pressure altitude, takeoff weight, and velocity.

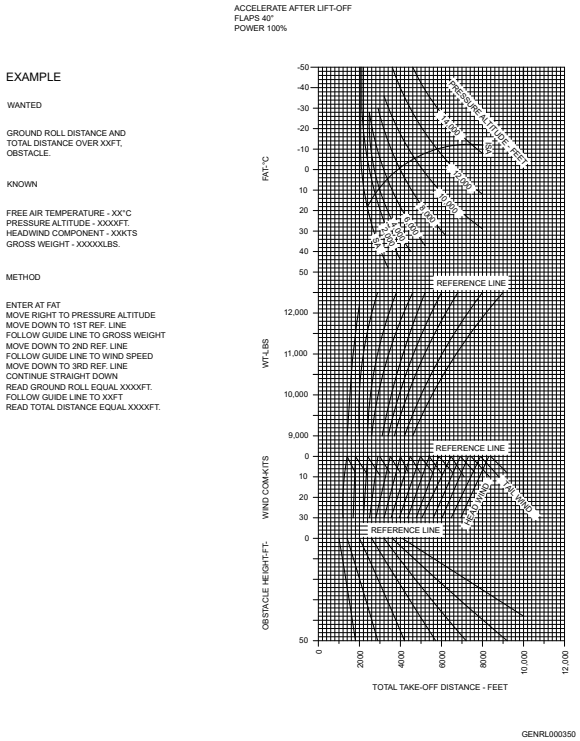


FIGURE H-21. Example accelerate after lift-off chart (FW).

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H.5.4.11 Minimum single engine control airspeed chart work package <minsglengchartwp>.

The minimum single engine control airspeed chart (FIGURE H-22) is applicable to multi-engine aircraft and shall show the minimum controllable airspeed (Vmc), with parameters of FAT, pressure altitude, and gross weight, following engine failure during takeoff. The chart shall be based on the operating engine's capability to produce full takeoff power. The primary use of the chart shall be to provide Vmc at takeoff, not to provide single engine rate of climb information. All applicable limits shall be shown and explained in the text. Conditions such as flap setting, landing gear position, etc. shall be included in the subheading or explained in the text. The effect of engine failure on takeoff, climb, and cruising performance, the effect of wind-milling and feathered propellers on aircraft drag, and other adverse factors shall be described.

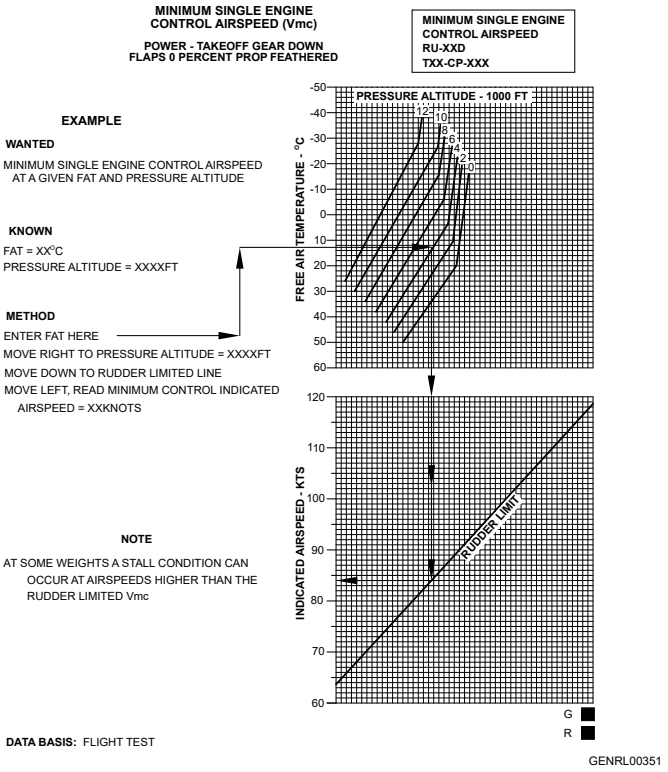


FIGURE H-22. Example minimum single engine control airspeed chart (FW).

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H.5.4.12 Single engine climb chart work package <singengclmbchartwp>.

This chart (FIGURE H-23 and FIGURE H-24) shall present single engine airspeeds and rate of climb data for various temperatures, altitudes, and gross weights. Single engine rate of climb shall be based on takeoff airspeeds to include gear-up and gear-down configurations. When alternate aircraft configurations change the validity of information being presented, additional charts shall be prepared with an explanation of the alternate configuration provided in the subheading and within the text when necessary. Information indirectly obtained from the chart that would help in the determination of the best course of action to be taken shall also be included in the text. Reference shall be made to other charts related to single engine operations.

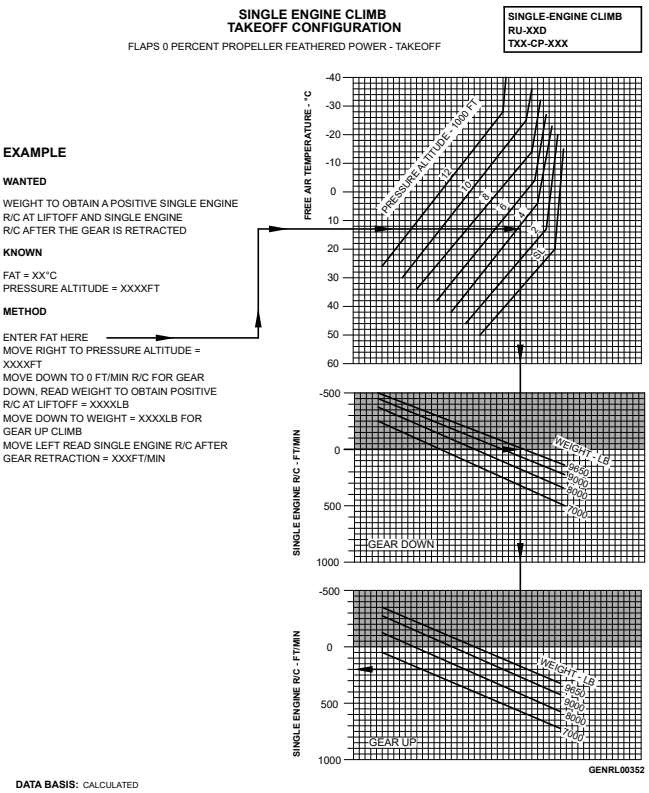


FIGURE H-23. Example single engine climb chart (FW) (sheet 1 of 2).

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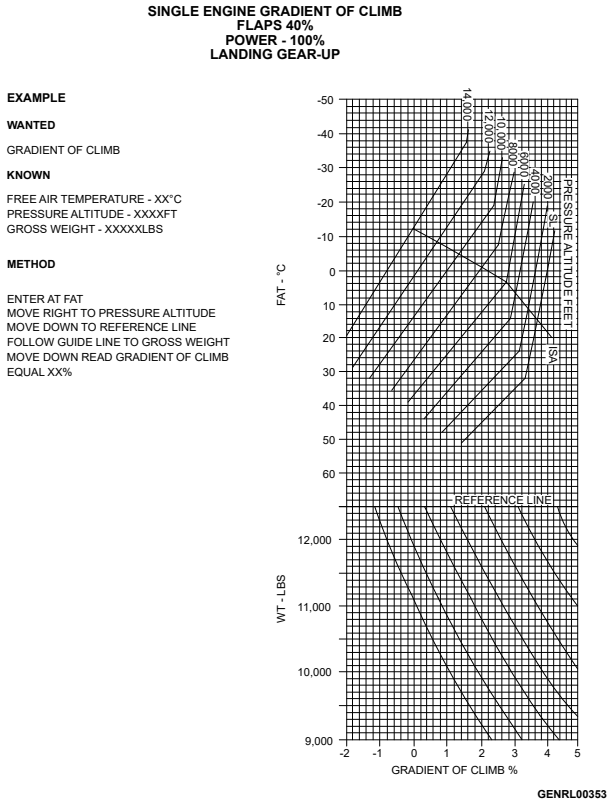


FIGURE H-24. Example single engine climb chart (FW) (sheet 2 of 2).

H.5.4.13 Service ceiling, one engine inoperative chart work package <servoneinopwp>.

This chart shall present single engine airspeeds and rate of climb data for various temperatures, altitudes, and gross weights. Single engine rate of climb shall be based on takeoff airspeeds to include gear-up and gear-down configurations. When alternate aircraft configurations change the validity of information being presented, additional charts shall be prepared with an explanation of the alternate configuration provided in the subheading and within the text when necessary. Information indirectly obtained from the chart that would help in the determination of the best course of action to be taken shall also be included in the text. Reference shall be made to other charts related to single engine operations. Data shall be presented as service ceiling and outside air temperature versus weight. Service ceiling shall be the pressure altitude where an airplane has the capability of climbing 50 ft/min with one engine propeller feathered.

H.5.4.14 Rate of climb chart work package <rateclimbwp>.

Data shall be presented as rate of climb versus outside air temperature, altitude and weight at the maximum power approved or as specified by the appropriate FAA requirements. Separate charts shall be included for the following:

a. Rate of climb - all engines operating with flaps set to the takeoff position and landing gear retracted.

b. Rate of climb - all engines operating with flaps set to the enroute position and landing gear retracted (if applicable).

c. Rate of climb - one engine inoperative with flaps set to the enroute position and landing gear retracted.

d. Rate of climb - Balked Landing. The climb speeds appropriate to each configuration shall be scheduled in IAS.

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H.5.4.15 Cruise climb chart work package <cruiseclmbchartwp>.

The cruise climb chart (FIGURE H-25) shall be used to find the time, fuel, and distance required to climb. Parameters shall include initial and final OAT, initial and final pressure altitude, and initial gross weight.

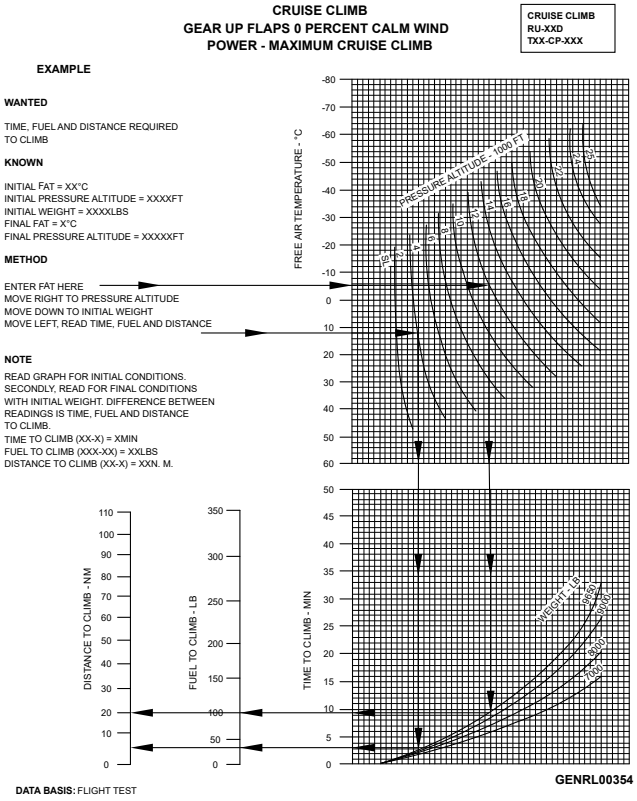


FIGURE H-25. Example cruise climb chart.

H.5.4.16 Drag chart work package <dragchartwp>.

The drag chart shall show additional shaft horsepower required at various airspeeds, altitudes, and temperatures due to drag increases caused by changes in external configuration. Additional shaft horsepower shall be given per engine for multi-engine aircraft. Charts used in connection with the drag chart shall be referenced in the text. Tabular data presenting each drag item and the drag area change in square feet shall be included in the text.

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H.5.4.17 Cruise chart work package <cruisechartwp>.

The cruise chart (FIGURE H-26) shall show the obtainable airspeed, required engine shaft horsepower, engine torque pressure, shaft horsepower increase required due to increases in drag, fuel flow and optimum propeller rpm for maximum range during cruise flight at various aircraft gross weights, altitudes, and temperatures. The particular altitudes, configurations, and temperatures at which cruise data are to be presented shall be specified by the acquiring activity. This information shall be presented for each engine when performance data pertain to multi-engine aircraft. When fuel flow variations exist due to alternate engine operations, fuel flow for each alternate condition shall be shown. Single engine data shall be placed on the same charts as multi-engine data only when approved by the acquiring activity. Maximum performance, precautionary, and limits data shall be shown on each chart and explained in the text. Additional charts showing one engine inoperative shall be presented when specified by the acquiring activity. Indicated and true airspeed for each altitude shall be shown. When an altitude limitation prevents safe single engine cruise for multi-engine aircraft, the single engine graph shall be omitted. Additional charts related to cruise performance shall be referenced in the text.

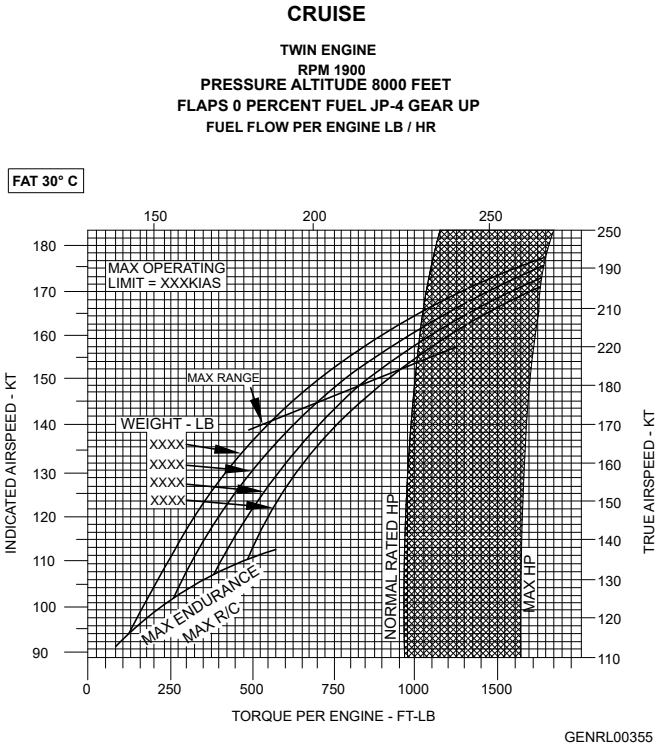


FIGURE H-26. Example cruise chart (FW).

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H.5.4.18 Climb-descent chart work package <clmbdeschartwp>.

The climb-descent chart (FIGURE H-27) shall show changes in torque and horsepower required to obtain a desired rate of climb or descent at a known gross weight and propeller rpm. For maximum rate of climb information, reference shall be made to the cruise charts. If the aircraft is other than baseline configuration, an increase in horsepower due to drag shall be computed from the drag chart and added to the horsepower required per engine. Charts used in connection with the climb-descent charts shall be referenced in the text and in the single engine climb chart.

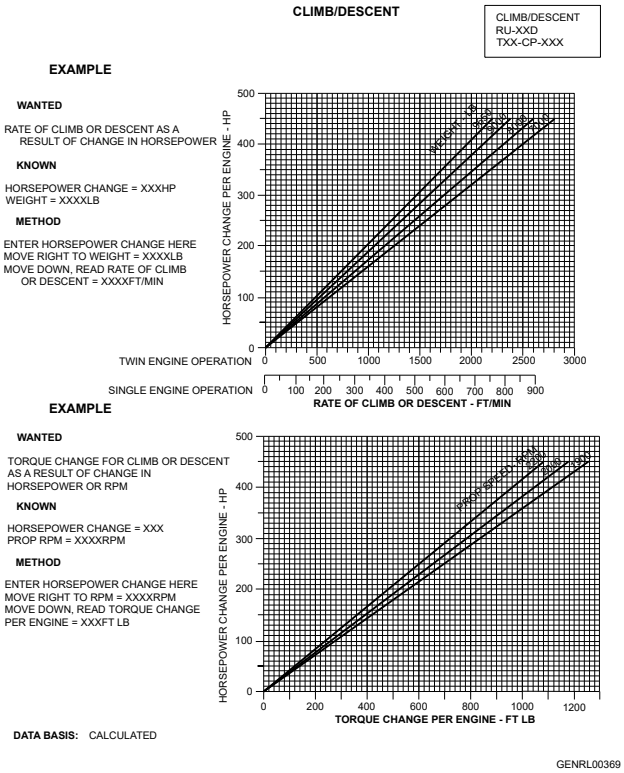


FIGURE H-27. Example climb/descent chart (FW).

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H.5.4.19 Approach speed chart work package <appspdchartwp>.

The approach speed chart (FIGURE H-28) shall present the recommended airspeeds during approach to landing for the full range of gross weights and flap settings for the aircraft. The chart shall be valid for all aircraft configurations, unless otherwise specified by the acquiring activity. Charts used in connection with the approach speed chart shall be referenced in the text.

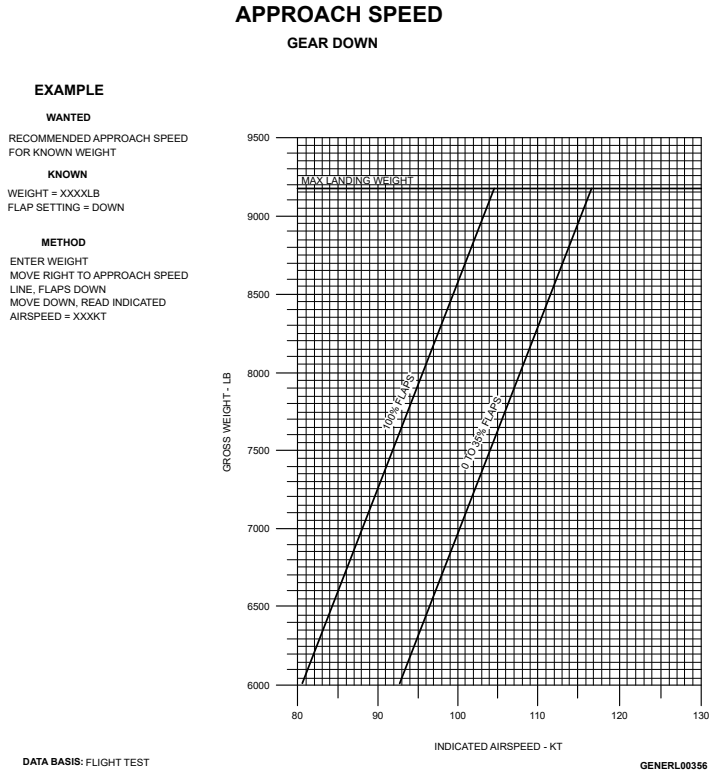


FIGURE H-28. Example approach speed chart (FW).

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H.5.4.20 Landing chart work package <landchartwp>.

The landing chart (FIGURE H-29) shall show the total ground roll distance for landing with no reverse thrust at known gross weight, pressure altitude, and ambient air temperature. Landing distance shall be based on touching down at the approach speed obtained from the approach speed chart, full braking with 0 degrees, and normal landing flap settings. The correct approach speed is obtained from the approach speed chart.

Landing performance shall be based on a dry, level, hard surface runway and calm wind conditions. This chart shall be valid for all stores configurations unless otherwise specified by the acquiring activity. The chart used in computing landing distances shall be described in the text.

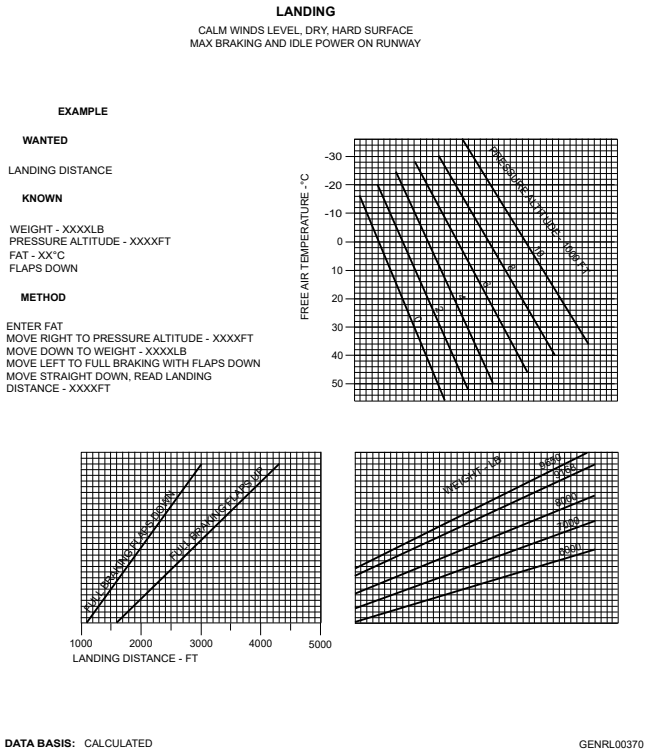


FIGURE H-29. Example landing chart (FW).

H.5.4.21 Optimum cruise work package <optcruisechartwp>.

When requested by the acquiring activity, data shall be provided to determine the altitude for maximum range and maximum endurance as a function of gross weight and ambient temperature. Information shall also be provided for optimum propeller rpm for maximum range and endurance. Where optimum rpm is different from that presented for the (normal) cruise data, information shall be provided to correct fuel flow for the different rpm. Optimum cruise speed (maximum range or endurance) presented on the cruise chart shall be referenced and used. Airspeed and power schedules for climb and descent to maximize total range or endurance shall be described. A means shall be provided for estimating ambient temperature at optimum altitude. Also, a means shall be provided for comparing the effects of varying winds with altitude with the change in aircraft performance with altitude. Data shall cover the range of gross weights and ambient temperatures presented on the cruise charts, and the limits of altitude on the cruise charts, if required. If corrections to optimum altitude for configuration variations are significant and capable of being done, this information shall be provided. When requested by the acquiring activity, the impact of headwinds and tailwinds on optimum cruise shall be provided.

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H.5.4.22 Stall chart work package <stallchartwp>.

Stalling airspeeds (with power-on and power-off configurations) for takeoff, landing, and cruise shall be presented showing the variations of bank angle and gross weight.

H.5.4.23 Additional performance charts work package <genchartwp>.

As determined by the acquiring activity, additional charts necessary for the completion of preflight and in-flight mission planning may be included when the specific type of performance chart is not covered as described in H.5.4 through H.5.4.22. Each additional chart shall be a separate work package and shall comply with all of the requirements of H.5.1 through H.5.1.10. If additional performance charts are used, the proponent shall submit to AMCOM the requirements for this performance chart type for possible incorporation within future revisions to this standard.

H.6 Notes.

The notes in Section 6 apply to this appendix.

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APPENDIX I

NORMAL PROCEDURES (MODULE 8)

I.1 Scope.

This appendix establishes the technical content requirements for the preparation of normal procedures for aviation weapon systems. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

I.2 Applicable publications.

The applicable publications in Section 2 apply to this appendix.

I.3 Definitions.

The definitions in Section 3 apply to this appendix.

I.4 General requirements.

Procedures from the time a flight is planned until the flight is completed and the aircraft is properly parked and secured shall be provided. The procedures shall include all steps necessary to ensure safe flight under normal, night, and instrument conditions.

I.4.1 Preparation of digital data for electronic delivery.

Data prepared and delivered digitally in accordance with this standard shall be XML tagged using the current Army DTD. Data shall be formatted for presentation using the Army stylesheets. Refer to I.4.2 for information on obtaining or accessing the Army DTD and stylesheets. XML tags used in the DTD are noted throughout the text of this standard in bracketed, bold characters (e.g., **<geninfo>**) as a convenience for the author and to ensure that the tags are used correctly when developing a document instance.

I.4.2 Use of the Document Type Definition (DTD)/stylesheet.

The DTD referenced in this appendix interprets the technical content and structure for the functional requirements contained in this standard. Development of the TMs shall be accomplished through the use of this standard, the Army DTD, and the Army stylesheets.

I.4.3 Content structure.

The examples provided herein are an accurate representation of the content structure requirements contained in this appendix and shall be followed to permit the effective use of the DTD.

I.4.4 Style and format.

This standard provides style and format requirements for the technical content requirements described in this appendix. These requirements are considered mandatory and are intended for compliance.

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I.4.5 Work package development.

Data developed in accordance with this appendix shall be divided into work packages. These work packages shall be stand alone and shall contain all information and references required to support the work package type.

I.4.6 Selective application and tailoring of content using Appendix A matrices.

This standard contains some requirements that may not be applicable to the preparation of all TMs. Selective application and tailoring of requirements contained in this standard are the responsibility of the acquiring activity and shall be accomplished using APPENDIX A. The applicability of some requirements is also designated by one of the following statements: unless specified otherwise by the acquiring activity; as specified by the acquiring activity; or when specified by the acquiring activity.

I.5 Detailed requirements.

Procedures from the time a flight is planned until the flight is completed and the aircraft is properly parked and secured shall be provided. The procedures shall include all steps necessary to ensure safe flight under normal, night, and instrument conditions. Only the duties of the minimum crew necessary for the actual operation of the aircraft shall be included.

a. The symbols and guidance in 5.2.2.1.7 shall be used in the checklist on every step applicable to identify certain conditions or duties.

b. Instructions for the operation of utility, avionic, mission equipment and controls are contained in Modules Section 2, Section 3, and Section 4 and shall be included in this module only if neglect would affect safety or efficiency of the flight or cause damage to the equipment. (This does not preclude the inclusion of utility equipment checklists in modules to which they pertain.)

c. Only unique feel, characteristics, and reaction of the aircraft during the various specified phases of operation, and the techniques or procedures used for operating the aircraft shall be described in detail.

d. All precautions to be observed during the various operations shall be covered.

e. Procedures for operation under all adverse environmental conditions shall be described.

f. Instrument flight procedures shall be integrated with normal procedures as much as possible.

g. For aircraft where no unique or abnormal techniques apply, reference shall be made to appropriate flight training publications.

I.5.1 Preparation of normal procedures work packages.

Normal procedure work packages shall be developed as detailed in I.5.1.1 through I.5.1.3.

I.5.1.1 Normal procedure work packages.

Normal procedures shall be divided into the following work package categories as determined by TABLE A-II in APPENDIX A:

a. Normal Procedures Introduction Work Package

b. Ground Equipment Procedures Work Package

c. Preflight Procedures Work Package

d. Flight Procedures Work Package

e. Post-flight Procedures Work Package

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f. Instrument Flight Work Package

g. Transfer Procedures Work Package

h. Mission Equipment Procedures Work Package

i. Communications Equipment Work Package

j. Flight Characteristics Work Package

k. Adverse Environmental Conditions Work Packages

l. General Normal Procedures Work Packages

I.5.1.2 Procedures.

Procedural steps shall be written so that crewmembers shall not be required to retrace steps. Insofar as possible, checks shall be grouped to keep control manipulation and ground operating time at a minimum. Phases shall be added or deleted to provide for special aircraft or special situations. However, the interpretation of the period of operation encompassed by a given phase shall be identical in all operators' manuals. In the checklists, the condition and response of a procedural step shall be separated by a long dash.

I.5.1.2.1 Sequence.

Sequence of phases and actions shall be arranged chronologically.

I.5.1.2.2 Checks.

All checks shall be made from left to right or top to bottom except where chronology shall take precedence.

I.5.1.2.3 Performance charts.

Where applicable, performance charts provided in Module 7 that are required to carry out specific flight procedures or maneuvers shall be referenced or linked as necessary.

I.5.1.2.4 Checklist changes.

The specific checks described in I.5.4 through I.5.6.3 may be deleted or new checks added when approved by the acquiring activity.

I.5.1.3 Amplified checklist.

Module 8's normal procedures, also referred to as the amplified checklist, shall consist of numbered items supplemented where necessary by explanatory material. Where required for emphasis, a brief explanation shall be provided as to why it is required. These amplified checklists shall be provided in the operator's checklist manuals for each aircraft, and they shall be the basis of all operators' checklist manuals. An amplified normal checklist shall be included for the pilot, pilot (not on the controls), and flight engineers, as applicable. The amplified checklist shall contain, at minimum, the requirements provided in I.5.2 through I.5.12.5.

I.5.2 Normal procedures introduction work package <norm_introwp>.

An introduction to the normal procedures shall be prepared as detailed in I.5.2.1 through I.5.2.3.

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I.5.2.1 Crew duties.

Unique crew responsibilities that result from the specific characteristics of the aircraft shall be described. When applicable, a description of mission planning shall also be included.

I.5.2.2 Mission planning statement.

The following boilerplate statement shall be included verbatim:

> "Mission planning begins when the mission is assigned and extends to the preflight check of the *[aircraft]*. It includes but is not limited to checks of operating limits and restrictions; weight, balance, and loading; performance; publications; flight plan; and crew and passenger briefings. The pilot in command shall ensure compliance with the contents of this manual that are applicable to the mission and all aviation support equipment required for the mission (e.g., helmets, gloves, survival vests, survival kits, etc)."

I.5.2.3 Amplified checklist statement.

The following boilerplate statement shall be included only in the amplified checklist:

> "Normal procedures are given primarily in checklist form and amplified as necessary in accompanying paragraph form when a detailed description of a procedure or maneuver is required. A condensed version of the amplified checklist, omitting explanatory text and most warnings, cautions, and notes, is contained in the Operator's Checklist Manual, *(enter TM number of applicable checklist)*. The procedural steps are numbered to coincide with the corresponding numbered steps in this manual."

I.5.2.4 Checklist symbols.

The following verbatim statement shall be included to direct the operator to the About this Manual section for the table of symbols used in the checklist.

> "Symbols used in the checklist are identified in *(applicable table number)* in About this Manual."

I.5.3 Ground equipment procedures work package <groundnormwp>.

Procedures involving ground equipment shall be developed. If the weapon system does not have ground equipment for operations, this WP is not required. To better organize and manage the data required in the WP, it may be separated into multiple Ground Equipment Procedures WPs or combined into one WP so long as the WP remains under 50 pages.

I.5.4 Preflight procedures work package <preflightnormwp>.

Preflight procedures shall be developed and contain at minimum the requirements of I.5.4.1 through I.5.4.2. To better organize and manage the data required in the WP, it may be separated into multiple Preflight Procedures WPs or combined into one WP so long as the WP remains under 50 pages.

I.5.4.1 Preflight check.

The amplified preflight check shall include a before exterior check, if required, and the exterior and interior checks. The amplified checklist shall emphasize that the preflight is not intended to be a detailed mechanical inspection and that the order is a recommended sequence only. In addition, the expanded sub-steps shall not need to be memorized or accomplished in a certain order. The preflight may be made as comprehensive as conditions warrant at the

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discretion of the pilot/UAS Mission Commander. UAS manuals shall include a thorough preflight of the ground control station and supporting equipment.

I.5.4.1.1 Before exterior check.

When required by the aircraft configuration, all necessary actions that shall be performed prior to starting the exterior check shall be included. Emphasis shall be placed on items that affect safety during the inspections to follow.

I.5.4.1.2 Exterior check.

Only those exterior points that significantly affect the flight shall be included avoiding needless repetition of items which are the normal responsibility of the maintenance crew. The criteria on which these checks shall be based are safety of flight, items that have previously been a problem or that are anticipated to be a problem, and ease of accomplishing the check. Inspections usually should proceed counter-clockwise (as viewed from the top) around the aircraft.

I.5.4.1.3 Interior check.

The complete interior check shall be described, including all necessary check items up to the point where the pilot is strapped/operator is in the seat. All necessary equipment including, but not limited to, a first aid kit, fire axes, pyrotechnic equipment, aircraft covers, tie downs, and control locks shall be stowed. A check of the headrest area of the ejection seat shall be included to determine that the face curtain handles are properly stowed, that the catapult pin is installed and connected to the removal mechanism, and that the catapult firing yoke is properly positioned and connected. Instructions shall be included to ensure that controls are positioned as necessary to facilitate the exterior check (only for those aircraft where the interior check is performed before the exterior check). On large aircraft, it may be necessary to include an interior check diagram. UAS ground equipment interior checks shall be described.

I.5.4.1.4 Crew/passenger briefing check.

Instructions shall be provided to ensure that crew and passenger briefings have been completed prior to starting engines.

I.5.4.2 Before starting engine(s).

Precautions to be observed and checks to be accomplished before starting engine(s) shall be included. Such checks as should be accomplished before starting engine(s), but which could not be properly accomplished during the interior check shall be included. Instructions for positioning important controls and checking important indicators shall be included. Insofar as is practicable, all controls shall be positioned as required for engine starting. Functional checks shall be included for those systems that can be checked before the engines are started. For those aircraft in which engine power is not necessary, flight controls shall be checked for free and correct movement. Instructions shall be provided on the use of external power or auxiliary power units and any necessary switching involved in its use.

I.5.5 Flight procedures work package <flightnormwp>.

Flight procedures shall be developed and contain at minimum the requirements of I.5.5.1 through I.5.5.15. To better organize and manage the data required in the WP, it may be separated into multiple Flight Procedures WPs or combined into one WP so long as the WP remains under 50 pages.

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I.5.5.1 Starting engine(s).

The complete procedure for starting the engine(s) shall be provided, including the order of starting for multi-engine aircraft. Except when significant differences in procedures are required for multi-engine aircraft, engine start procedures shall not be repeated. For jet and turbine powered aircraft, the means to avoid hot starts and procedures to follow when a hot start is experienced shall be included. Procedures for engaging rotors for rotary wing aircraft shall be given.

I.5.5.2 Engine ground options.

When required, warm-up and ground operation power setting shall be specified. Any special precaution or limitation shall be stated. For rotary wing aircraft a requirement for flight control checks before the rotor is engaged shall be included, if applicable.

I.5.5.3 Engine run-up.

Instructions shall include, but shall not be limited to, checking engine propeller/rotor operation, including power, ignition, and use of brakes.

I.5.5.4 Before taxiing.

All checks to be accomplished before taxiing, such as check flight controls for free and correct movement (for those aircraft which require engine power to perform this check), windows and doors, control locks, and hydraulic pressure checks shall be included.

I.5.5.5 Taxiing.

Any unusual taxiing characteristics or techniques shall be described, including special instructions for engine cooling, reverse pitch, and use of brakes. A requirement that flight instruments be checked during taxiing shall be included.

I.5.5.6 Before takeoff.

All checks, which shall be accomplished immediately prior to takeoff/departure, shall be listed.

I.5.5.7 Lineup check (FW only).

When aircraft configuration or mission requirements preclude performance of some of the takeoff checks before taxiing onto the active runway, a lineup check shall be provided. This may include activation of anti-icing/deicing system switches, transponder switches, setting or aligning gyros, and stabilizing power prior to starting takeoff.

I.5.5.8 Takeoff.

Takeoff techniques required to produce the results shown on the takeoff charts in the appropriate work packages in Module 7 shall be covered in detail. A requirement to reference the takeoff charts shall be included. When appropriate, manipulation of brakes and throttles/power levers, etc., shall be described. Detailed information shall be given regarding unique reactions of the aircraft during takeoff. Criteria for continuing a takeoff or aborting under various circumstances shall be included. Operational consideration and general rules contributing to hovering capability and power availability shall be stated. Unique hover/taxi, sideward and rearward flight techniques and

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power check shall be included. The necessity for a prepared runway shall be discussed for various conditions of altitude and weight of aircraft that may be required to operate from temporary or unfinished runways.

I.5.5.9 After takeoff.

All actions and techniques to be accomplished immediately after takeoff shall be listed. If flap retraction procedures differ under various conditions including, but not limited to, heavy weight and weather, it shall be so stated. When applicable, minimum airspeed and altitude for retracting flaps shall be covered. A minimum flap retraction airspeed chart shall be included for aircraft of highly variable gross weight. All actions needed to establish the required climb shall be covered, including the airspeed at which the climb should be started.

I.5.5.10 Climb.

A description of unique climb techniques required to produce the results stated in the climb charts in the appropriate work packages in Module 7 shall be included. Unusual characteristics of the aircraft in climb shall be described. Since the preceding paragraph includes the climb checklist, this paragraph shall contain discussion only.

I.5.5.11 Cruise.

An explanation shall be provided for all actions that shall be performed when the transition from climb to cruise is made. Any particular matters that shall be considered during cruise flight shall be described. Reference shall be made to the appropriate work packages in Module 2 and Module 7 concerning fuel system management and other actions that should be considered during flight. Actual procedures shall not be covered here.

I.5.5.12 Descent-arrival.

A checklist and discussion of this phase of operation shall be included as appropriate. The checklist shall include all checks that shall be made immediately before and during a descent preparatory to landing. Special instructions regarding various types of descent shall be included as applicable, including any special devices that may be provided to facilitate descent.

I.5.5.13 Before landing.

All checks that shall be made immediately before entering the traffic pattern until the aircraft is committed to landing shall be covered.

I.5.5.14 Landing.

A landing checklist and a narrative discussion of the landing problems and techniques shall be provided. The landing checklist shall include all actions to be performed from the time the landing is committed until it is affected. Landing techniques required to produce the results stated in the landing charts in the appropriate work packages in Module 7 shall be included. Braking techniques and devices used during the landing and after-landing roll shall be described. Approach and landing airspeed corrections required to compensate for gusts shall be covered. In addition, landing techniques from the viewpoint of recommended maximum and minimum approach and landing airspeeds as related to aircraft flight classification, aircraft strength, aircraft touchdown bounce characteristics, and other aircraft characteristics shall be included. Reference shall be made to the appropriate work packages in Module 7 for supplemental information provided by landing and approach speed charts. Coverage of approach and landing shall include cautions, when applicable, in the use of the engine during approach, performing a go-around, for the use of the angle-of-attack indicator in making an approach, etc. Shipboard landing techniques, when applicable, shall be included for rotary wing aircraft when unusual characteristics dictate.

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I.5.5.15 Touch and go landings/condition statement.

All instructions including, but not limited to, trim changes and flap settings for executing these procedures shall be included. Proper throttle/power lever technique shall be emphasized, when applicable.

I.5.6 Post-flight procedures work package <postflightnormwp>.

Post-flight procedures shall be developed and contain at minimum the requirements of I.5.6.1 through I.5.6.3. To better organize and manage the data required in the WP, it may be separated into multiple Post-flight Procedures WPs or combined into one WP so long as the WP remains under 50 pages.

I.5.6.1 After landing.

All checks and operations to be performed from immediately after landing until the parking area is reached shall be included.

I.5.6.2 Engine shutdown.

A checklist shall be provided covering proper procedures and precautions for stopping engines.

I.5.6.3 Before leaving the aircraft.

A checklist of settings of all controls, control locks, and safety devices for securing the aircraft shall be provided for pilots and crew, air vehicle operator, payload operator, external operator, and ground-crew personnel. The following statement shall be included verbatim:

> "In addition to established requirements for reporting any system defects or unusual and excessive operations such as hard landings, the flight crew shall also make entries on DA Form 2408-13-1 to indicate when any limits of the operator's manual have been exceeded."

I.5.7 Instrument flight work package <instrumentwp>.

Unique qualities and capabilities of the aircraft under instrument flight conditions shall be briefly described. Only those procedures and techniques that are used for instrument flight that are different from normal procedures in TC 3-04.5 shall be discussed. Instrument flight conditions to be considered shall include instrument takeoff, climb, cruise, descent, and approaches; holding; and automatic approaches.

I.5.8 Transfer procedures work package <xfer_normwp>.

UAS transfer procedures shall be developed. To better organize and manage the data required in the WP, it may be separated into multiple Transfer Procedures WPs or combined into one WP so long as the WP remains under 50 pages.

I.5.9 UAS mission equipment procedures work package <mis_wp>.

Specific procedures for UAS Mission Equipment shall be developed. To better organize and manage the data required in the WP, it may be separated into multiple Mission Equipment Procedures WPs or combined into one WP so long as the WP remains under 50 pages.

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I.5.10 Communications equipment work package <commequipwp>.

Information for communications equipment installed in the aircraft shall be developed. To better organize and manage the data required in the WP, it may be separated into multiple Communications Equipment Procedures WPs or combined into one WP so long as the WP remains under 50 pages.

I.5.11 Flight characteristics work package <flt_charwp>.

Detailed unique flight characteristics of the particular aircraft that are different from TC 3-04.5 shall be provided. Emphasis shall be placed on advantageous flight characteristics as well as on any dangerous tendencies. The extent of coverage shall depend principally on the type of aircraft being discussed. To better organize and manage the data required in the WP, it may be separated into multiple Flight Characteristics Procedures WPs or combined into one WP so long as the WP remains under 50 pages.

I.5.11.1 Stalls (fixed wing required, rotary wing and UAS optional).

The power-off and power-on stalling characteristics of the airplane in the takeoff, landing, and cruise configurations shall be described. Stalling characteristics shall also be included for the approach configuration if sufficiently different from landing. A definition of power-off and power-on as used in the discussion shall be included. Information about the stall warning shall also be included. Normal and accelerated stalls shall be covered, and recommended procedures for initiating stalls shall be included. Stall recovery technique shall be emphasized. For helicopters, appropriate information shall be included on blade stalls.

I.5.11.2 Stall chart (fixed wing only).

Stalling airspeeds (with power-on and power-off configurations) for takeoff, landing, and cruise shall be presented showing the variations of bank angle and gross weight.

I.5.11.3 Spins (fixed wing only).

Spin characteristics and limitations shall be given, including details of any special techniques recommended for recovery. Recovery techniques shall be given whether or not spins are permitted. Altitude lost in effecting a recovery and minimum altitude at which bailout shall be effected if aircraft has not been brought under control shall be stated.

I.5.11.4 Diving (fixed wing required, rotary wing and UAS optional).

The diving characteristics of the aircraft shall be described with particular emphasis on high speed diving and compressibility effects. Dive recovery techniques and precautions shall be given, including any special information regarding power plant operation and trim changes. For highly maneuverable aircraft, dive recovery charts shall be included for various G pullouts given varying parameters of altitude, airspeed, and dive angle.

I.5.11.5 Maneuvering flight.

Maneuvering flight shall be described, including characteristics under accelerated flight conditions. Stick forces shall be included, emphasizing conditions that may result in stick reversal.

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I.5.11.6 Flight controls.

Detailed coverage of the effectiveness and unusual reactions that may be encountered in the operation and use of the flight controls shall be included. All the various types of flight controls, such as ailerons, elevators, rudders, stipulators, trim tabs, speed brakes, slats, cyclic stick, and collective pitch shall be described. The text shall state when and how the various controls are used to achieve maximum benefits and what precautions shall be observed. The capabilities and limitations of power-boosted systems when power boost is inoperative shall be covered.

I.5.11.7 Level flight.

Characteristics of level flight under slow, cruising, and high speed conditions shall be described.

I.5.11.8 External loads.

Changes in flight characteristics due to external loads shall be described.

I.5.11.9 Asymmetrical loads.

Coverage of characteristics and techniques to be employed when operating with asymmetrical loads or configurations shall be presented.

I.5.12 Adverse environmental conditions work packages.

Information relative to operations that are unique to the specific aircraft under adverse environmental conditions (snow, ice, rain, turbulent air flight, extreme cold and hot weather, desert operations, and high altitude operations) for parameters including, but not limited to, gross weight and aircraft configuration shall be provided.

a. The information presented shall be primarily narrative in nature. Checklists shall be avoided; they shall be used only to cover specific procedures that are characteristic of all weather operations.

b. A description of equipment shall not be included.

c. An introductory paragraph shall be included explaining the function of this section.

d. Coverage of duties to be accomplished before leaving the aircraft, including, but not limited to, leaving the canopy slightly open, positioning of doors, battery care, and installing covers shall be included for applicable environmental conditions.

e. Adverse Environmental Conditions WPs are separated into the following types:

(1) Cold Weather Operations Work Package.

(2) Desert and Hot Weather Operations Work Package.

(3) Turbulence and Thunderstorm Operations Work Package.

(4) Rain Operations Work Package.

(5) General Adverse Environmental Conditions Work Package.

I.5.12.1 Content requirements for adverse environmental conditions work packages.

The content detailed in I.5.12.1.1 through I.5.12.1.10 shall be developed for all adverse environmental conditions work packages as applicable.

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I.5.12.1.1 Preparation for flight.

Special problems that occur during adverse conditions shall be adressed (e.g. application of heat, removal of ice and snow from the aircraft surfaces, fuel and oil tank vents, pitot tubes, props, and supplying external power).

I.5.12.1.2 Engine starting.

Any special precautions that shall be observed before starting the engines shall be included (e.g. cold weather starting techniques shall be explained including the use of special fuels and carburetor heat).

I.5.12.1.3 Warm-up and ground testing.

This shall include, but shall not be limited to, coverage of carburetor heat, cowl flap position, and technique of switching from a special starting fuel. If oil dilution is available, the fuel boil-off procedure shall be covered, including a reference to the oil dilution table. The importance of ground testing of systems that may be adversely affected by adverse conditions shall be included.

I.5.12.1.4 Taxiing and hovering instructions.

The unique techniques and precautions to be observed when taxiing in adverse conditions (e.g. on snow, ice, or slush covered water) shall be explained, as well as, instructions for operator/ground crew to visually check wheels to ensure they are turning.

I.5.12.1.5 Before takeoff.

Any unique checks required before takeoff shall be included (e.g. checks for ice and snow buildup on the aircraft).

I.5.12.1.6 Takeoff.

Unique techniques and precautions to be observed when taking off under adverse conditions shall be included (e.g. the effect of snow or ice covered runways on takeoff, of extremely cold weather on engine and aircraft performance, etc.).

I.5.12.1.7 During flight.

Any special precautions that shall be observed during flight in adverse conditions shall be described (e.g. cycling propeller governing systems and procedures for dealing with in-flight icing)

I.5.12.1.8 Descent.

Any special instructions regarding descent as may be applicable to adverse condition operation shall be included (e. g. switching on the auxiliary power unit early to ensure that it is sufficiently warmed up prior to landing).

I.5.12.1.9 Landing.

Unique techniques and precautions to be observed during landing in adverse conditions shall be included (e.g. the use of brakes and reverse pitch propellers when landing on snow or ice covered runways and any restrictions regarding the use of landing or dive flaps when landing on snow or slush covered runways or slush covered water where ice is suspected).

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I.5.12.1.10 Engine shutdown.

The proper method of shutting down the engine shall be given, including a table showing the required oil dilution time for various temperatures, and the techniques and precautions to be observed in using oil dilution shall be covered. Operation of systems depending on engine oil (including, but not limited to, supercharger clutch and propeller governor), to ensure that these systems are supplied with diluted oil, shall be included. Complete instructions for purging normal fuel from the system and replacing with special fuel shall be included. Time, speed or other requirements for turbine temperature stabilization prior to shutdown shall be stated.

I.5.12.2 Cold weather operations work package <adverse_envwp>.

A brief discussion of the general problems involved in maintaining satisfactory operations in extreme cold shall be included. The relationship of proper engine shutdown to subsequent engine starting shall be emphasized, and operations under icing conditions shall be covered. Any special problems resulting from operations when snow is present shall be included. All applicable sections from I.5.12.1 through I.5.12.1.10 shall be included.

I.5.12.3 Desert and hot weather operations work package <adverse_envwp>.

A brief discussion of the general problems involved in maintaining satisfactory operations in desert and hot weather shall be included. The relationship of proper engine shutdown to subsequent engine starting shall be emphasized. Any special problems resulting from operations in desert conditions shall be included. All applicable sections from I.5.12.1 through I.5.12.1.10 shall be included.

I.5.12.4 Turbulence and thunderstorm operations work package <adverse_envwp>.

A discussion on the general qualities of the aircraft in turbulence and thunderstorms shall be included. A description of the techniques to be used shall be given and all preparations to be made before entering turbulence or thunderstorms shall be included. All applicable sections from I.5.12.1 through I.5.12.1.10 shall be included.

I.5.12.5 Rain operations work package <adverse_envwp>.

General coverage of the problem of rain during each phase of flight, including before takeoff, takeoff, climb, and cruise, shall be included. Performance of the rain removal system shall be described. All applicable sections from I.5.12.1 through I.5.12.1.10 shall be included.

I.5.12.6 General adverse environmental conditions work package <adverse_envwp>.

When specified by the acquiring activity, additional adverse environmental conditions work packages may be developed to provide operational guidance and procedures for specific conditions that a particular aviation platform may encounter. These include, but are not limited to, sea water and volcanic activity. All applicable sections from I.5.12.1 through I.5.12.1.10 shall be included for each work package, and each adverse condition shall be a separate work package.

I.5.13 General normal procedures work packages <gennormwp>.

When specified by the acquiring activity, additional work packages may be developed when the specific type of normal procedure is not covered as described in I.1 through I.5. Each additional normal procedure shall be a separate work package and shall comply with all of the requirements of I.1 through I.5. If additional normal procedure work packages are used, the proponent shall submit to AMCOM the requirements for this normal procedure work package type for possible incorporation within future revisions to this standard.

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I.6 Notes.

The notes in Section 6 apply to this appendix.

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APPENDIX J

EMERGENCY PROCEDURES (MODULE 9)

J.1 Scope.

This appendix establishes the technical content requirements for the preparation of emergency procedure instructions for aviation weapon systems. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

J.2 Applicable publications.

The applicable publications in Section 2 apply to this appendix.

J.3 Definitions.

The definitions in Section 3 apply to this appendix.

J.4 General requirements.

Procedures to be followed in dealing with emergencies that could reasonably be encountered shall be provided. Minor malfunctions that do not adversely affect the continued safe operation of the aircraft and compound or multiple failure emergency procedures shall not be included.

J.4.1 Preparation of digital data for electronic delivery.

Data prepared and delivered digitally in accordance with this standard shall be XML tagged using the current Army DTD. Data shall be formatted for presentation using the Army stylesheets. Refer to J.4.2 for information on obtaining or accessing the Army DTD and stylesheets. XML tags used in the DTD are noted throughout the text of this standard in bracketed, bold characters (e.g., **<geninfo>**) as a convenience for the author and to ensure that the tags are used correctly when developing a document instance.

J.4.2 Use of the Document Type Definition (DTD)/stylesheet.

The DTD referenced in this appendix interprets the technical content and structure for the functional requirements contained in this standard. Development of the TMs shall be accomplished through the use of this standard, the Army DTD, and the Army stylesheets.

J.4.3 Content structure.

The examples provided herein are an accurate representation of the content structure requirements contained in this appendix and shall be followed to permit the effective use of the DTD.

J.4.4 Style and format.

This standard provides style and format requirements for the technical content requirements described in this appendix. These requirements are considered mandatory and are intended for compliance.

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J.4.5 Work package development.

Data developed in accordance with this appendix shall be divided into work packages. These work packages shall be stand alone and shall contain all information and references required to support the work package type.

J.4.6 Selective application and tailoring of content using Appendix A matrices.

This standard contains some requirements that may not be applicable to the preparation of all TMs. Selective application and tailoring of requirements contained in this standard are the responsibility of the acquiring activity and shall be accomplished using APPENDIX A. The applicability of some requirements is also designated by one of the following statements: unless specified otherwise by the acquiring activity; as specified by the acquiring activity; or when specified by the acquiring activity.

J.5 Detailed requirements.

Emergency procedure instructions shall be prepared and subdivided into individual work packages that provide the operator of the weapon system/equipment with procedures to be followed in dealing with emergencies that could reasonably be encountered.

J.5.1 Preparation of emergency procedure work packages.

Emergency procedures shall be developed into specific work package types containing an introduction/general information section **<geninfo>** and applicable emergency tasks. All precautions to be observed in coping with an emergency shall be included. Emergency procedure data shall be developed in emergency task **<emertsk>** or **<fwuas_emertsk>**, order format as detailed in J.5.1.1 through J.5.1.4.

J.5.1.1 Emergency procedure work packages.

Emergencies shall be divided into the following work package categories as applicable:

a. Emergency Introduction Work Package.

b. Engine Work Package.

c. Propeller/Rotor Work Package.

d. Fire Work Package.

e. Fuel Work Package.

f. Electrical Work Package.

g. Hydraulic Work Package.

h. Landing and Ditching Work Package (or for UAS: Takeoff, Landing and Ditching).

i. Flight Controls Work Package.

j. Bailout/Ejection Work Package (when applicable).

k. Mission Equipment Work Package.

l. Ground Control Support Equipment Malfunctions Work Package.

m. Other Emergency Procedures Work Package.

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J.5.1.2 Utility systems emergency procedures.

Emergency procedures in connection with the utility systems shall be described in Module 2 in the Utility Systems work package. Emergency operation of utility systems shall be included only if it affects safety of flight.

J.5.1.3 Corrective actions.

Within an emergency classification, emergencies that have identical corrective actions may be combined under one paragraph heading. Those checks that shall be performed immediately in an emergency procedure shall be underlined, and a statement that such underlined steps shall be performed immediately without reference to the checklist shall be included.

J.5.1.4 Emergency work packages general format.

Emergency procedures shall be developed into specific work package types containing an introduction/general information section **<geninfo>** and applicable emergency tasks.

a. General Information.

b. Emergency Tasks.

J.5.1.4.1 General information <geninfo>.

Descriptions and characteristics of the emergency situation shall be included.

J.5.1.4.2 Emergency tasks <emertsk> or <fwuas_emertsk>.

Emergency tasks shall be developed as detailed in J.5.1.4.2.1 through J.5.1.4.2.11.

J.5.1.4.2.1 Rotary wing emergency task order.

Rotary Wing aircraft shall develop all emergency task content in the order listed below:

a. Title **<title>**.

b. Indications **<indication>**.

c. Action Response **<immediate_action>** or **<action>**.

d. Urgency To Land **<urgencyland>**.

e. Warnings **<warning>**.

f. Cautions **<caution>**.

g. Subsequent Actions **<subaction>**.

h. Considerations **<consideration>**.

i. Notes **<note>**.

J.5.1.4.2.2 Fixed wing/UAS emergency task order.

Fixed Wing and UAS aircraft shall develop all emergency task content in the order listed below:

a. Title **<title>**.

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b. Indications **<indication>**.

c. Action Response **<av_proc>**.

J.5.1.4.2.3 Title <title>.

The title of the emergency procedure shall allow for easy identification by target audience personnel. Emergency procedure titles shall be based on how the pilot recognizes the emergency rather than what caused the emergency (for example, “Low RPM” not “Governor Control failure”). Complete coverage shall be required regarding the recognition, feel, characteristics, and reaction of the aircraft during various emergencies affecting flight.

J.5.1.4.2.4 Indications <indication>.

All indications that aid in successful identification of the emergency situation shall be included. This includes illuminated messages, voice warning systems, proprioceptive, audio, or visual cues associated with malfunctions; ie. smoke, fumes, MFD or MPD indicators, unusual vibrations, etc. See FIGURE J-1 for example of displayed indications.

a. Indications shall be presented to match the aircraft indications, to include color and shape of the indication.

b. Colors shall be discernible in all flight environments.

c. In-line graphics may be used.

d. Audible indications shall be represented by the term “audible,” use of an audible symbol, or appear with quotes.

e. Condition statements **<condition>**, for specific indications may be used.

f. Indications shall still be identified even if repeated from the title.



FIGURE J-1. Displayed indications example.

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J.5.1.4.2.5 RW action response <immediate_action> or <action>.

Every emergency procedure shall be developed with immediate actions or actions. The only exception is if the urgency to land is the only action for the procedure. Immediate action steps and actions steps shall not be mixed within the same action response section.

a. Immediate Actions **<immediate_action>**. Procedures shall be developed in the immediate action response when the urgency of the malfunction requires an immediate response and time does not permit the utilization of the emergency procedures section of the operator's checklist manual in order to ensure safe flight is not jeopardized.

(1) All immediate action steps shall be underlined.

(2) Steps shall be developed per 4.5.2.10.2.

(3) A determining factor is maintaining control of the aircraft and time does not permit use of the checklist.

(4) When required, condition statements **<condition>** for immediate actions may be used. Examples of qualifying statements are: “If control restored”, “If single engine capable”, “If required”, etc.

(5) Immediate actions are complete when the immediate danger to safe flight is over; the crew can then verify responses through use of the checklist.

b. Actions **<action>**. Procedures shall be developed in the action response when the urgency of the malfunction allows for time to reference the emergency procedures section of the operator's checklist manual.

(1) Action steps shall not be underlined.

(2) Steps shall be developed per 4.5.2.10.2.

(3) A determining factor is the aircraft remains under control and time permits the use of the emergency procedures section of the operator's checklist manual.

(4) When required, condition statements **<condition>** for actions may be used. Examples of qualifying statements are: “If control restored”, “If trouble remains”, etc.

J.5.1.4.2.6 FW and UAS action response <av_proc>.

Every emergency procedure shall be developed with immediate action steps or action steps. Warnings, cautions and notes should appear at the corresponding task or step level. Immediate action steps shall be developed when the urgency of the malfunction requires an immediate response and time does not permit the utilization of the emergency procedures section of the operator's checklist manual in order to ensure safe flight is not jeopardized. Action steps shall be developed when the urgency of the malfunction allows for time to reference the emergency procedures section of the operator's checklist manual.

a. All immediate action steps shall be underlined. A determining factor is maintaining control of the aircraft and time does not permit use of the checklist.

b. Action steps shall not be underlined.

c. Urgency to land shall be included as applicable.

d. Steps shall be developed per 4.5.2.10.2.

e. When required, condition statements **<condition>** for any actions may be used. Examples of qualifying statements are: “If control restored”, “If single engine capable”, “If required”, etc.

f. Immediate actions are complete when the immediate danger to safe flight is over; the crew can then verify responses through use of the checklist.

g. Warnings, cautions, and notes shall be placed prior to the corresponding task or step.

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J.5.1.4.2.7 RW urgency to land <urgencyland>.

The procedure's urgency to land shall appear immediately after the action response. This is determined by the nature of the failure and the prevailing situational conditions. Refer to TABLE J-I for options that shall be used in determining the urgency to land.

a. Condition statements **<condition>** for the urgency to land may be used and shall precede the landing option.

b. When conditions exist that change the urgency to land, a condition statement **<condition>** shall precede each landing option.

c. Urgency to land may be underlined as determined by the acquiring activity.

J.5.1.4.2.8 RW warnings <warning>.

Warnings pertinent to the emergency procedure may be included in the checklist emergency procedure as determined by the acquiring activity.

J.5.1.4.2.9 RW cautions <caution>.

Cautions pertinent to the emergency procedure shall be included in the checklist emergency procedure as determined by the acquiring activity.

J.5.1.4.2.10 RW subsequent actions <subaction>.

Subsequent actions may be developed only if the emergency procedure previously required immediate actions.

a. Subsequent action steps shall not be underlined.

b. A determining factor is the aircraft is under control, danger to aircraft and aircrew has passed, and time permits the use of the checklist to resolve the malfunction and danger.

c. When required, condition statements **<condition>** for subsequent actions may be used. Examples of qualifying statements are: "If control restored", "If trouble remains", etc.

J.5.1.4.2.11 RW considerations <consideration>.

Considerations are nonessential, optional procedural steps that assist in assessing longer term implications of an emergency. Considerations shall be presented as actionable steps.

J.5.1.4.2.12 RW notes <note>.

Notes pertinent to the emergency procedure shall be included in the checklist emergency procedure as determined by the acquiring activity. Notes from the applicable Operator's manual shall be used.

J.5.2 Aircraft systems <emergency_introwp>.

Emergency procedures to be performed in the event of an aircraft system malfunction under various conditions shall be provided. The work package shall also meet all requirements of J.5.1.

a. The following statement shall be included verbatim:

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"Emergency operation of mission equipment is provided insofar as its use affects safety of flight. Emergency procedures are presented in checklist form when applicable. A condensed version of these procedures is contained in the condensed checklist *(enter TM number of applicable checklist)*."

b. The following note shall be included verbatim:

"Note

The urgency of certain emergencies requires immediate and instinctive action by the pilot. The most important single consideration is aircraft control. All procedures are subordinate to this requirement."

c. A statement similar to the one provided shall also be included:

"Terms may be defined as necessary to simplify the procedural memory steps within the existing emergency procedures. Each term shall be used as an emergency procedure step instead of listing the individual steps used to define the term. For example, the term "EMER ENG SHUTDOWN" is defined as engine stoppage without delay and is accomplished as follows:

1. Throttle - off.
2. **FUEL** switches - **OFF**
3. **BAT** switch - **OFF**."

d. A table similar to TABLE J-I containing the term and definition shall be included tailoring the included definitions to the aircraft type. Refer to for the appropriate definition to use.

TABLE J-I. Urgency to land definitions.

Term	Platform	Definition
Land without Delay	RW	A landing in which the primary consideration is continued control of the aircraft and survival of the occupants. It is meant to be more urgent than Land As Soon As Possible. The situation may not permit the crew to continue to maneuver the aircraft to a suitable landing area (e.g., open field). If maneuvering to an open area is not practical, then the crew shall make the decision to land in an area that will have the least amount of negative impact on crew survivability. (e.g. Over dense forest, select an area with the smallest trees, in a mountainous area, choose an area with the least amount of slope). This term shall be underlined.
Land as Soon as Possible.	RW, FW	A landing at the nearest suitable landing area (e.g. open field). The primary consideration is to ensure the survival of occupants. This term may be underlined as determined by the acquiring activity.
	UAS	Landing at a surveyed launch and recovery site without delay. If landing at a surveyed launch and recovery site is not feasible, the crew shall make a decision to land in an area that will have the least amount of negative impact to the aircraft and personnel on the ground.
Land as Soon as Practicable	RW, FW	A landing at a suitable landing area. The primary consideration is the urgency of the emergency.

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TABLE J-I. Urgency to land definitions. (continued)

Term	Platform	Definition
	UAS	Landing at a surveyed launch and recovery site. The primary consideration is the urgency of the emergency versus the mission priority.

e. The following statement shall also be included for all rotary wing aircraft:

"AUTOROTATE (always underlined) is defined as adjusting the flight controls as necessary to establish an autorotational descent and landing."

J.5.2.1 Emergency equipment and exits.

The following emergency equipment and exits shall be illustrated:

a. The aircraft/ground station interior shall be illustrated showing Aviation Life Support Equipment (ALSE) permanently installed in the aircraft including, but not limited to, fire axes, flares, pyrotechnic pistols, and hand fire extinguishers (FIGURE J-2 and FIGURE J-3).

b. If the aircraft is large enough to permit movement of personnel, emergency stations and routes of egress to be followed in flight and after crash landing on land or water shall be indicated for all personnel. Coding shall be used to differentiate between routes and exits to be used in flight and those to be used after a crash landing (FIGURE J-2). This illustration shall be an interior view or as viewed by the occupants of the aircraft. It may be combined with the emergency equipment diagram and the emergency entrance diagram, unless the resulting illustration would be confusing.

c. A diagram shall be included to show points at which emergency personnel can enter into the aircraft after it has crash landed. This illustration may be combined with the routes of escape and exits diagram, unless the resulting illustration would be confusing.

d. For ground stations, an illustration detailing emergency stations and routes of ingress and egress to be followed in an emergency shall be indicated for all personnel. This may be presented in separate illustrations if combining the information into one illustration would be confusing.

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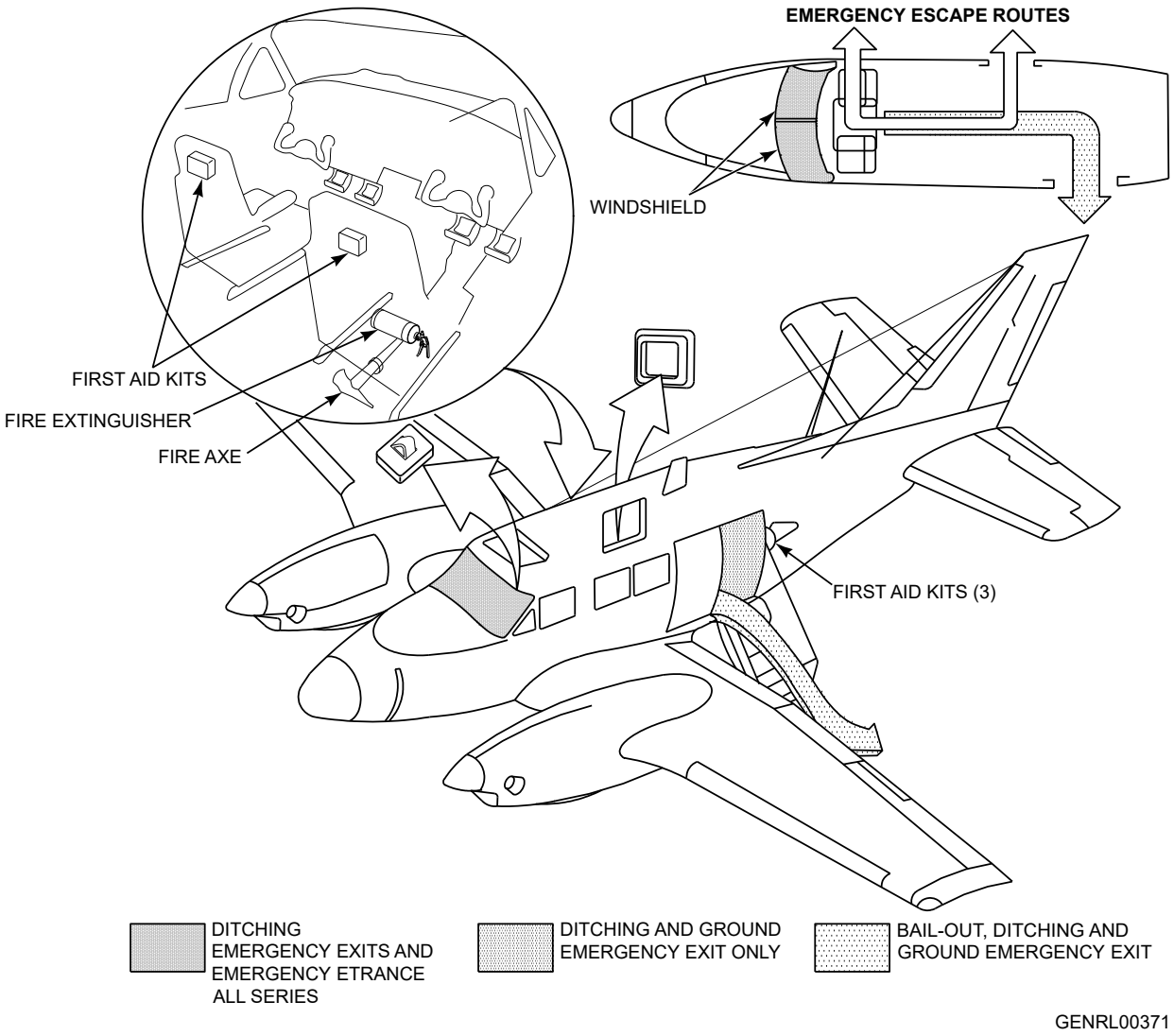


FIGURE J-2. Example emergency equipment and emergency exits 1.

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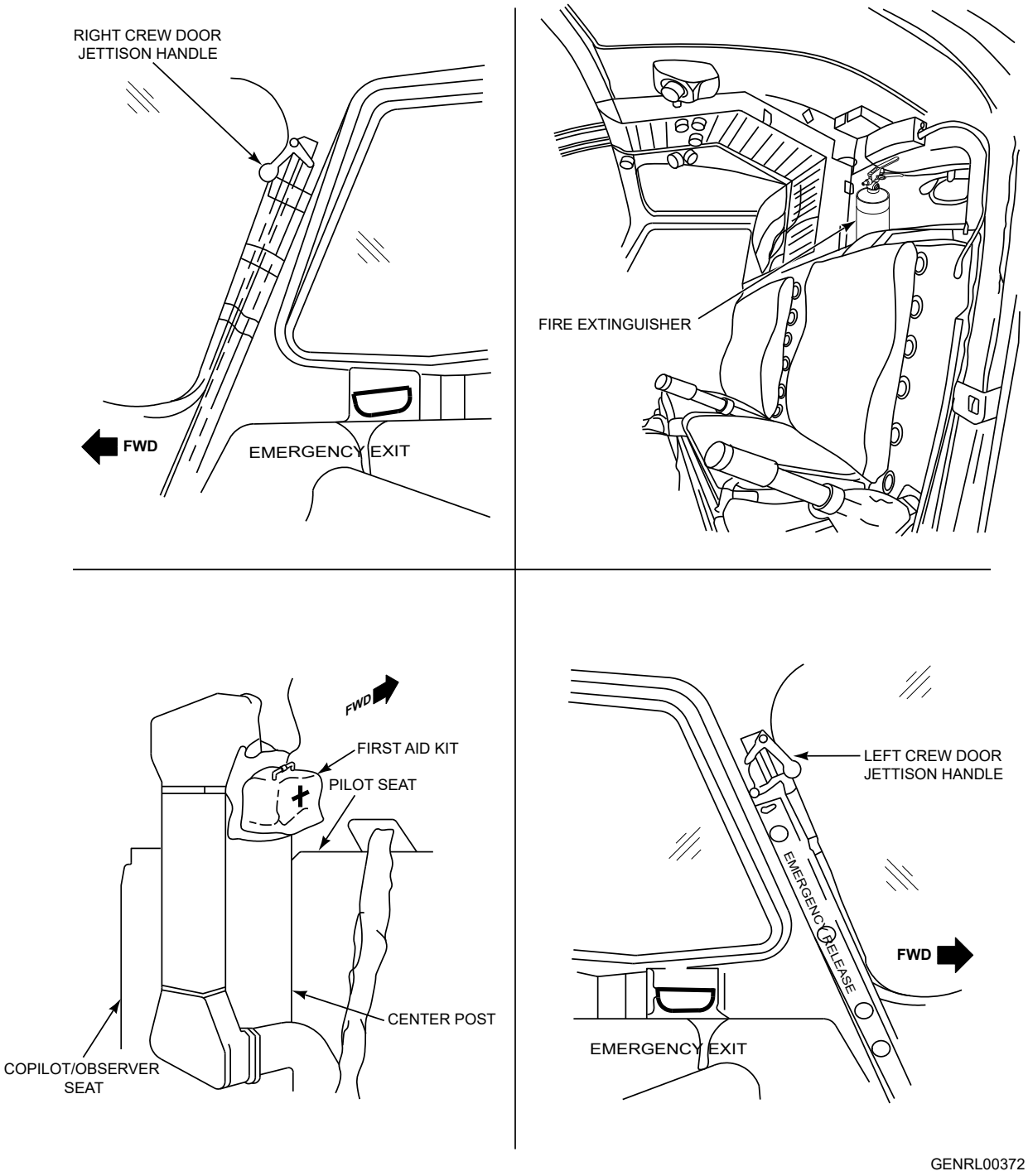


FIGURE J-3. Example emergency equipment and emergency exits 2.

J.5.3 Engine work package <emenginewp>.

Emergency procedures shall be described in the event of an engine malfunction under a variety of conditions. The work package shall also meet all requirements of J.5.1.

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J.5.3.1 Flight characteristics under partial power conditions.

A description of the characteristics and reactions of the aircraft when flying with one or more inoperative engines or with an engine having only partial power capability shall be included. Emphasis shall be given to any special precautions that shall be observed and any dangerous tendencies of the aircraft. Information shall be included on how to determine which engine is inoperative. The problems of maintaining altitude, directional control, and any other special considerations shall be discussed.

J.5.3.2 Engine malfunction under specific conditions.

Additional paragraphs shall be included as necessary to indicate action to be taken in the event of engine malfunction under various conditions. Partial engine malfunctions as well as complete engine failure shall be described. A complete checklist procedure to be followed in shutting down the malfunctioning engine and establishing continued flight shall be included. Insofar as possible, shutdown procedure shall be identical to that required in the event of engine fire. Recommended best techniques and procedures for crash landing while operating within avoidance areas shall be discussed.

J.5.3.3 Engine malfunction during takeoff and low altitude/low airspeed flight.

This shall include an abort during the takeoff run, immediately after liftoff and continued flight. Coverage shall be included for both complete engine failure and partial loss of power. For rotary wing aircraft, differentiation between engine malfunction while at a hover and engine malfunction after takeoff (in translational lift) shall be included. Information shall be included, but not limited to, jettisoning external stores, landing gear retraction, pilot techniques, and best airspeed for minimum power required (partial loss of power).

J.5.3.4 Engine malfunction during cruise.

Reference shall be made to the performance chart data in Module 7 covering cruise control with one or more engines inoperative. The effect of loss of each engine on the various aircraft systems and equipment shall be included. Procedures to be followed in the event of partial power loss as well as for the complete engine failure shall be included.

J.5.3.5 Engine malfunction during final approach.

For multi-engine aircraft, procedures shall be provided for loss of one engine while on final approach in the landing configuration. Information shall be included concerning application of maximum controllable power, jettisoning external stores if applicable, landing gear position, use of flaps, pilot techniques, and airspeed requirements.

J.5.3.6 Engine restart during flight.

Instructions for proper means for restarting an engine in flight and resuming normal flight shall be presented. Special emphasis shall be placed on parameters such as altitude, airspeed, and rpm. If considered advantageous, they may be presented in chart form. A warning shall be included that the engine should not be restarted unless it can be determined that it would be reasonably safe to do so.

J.5.3.7 Maximum glide (FW only).

Glide requirements that shall result in maximum range with no power available shall be provided. This information is required for all single engine and twin engine aircraft. A graph showing glide distance attainable from the service ceiling to sea level shall be included (FIGURE J-4).

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MAXIMUM GLIDE DISTANCE

POWER OFF (PROPELLERS FEATHERED)
GEAR AND FLAPS UP ZERO WIND

WEIGHT - LBS.	BEST GLIDE SPEED KIAS
XXXX	XXX
XXXX	XXX
XXXX	XXX
XXXX	XX
XXXX	XX

EXAMPLE

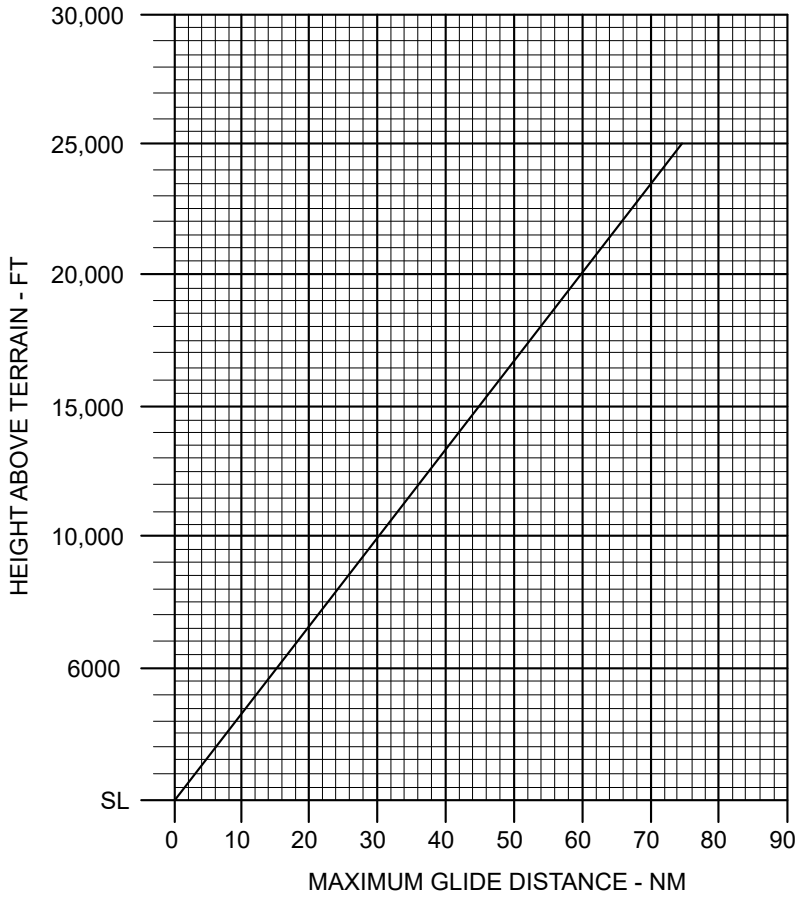
WANTED
MAXIMUM GLIDE DISTANCE

KNOWN

HEIGHT ABOVE TERRAIN = 7400 FT

METHOD

ENTER HEIGHT ABOVE TERRAIN
MOVE RIGHT TO MAXIMUM GLIDE DISTANCE LINE
MOVE DOWN, READ MAXIMUM GLIDE DISTANCE = XXNM



DATA BASIS: ESTIMATED

GENERL00358

FIGURE J-4. Maximum glide distance (FW only).

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J.5.3.8 Autorotational descent (RW only).

A chart that presents autorotational rate of descent versus indicated airspeed at normal rotor speed shall be provided (FIGURE J-5). The indicated airspeeds for minimum rate of descent and maximum glide distance shall be shown on the chart. Data and/or instructions for determining autorotational descent information for variations in aircraft configurations shall also be provided.

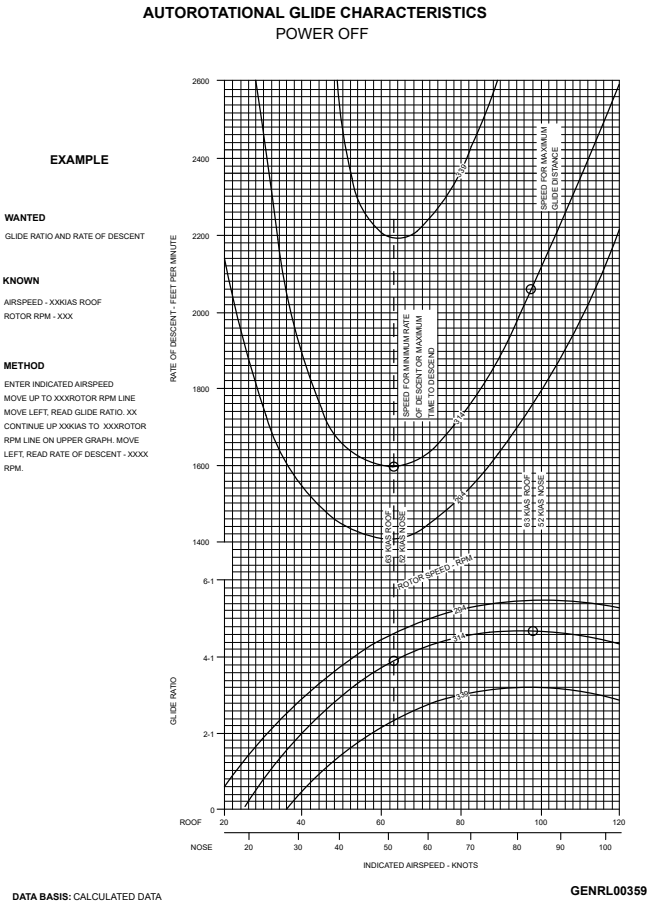


FIGURE J-5. Autorotational descent.

J.5.3.9 Landing with one or more engines inoperative.

The recommended procedure shall be described, including important precautions. A brief discussion of any changes that include, but are not limited to, the use of landing gear, wing flaps, and slats during such landing shall be included. For single engine and twin engine aircraft, proper landing procedures with no power shall be emphasized. For rotary wing aircraft, reference shall be made to the height velocity diagram.

J.5.3.10 Go-around with one or more engines inoperative (FW only).

Recommended procedures shall be described, including important precautions.

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J.5.3.11 Height velocity.

The minimum height for safe landing following loss of power for both single and multi-engine helicopters shall be provided. Plots of height required for safe auto-rotational landing after loss of power and initial engine failure shall be included as applicable. For a multi-engine helicopter, a recommended approach corridor with the critical engines inoperative shall be shown on the plot. Regions of caution, avoidance, and safe operation shall be shown (FIGURE J-6). The plots shall be based on initiation of the necessary manual collective pitch control motion after at least a two-second delay following loss of power, or as approved by the acquiring activity.

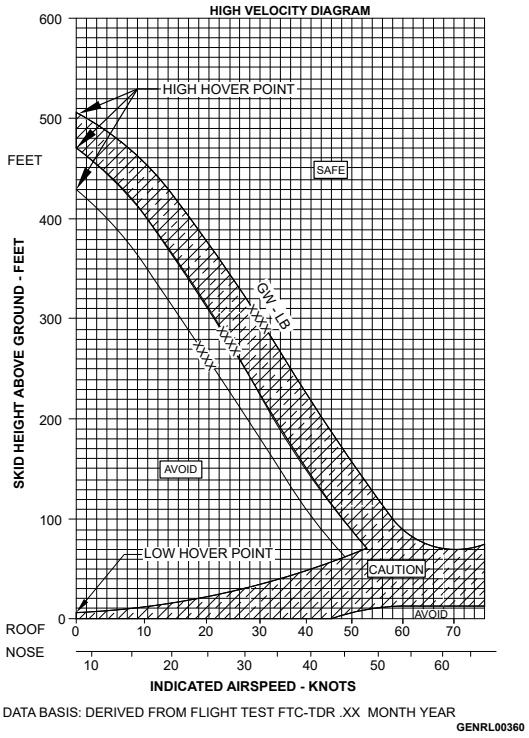


FIGURE J-6. Example high velocity diagram.

J.5.4 Propeller/rotor, transmissions, and drive systems work package <emproprotwp>.

Emergency procedures shall be described in the event of propeller/rotor, transmission, or drive system failure. The work package shall also meet all requirements of J.5.1.

J.5.4.1 Propeller failure (FW only).

Instructions shall be given regarding recommended procedures in the event of a runaway propeller and other types of propeller failure. Instructions shall be included regarding action to be taken if propeller does not feather properly.

J.5.4.2 Tail rotor failure and directional control malfunctions.

Instructions shall be given regarding all modes of directional control malfunctions and tail rotor failures. Coverage shall include emergency procedures to be used in the event of failures during takeoff, hovering, in flight, and while landing. Instructions for maintaining powered flight as opposed to autorotation shall be included.

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J.5.4.3 Malfunctions of main rotor transmission and drive systems.

Differentiation between malfunctions with the drive system between the engine and transmission, and malfunctions of the drive system between the transmission and main rotor shall be included. Actual and erroneous instrument/ warning light indications shall be discussed, including procedures for specific malfunctions.

J.5.4.4 Other emergencies.

Other emergencies such as ground resonance and mast bumping shall be described. Restrictions and preventive actions shall be described.

J.5.5 Fire work package <emfirewp>.

Emergency procedures shall be included for fires as directed in the following paragraphs. The work package shall also meet all requirements of J.5.1.

J.5.5.1 Engine fire.

Instructions shall be included regarding the recommended method of dealing with engine fires on the ground and during flight. Insofar as possible, engine shutdown procedures shall be identical to those used during engine failure.

J.5.5.2 Fuselage fire.

Instructions shall be included regarding procedures to be followed when a fuselage fire breaks out. Warnings regarding dangers involved in using fire extinguishing agents shall be included.

J.5.5.3 Wing fire.

Instructions shall be included on means of dealing with wing fires, including shutting down systems which may be feeding the fire.

J.5.5.4 Electrical fire.

Instructions for dealing with an electrical fire shall be included. If certain aircraft fire extinguishers are not to be used for electrical fires, that information shall be included.

J.5.5.5 Smoke and fume elimination.

Instructions shall be given for most rapid means of dissipating smoke and toxic fumes.

J.5.6 Fuel system work package <emfuelwp>.

Procedures shall be given for dealing with fuel system failures and shall include a description of metering system failures, fuel pump failures, and control linkage failures (loss of fuel control with fuel input in a fixed position). Emergency procedures shall be included for each condition. The work package shall also meet all requirements of J.5.1.

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J.5.7 Electrical system work package <emelectricwp>.

Instructions shall be given for methods of dealing with electrical system failures. Procedures shall be expressed as actions to be taken involving circuit breakers. For push-pull types, procedures shall indicate in or out. Where the circuit breakers are a switch type, procedures shall indicate off or on. The work package shall also meet all requirements of J.5.1.

J.5.8 Hydraulic system work package <emhydraulicwp>.

Instructions shall be given for dealing with hydraulic system component failures. The work package shall also meet all requirements of J.5.1.

J.5.9 Landing and ditching work package <emlandwp>.

Instructions shall be given regarding landing and ditching emergency procedures as described in the following paragraphs as applicable. UAS shall title the WP as Takeoff, Landing, and Ditching Work Package if takeoff emergency procedures are included. The work package shall also meet all requirements of J.5.1.

J.5.9.1 Emergency descent.

The means of accomplishing an emergency descent shall be provided. Emergency descent is a maximum effort in which damage to the aircraft or power plant is considered secondary to getting the aircraft on the ground.

J.5.9.2 Landing emergencies.

Preparation, warning signals to crew, approach, crew/passenger positions, harness locks, landing technique, routes, and methods of crew exits shall be included for both hard and soft ground. Landings with one or more landing gears retracted, flat tires, no wing flaps, and landing on unprepared runways shall also be covered. Information regarding pilot techniques for forced landings in trees or wooded areas shall also be included as applicable. UAS shall title the WP as Takeoff, Landing, and Ditching Work Package if takeoff emergency procedures are included.

J.5.9.3 Body positions.

The body positions to be used by passengers and crew in emergency landings shall be illustrated.

J.5.9.4 Ditching.

Instructions shall be included for ditching the aircraft. The ditching capabilities of the aircraft and the advantages of ditching versus bailout shall be included. The following shall be described: preparation; warning signals to crew; approach; crew/passenger positions; ditching equipment, such as ditching belts and bulkheads; landing techniques; duties of each crewmember immediately after ditching; and methods of crew exits. As applicable, an illustration shall be included showing the position of each crewmember during ditching and crash landing.

J.5.9.5 Takeoff (UAS only).

Instructions shall be given for emergencies that occur during takeoff for UAS.

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J.5.10 Flight controls work package <emfligcontwp>.

Procedures to be employed in event of flight control failure shall be provided. The work package shall also meet all requirements of J.5.1.

J.5.11 Bailout/eject work package <embailejectwp>.

For all aircraft with established crew bailout or ejection procedures, the techniques, precautions, and warning signals for leaving the aircraft in flight shall be described, including instructions for separation from the seat. Bailout procedures to be used when seat ejection fails shall be included. The proper method of preparing the aircraft for bailout and method of jettisoning pilot's compartment enclosures and doors shall be described. A pictorial sequence of operations for ejection shall be provided, including alternate methods of removing safety pins where applicable. The work package shall also meet all requirements of J.5.1.

J.5.12 Mission wquipment work package <emmissequipwp>.

Emergency procedures shall be outlined for malfunctioning mission equipment that constitutes a safety hazard. The work package shall also meet all requirements of J.5.1.

J.5.12.1 Emergency jettisoning.

All means of accomplishing emergency jettisoning of fuel, cargo, and equipment shall be covered. Appropriate cautions relative to possible damage, (sudden shifting of CG, etc). shall be included.

J.5.13 Ground control malfunctions work package <emshelterwp>.

Emergency procedures shall be outlined for malfunctioning UAS Ground Control equipment that constitutes a safety hazard as described in the following paragraphs. The work package shall also meet all requirements of J.5.1.

J.5.13.1 Ground control station malfunctions.

Procedures to be employed in event of ground control equipment failure shall be provided.

J.5.13.2 Data link malfunctions.

Procedures to be employed in event of primary or secondary data link failure shall be provided.

J.5.13.3 Ground control support equipment malfunctions.

Procedures to be employed in event of UAS system support equipment failure that constitutes a safety hazard during flight/system operation shall be provided (Launcher/RATO/Arresting gear failure, automated landing systems failure, etc.).

J.5.14 Other emergency procedures <emgenwp>.

When specified by the acquiring activity, additional work packages may be developed when the specific type of emergency procedure is not covered as described in J.5.2 through J.5.13.3. Each additional emergency procedure shall be a separate work package and shall comply with all of the requirements of J.5.1 through J.5.1.4.2.12. If

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additional emergency procedure work packages are used, the proponent shall submit to AMCOM the requirements for this emergency procedure work package type for possible incorporation within future revisions to this standard.

J.6 Notes.

The notes in Section 6 apply to this appendix.

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APPENDIX K

SUPPORTING INFORMATION (MODULE 10)

K.1 Scope.

This appendix establishes the technical content requirements for the preparation of supporting information for aviation weapon systems. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

K.2 Applicable publications.

The applicable publications in Section 2 apply to this appendix.

K.3 Definitions.

The definitions in Section 3 apply to this appendix.

K.4 General requirements.

A complete list all publications referenced in the TM that are required by the user to operate the equipment shall be provided. Any additional supporting information required by the user to operate the equipment shall be included as determined by the acquiring activity.

K.4.1 Preparation of digital data for electronic delivery.

Data prepared and delivered digitally in accordance with this standard shall be XML tagged using the current Army DTD. Data shall be formatted for presentation using the Army stylesheets. Refer to K.4.2 for information on obtaining or accessing the Army DTD and stylesheets. XML tags used in the DTD are noted throughout the text of this standard in bracketed, bold characters (e.g., **<geninfo>**) as a convenience for the author and to ensure that the tags are used correctly when developing a document instance.

K.4.2 Use of the Document Type Definition (DTD)/stylesheet.

The DTD referenced in this appendix interprets the technical content and structure for the functional requirements contained in this standard. Development of the TMs shall be accomplished through the use of this standard, the Army DTD, and the Army stylesheets.

K.4.3 Content structure.

The examples provided herein are an accurate representation of the content structure requirements contained in this appendix and shall be followed to permit the effective use of the DTD.

K.4.4 Style and format.

This standard provides style and format requirements for the technical content requirements described in this appendix. These requirements are considered mandatory and are intended for compliance.

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K.4.5 Work package development.

Data developed in accordance with this appendix shall be divided into work packages. These work packages shall be stand alone and shall contain all information and references required to support the work package type.

K.4.6 Selective application and tailoring of content using Appendix A matrices.

This standard contains some requirements that may not be applicable to the preparation of all TMs. Selective application and tailoring of requirements contained in this standard are the responsibility of the acquiring activity and shall be accomplished using APPENDIX A. The applicability of some requirements is also designated by one of the following statements: unless specified otherwise by the acquiring activity; as specified by the acquiring activity; or when specified by the acquiring activity.

K.5 Detailed requirements.

A complete list all publications referenced in the TM that are required by the user to operate the equipment shall be provided. Any additional supporting information required by the user to operate the equipment shall be included as determined by the acquiring activity.

K.5.1 Preparation of supporting information.

Supporting information shall be developed as work packages. Supporting information work packages are described in MIL-STD-40051 unless otherwise stated in this standard. Supporting information work packages shall be contained in a module titled “Supporting Information” within the TM. These work packages shall be placed in the ETM or IETM in the order in which they are presented herein, as applicable.

K.5.2 Reference work package <refwp>.

Refer to MIL-STD-40051 for preparation of the reference work package.

K.5.3 Cockpit indications index work package <cockpitwp>.

As required by the acquiring activity, a table of all cockpit indications, warnings, cautions, advisories, and their explanations may be provided.

K.5.4 General supporting information work packages <genwp>.

When specified by the acquiring the activity, additional work packages shall be prepared when the work packages previously described herein do not support the data/information to be presented. Refer to MIL-STD-40051.

K.6 Notes.

The notes in Section 6 apply to this appendix.

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APPENDIX L

CONTENT REQUIREMENTS FOR OPERATOR'S AND CREWMEMBER'S CHECKLISTS (-CL)

L.1 Scope.

This appendix establishes the technical content requirements for the preparation of the operator and crewmember checklists for aviation weapon systems. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

L.2 Applicable publications.

The applicable publications in Section 2 apply to this appendix.

L.3 Definitions.

The definitions in Section 3 apply to this appendix.

L.4 General requirements.

A supplemental manual containing a simplified version of the normal procedures and emergency procedures shall be developed in conjunction with the -10 operator's manual to serve as the operator's and crewmember's checklist.

L.4.1 Preparation of digital data for electronic delivery.

Data prepared and delivered digitally in accordance with this standard shall be XML tagged using the current Army DTD. Data shall be formatted for presentation using the Army stylesheets. Refer to L.4.2 for information on obtaining or accessing the Army DTD and stylesheets. XML tags used in the DTD are noted throughout the text of this standard in bracketed, bold characters (e.g., **<geninfo>**) as a convenience for the author and to ensure that the tags are used correctly when developing a document instance.

L.4.2 Use of the Document Type Definition (DTD)/stylesheet.

The DTD referenced in this appendix interprets the technical content and structure for the functional requirements contained in this standard. Development of the TMs shall be accomplished through the use of this standard, the Army DTD, and the Army stylesheets.

L.4.3 Content structure.

The examples provided herein are an accurate representation of the content structure requirements contained in this appendix and shall be followed to permit the effective use of the DTD.

L.4.4 Style and format.

This standard provides style and format requirements for the technical content requirements described in this appendix. These requirements are considered mandatory and are intended for compliance.

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L.4.5 Work package development.

Data developed in accordance with this appendix shall be divided into work packages. These work packages shall be stand alone and shall contain all information and references required to support the work package type.

L.4.6 Selective application and tailoring of content using Appendix A matrices.

This standard contains some requirements that may not be applicable to the preparation of all TMs. Selective application and tailoring of requirements contained in this standard are the responsibility of the acquiring activity and shall be accomplished using APPENDIX A. The applicability of some requirements is also designated by one of the following statements: unless specified otherwise by the acquiring activity; as specified by the acquiring activity; or when specified by the acquiring activity.

L.5 Detailed requirements.

A supplemental manual containing a simplified version of the normal procedures and emergency procedures shall be developed in conjunction with the -10 Operator's Manual to serve as the operator's and crewmember's checklist. Performance data and limitations information shall also be included. The checklist shall be developed as detailed in L.5.1 through L.5.5.4.6.

L.5.1 Rotary wing checklist content requirements.

For Rotary Wing (RW) aircraft, the Checklist shall be developed into two distinct volumes: Volume 1-Normal and Detailed Procedures and Volume 2-Emergency Procedures and Limitations.

L.5.1.1 Rotary wing CL volume 1: normal and detailed procedures.

The Rotary Wing CL Volume 1 shall consist of the following content:

a. Front Matter.

b. Normal Procedures.

c. Detailed Procedures.

d. Charts.

e. Rear Matter.

L.5.1.2 Rotary wing CL volume 2: emergency procedures and limitations.

The Rotary Wing CL Volume 2 shall consist of the following content:

a. Front Matter.

b. Warning Section.

c. Caution Section.

d. Advisory Section.

e. Mission Equipment Section.

f. Limitations.

g. Rear Matter.

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L.5.2 Fixed wing and uncrewed systems checklist content requirements.

For Fixed Wing (FW) and Uncrewed Systems Aircraft (UAS), the Checklist shall be developed as follows:

a. Front Matter.

b. Normal Procedures.

c. Detailed Procedures.

d. Performance Charts.

e. Emergency Section.

f. Abnormal/Caution Section.

g. Advisory Section (optional).

h. Mission Equipment Section (optional).

i. Limitations.

j. Rear Matter.

L.5.3 Preparation of checklist (-CL) manuals.

All aviation checklist manuals will be developed as detailed in L.5.3.1 through L.5.3.7.

L.5.3.1 Front matter <av_paperfrnt>.

The Front Matter shall be developed in accordance with 5.2.1.

L.5.3.2 Normal procedures.

A condensed version of the normal procedures or crew duties' portion of the applicable operator's manual (-10) shall be developed. Normal procedures shall be developed into sections, work packages, and tasks.

L.5.3.2.1 Normal procedure sections <normsect>.

Normal sections shall be developed to divide procedural content. The acquiring activity shall determine how content is presented within the sections.

a. Each section shall begin with a tabbed page.

b. The tabbed page shall contain a section title and index of the tasks found within the section.

c. A title shall be displayed on the tab of the section divider; the title may be abbreviated to fit within the tab as determined by the acquiring activity.

d. When only a single normal section is used, it shall be titled "Normal Procedures" to include the short title appearing on the tab.

L.5.3.2.2 Normal procedure work packages <normalwp>.

Normal procedure content shall be developed within work packages. Work packages are intended to create natural breaks in content to make publication updates efficient. Due to the abbreviated nature of the checklist, and to facilitate efficient use of page space, the acquiring activity may determine the content within each work package. The following are considerations to assist in work package development:

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a. When an update is made to a work package, the entire work package shall be replaced.

b. Work package size for -CL manuals shall not exceed 10 pages or 5 sheets.

c. A work package shall consist of, at minimum, one page front and back, regardless of content to include blank pages.

d. A work package title may be used.

L.5.3.2.3 Normal procedure tasks <normaltsk>.

Normal procedure tasks shall be developed as found in the applicable Operator's manual, -10 in accordance with APPENDIX I.

a. When the star symbol "⋆" is used to indicate additional details, the step shall contain a linked reference **<xref>** to the appropriate detail procedure found within the checklist.

b. References **<xref>** shall also be made to applicable figures, tables, and charts found within the checklist.

c. A task may have more than one procedure. All procedures within the task shall be related. When more than one procedure is present, the procedures shall contain titles.

L.5.3.3 Detailed procedures.

Detailed procedures shall be developed as directed by the acquiring activity. Detailed procedures may be developed as expanded procedures from the normal procedures, "Star" steps, or procedures for specific mission functions. Detailed procedures shall be developed into sections, work packages, and tasks.

L.5.3.3.1 Detailed procedure section <detailsect>.

When required by the acquiring activity, one or more detailed sections shall be developed to divide procedural content. The acquiring activity shall determine how content is presented within the sections.

a. Each section shall begin with a tabbed page.

b. The tabbed page shall contain a section title and index of the tasks found within the section.

c. A title shall be displayed on the tab of the section divider; the title may be abbreviated to fit within the tab as determined by the acquiring activity.

d. When only a single detailed section is used, it shall be titled "Detailed Procedures" to include the short title appearing on the tab.

L.5.3.3.2 Detailed procedure work packages <detailwp>.

Detailed procedure content shall be developed within work packages. Work packages are intended to create natural breaks in content to make publication updates efficient. Due to the abbreviated nature of the checklist, and to facilitate efficient use of page space, the acquiring activity may determine the content within each work package. The following are considerations to assist in work package development:

a. When an update is made to a work package, the entire work package shall be replaced.

b. Work package size for -CL manuals shall not exceed 10 pages or 5 sheets.

c. A work package shall consist of, at minimum, one page front and back, regardless of content to include blank pages.

d. A work package title may be used.

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L.5.3.3.3 Detailed procedure task <detailtsk>.

Detailed procedure tasks shall be developed as found in the applicable Operator's manual, -10 in accordance with APPENDIX I.

a. When the detailed procedure is referenced from the normal procedures, it shall include a return reference **<return>** to the applicable normal procedure found within the checklist.

b. If more than one return reference is required, it shall be clearly identified, for example:

- PILOT: RETURN TO: Page N3-1, Step 5.
- COPLIOT: RETURN TO: Page N4-1, Step 6.

c. References **<xref>** shall also be made to applicable figures, tables, and charts found within the checklist.

d. A task may have more than one procedure. All procedures within the task shall be related. When more than one procedure is present, the procedures shall contain titles.

L.5.3.4 Performance data.

Performance data shall be included as directed by the acquiring activity. Performance data shall be derived from the applicable Operator's manual, -10 performance data module. Performance data shall be presented in sections and work packages. Charts, tables, and checklists used during preflight, takeoff, cruise, landing, and shutdown shall be included.

L.5.3.4.1 Performance data section <perfdatasect>.

When required by the acquiring activity, one or more performance sections shall be developed. The acquiring activity shall determine how content is presented within the sections.

a. Each section shall begin with a tabbed page.

b. The tabbed page shall contain a section title and index of the performance data charts found within the section.

c. A title shall be displayed on the tab of the section divider; the title may be abbreviated **<shorttitle>** to fit within the tab as determined by the acquiring activity.

d. When only a single performance data section is used, it shall be titled "Performance Data Charts" to include the short title appearing on the tab.

L.5.3.4.2 Performance data work packages <perfdatawp>.

Performance data shall be developed as found in the applicable Operator's manual, -10 in accordance with APPENDIX H.

a. Performance data work packages shall be titled as applicable to the data.

b. Each work package shall contain a single figure **<figure>** or table **<table>**.

c. A brief introduction may be developed.

d. Tables and charts within this WP shall display within the standard page size. If greater detail is desired, fold out pages shall be utilized in the rear matter.

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L.5.3.5 Emergency procedures.

Emergency procedure format shall be developed based off aircraft type. Rotary Wing aircraft shall follow the requirements described in L.5.4. Fixed Wing and UAS aircraft shall follow the requirements described in L.5.5. The content of the emergency procedures shall be developed as found in the applicable Operator's manual, -10 in accordance with APPENDIX J. No emergency procedures shall be included that do not already exist in the applicable operator's -10 manual.

L.5.3.6 Limitations <limitsect>.

The limitations section shall be developed as follows:

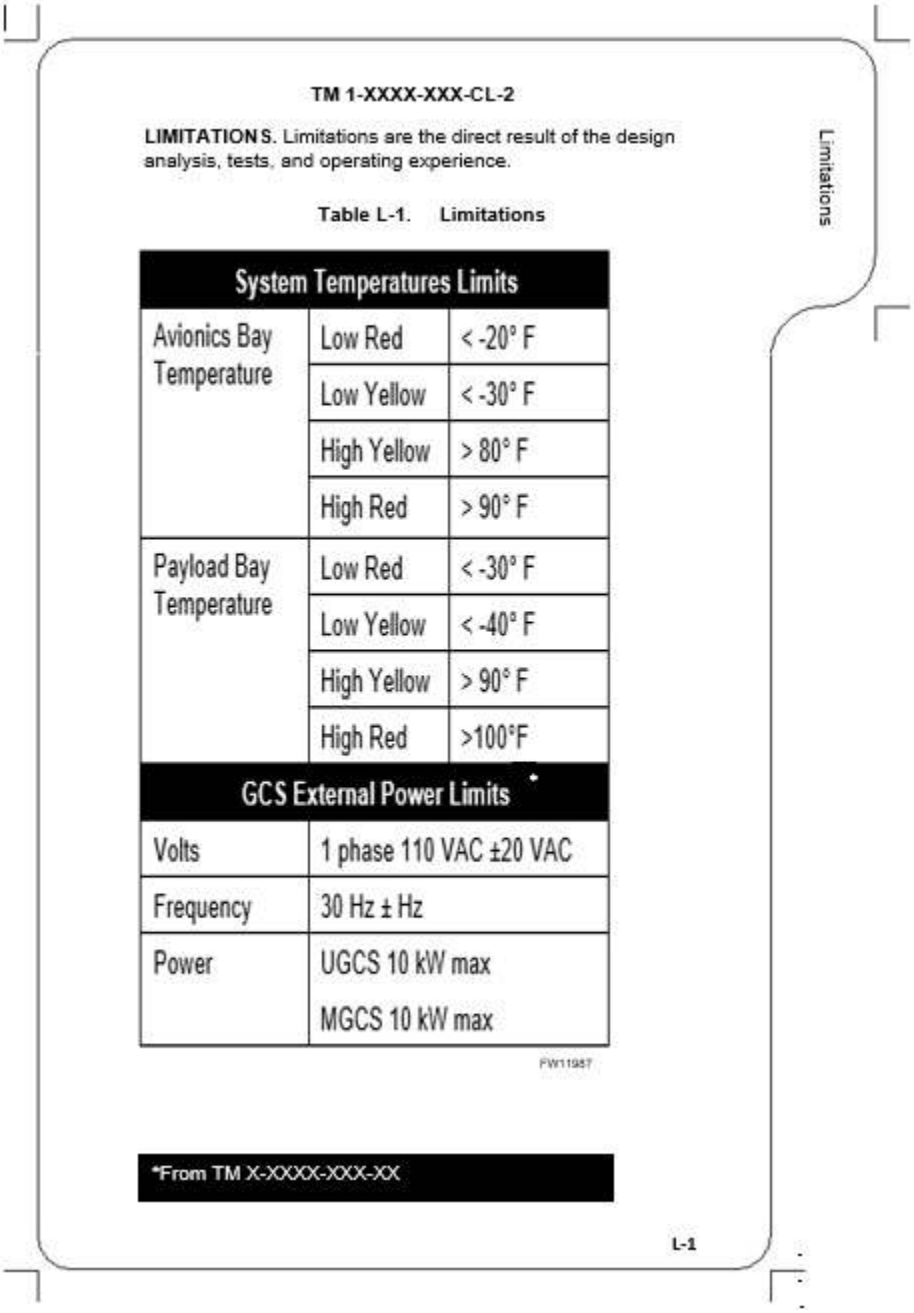
a. Limitations shall be presented in tabular format and derived from the applicable operator or maintenance manuals.

b. When limitations are derived from sources other than the applicable operator's manuals, a note shall be included indicating the source.

c. Limits headers shall have black background with white text (FIGURE L-1).

d. Limitations for model/software version shall be limited to a single page, front and back.

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TM 1-XXXX-XXX-CL-2

LIMITATIONS. Limitations are the direct result of the design analysis, tests, and operating experience.

Limitations

Table L-1. Limitations

System Temperatures Limits		
Avionics Bay Temperature	Low Red	< -20° F
	Low Yellow	< -30° F
	High Yellow	> 80° F
	High Red	> 90° F
Payload Bay Temperature	Low Red	< -30° F
	Low Yellow	< -40° F
	High Yellow	> 90° F
	High Red	>100°F
GCS External Power Limits		
Volts	1 phase 110 VAC ±20 VAC	
Frequency	30 Hz ± Hz	
Power	UGCS 10 kW max MGCS 10 kW max	

FW11987

*From TM X-XXXX-XXX-XX

L-1

FIGURE L-1. Limitations example.

L.5.3.7 Rear matter <rear>.

Rear matter shall consist of the requirements listed in APPENDIX A, TABLE A-III, TABLE A-IV, and TABLE A-V of this standard and shall follow the requirements of 5.2.3 for developing it. Fixed Wing -CL manuals shall also include the requirements of L.5.3.7.1.

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L.5.3.7.1 Annunciator panel <annunciator>.

The rear matter of Fixed Wing -CL manuals shall contain a work package that has graphics of the warning annunciator panel and/or caution annunciator panel. As determined by the acquiring activity, RW and UAS may develop annunciator work packages.

L.5.4 Rotary wing emergency procedures.

Rotary Wing emergency procedures shall be developed in a flight reference card format. These cards are designed to differentiate between the Warning, Caution, and Advisory messages displayed to the flight crew. Content shall be derived from the applicable operator's manual. The cards are sorted in sections, and each section is developed with an introduction followed by applicable emergency tasks as detailed in L.5.4.1 through L.5.4.4.4.

L.5.4.1 Intro section <intro>.

An index of procedures shall be developed for every emergency procedures section. As determined by the acquiring activity, the intro section can contain a brief explanatory paragraph prior to the index of procedures. Refer to FIGURE L-2.

a. In the warning section, the index of procedures shall be arranged by the associated cockpit indications in priority order as determined by the acquiring activity.

b. The warning section shall have an additional index for fires and emergencies without W/C/A indications.

c. In the caution, advisory, and mission equipment sections, the index of procedures shall be arranged alphabetically by the associated cockpit indications.

d. Index entries to emergency procedures that contain immediate action steps shall be bolded within the index.

e. Each entry within the index shall reference to a single card. Spacing between index entries should be a major consideration to accommodate discernible links when an Enhanced Version (EV) version will also be developed.

f. Entries within the index that do not contain an associated card shall have a distinct symbol indicating as such, and the symbol shall be defined in the checklist "About this Manual".

The sections of the emergency procedures shall be used to differentiate any warnings, cautions, or advisory messages displayed to the flight crew. Emergency procedures for mission equipment shall be developed in the mission equipment section when applicable.

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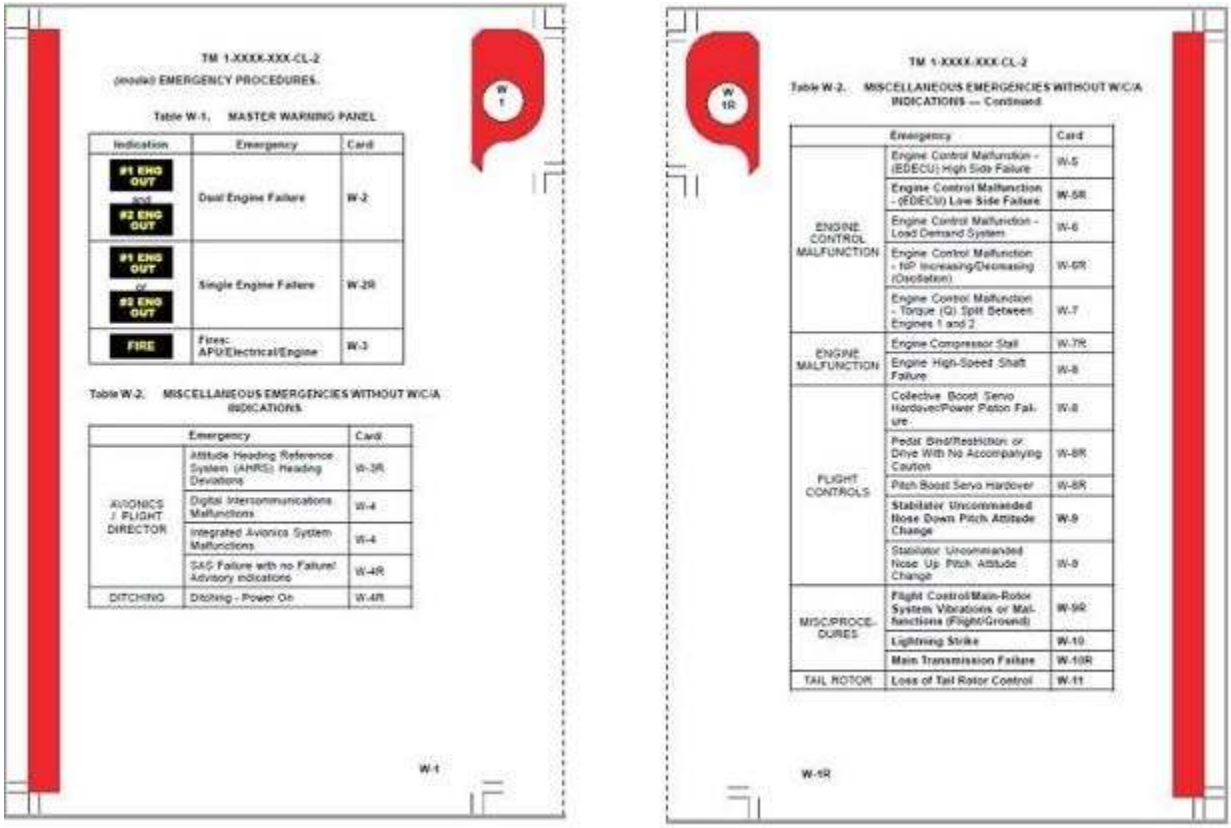


FIGURE L-2. Example of section index.

L.5.4.2 Emergency tasks <emertsk>.

Emergency procedure tasks shall be developed as found in the applicable Operator's manual, -10 in accordance with APPENDIX J.

L.5.4.3 Example emergencies.

Emergencies are defined as critical situations that almost always involve failures that require immediate and rapid response. Examples are:

a. Total powerplant failure.

b. Fire or smoke.

c. Rapid cabin decompression.

L.5.4.4 Sections and section indexes.

The emergency procedures of the checklist shall be developed into the following sections, as applicable:

a. Warnings.

b. Cautions.

c. Advisories.

d. Mission Equipment.

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e. Limitations.

L.5.4.4.1 Warnings <warnsect>.

The warning section shall contain the following types of procedures:

a. Messages displayed to the flight crew as warnings as defined in the Operator's manual.

b. All fire procedures.

c. All emergency procedures that do not have associated W/C/A messages, excluding mission equipment.

L.5.4.4.2 Cautions <cautsect>.

The caution section shall include all procedures from messages displayed to the flight crew as cautions as defined in the Operator's manual. The acquiring activity may use the section index to define cautions that may appear to the flight crew, but do not have an associated emergency procedure.

L.5.4.4.3 Advisories <advssect>.

The advisories section shall include all procedures from messages displayed to the flight crew as advisories or annunciations as defined in the Operator's manual. The acquiring activity may use the section index to define advisories that may appear to the flight crew but do not have an associated emergency procedure.

L.5.4.4.4 Mission equipment <missnsect>.

The mission equipment section shall include all emergency procedures for mission equipment as defined in the operator's manual.

L.5.5 Fixed wing and UAS emergency procedures.

FIXED WING: Emergency procedures shall be divided into groups of colored tabs: Emergency Procedures (RED tabs), Abnormal Procedures (YELLOW tabs), Advisory Procedures (GREEN tabs), and Mission Equipment Procedures (CYAN tabs). These groupings will be further divided into sections as found in Appendix A. These sections will be divided into work packages and tasks written in checklist format.

UAS: Emergency procedures shall be divided into groups of colored tabs: Warning Procedures (RED tabs), Caution Procedures (YELLOW tabs), Advisory Procedures (GREEN tabs), and Mission Equipment Procedures (CYAN tabs) based on urgency and messages displayed to the flight crew. These groupings will be further divided into tabbed sections as directed by the acquiring activity. These sections will be divided into work packages and tasks written in checklist format. Tabbed sections shall be titled. A short title shall appear on the tab. Each tabbed section shall contain an index of the procedures found within that section on the tabbed page. Each section is developed with an introduction followed by applicable emergency tasks as detailed in L.5.5.1 through L.5.5.4.6.

L.5.5.1 Section divisions.

An index of procedures shall be developed for every emergency procedures section. Refer to FIGURE L-2. Fixed Wing shall follow the further requirements of L.5.5.4.1 and UAS shall follow the further requirements of L.5.5.4.3. All sections shall be divided into work packages **<emergwp>**.

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L.5.5.2 Emergency tasks <fwuas_emertsk>.

Emergency procedure tasks shall be developed as found in the applicable Operator's manual, -10 in accordance with APPENDIX J.

L.5.5.3 Example emergencies.

Emergencies are defined as critical situations that almost always involve failures that require immediate and rapid response. Examples are:

a. Total powerplant failure.

b. Fire or smoke.

c. Rapid cabin decompression.

L.5.5.4 Sections and section indexes.

The emergency procedure section of the checklist shall be divided into the following sections:

a. Emergencies.

b. Abnormal/Caution Procedures.

c. Advisories.

d. Mission Equipment.

e. Limitations.

L.5.5.4.1 Fixed wing emergency section <warnsect>.

Emergency procedures shall be divided into functional group sections **<warnsect>** in accordance with APPENDIX A. These procedures shall clearly and precisely describe the recommended procedures for coping with various types of emergencies or critical situations. Procedures for handling malfunctions or other abnormalities that are not of emergency nature, or involve a potential emergency that may be deferred, shall be found in the Abnormal Section of the checklist. The difference between an "emergency" and "abnormal" situation may be dependent on the circumstances of flight, i.e., night or Instrument Flight Rules (IFR) versus day Visual Flight Rules (VFR), the presence of icing conditions, or the occurrence of multiple malfunctions.

L.5.5.4.2 Fixed wing abnormal/caution section <abnormsect>.

Abnormal procedures shall be divided into functional group sections **<abnormsect>** in accordance with APPENDIX A. This section shall clearly and precisely describe the recommended procedures for handling malfunctions of equipment, or other abnormalities, that are not of an emergency nature or involve a potential emergency that may be deferred. Examples are:

a. A landing gear that does not respond properly to normal gear switch actions when there are no other malfunctions in the powerplant or other systems.

b. A gear up emergency landing, if necessary, can be deferred until after all other methods of lowering the gear have been unsuccessfully tried and a suitable landing area has been selected.

c. Failure of some portion of the electrical system in day VFR conditions.

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L.5.5.4.3 UAS emergency section <warnsect>.

The warning section **<warnsect>** shall contain the following types of procedures:

a. Messages displayed to the flight crew as warnings as defined in the Operator's manual.

b. All fire procedures.

c. All emergency procedures that do not have associated W/C/A messages, excluding mission equipment.

L.5.5.4.4 UAS caution section <cautsect>.

The caution section shall include all procedures from messages displayed to the flight crew as cautions as defined in the Operator's manual.

L.5.5.4.5 Advisories <advssect>.

The advisories section shall include all procedures from messages displayed to the flight crew as advisories or annunciations as defined in the Operator's manual. The acquiring activity may use the section index to define advisories that may appear to the flight crew but do not have an associated emergency procedure.

L.5.5.4.6 Mission equipment <missnsect>.

The mission equipment section shall include all emergency procedures for mission equipment as defined in the operator's manual.

L.6 Alternate checklist <alternatecl>.

The acquiring activity shall have the option to specify that a one or two sheet alternate operator's checklist (FIGURE L-3 and FIGURE L-4) be prepared instead of the standard operator's checklist following the outline indicated below:

a. Front matter.

b. General information and scope.

c. Normal procedures.

d. Authentication Block.

e. Emergency procedures.

f. Other Procedures, if applicable.

g. Performance Data, if applicable.

L.6.1 Front matter <alternatefront>.

The alternate operator's checklist front matter shall contain the TM title and additional subtitles, to include nomenclature, model number, national stock number and/or part number, and end item code; notices; and the publication date.

L.6.1.1 Notices.

Following the TM title and additional required subtitles, the following notices shall be provided in the order listed below, as applicable. Refer to MIL-STD-40051 for explanations of these items.

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a. CUI designation indicator box **<cui_set>** (as applicable).

b. Export control notice **<export>** (as applicable).

c. Distribution statement **<dist>**.

d. Destruction notice **<destr>** (as applicable).

e. Supersedure notice **<super>** (as applicable).

L.6.2 General information and scope <geninfo>.

The general information shall contain a usage statement and any information aiding to the use of the TM to include symbols used within the TM.

L.6.3 Normal procedures <normal>.

A condensed version of the normal procedures or crew duties portion of the applicable operator's manual (-10) shall be developed according to APPENDIX I.

L.6.4 Emergency procedures <emergency>.

A condensed version of the emergency procedures or crew duties portion of the applicable operator's manual (-10) shall be developed according to APPENDIX J.

L.6.5 Other procedures <otherprocs>.

As determined by the acquiring activity, other information and procedures may be included in the alternate checklist. See FIGURE L-3 and FIGURE L-4 for examples of additional information that can be included.

L.6.6 Performance data <perfdata>.

As determined by the acquiring activity, performance data may be included in the alternate checklist as long as the data fits into the 2 page (1 sheet front and back) format of the alternate checklist.

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***TM 1-1520-228-CL**

OPERATOR'S AND CREWMEMBER'S CHECKLIST
OH-58A/C HELICOPTER
10 May 2006

DISTRIBUTION STATEMENTS A: Approved for public release; distribution is unlimited.

***** This manual supersedes TM 1-1520-228-CL, 29 July 2005.

HELICOPTER AND SYSTEMS

BEFORE EXTERIOR CHECK
1. Publications – Check.
2. Covers, locking devices, tiedowns (except main rotor), and grounding cables – Removed and secured.
3. Ignition Switch – On.
4. Cockpit – Check.

EXTERIOR CHECK
1. Cabin interior – Check.
2. Fuselage (Area 1) – Check.
3. Tailboom (Area 2) – Check, tiedown removed.
4. Main rotor blade (Area 2) – Check, tiedown removed.
5. Tail rotor gear box – Check.
6. Tail rotor – Check.
7. Tailboom (Area 3) – Check.
8. Fuselage (Area 4) – Check.
9. Main rotor blade (Area 4) – Check.
10. Fuselage top – Check.
11. Main rotor system – Check.
12. Fuselage (Area 5) – Check.
13. Cockpit – Check.
14. Fuselage (Area 6) – Check.
15. Crew and passenger briefing – Complete.

BEFORE STARTING ENGINES
1. Shoulder harness lock(s) – Check.
2. Overhead switches and circuit breakers – Set.
3. GPU – Connect for GPU – start.
4. Avionics – Off and set.
5. Instrument panel instruments & switches – Check and set.
6. Flight controls – Check and set.
7. Throttle – Check. Move to open, then to idle stop; press idle release and close.

STARTING ENGINE
1. Fireguard – Post if available.
2. Rotor Blades – Check clear and untied.
3. Engine – Start.
4. GPU – Disconnect; then BAT switch – BAT, if required.
5. N2 – Stabilized.
6. THROTTLE ADJUST – 70 percent N1.
7. GEN switch – GEN.
8. DC amps – check 60 or less.
9. INV switch – INV.
10. Avionics – On.

ENGINE RUNUP
1. Engine and transmission instruments – Check.
2. Throttle – Slowly increase to open. Set N2 to 100 percent.
3. HEAT and DEICE systems – Check if use is anticipated; then set as required.
4. Avionics – Check as required.
5. Flight Instruments – Check and set.
6. Doors, armor side panels, and seat belts – Secure.
7. Health Indicator Test (HIT) – Check.
8. Deceleration check – Perform if required.

BEFORE TAKEOFF
1. N2 – 100 percent.
2. Systems – Check.
3. Crew, passengers, mission equipment, and seat belts – Check.
4. Avionics – As required.

HOVER CHECK
1. Flight controls – Check.
2. Engine and transmission instruments – Check.
3. Flight instruments – Check as required.
4. Power – Check.

BEFORE LANDING
1. Crew, passenger, and mission equipment – Check.
2. Landing light – As required.

ENGINE SHUTDOWN
1. Throttle – Engine idle: stabilize TOT for two minutes.
2. FORCE TRIM switch – FORCE TRIM.
3. FUEL BOOST switch – OFF.
4. NGV/LDG LTS – OFF.
5. Control frictions – On.
6. Avionics – OFF.
7. Overhead switches and circuit breakers – Set.
8. Battery charge – Check.
9. Throttle – Close.
10. Overhead switches – OFF as required.
11. Ignition switches – OFF (Keys as required).
O12. Doors – Close immediately after exiting.

BEFORE LEAVING THE HELICOPTER
1. Main rotor blades – Tie down as required.
2. Walk-around – Complete.
3. DA Form 2408-12 and 2408-13-1 – Complete.
4. Secure helicopter – As required.

THRU-FLIGHT CHECKLIST

PREFLIGHT
1. Covers, locking devices, tiedowns (except main rotor), and grounding cables – Removed and secured.
2. Ignition switch – On.
O 3. Cargo/loose equipment – Check.
O 4. Auxiliary fuel cell – Check.
5. Passenger and seat belts – Check.
6. Hydraulic reservoirs/servos and flight controls – Check.
7. Transmission compartment – Check.
8. Fuel – Check quantity.
9. Main rotor blade – Check, tiedown removed.
10. Tail rotor gearbox – Check.
11. Tail rotor – Check.
12. Engine Oil Level – Check.
13. Main rotor system – Check condition.
14. Crew and passenger briefing – Complete.

BEFORE STARTING ENGINE
1. ANTI COLL LTS switch – As required.
2. POS LTS switch – As required.
3. BAT switch – As required.
4. GPU – Connect for GPU start.
5. Flight control – Check and set.
6. Throttle – Check. Move to open, then to idle stop; press idle release and close.

STARTING ENGINE
1. Fireguard – Post if available.
2. Rotor Blades – Check clear and untied.
3. Engine – Start.
4. GPU – Disconnect; then BAT switch – BAT, if required.
5. N2 – Stabilized.
6. THROTTLE ADJUST – 70 percent N1.
7. GEN switch – GEN.
8. DC amps – Check 60 or less.
9. INV switch – INV.
10. Avionics – On.

ENGINE RUNUP
1. Engine and transmission instruments – Check.
2. Throttle – Slowly increase to open. Set N2 to 100 percent.
3. Systems – Check.
4. Doors, armor side panels; and seat belts – Secure.

BEFORE TAKEOFF
1. N2 – 100 percent.
2. Systems – Check.
3. Crew, passengers, mission equipment, and seat belts – Check.
4. Avionics – As required.

By Order of the Secretary of the Army:

PETER J. SCHOOMAKER
General, United States Army
Chief of Staff

Official:

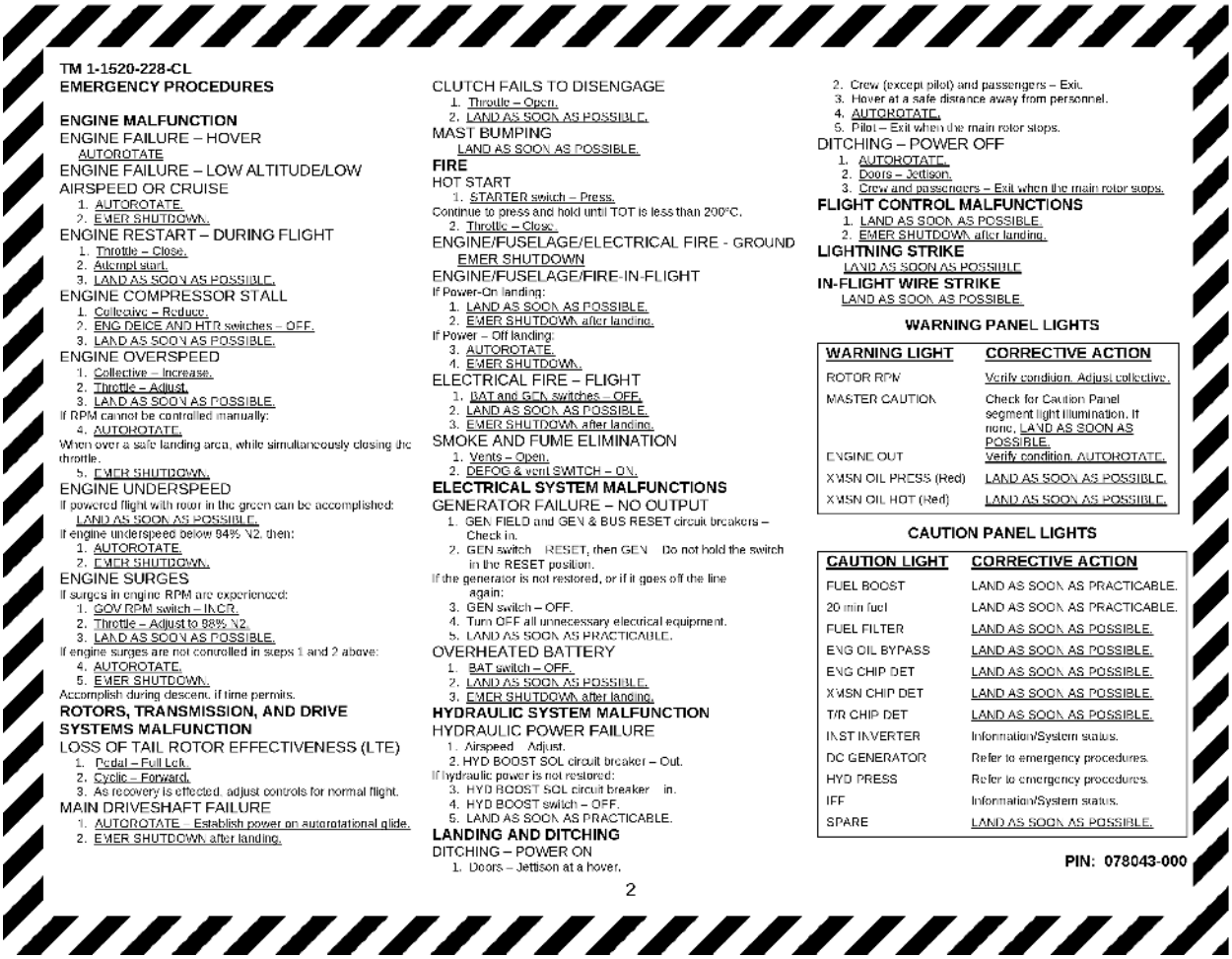
JOYCE E MORROW
Administrative Assistance to the Secretary of the Army
0612303

To be distributed in accordance with initial Distribution Number (IDN) 310239, requirements for TM 1-1520-228-CL.

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FIGURE L-3. Example 1 of an alternate operator's and crewmember's checklist. (This example has been reduced to fit on the page.).

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TM 1-1520-228-CL
EMERGENCY PROCEDURES

ENGINE MALFUNCTION
ENGINE FAILURE – HOVER
AUTOROTATE
ENGINE FAILURE – LOW ALTITUDE/LOW AIRSPEED OR CRUISE
1. AUTOROTATE.
2. EMER SHUTDOWN.
ENGINE RESTART – DURING FLIGHT
1. Throttle – Close.
2. Attempt start.
3. LAND AS SOON AS POSSIBLE.
ENGINE COMPRESSOR STALL
1. Collective – Reduce.
2. ENG DEICE AND HTR switches – OFF.
3. LAND AS SOON AS POSSIBLE.
ENGINE OVERSPEED
1. Collective – Increase.
2. Throttle – Adjust.
3. LAND AS SOON AS POSSIBLE.
If RPM cannot be controlled manually:
4. AUTOROTATE.
When over a safe landing area, while simultaneously closing the throttle.
5. EMER SHUTDOWN.
ENGINE UNDERSPEED
If powered flight with rotor in the green can be accomplished:
LAND AS SOON AS POSSIBLE.
If engine underspeed below 94% N2, then:
1. AUTOROTATE.
2. EMER SHUTDOWN.
ENGINE SURGES
If surges in engine RPM are experienced:
1. GOV RPM switch – INCR.
2. Throttle – Adjust to 88% N2.
3. LAND AS SOON AS POSSIBLE.
If engine surges are not controlled in steps 1 and 2 above:
4. AUTOROTATE.
5. EMER SHUTDOWN.
Accomplish during descent, if time permits.
ROTORS, TRANSMISSION, AND DRIVE SYSTEMS MALFUNCTION
LOSS OF TAIL ROTOR EFFECTIVENESS (LTE)
1. Pedal – Full Left.
2. Cyclic – Forward.
3. As recovery is effected, adjust controls for normal flight.
MAIN DRIVESHAFT FAILURE
1. AUTOROTATE. Establish power on autorotational glide.
2. EMER SHUTDOWN after landing.

CLUTCH FAILS TO DISENGAGE
1. Throttle – Open.
2. LAND AS SOON AS POSSIBLE.
MAST BUMPING
LAND AS SOON AS POSSIBLE.
FIRE
HOT START
1. STARTER switch – Press.
Continue to press and hold until TOT is less than 200°C.
2. Throttle – Close.
ENGINE/FUSELAGE/ELECTRICAL FIRE - GROUND
EMER SHUTDOWN
ENGINE/FUSELAGE/FIRE-IN-FLIGHT
If Power-On landing:
1. LAND AS SOON AS POSSIBLE.
2. EMER SHUTDOWN after landing.
If Power – Off landing:
3. AUTOROTATE.
4. EMER SHUTDOWN.
ELECTRICAL FIRE – FLIGHT
1. BAT and GEN switches – OFF.
2. LAND AS SOON AS POSSIBLE.
3. EMER SHUTDOWN after landing.
SMOKE AND FUME ELIMINATION
1. Vents – Open.
2. DEFOG & vent SWITCH – ON.
ELECTRICAL SYSTEM MALFUNCTIONS
GENERATOR FAILURE – NO OUTPUT
1. GEN FIELD and GEN & BUS RESET circuit breakers – Check in.
2. GEN switch – RESET, then GEN – Do not hold the switch in the RESET position.
If the generator is not restored, or if it goes off the line again:
3. GEN switch – OFF.
4. Turn OFF all unnecessary electrical equipment.
5. LAND AS SOON AS PRACTICABLE.
OVERHEATED BATTERY
1. BAT switch – OFF.
2. LAND AS SOON AS POSSIBLE.
3. EMER SHUTDOWN after landing.
HYDRAULIC SYSTEM MALFUNCTION
HYDRAULIC POWER FAILURE
1. Airspeed – Adjust.
2. HYD BOOST SOL circuit breaker – Out.
If hydraulic power is not restored:
3. HYD BOOST SOL circuit breaker – In.
4. HYD BOOST switch – OFF.
5. LAND AS SOON AS PRACTICABLE.
LANDING AND DITCHING
DITCHING – POWER ON
1. Doors – Jettison at a hover.
2. Crew (except pilot) and passengers – Exit.
3. Hover at a safe distance away from personnel.
4. AUTOROTATE.
5. Pilot – Exit when the main rotor stops.
DITCHING – POWER OFF
1. AUTOROTATE.
2. Doors – Jettison.
3. Crew and passengers – Exit when the main rotor stops.
FLIGHT CONTROL MALFUNCTIONS
1. LAND AS SOON AS POSSIBLE.
2. EMER SHUTDOWN after landing.
LIGHTNING STRIKE
LAND AS SOON AS POSSIBLE
IN-FLIGHT WIRE STRIKE
LAND AS SOON AS POSSIBLE.

WARNING PANEL LIGHTS

WARNING LIGHT	CORRECTIVE ACTION
ROTOR RPM	Verify condition. Adjust collective.
MASTER CAUTION	Check for Caution Panel segment light illumination. If none, LAND AS SOON AS POSSIBLE.
ENGINE OUT	Verify condition. AUTOROTATE.
XMSN OIL PRESS (Red)	LAND AS SOON AS POSSIBLE.
XMSN OIL HOT (Red)	LAND AS SOON AS POSSIBLE.

CAUTION PANEL LIGHTS

CAUTION LIGHT	CORRECTIVE ACTION
FUEL BOOST	LAND AS SOON AS PRACTICABLE.
20 min fuel	LAND AS SOON AS PRACTICABLE.
FUEL FILTER	LAND AS SOON AS POSSIBLE.
ENG OIL BYPASS	LAND AS SOON AS POSSIBLE.
ENG CHIP DET	LAND AS SOON AS POSSIBLE.
XMSN CHIP DET	LAND AS SOON AS POSSIBLE.
T/R CHIP DET	LAND AS SOON AS POSSIBLE.
INST INVERTER	Information/System status.
DC GENERATOR	Refer to emergency procedures.
HYD PRESS	Refer to emergency procedures.
IFF	Information/System status.
SPARE	LAND AS SOON AS POSSIBLE.

PIN: 078043-000

2

FIGURE L-4. Example 2 of an alternate operator's and crewmember's checklist. (This example has been reduced to show bleed-borders.).

L.7 Notes.

The notes in Section 6 apply to this appendix.

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APPENDIX M

CONTENT REQUIREMENTS FOR MAINTENANCE TEST FLIGHT MANUALS (-MTF)

M.1 Scope.

This appendix establishes the technical content requirements for the preparation of operator maintenance test flight manuals for aviation weapon systems. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

M.2 Applicable publications.

The applicable publications in Section 2 apply to this appendix.

M.3 Definitions.

The definitions in Section 3 apply to this appendix.

M.4 General requirements.

A supplemental manual containing a maintenance test flight checklist, troubleshooting guides, and a simplified version of the detailed procedures, as applicable, shall be developed in conjunction with the -10 operator's manual to serve as the operator's and crewmember's maintenance test flight manual.

M.4.1 Preparation of digital data for electronic delivery.

Data prepared and delivered digitally in accordance with this standard shall be XML tagged using the current Army DTD. Data shall be formatted for presentation using the Army stylesheets. Refer to M.4.2 for information on obtaining or accessing the Army DTD and stylesheets. XML tags used in the DTD are noted throughout the text of this standard in bracketed, bold characters (e.g., **<geninfo>**) as a convenience for the author and to ensure that the tags are used correctly when developing a document instance.

M.4.2 Use of the Document Type Definition (DTD)/stylesheet.

The DTD referenced in this appendix interprets the technical content and structure for the functional requirements contained in this standard. Development of the TMs shall be accomplished through the use of this standard, the Army DTD, and the Army stylesheets.

M.4.3 Content structure.

The examples provided herein are an accurate representation of the content structure requirements contained in this appendix and shall be followed to permit the effective use of the DTD.

M.4.4 Style and format.

This standard provides style and format requirements for the technical content requirements described in this appendix. These requirements are considered mandatory and are intended for compliance.

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M.4.5 Work package development.

Data developed in accordance with this appendix shall be divided into work packages. These work packages shall be stand alone and shall contain all information and references required to support the work package type.

M.4.6 Selective application and tailoring of content using Appendix A matrices.

This standard contains some requirements that may not be applicable to the preparation of all TMs. Selective application and tailoring of requirements contained in this standard are the responsibility of the acquiring activity and shall be accomplished using APPENDIX A. The applicability of some requirements is also designated by one of the following statements: unless specified otherwise by the acquiring activity; as specified by the acquiring activity; or when specified by the acquiring activity.

M.5 Detailed requirements.

A supplemental manual containing a maintenance test flight checklist, troubleshooting guides, and a simplified version of the detailed procedures shall be developed in conjunction with the -10 operator's manual to serve as the operator's and crewmember's maintenance test flight manual. The maintenance test flight manual shall be developed as detailed in M.5.1 through M.5.6.2. In addition to the draft TM, the acquiring activity may require submission of an aerodynamic report illustrating the derivation of the data entered on the charts included in the TM. The report should include an analysis leading to the establishment of lift and drag values used in the calculations, aircraft efficiency and compressibility correction factors, methods of computing power or thrust required and available, a discussion of duct loss and propeller efficiencies, and adequate references to appropriate wind tunnel or flight test data. Calculation methods need to be fully explained and a sample calculation given. The calculations should be presented in sufficient detail to permit ready review and check of conclusions and to enable additional calculations to be made.

M.5.1 Maintenance test flight manual contents.

MTF requirements for specific Army aircraft shall be provided.

a. This includes specific aircraft test weight and center of gravity limits required for completion of individual tests.

b. Criteria for performing MTFs shall be in accordance with TM 1-1500-328-23.

c. Requirements shall ensure a thorough inspection on the aircraft before flight during flight, and upon completion of the MTF.

d. Unless otherwise specified by the acquiring activity, checklist items shall include those that are contained in the applicable aircraft operator’s checklist plus those MTF checks peculiar to the aircraft in question.

An MTF manual shall be prepared in accordance with the following outline indicated below:

a. Front matter.

b. Maintenance test flight checklist.

c. Troubleshooting guides (as applicable).

d. Special/detailed procedures (as applicable).

e. Performance Charts (as applicable).

f. Forms.

g. Rear matter.

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M.5.2 Front matter <av_paperfrnt>.

The Front Matter shall be developed IAW 5.2.1.

M.5.3 MTF checklist <normsect>.

MTF Checklists shall be developed into sections, work packages, and tasks. MTF Checklist sections shall be developed to divide procedural content. The acquiring activity shall determine how content is presented within the sections.

a. Each section shall begin with a tabbed page.

b. The tabbed page shall contain a section title and index of the tasks found within the section.

c. A title shall be displayed on the tab of the section divider; the title may be abbreviated to fit within the tab as determined by the acquiring activity.

d. When only a single MTF Checklist section is used, it shall be titled "MTF Checklist" to include the short title appearing on the tab.

M.5.3.1 MTF checklist work packages <normalwp>.

MTF Checklist content shall be developed within work packages. Work packages are intended to create natural breaks in content to make publication updates efficient. Due to the abbreviated nature of the checklist, and to facilitate efficient use of page space, the acquiring activity may determine the content within each work package. The following are considerations to assist in work package development:

a. When an update is made to a work package, the entire work package shall be replaced.

b. Work package size for -MTF manuals shall not exceed 10 pages or 5 sheets.

c. A work package shall consist of, at minimum, one page front and back, regardless of content to include blank pages.

d. A work package title may be used.

M.5.3.2 MTF checklist tasks <normaltsk>.

MTF Checklist tasks shall be developed as found in the applicable Operator's manual, -10 in accordance with APPENDIX I, plus those MTF checks peculiar to the aircraft in question.

a. When the star symbol "⋆" is used to indicate additional details, the step shall contain a linked reference **<xref>** to the appropriate detail procedure found within the checklist.

b. References **<xref>** shall also be made to applicable figures, tables, and charts found within the checklist.

c. A task may have more than one procedure. All procedures within the task shall be related. When more than one procedure is present, the procedures shall contain titles.

M.5.4 Troubleshooting guides <troublesect>.

When required by the acquiring activity, troubleshooting guides shall be developed into sections, work packages, and tasks. Troubleshooting guides sections shall be developed to divide procedural content. The acquiring activity shall determine how content is presented within the sections.

a. Each section shall begin with a tabbed page.

b. The tabbed page shall contain a section title and index of the tasks found within the section.

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c. A title shall be displayed on the tab of the section divider; the title may be abbreviated to fit within the tab as determined by the acquiring activity.

d. When only a single MTF Checklist section is used, it shall be titled "Troubleshooting Guide" to include the short title appearing on the tab.

M.5.4.1 Troubleshooting guide work packages <troublewp>.

Troubleshooting guide content shall be developed within work packages. Work packages are intended to create natural breaks in content to make publication updates efficient. Due to the abbreviated nature of the -MTF, and to facilitate efficient use of page space, the acquiring activity may determine the content within each work package. The following are considerations to assist in work package development:

a. When an update is made to a work package, the entire work package shall be replaced.

b. Work package size for MTF manuals shall not exceed 10 pages or 5 sheets.

c. A work package shall consist of, at minimum, one page front and back, regardless of content to include blank pages.

d. A work package title may be used.

M.5.4.2 Troubleshooting guide tasks <troubletsk>.

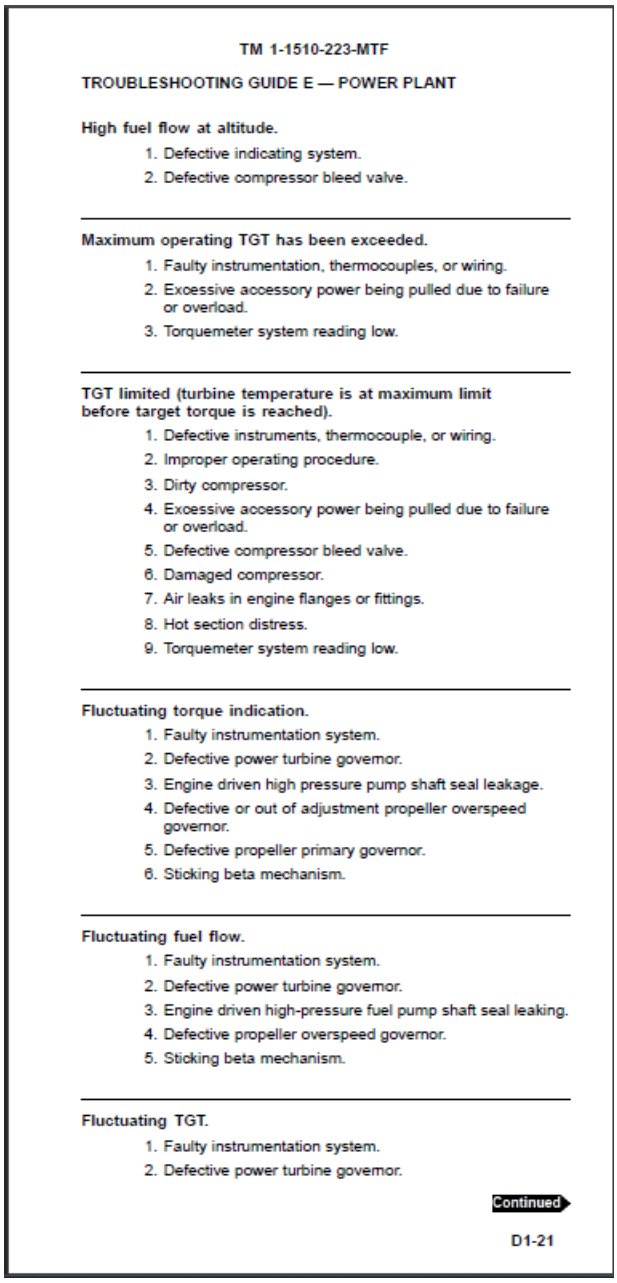
Refer to FIGURE M-1. Troubleshooting guide tasks shall be developed as follows:

a. When the star symbol "⋆" is used to indicate additional details, the step shall contain a linked reference **<xref>** to the appropriate detail procedure found within the checklist.

b. References **<xref>** shall also be made to applicable figures, tables, and charts found within the checklist.

c. A task may have more than one procedure. All procedures within the task shall be related. When more than one procedure is present, the procedures shall contain titles.

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TM 1-1510-223-MTF

TROUBLESHOOTING GUIDE E — POWER PLANT

High fuel flow at altitude.

1. Defective indicating system.
2. Defective compressor bleed valve.

Maximum operating TGT has been exceeded.

1. Faulty instrumentation, thermocouples, or wiring.
2. Excessive accessory power being pulled due to failure or overload.
3. Torquemeter system reading low.

TGT limited (turbine temperature is at maximum limit before target torque is reached).

1. Defective instruments, thermocouple, or wiring.
2. Improper operating procedure.
3. Dirty compressor.
4. Excessive accessory power being pulled due to failure or overload.
5. Defective compressor bleed valve.
6. Damaged compressor.
7. Air leaks in engine flanges or fittings.
8. Hot section distress.
9. Torquemeter system reading low.

Fluctuating torque indication.

1. Faulty instrumentation system.
2. Defective power turbine governor.
3. Engine driven high pressure pump shaft seal leakage.
4. Defective or out of adjustment propeller overspeed governor.
5. Defective propeller primary governor.
6. Sticking beta mechanism.

Fluctuating fuel flow.

1. Faulty instrumentation system.
2. Defective power turbine governor.
3. Engine driven high-pressure fuel pump shaft seal leaking.
4. Defective propeller overspeed governor.
5. Sticking beta mechanism.

Fluctuating TGT.

1. Faulty instrumentation system.
2. Defective power turbine governor.

Continued

D1-21

FIGURE M-1. Example MTF troubleshooting guide.

M.5.5 Special/detailed procedures <detailsect>.

Those special/detailed procedures that are referenced in Section M.5.3 shall be included and any additional special/detailed procedures shall be specified by the acquiring activity. Complete instructions for each procedure shall be listed. Examples of special/detailed procedures shall include rotor smoothing techniques, speed trim checks, engine conditioning, engine starting, etc.. Detailed procedures shall be developed into sections, work packages, and tasks. When required by the acquiring activity, one or more detailed sections shall be developed to divide procedural content. The acquiring activity shall determine how content is presented within the sections.

a. Each section shall begin with a tabbed page.

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b. The tabbed page shall contain a section title and index of the tasks found within the section.

c. A title shall be displayed on the tab of the section divider; the title may be abbreviated to fit within the tab as determined by the acquiring activity.

d. When only a single detailed section is used, it shall be titled “Detailed Procedures” to include the short title appearing on the tab.

M.5.5.1 Detailed procedure work packages <detailwp>.

Detailed procedure content shall be developed within work packages. Work packages are intended to create natural breaks in content to make publication updates efficient. Due to the abbreviated nature of the MTF, and to facilitate efficient use of page space, the acquiring activity may determine the content within each work package. The following are considerations to assist in work package development:

a. When an update is made to a work package, the entire work package shall be replaced.

b. Work package size for -MTF manuals shall not exceed 10 pages or 5 sheets.

c. A work package shall consist of, at minimum, one page front and back, regardless of content to include blank pages.

d. A work package title may be used.

M.5.5.1.1 Detailed procedure task <detailtsk>.

Detailed procedure tasks shall be developed as found in the applicable Operator’s manual, -10 in accordance with APPENDIX I.

a. When the detailed procedure is referenced from the MTF Checklist section, it shall include a return reference **<return>** to the applicable procedure found within the MTF Checklist.

b. If more than one return reference is required, it shall be clearly identified, for example:

- PILOT: RETURN TO: Page N3-1, Step 5.
- COPLIOT: RETURN TO: Page N4-1, Step 6.

c. References **<xref>** shall also be made to applicable figures, tables, and charts found within the Checklist.

d. A task may have more than one procedure. All procedures within the task shall be related. When more than one procedure is present, the procedures shall contain titles.

M.5.6 Charts and forms.

Charts and forms (FIGURE M-2) shall be prepared, as necessary, to help perform and record MTFs. Charts and forms shall be presented in sections and work packages.

M.5.6.1 Charts <perfdatasect>.

When required by the acquiring activity, one or more charts sections shall be developed. The acquiring activity shall determine how content is presented within the sections. Charts shall be prepared that shall include, but not be limited to, bleed band opening envelope, Turbine Entire Analysis Check (TEAC), and power adjusting. A list of required charts, including the contents, size, and format, shall be provided by the acquiring activity. The number of foldouts shall be kept to a minimum. Fold-up charts shall not be used. Placement and identification of foldouts are provided in Section 4.5.6.

a. Each section shall begin with a tabbed page.

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b. The tabbed page shall contain a section title and index of the charts found within the section.

c. A title shall be displayed on the tab of the section divider; the title may be abbreviated **<shorttitle>** to fit within the tab as determined by the acquiring activity.

d. When only a single chart section is used, it shall be titled “Charts” to include the short title appearing on the tab.

M.5.6.1.1 Charts work package <perfdatawp>.

Charts shall be developed as found in the applicable Operator’s manual, -10 in accordance with APPENDIX H.

a. Charts work packages shall be titled as applicable to the data.

b. Each work package shall contain a single figure **<figure>** or table **<table>**.

c. A brief introduction may be developed.

d. Tables and charts within this WP shall display within the standard page size. If greater detail is desired, fold out pages shall be utilized in the rear matter.

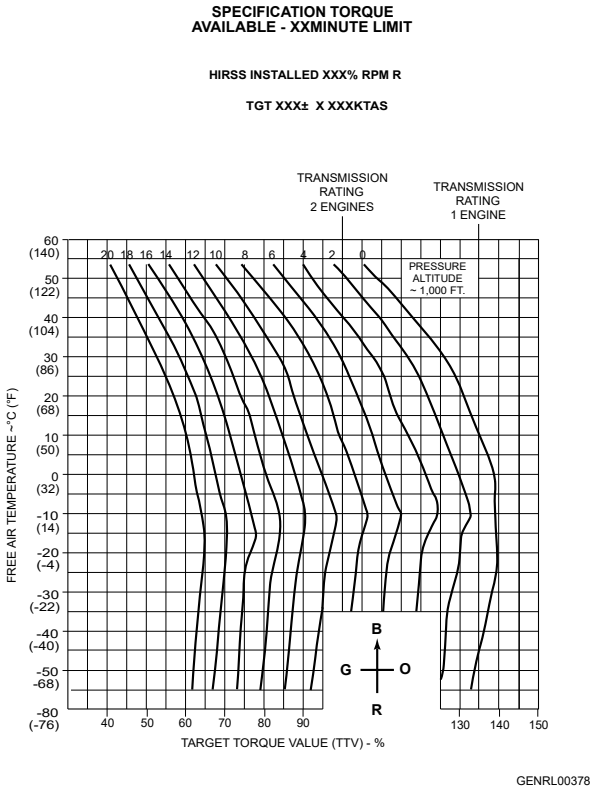


FIGURE M-2. Example of MTF chart.

M.5.6.2 Forms <perfdatasect>.

One or more forms sections shall be developed. The acquiring activity shall determine how content is presented within the sections.

a. Each section shall begin with a tabbed page.

b. The tabbed page shall contain a section title and index of the forms found within the section.

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c. A title shall be displayed on the tab of the section divider; the title may be abbreviated **<shorttitle>** to fit within the tab as determined by the acquiring activity.

d. When only a single form section is used, it shall be titled "Forms" to include the short title appearing on the tab.

M.5.6.2.1 Forms work package <perfdatawp>.

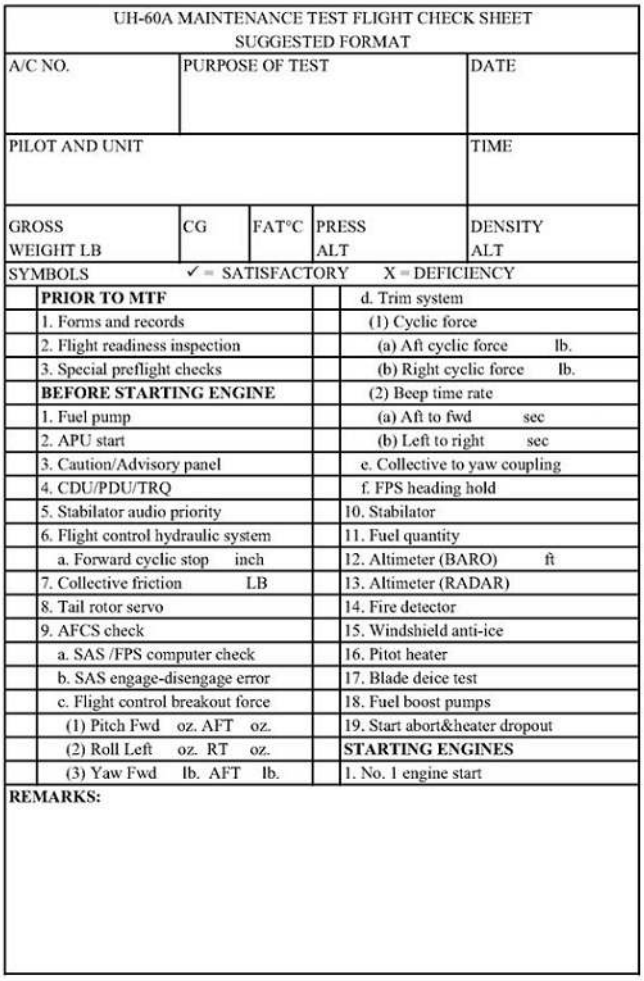
The forms shall be used to record readings, pressures, rpm, etc. obtained during MTFs. A checksheet, similar to the one in FIGURE M-3, shall be provided for use by the person(s) conducting the checks.

a. Forms work packages shall be titled as applicable to the data.

b. Each work package shall contain a single figure **<figure>** or table **<table>**.

c. A brief introduction may be developed.

d. Forms within this WP shall display within the standard page size.



UH-60A MAINTENANCE TEST FLIGHT CHECK SHEET
SUGGESTED FORMAT

A/C NO.	PURPOSE OF TEST	DATE

PILOT AND UNIT	TIME

GROSS WEIGHT LB	CG	FAT°C	PRESS ALT	DENSITY ALT

SYMBOLS ✓ = SATISFACTORY X = DEFICIENCY

	PRIOR TO MTF		d. Trim system
	1. Forms and records		(1) Cyclic force
	2. Flight readiness inspection		(a) Aft cyclic force lb.
	3. Special preflight checks		(b) Right cyclic force lb.
	BEFORE STARTING ENGINE		(2) Beep time rate
	1. Fuel pump		(a) Aft to fwd sec
	2. APU start		(b) Left to right sec
	3. Caution/Advisory panel		e. Collective to yaw coupling
	4. CDU/PDU/TRQ		f. FPS heading hold
	5. Stabilator audio priority		10. Stabilator
	6. Flight control hydraulic system		11. Fuel quantity
	a. Forward cyclic stop inch		12. Altimeter (BARO) ft
	7. Collective friction LB		13. Altimeter (RADAR)
	8. Tail rotor servo		14. Fire detector
	9. AFCS check		15. Windshield anti-ice
	a. SAS /FPS computer check		16. Pitot heater
	b. SAS engage-disengage error		17. Blade deice test
	c. Flight control breakout force		18. Fuel boost pumps
	(1) Pitch Fwd oz. AFT oz.		19. Start abort&heater dropout
	(2) Roll Left oz. RT oz.		**STARTING ENGINES**
	(3) Yaw Fwd lb. AFT lb.		1. No. 1 engine start

REMARKS:

FIGURE M-3. Example of MTF checksheet (reduced in size).

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M.5.7 Rear matter <rear>.

Rear matter shall consist of the requirements listed in APPENDIX A, TABLE A-VII of this standard and shall follow the requirements of 5.2.3 for developing it. Fixed Wing -MTF manuals shall also include the requirements of L.5.3.7.1.

M.5.8 Annunciator panel <annunciator>.

The rear matter of Fixed Wing -MTF manuals shall contain a work package that has graphics of the warning annunciator panel and/or caution annunciator panel. As determined by the acquiring activity, RW and UAS may develop annunciator work packages.

M.6 Notes.

The notes in Section 6 apply to this appendix.

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APPENDIX N

STYLE AND FORMAT REQUIREMENTS FOR ENHANCED VERSION (-EV) CHECKLIST OR MAINTENANCE TEST FLIGHT MANUAL

N.1 Scope.

This appendix provides guidance for the enhanced version (-EV) of the Operator's checklist (-CL) or maintenance test flight (-MTF) manual. The EV is intended to be used in the exact same manner an operator would use the printed version of the CL or MTF.

N.2 Requirements.

Note

Interactive Authoring and Display Software (IADS) is the required viewer for all Army electronic technical manuals in accordance with AR 25-30. This appendix shall be used as guidance until IADS is fully capable of rendering the Aviation CL or MTF enhanced version in the desired form.

The Enhanced Version (-EV) is defined as building additional function and interactivity into the ETM document. A Portable Document Format (PDF) is the only authorized deliverable for EV publishing. The EV of the operator's CL or MTF shall use the same look and feel of the printed version of the CL or MTF. All interactive functionality added for the EV will be verified hands-on by the user representative on a tablet device. The EV will utilize the exact same data as found in the CL or MTF and be published concurrently. The same source data (xml) shall be used to produce the EV and printed version.

N.2.1 Content and format.

Content and format shall match the authenticated printed product of the CL or MTF exactly, but in digital form. For example, page N2-1 shall have the exact same content in the EV and in the authenticated printed version. This shall ensure primacy of the printed version and reduce training needs of the operator.

N.2.2 Normal and detailed procedure static menu.

A static menu shall be developed to aid in navigation on a tablet device.

a. The static menu shall appear on every page, and be clearly distinguishable from the content.

b. The static menu shall appear on the right side of every page and right justified to the edge of the frame.

c. At minimum, the static menu shall consist of the section tab titles included in the printed version of the checklist.

d. Additional entries may be added as directed by the acquiring activity.

e. The static menu entries shall be a minimum of 1" below the emergency tabs and shall fit between the right side border and edge of the document.

Refer to FIGURE N-1.

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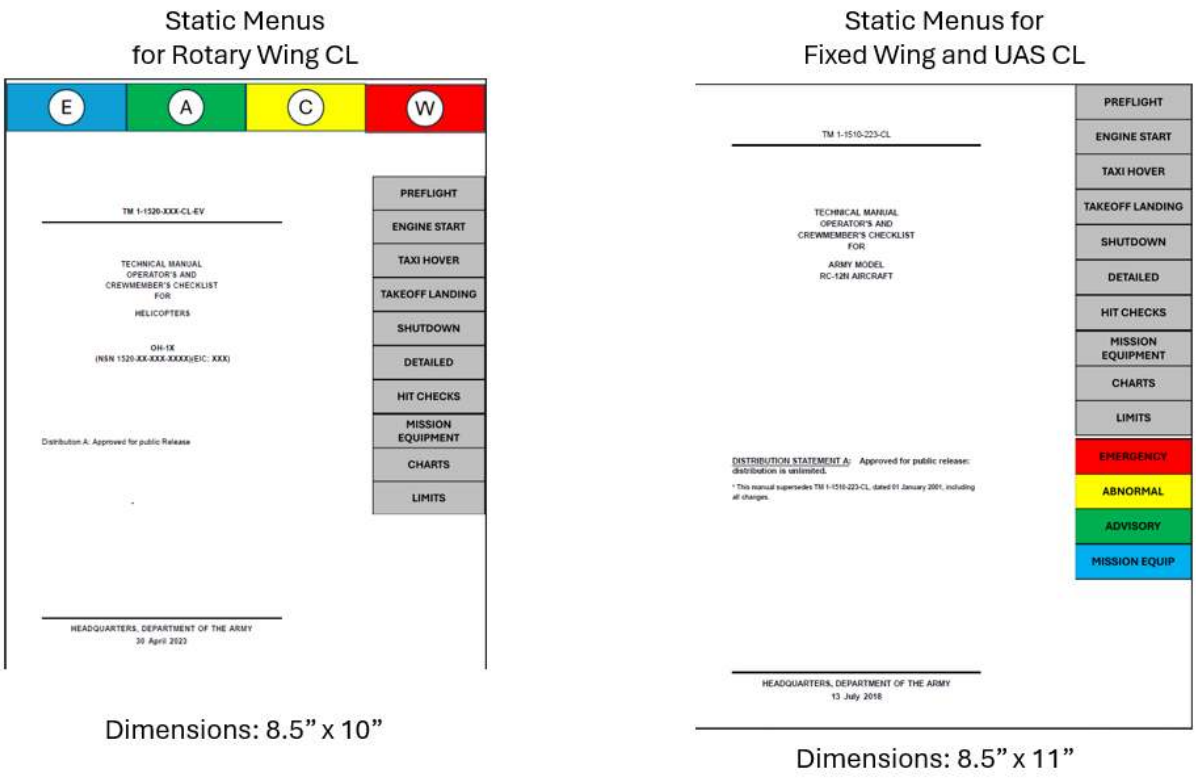


FIGURE N-1. Example of a static menu.

N.2.3 Emergency procedure static menu.

a. For Rotary Wing: Emergency procedure tabs representative of those used in the printed version, to include color, shall be displayed at the top of every page within the checklist. Each tab will link to the applicable first index page of the emergency section.

b. For Fixed Wing and UAS: Emergency and abnormal entries shall be placed in the static menu as found in the authenticated printed version.

Refer to FIGURE N-1.

N.2.4 Interactive referencing.

All references provided within the EV should take into consideration the operational environment and be easily accessible with gloved fingers. Care should be taken especially within index sections that links are spaced appropriately for accurate selection.

N.2.4.1 Display of interactive references.

All interactive links found within the pages of the checklist shall be distinguishable from the text by including a red box around the reference. Refer to FIGURE N-2.

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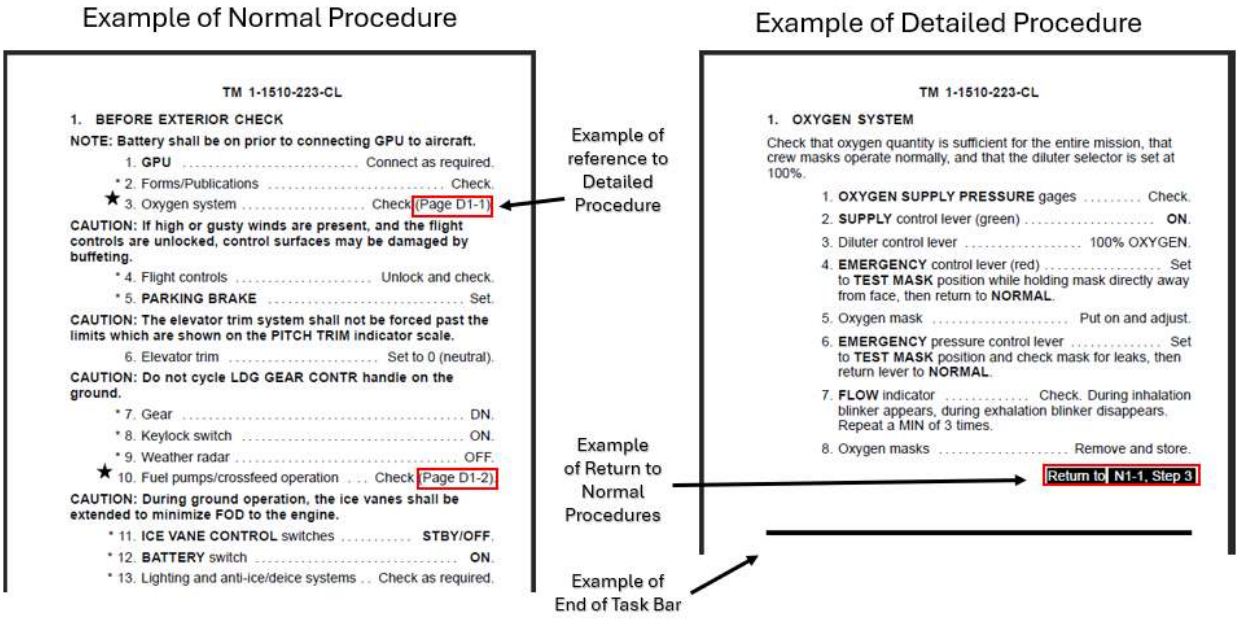


FIGURE N-2. Display of interactive references.

N.2.4.2 Additional interactivity.

When required by the acquiring activity, additional interactivity may be added to the enhanced version of the CL or MTF.

N.2.5 EV size.

The EV page size will be expanded from the printed version:

a. The page size will be increased vertically one inch at the top of the document to accommodate the emergency procedure tabs.

b. The page size will be increased horizontally two inches on the right side to accommodate the static menu.

Refer to FIGURE N-1.

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MIL-STD-3083(AV)

CONCLUDING MATERIAL

Custodians:
Army – AV

Preparing activity:
Army – AV
(Project TMSS-2025-014)

Review activities:
Army – CR, TM

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST at https://assist.dla.mil.

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